Solomonoff Function Induction:
Definitions, Versions and Basic Theorems

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Abstract

Solomonoff function induction (SFI) is Bayesian prediction of labels from input–output examples under a universal prior over random functions: a random program, reading random bits shared by all inputs, labels every input. A universal function machine exists; its law dominates the law of every function machine and does not depend, up to a constant factor, on the universal machine or on computable encodings. The shortest consistent program, the program mixture, the mixture of product laws, SFI and the universal weak semimeasure agree up to $K(\operatorname{dom} D)$ bits, but no two of the first three agree up to a constant: there is no coding theorem for function programs. On every consistent data sequence, SFI's cumulative log loss exceeds that of every computable random labeller by at most the labeller's complexity plus a constant; with i.i.d. inputs from any distribution and computable labels, the averaged predictor has risk $O(K/n)$. SFI cannot model fresh label noise on repeated inputs; a per-query mixture can. SFI cannot use the order of the data: on a random cyclic permutation revealed along its cycle it pays about $(d - \log_2 e)2^d$ bits in the semimeasure convention, while Solomonoff induction on the interleaved sequence pays $O(1)$ in expectation. Hypotheses that see the list of previous inputs give the universal chronological semimeasure. A dominance theorem for computable reveal orders holds in its intended reading, though an earlier proof sketch fails. Further sections treat Sam Eisenstat's coin example, out-of-distribution behaviour, computable versions and neural-network Bayes. A companion paper treats steeper simplicity priors.

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1 Introduction

In supervised learning a learner sees input–output pairs $(x_1, y_1), (x_2, y_2), \ldots$ and predicts the output at a new input. For predicting sequences, Solomonoff induction (SI) is the standard ideal reference: a Bayes mixture over all computable sources whose cumulative log loss exceeds that of each computable source by at most a constant that depends on the source (Li and Vitányi, 2008). Hänni (2026b) proposes the analogous object for supervised learning, *Solomonoff function induction* (SFI): "you think of the input-output pairs as being generated by picking a random program from this prior and generating using it, and so you update on seen pairs as you should". This paper defines SFI carefully, proves its basic properties, checks the claims about it made in those notes, and works out examples.

Why SFI. Supervised learning is central to machine learning, and SFI is a candidate for its ideal form. It is the Bayes predictor for the prior "a random lower-semicomputable labeller". On every sequence of pairs that some function could produce, its cumulative log loss exceeds that of every computable random labeller by at most that labeller's complexity plus a constant (Theorem 4.1). Since SFI never predicts the inputs, this holds for every input distribution, including adversarial and non-computable ones; with i.i.d. inputs it gives the Occam rate $O(K/n)$ for the averaged predictor, without a union bound (Theorem 4.5). The analogy has limits, which we make precise: the guarantees are for log loss and for averaged predictors; a random function cannot produce fresh label noise on a repeated input; SFI is not computable; and SFI cannot use the order in which the data arrive (Remark 4.16).

Relation to Solomonoff induction. Up to constant factors, SI is SFI on the inputs $0, 1, \ldots, n-1$ (Proposition 2.23). In general, though, SFI sees the data as a set: each hypothesis answers each query without seeing the other inputs, and SFI's cumulative loss in the semimeasure convention does not depend on the order of the data (Proposition 2.16). SI applied to the interleaved sequence $x_1 y_1 x_2 y_2 \cdots$ sees the order and can do much better (Sections 5 and 6). Giving SFI's hypotheses the list of previous inputs closes this gap: the result, $\mathrm{SFI}^{\mathrm{hist}}$, is the universal chronological semimeasure (Corollary 5.18). With inputs as actions and labels as percepts this is the kind of object that underlies Hutter's AIXI (Hutter, 2000, 2005); we have not checked the identification against Hutter's texts.

Relation to neural-network Bayes. Bayesian learning with a neural network is function-space Bayes with the prior that the parameter prior induces on functions (Proposition 9.1). Vaintrob and Hänni (2026b) argue that under a strong weight prior this prior roughly tracks a circuit complexity of the function, so that NN Bayes behaves like a circuit version of SFI. Sections 8 and 9 prove what can be proved in general about circuit SFI and about this link, and quote the slides' main claims as theirs.

1.1 Contributions

- *Definitions and universality* (Section 2). Function machines read one random tape shared by all inputs. A universal function machine exists, and its law M_F (SFI) dominates every machine law, including lower-semicomputable product laws, partial computable functions and laws of random functions with computable cylinder probabilities (Theorem 2.9 and Proposition 2.10). The one-point inequality alone defines a strictly larger class, whose universal

element exceeds M_F by a number of bits linear in the size of some data sets (Theorem 2.13). We prove invariance under the universal machine and under computable encodings (Theorem 2.20), coding-theorem bounds (Theorem 2.21) and non-computability (Proposition 2.24), and we separate the semimeasure convention for losses from the normalised one (Proposition 2.17 and Remark 2.18).

- *Kolmogorov and semimeasure versions* (Section 3). The shortest consistent program K_F, the program mixture W, the product mixture M_{prod}, SFI and the weak universal semimeasure agree up to $K(\operatorname{dom} D) + O(1)$ bits (Theorem 3.4), but no two of K_F, W, M_F agree up to a constant. There is no coding theorem for function programs: on some two-point data sets $K_F \geq \log d - 2$ while $-\log W$ is $O(\log \log d)$ (Theorem 3.9 and Corollary 3.10). W falls short of M_F only by the padded complexity of the needed tape length (Proposition 3.6). Table 2 lists every comparison.

- *Learning theorems* (Section 4). Pathwise dominance (Theorem 4.1), merging (Theorem 4.3), a mistake bound (Corollary 4.4) and online-to-batch bounds (Theorem 4.5). Under fresh label noise with repeated inputs, random-function SFI has infinite expected loss (Proposition 4.6); time stamps, frozen noise or a per-query mixture repair this (Theorem 4.5 and Proposition 4.7). The expected risk of a Bayes predictor need not decrease with more data (Proposition 4.8); per-step in-distribution error bounds are in Corollary 4.14.

- *History-aware prediction* (Section 5). On Hänni's chain and skipped-string examples SFI's loss grows quadratically in the number of blocks when their lengths are $1, 2, 3, \ldots$, in both conventions up to lower-order terms, while a constant-size set-based predictor has loss 0, and every order-independent predictor pays the same lower bounds (Theorems 5.3 and 5.4 and Propositions 5.5 and 5.6). Hänni's dominance theorem for computable reveal laws is true in the reading his text supports, under finiteness hypotheses; its proof sketch fails, and a stronger reading is false (Proposition 5.8, Corollary 5.14, and Theorem 5.12). SFI^{hist} is the universal chronological semimeasure (Theorem 5.17 and Corollary 5.18); its exact relation to SI on the interleaved sequence is in Theorem 5.20 and Proposition 5.21.

- *Worked examples* (Section 6). For a random cyclic permutation of $\{0,1\}^d$ revealed along its cycle, SFI pays $\log((N-1)!) \approx (d - \log e)2^d$ bits with high probability in the semimeasure convention, and in the normalised convention at least $\delta d - O(\log d)$ bits per step in expectation for the first $2^{(1-\delta)d}$ steps, while SI pays $O(1)$ in expectation and SFI^{hist} pays $O(1)$ on every instance, uniformly in d (Theorem 6.3 and Propositions 6.6, 6.7 and 6.9). In Sam Eisenstat's coin example SFI pays $\frac{H(1/3)}{2} n \pm O(\log n)$ bits in the semimeasure convention, which is optimal among order-independent predictors up to $O(\log n)$, while SI pays at most 2.4 bits plus a constant in expectation (Theorems 6.15 and 6.19 and Proposition 6.16). Positive examples are in Section 6.4.

- *Out-of-distribution behaviour* (Section 7). An unseen simple property gives every simple behaviour on it a posterior probability bounded below, with no logarithmic term and independently of the amount of data (Proposition 7.1), and the bound depends on the universal machine (Proposition 7.4). Over any sequence of distinct inputs, the total posterior error on a computable target f is at most $(\ln 2)(K(f) + c)$ (Theorem 7.7). The naive converse fails, and a converse given the labelled sample holds (Propositions 7.10 and 7.12).

- *Computable versions and networks* (Sections 8 and 9). Time-bounded mixtures have constant regret against all predictors of a given running time (Theorem 8.2, Proposition 8.4,

and Corollary 8.5). For circuit Bayes, mistakes and circuit size are polynomially related on every ordering of the data (Theorem 8.10), while simplest-consistent (Occam) learning on the same prior makes $2^{\Omega(\ell/\log \ell)}$ mistakes on some ordering for a target of description length ℓ (Theorem 8.14). For networks we prove the general facts about the induced prior and a separation between Gaussian-process limits and SFI on parities (Propositions 9.2, 9.4 and 9.5, Corollary 9.8, and Remark 9.9).

- *Steeper simplicity priors* (Section 10) summarises the companion paper (Hänni, 2026a): when a steeper penalty on program length keeps a learner from reproducing a teacher's systematic errors, and when it prefers a simple sampler to a model with filled-in random bits.

Corrections to earlier claims. Several claims of Hänni (2026b) are corrected here; the text says so where each appears. The proof sketch of the dominance theorem for reveal laws fails, although the theorem holds in its intended reading (Proposition 5.8 and Corollary 5.14). The two-sided bounds on out-of-distribution generalisation in the PDF draft are false as literally stated; the intended statement holds (Proposition 7.6 and Theorem 7.5). The converse asked about in the notes, "no simple separating property implies good generalisation", fails unless the bound may depend on $K(f)$ (Propositions 7.10 and 7.12). The Kolmogorov and semimeasure versions, which the draft hoped to relate by a coding theorem, are not equivalent (Remark 3.14). The "physicist's proof" of few online mistakes is a proof for weighted majority, not for Occam learning (Remark 8.13 and Theorem 8.14). The e-mail argument for Eisenstat's example is correct in the semimeasure convention after two repairs (Remark 6.17). Table 1 lists the ten working hypotheses with which this study began and what became of each.

Credit. The notion of SFI, many of the claims examined here and most of the examples come from Hänni (2026b) and Hänni (2026c): the dominance template for samplers that see only the input, the remark on input encodings, the chain and skipped-string examples, the dominance theorem for reveal laws, the conjecture on hypotheses that see the previous inputs, the cyclic-permutation example, the out-of-distribution proposition and examples, and the time-bounded and circuit versions. The coin example of Section 6.3 is due to Sam Eisenstat, as reported in Hänni (2026c). The material on neural-network Bayes, including the conjunction ladder of Theorem 8.14, follows the slides of Vaintrob and Hänni (2026b). Hutter's universal chronological semimeasure and his treatment of supervised learning as a problem class for AIXI (Hutter, 2000, 2005, 2007) are prior work in substance for Section 5; we have checked them only at the level of abstracts and tables of contents (Remark 4.17 and Section 11).

What is not achieved. Several natural questions stay open: whether M_F dominates every lower-semicomputable law of a random partial function (Section 2.3); the exact size of the gap between W and M_F (Section 3.4); several lower bounds in the normalised convention (Sections 2.4 and 6.3); whether the $K(n) + K(A)$ term in the dominance theorem for reveal laws can be removed (Remark D.6); an $O(K/n)$ bound for the last iterate rather than the average (Remark 4.9 and Conjecture 4.12); efficient time-bounded learning against samplers (Section 8.1); and a two-sided quantitative bridge between NN Bayes and circuit SFI, of which one half is proved for uniform circuit priors (Proposition 9.5 and Section 9). The main theorems of the slides are not checked here. The constants in our bounds depend on the universal machine and can be large, and SFI itself is not computable, so the results describe an ideal, not an algorithm. Much of the literature was seen only through abstracts and secondary sources (Section 11).

	hypothesis (paraphrased)	verdict and where
H1(a)	Universal lsc function semimeasures exist, equivalently given by function machines	Universal elements exist in both senses (Thms. 2.9 and 2.12). The equivalence is refuted for weak semimeasures, which can exceed M_F by linearly many bits (Thm. 2.13 and Rem. 2.14); open for laws of random functions (§ 2.3).
H1(b)	The program mixture loses about $K(\operatorname{dom} D)$, or a log factor via hashing, against SFI	All versions agree up to $K(\operatorname{dom} D) + O(1)$ (Thm. 3.4). W loses only a padded tape-length cost, with no log factor (Prop. 3.6); the shortest program loses up to $\log d \pm O(\log \log d)$ on two points (Thm. 3.9 and Cor. 3.10); a log factor of the hashing kind describes K_F (Prop. 3.11).
H1(c)	The mixture of product laws is a third object	Strictly between W and M_F, with unbounded gaps (Props. 3.5 and 3.7); the per-query mixture is a fourth (Def. 3.1).
H1	SFI's cumulative loss does not depend on the order; input encoding matters little	Confirmed in the semimeasure convention (Prop. 2.16); normalised losses can depend on the order, open for M_F (Prop. 2.17). Invariance up to $K(\alpha, \beta) + O(1)$ (Thm. 2.20).
H2	Dominance over computable predictors that see only the input, on every input sequence	Confirmed for machine laws and consistent label sequences (Thm. 4.1); weak comparators need a $K(\operatorname{dom} D)$ term; open for general random-function laws (Rem. 4.2).
H3	Hänni's counterexamples; his Theorem 1 for computable reveal laws	Counterexamples proved, also normalised (Thms. 5.3 and 5.4 and Prop. 5.6). Theorem 1: proof sketch fails (Prop. 5.8); true in the intended reading with $K(n) + K(A)$ (Cor. 5.14); false in the oracle reading (Thm. 5.12).
H4	With the list of previous inputs, SFI is the universal chronological semimeasure and as good as SI	First part confirmed (Cor. 5.18); the set of previous inputs is not enough (Prop. 5.19). Against SI's joint loss confirmed (Thm. 5.20); against its conditional label loss the difference is SI's loss on the inputs (Prop. 5.21).
H5	Cyclic permutation: high loss under SFI, low under SI	Confirmed (Thms. 6.3 and 6.4 and Prop. 6.9); normalised for $t \leq 2^{(1-\delta)d}$ (Prop. 6.6); over the full cycle normalised: open.
H6	Eisenstat's example: SFI pays about $\frac{H(1/3)}{2} n$, SI a bounded amount	Confirmed up to $O(\log n)$ in the semimeasure convention (Thms. 6.13, 6.15 and 6.19); normalised: open (§ 6.3).
H7	For every input law and computable conditional law, expected regret $K(P) + O(1)$ and risk $K(P)/n$	Confirmed for deterministic labels, time-stamped and frozen noise (Thm. 4.5); refuted for fresh noise with repeated inputs (Prop. 4.6); holds for the per-query mixture (Prop. 4.7); last iterate open (Rem. 4.9).
H8	An unseen simple property gives constant probability of bad behaviour; a converse?	Confirmed with no log term (Prop. 7.1); machine-dependent (Prop. 7.4); naive converse refuted (Prop. 7.10); converse given the labelled sample (Prop. 7.12).
H9	A steeper simplicity prior with private randomness avoids a teacher's mistakes and splits model from randomness	Companion paper (§ 10): confirmed in a window of λ/n for incompressible error rules; the extension to all error rules of that density and description length is refuted; λ must grow linearly in n.
H10	Time-bounded and circuit versions; NN Bayes approximates circuit SFI	Time-bounded regret (Thm. 8.2 and Prop. 8.4); circuit mistakes versus size (Thm. 8.10); Occam fails out of distribution (Thm. 8.14); NN link partly proved, partly claimed in the slides (§ 9).

Table 1: The working hypotheses H1–H10, paraphrased, with verdicts. Most come from Hänni (2026b,c); H1, H2 and H7 formalise natural expectations. Lower bounds are in the semimeasure convention unless stated.

Reader's guide. Section 2 is the base for everything else. Section 3 answers the question of the Kolmogorov and semimeasure versions and can be skipped on a first reading. Section 4 gives the learning theorems. Sections 5 and 6 show where SFI and SI differ and how history-aware hypotheses remove the difference. Section 7 treats out-of-distribution behaviour, Sections 8 and 9 computable versions and networks, Section 10 the companion paper, and Sections 11 and 12 related work and open problems. Each main section states its results with short proof ideas; full proofs are in Appendices A to H. Two conventions for losses recur: the *semimeasure convention* scores SFI's raw predictive sub-probabilities, and the *normalised convention* rescales them to sum to one at each step. Upper bounds in the first hold in both; every lower bound says which convention it is for (Remark 2.18). Logarithms are to base 2, losses are in bits, and c denotes a constant that depends only on the fixed universal machines and encodings.

2 Function semimeasures, function machines and SFI

Hänni's definition of Solomonoff function induction (SFI) reads: "you have a bunch of input-output pairs. you have the usual prior on all programs. solomonoff function induction := you think of the input-output pairs as being generated by picking a random program from this prior and generating using it, and so you update on seen pairs as you should" (Hänni, 2026b). A random program may be deterministic or read random bits, private to each input or shared by all inputs. We take the most general reading that still yields a random function: one random tape read by all inputs (Definition 2.7); Section 3 compares the other readings. The notion and several of the claims checked below are Hänni's. Proofs are in Appendix A.

2.1 Data and conventions

Definition 2.1 (Data)**.** foundations §1 Inputs X, labels Y; by default $X = Y = \{0,1\}^*$, but any infinite decidable X and countable Y, $|Y| \geq 2$, will do unless stated. $\perp$ means "undefined", $Y_\perp := Y \cup \{\perp\}$. A *data set* is a finite partial function $D : X \rightharpoonup Y$; $D + (x,y) := D \cup \{(x,y)\}$ for $x \notin \operatorname{dom} D$; $D|_T$ is a restriction. A partial function f *extends* D, $f \supseteq D$, if $f(x) = y$ for all $(x,y) \in D$. A sequence $(x_t, y_t)_{t \leq n}$ is *consistent* if $D_n := \{(x_t, y_t) : t \leq n\}$ is a partial function; $h_t := (x_1, y_1, \ldots, x_t, y_t)$ is the history. For ν from data sets to $[0,1]$ the *conditional* is $\nu(y \mid D, x) := \nu(D + (x,y))/\nu(D)$ for $x \notin \operatorname{dom} D$ (0 if $\nu(D) = 0$), and $\nu(y \mid D, x) := \mathbf{1}[y = D(x)]$ for $x \in \operatorname{dom} D$ (the *repeated-input convention*).

Complexity and constants. V is a fixed universal prefix machine, K prefix complexity, φ_e the e-th partial computable function $X \rightharpoonup Y$, $\mathbf{m}(i) := \sum_{V(p)=i} 2^{-|p|} \geq 2^{-K(i)}$, and $K(\nu) := \min\{K(i) : \nu_i = \nu\}$ in an effective enumeration (ν_i); "lsc" means lower semicomputable. We use, without re-checking the sources (Levin, 1974; Gács, 1974; Chaitin, 1975), (Li and Vitányi, 2008, §4.3), that $\mathbf{m}$ dominates every lsc discrete semimeasure up to a constant factor, and the conditional coding theorem: if $P(a \mid z)$ is lsc uniformly in (a,z) with $\sum_a P(a \mid z) \leq 1$, then $K(a \mid z) \leq -\log P(a \mid z) + K(P) + c$; both relativise. $\log = \log_2$; losses are in bits; c and $O(1)$ depend only on the fixed machines and codings. $\mathbb{U}$ is the fair-coin measure on tapes $\omega \in \{0,1\}^\infty$, and $\omega_{1:L}$ the first L tape bits.

Two loss conventions. The predictors below issue sub-probabilities; the missing mass is the probability of "undefined". The *semimeasure convention* scores the raw conditionals, the *normalised convention* divides each by its sum. Statements about losses name their convention; up-

per bounds in the semimeasure convention hold in both, lower bounds need not (Proposition 2.17 and Remark 2.18).

2.2 Three notions of function semimeasure

Definition 2.2 (Weak function semimeasure)**.** foundations Def 2.1 A map ν from data sets to $[0,1]$ with $\nu(\emptyset) \le 1$ and $\nu(D) \ge \sum_{y \in Y} \nu(D + (x,y))$ for every D and $x \notin \operatorname{dom} D$.

Then ν is monotone, and $\sum_{D:\operatorname{dom} D = S} \nu(D) \le \nu(\emptyset)$ for finite S (sum out one input at a time). Call a predictor q (sub-probabilities $q(\cdot \mid D, x)$, with the repeated-input convention) *order-independent* if, for every D, the product $\nu_q(D)$ of its conditionals along a reveal order of D is the same for all orders.

Proposition 2.3 (Predictor reading)**.** [proved] foundations Prop 2.1a (= context Prop 5.1(a)) *(a) If ν is weak with $\nu(\emptyset) = 1$, its conditionals form an order-independent predictor q_ν with $\nu_{q_\nu} = \nu$. (b) If q is order-independent, ν_q is weak with $\nu_q(\emptyset) = 1$, and $q(y \mid D, x) = \nu_q(D + (x,y))/\nu_q(D)$ when $\nu_q(D) > 0$. So weak semimeasures with $\nu(\emptyset) = 1$ correspond one-to-one to order-independent predictors, identified where $\nu(D) > 0$.*

Definition 2.4 (Random partial function)**.**
foundations Def 2.2 A *random partial function* is a random element F of $Y_\perp^X$ (product σ-algebra) with law P; its *containment functional* is $\nu_P(D) := P(F \supseteq D)$.

Every containment functional is weak, since the events $\{F \supseteq D + (x,y)\}$, $y \in Y$, are disjoint subsets of $\{F \supseteq D\}$. The converse fails (Example 2.6). If every conditional of q sums to 1, ν_q is the law of a random total function, and M_F dominates every such lsc law (Proposition A.1 and Corollary A.3, Appendix A.1). For finite $T \subseteq X \setminus \operatorname{dom} D$ put $\nabla_T \nu(D) := \sum_{E:\operatorname{dom} E \subseteq T} (-1)^{|E|} \nu(D \cup E)$.

Proposition 2.5 (Möbius characterisation)**.** [proved] foundations Prop 2.3 (= context Prop 5.1(c)); checks c6, r4 *A weak ν with $\nu(\emptyset) = 1$ is the containment functional of a random partial function if and only if $\nabla_T \nu(D) \ge 0$ for all D and all finite $T \subseteq X \setminus \operatorname{dom} D$. The law is then unique, and $\nabla_T \nu(D) = P(F \supseteq D,\ F|_T \equiv \perp)$.*

The weak inequality is the case $|T| = 1$. The proposition is the function-space analogue of the correspondence between completely monotone set functions and random sets (Choquet, 1954; Matheron, 1975; Shafer, 1976) [known; unverified]; we have not checked the analogy against these sources.

Example 2.6 (Weak but not a random function)**.** [proved; computed] foundations Ex 2.4; checks c1, c6, c9(b), r1, r4 (a) Inputs a, b: $\nu(\emptyset) = \nu(\{(a,0)\}) = \nu(\{(b,0)\}) = 1$, all other values 0. It is weak, but $\nabla_{\{a,b\}} \nu(\emptyset) = -1$: "$F(a) = 0$ surely and $F(b) = 0$ surely, but never both". (b) Inputs a, b, c, labels $\{0,1\}$: $\nu_\triangle(\emptyset) = 1$, and $\nu_\triangle = \frac{1}{2}$ on the six one-point data sets and on $E_1 = \{(a,0),(b,1)\}$, $E_2 = \{(b,0),(c,1)\}$, $E_3 = \{(c,0),(a,1)\}$, and 0 elsewhere. $\nu_\triangle$ is weak (c1), and E_1, E_2, E_3 are pairwise inconsistent with total mass $\frac{3}{2}$; so no law P of a random partial function has $P(F \supseteq \cdot) \ge c\,\nu_\triangle$ with $c > \frac{2}{3}$, and the uniform law on $(F(a), F(b), F(c)) \in \{(0,1,0),(1,0,1),(1,0,0)\}$ attains $c = \frac{2}{3}$.

2.3 Function machines and universality

Definition 2.7 (Function machine)**.**
foundations Def 2.5 A *function machine* is a c.e. set $T \subseteq X \times \{0,1\}^* \times Y$ of triples that is *consistent*: if $(x,w,y),(x,w',y') \in T$ and w, w' are comparable, then $y = y'$. For a tape ω put $f_\omega^T(x) := y$

if some prefix w of ω has $(x, w, y) \in T$, and $f^T_\omega(x) := \perp$ otherwise. The *machine law* of T is $\nu_T(D) := \mathbb{U}\{\omega : \ f^T_\omega \supseteq D\}$.

Equivalently, T is a Turing machine with a one-way random tape which on query x reads finitely many tape bits and halts with a label or diverges; all queries read the same tape.

Proposition 2.8 (Machine laws)**.** [proved]
foundations Prop 2.6 and the enumeration remark *Every machine law is the containment functional of a random partial function, with $\nu_T(\emptyset) = 1$, and is lsc uniformly in D and an index of T. Function machines have an effective enumeration (T_i), to which $K(\nu)$ for machine laws refers.*

Theorem 2.9 (Universal function machine)**.** [proved]
foundations Thm 3.2 *$T_U := \{(x, pw, y) : \ p \in \operatorname{dom} V, \ (x, w, y) \in T_{V(p)}\}$ is a function machine. Its law $M_F := \nu_{T_U}$ has $M_F(\emptyset) = 1$ and, for every $D \neq \emptyset$,*

$$M_F(D) = \textstyle\sum_i \mathbf{m}(i)\, \nu_{T_i}(D), \qquad \text{hence} \qquad M_F \ \geq \ 2^{-K(\nu)}\, \nu \quad \text{for every machine law } \nu. \tag{1}$$

So M_F is the Bayes mixture "pick machine i with probability $\mathbf{m}(i)$, then a random tape"; the posterior weight of i after D is $\mathbf{m}(i)\nu_{T_i}(D)/M_F(D)$.

Proposition 2.10 (What machine laws include)**.** [proved]
foundations Prop 3.3 *(a)* Product laws (Hänni's samplers that see only the input). *If $P(y \mid x)$ is lsc with $\sum_y P(y \mid x) \leq 1$ (a* conditional semimeasure*), then $\nu_P(D) := \prod_{(x,y)\in D} P(y \mid x)$ is a machine law with $K(\nu_P) \leq K(P) + O(1)$; probabilistic machines that read x and output y (halting or not) define exactly these P. (b)* Deterministic programs. *For partial computable φ, $D \mapsto \mathbf{1}[\varphi \supseteq D]$ is a machine law of complexity at most $K(\varphi)+O(1)$. (c)* Computable laws. *A law of a random partial function whose cylinder probabilities $P(F|_S = v)$, $v \in Y^S_\perp$, are computable uniformly in (S, v) is a machine law; if Y is finite or the function is total, computable ν_P suffices. (d)* Mixtures. *If (ν_i) are machine laws with computable indices and w is an lsc discrete semimeasure, $D \mapsto \sum_i w_i\nu_i(D)$ $(D \neq \emptyset)$ is a machine law.*

In (a) each input samples from a private stream of tape bits, as Hänni proposed: "use different bits of randomness from the universal semimeasure's infinite read tape for sampling on different inputs; one can do this using dovetailing" (Hänni, 2026b). With Theorem 2.9 this gives his dominance theorem for such samplers (Theorem 4.1).

Remark 2.11 (Per input, not per query). foundations remark after Prop 3.3 In (a) a repeated query returns the same label; a sampler run afresh at each query would not. This choice fits Hänni's setting of "input-output pairs from a function" and makes ν_P a random function; the per-query reading gives a different object, M_{pq} (Definition 3.1 and Proposition 4.7).

Theorem 2.12 (Universal weak semimeasure)**.** [proved; trimming computed] foundations Thm 3.4; check r3 *There is an lsc weak semimeasure M^w with $M^w \geq 2^{-K(\nu)}\nu$ for every lsc weak semimeasure ν.*

Theorem 2.13 (Weak versus machine universality)**.** [proved; computed]
foundations Thm 3.5 and the remark after it; checks c1, c6, c9, r1 *(i) $M_F \leq 2^{O(1)}M^w$. (ii) For every lsc weak ν and every D, $M_F(D) \geq 2^{-K(\nu)-K(\operatorname{dom} D)-O(1)}\nu(D)$; so $M^w(D) \leq 2^{K(\operatorname{dom} D)+O(1)}M_F(D)$. (iii) For every k some D_k with $|\operatorname{dom} D_k| = 2k$ has $\log(M^w(D_k)/M_F(D_k)) \geq k \log \frac{3}{2} - K(k) - O(1)$. (iv) This bound holds on average over the 3^k explicit data sets $E_{\vec{\imath}}$ of the proof. (v) The constant $(2/3)^k$ behind (iii) is exact for $\nu^{\otimes k}_{\triangle}$.*

In (ii) the labels of $\operatorname{dom} D$, read in a fixed order, form a semimeasure on strings that a monotone machine realises exactly (Lemma A.5). In (iii) k blocks of $\nu_\triangle$ give 3^k pairwise inconsistent data sets of M^w-mass $\geq 2^{-k-K(k)-O(1)}$ each, while M_F gives them total mass ≤ 1; there $K(\operatorname{dom} D_k) \leq k \log 3 + K(k) + O(1)$, so at least a fraction $\log \frac{3}{2} / \log 3 \approx 0.37$ of the term $K(\operatorname{dom} D)$ is needed.

Remark 2.14 (Weak and machine universality differ). [refuted (weak definition); open (random-function laws)] foundations §3, claim F10 A Levin-style equivalence "lsc function semimeasures, or equivalently monotone function machines" (a natural conjecture, by analogy with sequences (Zvonkin and Levin, 1970)) is refuted for the weak definition: M^w and M_F agree up to $2^{K(\operatorname{dom} D)+O(1)}$, not up to a constant. For random-function laws it is open (Section 2.3).

Open question. Is every lsc law of a random partial function dominated by M_F up to a constant factor? Yes for the classes of Proposition 2.10 and for computable filling-in processes (Remark A.4); up to $2^{K(\operatorname{dom} D)+O(1)}$ always. It would suffice (proof sketch, Appendix A.6) that every such law has lower approximations increasing on all up-sets of the extension order, not only on the sets $\{F \supseteq D\}$; Strassen's theorem (Strassen, 1965) then gives couplings that fill in values.

2.4 The definition of SFI and its conventions

Definition 2.15 (Solomonoff function induction)**.**
foundations Def 5.1 *SFI* is the random partial function $f_\omega := f_\omega^{T_U}$, $\omega \sim \mathbb{U}$, with law M_F. After data D the *posterior* is the law of f_ω given $f_\omega \supseteq D$, with containment functional $E \mapsto M_F(D \cup E)/M_F(D)$. The *predictive conditional* at $x \notin \operatorname{dom} D$ is $M_F(y \mid D, x) = M_F(D + (x, y))/M_F(D)$; its defect $1 - \sum_y M_F(y \mid D, x)$ is the posterior probability that $f_\omega(x) = \bot$; at $x \in \operatorname{dom} D$ it is the stored label with probability 1. The *normaliser* is $\mathcal{Z}(D, x) := \sum_y M_F(D + (x, y))$, and the *normalised conditional* is $\bar{M}_F(y \mid D, x) := M_F(D + (x, y))/\mathcal{Z}(D, x)$. For a weak ν, $L_\nu(D) := -\log \nu(D)$, and SFI's loss is $L_{\mathrm{SFI}}(D) := L_{M_F}(D)$.

$M_F(D) > 0$ for every D (the lookup table of D is a program).

Proposition 2.16 (Order independence)**.** [proved; computed]
foundations Prop 5.2 (= context Fact A, history Prop 5.1); check c4 *Semimeasure convention. For every weak ν and consistent $(x_t, y_t)_{t \leq n}$, $\sum_{t \leq n} -\log \nu(y_t \mid D_{t-1}, x_t) = L_\nu(D_n) + \log \nu(\emptyset) \leq L_\nu(D_n)$, which depends only on the set D_n, not on the order or on repetitions. For SFI it is $L_{\mathrm{SFI}}(D_n)$. On an inconsistent sequence it is infinite.*

(A repeated input contributes the factor 1 or 0; the other factors telescope.)

Proposition 2.17 (Normalisation)**.** [proved; computed] foundations Prop 5.3; checks c3, r8, r13; context c3 *Let $\bar{L}_{\mathrm{SFI}} := \sum_{t \leq n} -\log \bar{M}_F(y_t \mid D_{t-1}, x_t)$ along a given consistent sequence. (a) $\bar{L}_{\mathrm{SFI}} \leq L_{\mathrm{SFI}}(D_n)$, so upper bounds on L_{SFI} hold in both conventions. (b) For laws of random partial functions the normalised loss can depend on the order. (c) For random total functions it equals the semimeasure loss.*

For (b), let $(F(a), F(b))$ be uniform on $\{(0,0), (0,\bot), (1,1), (\bot,\bot)\}$: $D = \{(a,0),(b,0)\}$ has normalised probability $\frac{2}{3}$ if a comes first and $\frac{1}{2}$ if b does, semimeasure probability $\frac{1}{4}$ in both orders (c3). On a random six-point mixture the semimeasure loss is 10.919980315195 bits in all 720 orders, the normalised loss between 7.33 and 10.64 bits (context c3).

What does transfer is collected in Proposition A.7 (Appendix A.7): with Z_t the posterior probability that $f_\omega(x_t)$ is defined, $\bar{L}_{\mathrm{SFI}} = L_{\mathrm{SFI}}(D_n) - \sum_t \log(1/Z_t)$, so single-point lower bounds

transfer; and two general lower bounds, $K^{\emptyset'}(y_{1:n} \mid x_{1:n}) - O(1)$ and the chronological Bayes loss, which are $O(1)$ in the chain, the skipped strings and the cyclic permutation (Sections 5 and 6).

Remark 2.18 (Which lower bounds are semimeasure-only). [proved (as a list); transfer open where stated] foundations Rem 5.6, updated by history Prop 2.4, examples Prop 2.9, context §1.2 Lower bounds on $L_{\mathrm{SFI}}(D_n)$, or for every order-free predictor, do not transfer automatically to $\bar{L}_{\mathrm{SFI}}$. In this paper they hold *in the semimeasure convention only* for: Theorem 2.21(a) beyond single points; Example 2.22; the $\mathrm{GF}(2^q)$ example (Theorem 5.26); the separations of Propositions 5.19 and D.11; the cyclic permutation over the full cycle (Theorem 6.3); Eisenstat's example (Theorems 6.13 and 6.15). *Also normalised:* the chain and the skipped strings, up to $O(n \log n)$ and $O(\ell \log \ell)$ bits (Proposition 5.6); the cyclic permutation at steps $t \le 2^{(1-\delta)d}$, at $\delta d - O(\log d)$ bits per step (Proposition 6.6). *In both conventions:* the out-of-distribution lower bounds of Section 7, which bound the posterior mass that is defined and wrong.

Open question. Is the normalised loss of M_F order-dependent, and which semimeasure-only lower bounds hold for $\bar{M}_F$? On two inputs the exact condition is the criterion $(\star)$ of Proposition A.6: the normaliser leaks order information exactly when whether one input is defined says something about another input's label. A failure of $(\star)$ for M_F is expected but not proved.

Remark 2.19 (Recommendation).
foundations §5 "Recommendation" We call M_F "SFI": it is the object Hänni describes; it has a posterior over functions (M^w has none); it dominates every machine law with constant $2^{-K(\nu)}$; it is invariant (Theorem 2.20); its semimeasure-convention loss is order-independent. The variants W, K_F, M_{prod}, M_{pq}, M^w (Section 3) agree with it up to $K(\mathrm{dom}\, D) + O(1)$ bits, differ by unbounded amounts, and should be kept distinct.

2.5 Basic properties

Theorem 2.20 (Invariance). [proved]
foundations Thm 6.2 *(a) If M_F, M'_F are built from two universal prefix machines and two enumerations of function machines, $|\log M_F(D) - \log M'_F(D)| \le c$ for all D. (b) For computable injections $\alpha : X \to X'$, $\beta : Y \to Y'$, with $D^{\alpha\beta} := \{(\alpha(x), \beta(y)) : (x, y) \in D\}$ and M'_F SFI over X', Y': $|\log M_F(D) - \log M'_F(D^{\alpha\beta})| \le K(\alpha, \beta) + O(1)$ for all D. So cumulative losses on original and re-encoded data differ by at most $K(\alpha, \beta) + O(1)$ bits, uniformly over data sets and reveal orders. (c) The same holds for W, M_{prod} and M^w (Definition 3.1).*

This makes precise Hänni's heading "input encoding doesn't matter much" (Hänni, 2026b). The constant is not small for complex encodings, which can change losses by up to that much (only the upper bound is proved); non-computable encodings are not covered.

Theorem 2.21 (Coding-theorem bounds). [proved] foundations Thm 6.3 (= history Lemma 1.1, examples Lemma 1.1, context Prop 5.4(1)) *For every data set D: (a) $L_{\mathrm{SFI}}(D) \ge K(D|_T \mid T) - O(1)$ for every $T \subseteq \mathrm{dom}\, D$; in particular $M_F(D) \le 2^{-K(D|\mathrm{dom}\, D)+O(1)}$, and $M_F(D) \le 2^{-K(y|x)+O(1)}$ for every $(x, y) \in D$; (b) $L_{\mathrm{SFI}}(D) \le \min\{K_F(D),\ \sum_{(x,y)\in D} K(y \mid x)\} + O(1) \le K(\mathrm{dom}\, D) + K(D \mid \mathrm{dom}\, D) + O(1)$, with K_F the shortest consistent program (Definition 3.1); (c) $L_{\mathrm{SFI}}(\{(x, y)\}) = K(y \mid x) \pm O(1)$.*

Example 2.22 (Swap data). [proved; computed]
foundations Ex 6.4; checks c9(c), r8 For distinct $x_1, x_2 \in \{0,1\}^d$ let $D := \{(x_1, x_2), (x_2, x_1)\}$. Then $K(D \mid \mathrm{dom}\, D) = O(1)$, but $L_{\mathrm{SFI}}(D) \ge \max\{K(x_2 \mid x_1), K(x_1 \mid x_2)\} - O(1)$. Conversely $L_{\mathrm{SFI}}(D) \le d + O(1)$ for every pair: the machine $F(x) := x \oplus \omega_{1:|x|}$ has complexity $O(1)$ and extends D exactly

when $\omega_{1:d} = x_1 \oplus x_2$, with probability 2^{-d} (c9(c)). For every x_1 some x_2 has $K(x_2 \mid x_1) \geq d$, and then $L_{\mathrm{SFI}}(D) = d \pm O(1)$; also $K_F(D) \leq d + 2\log d + O(1)$.

This is *not* a separation between SFI and history-aware prediction: a history-aware predictor shown x_1 first must predict x_2 from x_1 alone and pays $K(x_2 \mid x_1) - O(1)$ bits. Such separations are Theorems 5.3, 5.4 and 6.3.

Open question. A two-sided characterisation of $L_{\mathrm{SFI}}(D)$ up to $O(1)$ by standard complexities, other than the tautology $L_{\mathrm{SFI}}(D) = \min_\nu[K(\nu) - \log \nu(D)] \pm O(1)$ over machine laws ν.

Proposition 2.23 (Sequence induction is SFI on initial segments)**.** [proved] foundations Prop 7.1 *Let $X = \mathbb{N}$, $\{0,1\} \subseteq Y$, and M Solomonoff's universal monotone semimeasure (Zvonkin and Levin, 1970). For $x \in \{0,1\}^n$ put $D_x := \{(i, x_{i+1}) : i < n\}$. Then $2^{-O(1)} M(x) \leq M_F(D_x) \leq 2^{O(1)} M(x)$.*

So SFI generalises Solomonoff induction, and is in turn dominated by universal prediction that sees the history (Lemma 5.2 and Corollary 5.18).

Proposition 2.24 (Computability)**.** [proved] foundations Prop 7.3 *(a) $M_F(D)$ is lsc uniformly in D, and positive. (b) M_F is not computable. (c) Neither $M_F(y \mid D, x)$ nor $\bar{M}_F(y \mid D, x)$ is computable as a function of (D, x, y). (d) Both are computable relative to the halting problem $\emptyset'$.*

(c) is diagonal: a computable f that always takes a label of conditional below 0.6 would have $M_F(D_n) < 0.6^n$. So SFI is an ideal reference, not an algorithm (computable versions: Section 8).

Open question. The exact arithmetical level of the conditionals; for Solomonoff's M see Leike and Hutter (2015b) (not checked against the source).

3 Kolmogorov and semimeasure versions

Hänni's draft has a section headed "Equivalence between Kolmogorov version and universal semimeasure version", whose whole text is "need coding theorem" (Hänni, 2026b). He later asked that the paper "include an equivalence between the kolmogorov complexity version and the universal semimeasure version, if these turn out to be equivalent (if they don't, then it should explain that)" (Hänni, 2026c). They are not equivalent: Table 2 summarises what is equal, what differs and by how much, and Remark 3.14 answers in plain words. Proofs are in Appendix B.

3.1 The variants

Definition 3.1 (Variants)**.**
foundations Def 4.1 Let (P_i) be an effective enumeration of all lsc conditional semimeasures (it exists by trimming each $P(\cdot \mid x)$ as in Theorem 2.12).

- *Program mixture* (deterministic programs):
  $W(D) := \sum_{p \in \mathrm{dom}\, V} 2^{-|p|}\, \mathbf{1}[\varphi_{V(p)} \supseteq D] = \sum_e \mathbf{m}(e)\, \mathbf{1}[\varphi_e \supseteq D]$.
- *Kolmogorov version* (single shortest program): $K_F(D) := \min\{|p| : p \in \mathrm{dom}\, V,\ \varphi_{V(p)} \supseteq D\} = \min\{K(e) : \varphi_e \supseteq D\}$.
- *Product mixture* (private randomness per input, on data sets):
  $M_{\mathrm{prod}}(D) := \sum_i \mathbf{m}(i) \prod_{(x,y) \in D} P_i(y \mid x)$.

- *Per-query mixture* (private randomness per query, on sequences): $M_{\mathrm{pq}}(y_{1:n}\|x_{1:n}) := \sum_i \mathbf{m}(i)\prod_{t\le n} P_i(y_t \mid x_t)$, for every finite sequence of pairs, consistent or not.

W reads Hänni's random program as a deterministic program; M_F lets it read random bits. $W(\emptyset)$, $M_{\mathrm{prod}}(\emptyset)$, $M_{\mathrm{pq}}(\varepsilon\|\varepsilon)$ are below 1, which changes losses by a constant only. W's programs delimit themselves before reading the input, so W is a mixture of fixed functions (Remark B.1).

Remark 3.2 (Operator induction). [known; unverified; equality on distinct inputs proved] foundations Def 4.1, claim F36; context §6 item 1; history Prop 4.9 Solomonoff's third kind of induction takes "an unordered set of ordered pairs" and predicts the second element of a new pair from the first (Solomonoff, 2008) (abstract only seen). Secondary sources write its prediction as $\sum_j a_j \prod_{i\le n} O^j(A_i \mid Q_i)$, the product running over all question–answer pairs, repeats included (Özkural, 2017) (search snippet only). On that description operator induction is M_{pq}: equal to $M_{\mathrm{prod}}(D_n)$ on distinct inputs, different with repeats, where M_{prod} gives a conflicting label probability 0 and M_{pq} allows fresh noise (Proposition 4.7). So operator induction and M_{prod} coincide only on distinct inputs.

Remark 3.3 (Where the randomness lives).
foundations Rem 4.11 Nowhere in W and K_F (fixed functions); privately per input in M_{prod}; privately per query in M_{pq} (a repeated input is resampled); on one tape shared by all inputs in M_F, so one hypothesis may correlate labels across inputs, as $F(x) = \omega_{|x|+1}$ does. Every mixture also correlates labels through its posterior; the classes still differ by more than constants.

3.2 The chain and the gaps

Theorem 3.4 (The chain). [proved] foundations Thm 4.2 *For every data set $D \neq \emptyset$,*

$$2^{-K_F(D)} \le W(D) \le 2^{O(1)} M_{\mathrm{prod}}(D) \le 2^{O(1)} M_F(D) \le 2^{O(1)} M^w(D) \le 2^{K(\mathrm{dom}\, D)+O(1)}\, 2^{-K_F(D)}.$$

So the five quantities agree up to $K(\mathrm{dom}\, D) + O(1)$ bits.

The last step is the conditional coding theorem, $K(D \mid \mathrm{dom}\, D) \le -\log M^w(D) + O(1)$ (Appendix B.1). Let $\Omega_V^{\ge \ell} := \sum\{2^{-|p|} : p \in \mathrm{dom}\, V,\ |p| \ge \ell\}$, which tends to 0.

Proposition 3.5 (Deterministic programs cannot separate inputs cheaply). [proved]
foundations Prop 4.3 ((b) = context Prop 5.4(2)) *(a)* Two points, binary labels *($\{0,1\} \subseteq Y$). For every $\varepsilon > 0$ and all large d there are distinct $x_1, x_2 \in \{0,1\}^d$ with $W(\{(x_1,0),(x_2,1)\}) \le \varepsilon$, while $M_{\mathrm{prod}}(D) \ge 2^{-O(1)}/4$ and $M_F(D) \ge 2^{-O(1)}/4$ for every two-point D with labels in $\{0,1\}$. (b)* One point, long labels *($Y = \{0,1\}^*$). For every d some $y \in \{0,1\}^d$ has $W(\{(0^d, y)\}) \le 2^{1-d}\,\Omega_V^{\ge d-1}$, while $M_{\mathrm{prod}}(\{(0^d,y)\}) \ge 2^{-d-O(1)}$. So M_{prod}/W and M_F/W are unbounded.*

(Pigeonhole: the fewer than 2^ℓ programs shorter than ℓ give the 2^d inputs at most 3^{2^ℓ} output patterns, so two inputs are never separated by them.) The two-point gap of W against M_{prod} and M_F lies between $\log(1/\Omega_V^{\ge \ell_d})$, $\ell_d := \lceil \log(d/\log 3)\rceil - 1$, and $K^{\mathrm{pad}}(2^{d+1}) + O(1)$ (Remark B.2), where $K^{\mathrm{pad}}(N) := \min_{L \ge N} K(L)$ is the *padded complexity*, nondecreasing, unbounded and at most $K(N)$.

Proposition 3.6 (Tape-length bound). [proved; finite analogue computed]
foundations Prop 4.4 (= its referee's Prop R1); checks c9(a), r5 *(a) Let T be a function machine with law ν, and for $L \ge 0$ let $\nu^{(L)}(D) := \mathbb{U}\{\omega :$ every $(x,y) \in D$ has a triple $(x,w,y) \in T$ with w a prefix of $\omega_{1:L}\}$. Then $\nu^{(L)}(D) \uparrow \nu(D)$ as $L \to \infty$, and for every $D \neq \emptyset$ and every L,*

$$W(D) \;\ge\; 2^{-K(T)-K^{\mathrm{pad}}(L)-c}\,\nu^{(L)}(D),$$

with c independent of T, D and L. With $L_{1/2}(\nu, D) := \min\{L : \nu^{(L)}(D) \geq \nu(D)/2\}$ this gives $\log\left(M_F(D)/W(D)\right) \leq K^{\mathrm{pad}}(L_{1/2}(M_F, D)) + O(1)$. *(b)* Fair-coin labels. *For every D with labels in $\{0,1\}$ and* $\operatorname{dom} D \subseteq \{0,1\}^{\leq d}$*: $W(D) \geq 2^{-|D|-K^{\mathrm{pad}}(2^{d+1})-c'} \geq 2^{-|D|-K(\lceil\log(d+1)\rceil)-c''}$. (c) $K^{\mathrm{pad}}(2^{2^s}) \leq K(s)+O(1)$, and $K^{\mathrm{pad}}(N) \leq 2\log(g(N)+2)+c_g$ for every computable, nondecreasing, unbounded $g : \mathbb{N} \to \mathbb{N}$.*

The $2^{L'}$ programs "simulate T with stored tape prefix w", $|w| = L' \geq L$, $K(L') = K^{\mathrm{pad}}(L)$, carry weight $2^{-K(T)-K^{\mathrm{pad}}(L)-O(1)}\nu^{(L')}(D)$. So W imitates any shared-tape law, paying only for a self-delimiting bound on the number of tape bits used: a random program in Hänni's sense can carry the bits it needs. (b) applies (a) to $F(x) := \omega_{j(x)+1}$.

Proposition 3.7 (Shared randomness is not a mixture of private randomness)**.** [proved; counting step computed] foundations Prop 4.5; check r9(c) *Let $\{0,1\} \subseteq Y$ and let ν_{len} be the law of $F(x) := \omega_{|x|+1}$: one fair coin per input length, shared by all inputs of that length. ν_{len} is a machine law of complexity $O(1)$, but $\sup_D \nu_{\mathrm{len}}(D)/M_{\mathrm{prod}}(D) = \infty$. Hence M_F/M_{prod} is unbounded.*

(A product hypothesis labels 0^n and 1^n independently; a compactness argument shows that M_{prod} covers only a small part of the non-constant patterns of k shared coins.)

Remark 3.8 (Size of this gap)*.* [proved]
foundations Rem 4.5a The witnesses of Proposition 3.7 are not explicit, and their gap is small: on data with inputs of length at most d, $M_{\mathrm{prod}}(D) \geq 2^{-O(1)}W(D) \geq 2^{-K^{\mathrm{pad}}(d+1)-O(1)}\nu_{\mathrm{len}}(D)$ by Proposition 3.6. A lower bound on the rate is open.

3.3 No coding theorem for function programs

For strings, shortest description and algorithmic probability agree up to a constant (the coding theorem). For functions they do not.

Theorem 3.9 (No coding theorem for function programs)**.** [proved; arithmetic and covering numbers computed] foundations Thm 4.7 (= its referee's Prop A); checks c8(a), c8(b), r11 *Let $\{0,1\} \subseteq Y$ and $d \geq 2$. There are distinct $x_1, x_2 \in \{0,1\}^d$ such that $D := \{(x_1, 0), (x_2, 1)\}$ satisfies*

(i) $K_F(D) \geq \log d - 2$; *(ii)* $-\log W(D) \leq 2 + K(\lceil\log(d+1)\rceil) + c$; *(iii)* $L_{\mathrm{SFI}}(D) \leq 2 + c$,

with c independent of d. Hence $K_F(D) + \log W(D) \geq \log d - K(\lceil\log(d+1)\rceil) - c - 4 \to \infty$, and on these data sets $-\log W(D) \leq \log K_F(D) + O(\log\log K_F(D))$.

(i) is the colouring argument with $\ell = \ell_d$, (ii) is Proposition 3.6(b), (iii) the fair-coin product law. So a c.e. set of programs, $\{e : \varphi_e \supseteq D\}$, can have algorithmic probability $2^{-O(\log\log d)}$ while its simplest member has complexity $\log d - O(1)$ (related literature: Appendix B.5).

Corollary 3.10 (Two-point rate)**.** [proved]
foundations Cor 4.8 *Let $\rho_K(d)$ be the largest value of $K_F(D) - L_{\mathrm{SFI}}(D)$ over two-point data sets D with binary labels and d-bit inputs. Then* $\log d - O(1) \leq \rho_K(d) \leq \log d + K(\lceil\log(d+2)\rceil) + O(1) = \log d + O(\log\log d)$.

Proposition 3.11 (Covering bound for the single shortest program)**.** [proved; computed] foundations Prop 4.9 (its referee's Prop A3, sharpened); checks c8(c,d), r11(d) *For every D with labels in $\{0,1\}$, $|D| = k$ and* $\operatorname{dom} D \subseteq \{0,1\}^{\leq d}$*,*

$$K_F(D) \;\leq\; k + \log\left(k(d+2)\right) + K\left(\lceil\log(d+2)\rceil, k\right) + O(1).$$

Consequently, with $\delta_D := k - K(D \mid \operatorname{dom} D)$ *(the deficiency of randomness of the labels given the inputs),* $K_F(D) \le L_{\mathrm{SFI}}(D) + \delta_D + \log(k(d+2)) + K(\lceil \log(d+2) \rceil, k) + O(1)$.

(About $2^k k(d+2) \ln 2$ random truth tables cover all such data sets; a program indexes the first covering family.) So a logarithmic cost is real for the single shortest program, while W pays no factor depending on $|D|$.

Remark 3.12 (Witnesses of large gaps cannot be computable). [proved] foundations Rem 4.10 If $k \mapsto D_k$ is computable, then $K_F(D_k) \le K(k) + O(1)$, so the five quantities of Theorem 3.4, all between $2^{-K_F(D_k)-O(1)}$ and 1, agree on D_k up to $K(k) + O(1)$ bits; if D_d has d-bit inputs and $d \mapsto D_d$ is computable, they agree up to $O(1)$. So the witnesses of Proposition 3.5 and Theorem 3.9 and Theorem 2.13(iii) are non-constructive by necessity; Theorem 2.13(iv) is an explicit family with a linear gap on average.

Open question. Is there a computable family $k \mapsto D_k$ along which some gap of Table 2 tends to infinity (necessarily no faster than $K(k)$)?

3.4 Summary and meaning for prediction

Remark 3.13 (What the gaps mean for prediction). [proved] foundations §4 "What the differences mean for prediction" (1) *Gaps are cumulative losses:* by Proposition 2.16, a gap of g bits between two of $W, M_{\mathrm{prod}}, M_F, M^w$ on D is a difference of g bits in total log loss on D in every order (semimeasure convention, up to the constants $\log W(\emptyset)$, $\log M_{\mathrm{prod}}(\emptyset)$); single steps can differ more or less. (2) *Random tape or not:* W predicts as well as SFI up to $K^{\mathrm{pad}}(L_{1/2}(M_F, D)) + O(1)$ bits in total, and up to $K(\lceil \log(d+1) \rceil) + O(1)$ bits on fair-coin labels however many are predicted. So whether Hänni's random program has a random tape matters little, unless SFI's explanation of the data uses a number of tape bits with no simple bound; likewise private versus shared randomness, as M_{prod} lies between W and M_F. (3) *The shortest program:* $K_F(D)$ is a description length (MDL with one hypothesis; the Occam learner of Proposition 4.15), not a predictor's loss. It can exceed SFI's loss by $\log d - O(1)$ bits on two points and by about $\log(k(d+2))$ on k incompressible binary labels; for deterministic targets it has SFI's learning rate (Proposition 4.15). (4) M^w can beat M_F by linearly many bits on data no random function produces (blocks of $\nu_\triangle$), but has no posterior over functions. (5) For a deterministic computable f all five are at least $2^{-K(f)-O(1)}$ on its data, so the basic learning bounds (Theorem 4.1(b), Corollary 4.4, Theorem 4.5(a)) hold for all. (6) Fresh label noise with repeated inputs separates M_{pq} from all set-based objects: finite against infinite loss (Propositions 4.6 and 4.7). (7) On a single query SFI is the conditional universal distribution, $2^{-K(y|x)\pm O(1)}$; it commits to where the output ends, unlike sequence induction on an undelimited output, and with an end marker the two agree up to constants (Proposition B.4).

Remark 3.14 (Answer to Hänni's request). [proved] foundations §4 "Answer to the question in Hänni's draft"; claims F16, F17 There are three objects, and no two of them agree up to a constant.

- The *single shortest program* K_F falls short of M_F by $\log d \pm O(\log \log d)$ bits at worst on two points with d-bit inputs (Corollary 3.10): one program must be chosen, and short programs cannot separate every pair of inputs, while a random tape separates any two at constant cost.
- The *program mixture* W falls short of M_F unboundedly (Proposition 3.5), but only by the cost of naming a padded tape length, since the 2^L programs storing L-bit tape prefixes together weigh $2^{-K(L)}$ (Proposition 3.6).

comparison	equal up to $O(1)$?	gap: lower bound (witness)	gap: upper bound	open
2^{-K_F} vs W	no (Thm. 3.9)	$\log d - K(\lceil \log(d+1) \rceil) - O(1)$ on some two-point binary D (non-computable witness)	$K(\operatorname{dom} D) + O(1)$ (Thm. 3.4); binary labels, k points: $\delta_D + \log(k(d+2)) + K(\lceil \log(d+2) \rceil, k) + O(1)$ (Prop. 3.11; a bound on the gap to M_F, hence to W)	rate in general
2^{-K_F} vs M_F	no	$\log d - O(1)$ on two points (Thm. 3.9)	two points: $\log d + O(\log \log d)$ (Cor. 3.10); general: $K(\operatorname{dom} D) + O(1)$	general rate
W vs M_{prod} (and vs M_F)	no (Prop. 3.5)	two points: $\log(1/\Omega_V^{\geq \ell_d}) - O(1)$; one point, $Y = \{0,1\}^*$: $\log(1/\Omega_V^{\geq d-1}) - O(1)$ (non-computable witnesses)	two points, binary: $K^{\text{pad}}(2^{d+1}) + O(1)$; against a machine law ν: $K(\nu) + K^{\text{pad}}(L_{1/2}(\nu, D)) + O(1)$ (Prop. 3.6); general: $K(\operatorname{dom} D) + O(1)$	§ 3.4; rates
M_{prod} vs M_F	no (Prop. 3.7)	unbounded, no rate (compactness witnesses)	ν_{len}-type data: $K^{\text{pad}}(d+1) + O(1)$ (Rem. 3.8); general: $\min\{K^{\text{pad}}(L_{1/2}(M_F, D)), K(\operatorname{dom} D)\} + O(1)$	lower rate
M_F vs M^w	no (Thm. 2.13)	$k \log \frac{3}{2} - K(k) - O(1)$ on some $2k$-point D; on average over an explicit family	$K(\operatorname{dom} D) + O(1)$	how close to $K(\operatorname{dom} D)$
M_F vs every lsc random-function law	unknown	—	$K(\operatorname{dom} D) + O(1)$	§ 2.3
M_{prod} vs M_{pq}	equal on sequences with distinct inputs	with repeats $M_{\text{pq}} \leq M_{\text{prod}}$; $M_{\text{pq}} > 0 = M_{\text{prod}}$ on inconsistent sequences (Prop. 4.7)	—	—

Table 2: The Kolmogorov and semimeasure versions compared (Hänni's later request). Data sets D with inputs of length at most d; gaps in bits (log of the ratio of the two quantities, or K_F plus log of the other). Every row is proved as cited; by Rem. 3.12 no computable family exhibits a gap larger than $K(k) + O(1)$ in any row. Source: foundations Table 4.12.

- There is *no coding theorem for function programs*: $-\log W$ and K_F differ unboundedly, not even within a constant factor (Theorem 3.9); the same holds for history-aware SFI (Remark B.5).

The coding theorem, which Hänni's draft names as the tool, gives agreement of all five versions up to $K(\operatorname{dom} D) + O(1)$ bits (Theorem 3.4), not up to a constant. For deterministic computable targets all have loss at most $K(f) + O(1)$ on the data of f.

Open question. Is $\log(M_F(D)/W(D)) = O(K(d, |D|))$ for inputs of length at most d, or even K^{pad} of a computable function of $(d, |D|)$? Proved: $\min\{K^{\text{pad}}(L_{1/2}(M_F, D)), K(\operatorname{dom} D)\} + O(1)$, and $K^{\text{pad}}(2^{d+1}) + O(1)$ against fair coins; $L_{1/2}$ is not computably bounded in $(d, |D|)$. Also open: the exact rate of the two-point gap of W (between $\log(1/\Omega_V^{\geq \ell_d})$ and $K^{\text{pad}}(2^{d+1})$), a lower rate for

M_F/M_{prod}, and how close $\log(M^w/M_F)$ comes to $K(\operatorname{dom} D)$.

4 Learning theorems: SFI as supervised learning

Hänni describes SFI as "just ideal supervised learning" (Hänni, 2026c). This section states the guarantees behind that description and where they stop. Throughout, $D_n = \{(x_t, y_t) : t \le n\}$ and $L_{\text{SFI}}(D_n) = -\log M_F(D_n)$ is SFI's cumulative log loss in the semimeasure convention, independent of order and repetitions and infinite on inconsistent sequences (Proposition 2.16); upper bounds in it also hold normalised (Proposition 2.17). We write μ for the input distribution, c for a machine constant, and $L_\nu(D) := -\log \nu(D)$. For an lsc conditional semimeasure P (Proposition 2.10(a)), $K(P)$ is the least $K(i)$ over its indices in the enumeration of Definition 3.1. For a partial computable φ, $K(\varphi)$ is the least $K(e)$ with $\varphi_e = \varphi$.

4.1 Dominance

Theorem 4.1 (dominance)**.** [proved] foundations Thm 6.1 *Let ν be a machine law (Definition 2.7). Then $M_F(D) \ge 2^{-K(\nu)}\nu(D)$ for every data set D. Hence for every consistent sequence, however its inputs are chosen (in advance, adversarially, adaptively, non-computably),*

$$L_{\text{SFI}}(D_n) \;\le\; -\log \nu(D_n) + K(\nu). \tag{2}$$

In particular: (a) x-only samplers: *for every lsc conditional semimeasure P, $L_{\text{SFI}}(D_n) \le \sum_{t\le n} -\log P(y_t \mid x_t) + K(P) + c$, where the right side counts a repeated input each time it appears and the left side once; (b)* deterministic programs: *if $y_t = \varphi(x_t)$ for all t, with φ partial computable, then $L_{\text{SFI}}(D_n) \le K(\varphi) + c$; (c)* weak comparators: *for every lsc weak semimeasure ν (Definition 2.2), $L_{\text{SFI}}(D_n) \le -\log \nu(D_n) + K(\nu) + K(\operatorname{dom} D_n) + c$. The term $K(\operatorname{dom} D_n)$ cannot be replaced by a constant (semimeasure convention): on the $2k$-point data sets of Theorem 2.13(iii), L_{SFI} exceeds $-\log M^w$ by at least $k \log \frac{3}{2} - K(k) - c$, while $K(\operatorname{dom} D) \le k \log 3 + K(k) + c$.*

Proof. Theorem 2.9 gives $M_F \ge 2^{-K(\nu)}\nu$, and Proposition 2.16 gives $L_{\text{SFI}}(D_n) = -\log M_F(D_n)$ in every order. (a) The product law $\nu_P(D) = \prod_{(x,y)\in D} P(y \mid x)$ is a machine law with $K(\nu_P) \le K(P) + c$ (Proposition 2.10(a)); it counts each input once, and the extra factors of repeats are at most 1. (b) Proposition 2.10(b). (c) Theorem 2.13(ii),(iii); each domain of that family is determined by k and a vector in $\{1, 2, 3\}^k$. □

Taking expectations gives the bound under every data distribution. Part (a) is Hänni's "Thm 1 (template)" for computably samplable x-only predictors (Hänni, 2026b): each "loses to 'the solomonoff inductor for C'" by "at most a constant (depending on P but not the distribution)". It holds in his setting of "input-output pairs from a function" (consistent sequences), with randomness read once per input, as his remark on "different bits of randomness ... for sampling on different inputs" suggests.

Remark 4.2 (scope of dominance). [proved; open part] foundations §6.1, F20–F23 (i) The comparators include all computable laws of random total functions, for every countable Y (Proposition 2.10 and Lemma A.2). (ii) The label sequence must be consistent: if an input reappears with another label, $L_{\text{SFI}} = \infty$, while a sampler with fresh randomness per *query* has finite loss. Randomness per input is the right reading for data from a function (Remark 2.11); per query it gives M_{pq} (Proposition 4.7). (iii) For general lsc laws of random partial functions the statement is open (Section 2.3).

4.2 Merging and mistakes

Theorem 4.3 (merging)**.** [proved; finite analogue computed (foundations c4, r6; integrability inequality c11(c))] foundations Thm 6.5 and remarks *Let ν be a machine law of a random total function, equivalently a computable law of a random total function (Lemma A.2), and $F \sim \nu$. Let the inputs follow any measurable rule $x_t = s_t(h_{t-1}, \xi_t)$, with h_{t-1} the history, (ξ_t) independent of F, and s_t arbitrary (possibly non-computable); $y_t := F(x_t)$. With the one-step conditionals of Definition 2.1 let*

$$d_t := \mathrm{KL}\left(\nu(\cdot \mid D_{t-1}, x_t) \,\|\, M_F(\cdot \mid D_{t-1}, x_t)\right) \text{ bits} \qquad (d_t = 0 \text{ at a repeated input}).$$

Then $\sum_{t \le n} \mathbb{E} d_t = \mathbb{E} \log(\nu(D_n)/M_F(D_n)) \le K(\nu)$ for every n. Hence $\sum_{t \ge 1} \mathbb{E} d_t \le K(\nu)$, $d_t \to 0$ almost surely, and $\sum_{t \ge 1} \mathbb{E} \|\nu(\cdot \mid D_{t-1}, x_t) - M_F(\cdot \mid D_{t-1}, x_t)\|_1^2 \le 2 \ln 2 \cdot K(\nu)$.

Proof idea. Given the past, y_t has law $\nu(\cdot \mid D_{t-1}, x_t)$, so the expected log ratio at step t is d_t; the ratios telescope to $\log(\nu(D_n)/M_F(D_n)) \le K(\nu)$ (Theorem 4.1; integrability in Appendix C.1).

This is the function analogue of Solomonoff's convergence theorem (Solomonoff, 1978) (source not checked). If the rule may see F, e.g. picks inputs by their label, the expected bound can fail (summed KL 12.5 bits against a bound of 1 in a finite analogue, foundations r6); (2) still holds, and Theorem 5.11 bounds the excess by directed information.

Corollary 4.4 (mistake bound)**.** [proved; known (Littlestone 1988; Barzdin–Freivalds 1972, content unverified)] foundations Cor 6.6 *Let φ be partial computable, $x_1, x_2, \ldots \in \operatorname{dom} \varphi$ any inputs, and $y_t = \varphi(x_t)$. At most $K(\varphi) + c$ steps have $M_F(\varphi(x_t) \mid D_{t-1}, x_t) \le \frac{1}{2}$. So $\hat{y}_t := \arg\max_y M_F(y \mid D_{t-1}, x_t)$ makes at most $K(\varphi) + c$ mistakes on every input sequence.*

Proof. Each such step adds at least one bit to $L_{\mathrm{SFI}}(D_n) \le K(\varphi) + c$ (Theorem 4.1(b)); above $\frac{1}{2}$, $\varphi(x_t)$ is the unique maximiser. □

This is the halving bound (Littlestone, 1988; Barzdin and Freivalds, 1972) with $K(\varphi)$ for $\log |H|$; the predictor is not computable (Proposition 2.24); examples: Proposition 6.22.

4.3 Online-to-batch and noise

Theorem 4.5 (online-to-batch)**.** [proved; finite analogues computed (foundations r9; c7, r7)] foundations Thm 6.7 *Let $x_1, x_2, \ldots$ be i.i.d. from any distribution μ on X, computable or not.*

(a) Deterministic labels. *Let φ be partial computable, $\mu(\operatorname{dom} \varphi) = 1$, $y_t = \varphi(x_t)$. Put $Q_t(\cdot \mid x) := M_F(\cdot \mid D_t, x)$, $\bar{Q}_n := \frac{1}{n} \sum_{t<n} Q_t$, $R(Q) := \mathbb{E}_{x \sim \mu}[-\log Q(\varphi(x) \mid x)]$. Then $\sum_{t<n} \mathbb{E} R(Q_t) \le K(\varphi) + c$, $\mathbb{E} R(\bar{Q}_n) \le (K(\varphi) + c)/n$, and the expected 0–1 risk $\mathbb{E}\, \mu\{x : \arg\max_y \bar{Q}_n(y \mid x) \ne \varphi(x)\}$ is at most $(K(\varphi) + c)/n$.*

(b) Noisy labels, stamped inputs. *Let $y_t \sim P(\cdot \mid x_t)$ independently given the inputs, for any conditional probability P (possibly non-computable). Run SFI on $\tilde{x}_t := \langle x_t, t \rangle$, for a computable bijection $\langle \cdot, \cdot \rangle : X \times \mathbb{N} \to X$, with data sets $\tilde{D}_t$; put $q_t(\cdot \mid x) := M_F(\cdot \mid \tilde{D}_{t-1}, \langle x, t \rangle)$ and $\bar{q}_n := \frac{1}{n} \sum_{t \le n} q_t$. Then for every lsc conditional semimeasure P',*

$$\mathbb{E}\, \mathbb{E}_{x \sim \mu} \mathrm{KL}\left(P(\cdot \mid x) \,\|\, \bar{q}_n(\cdot \mid x)\right) \le \tfrac{1}{n}\big(K(P') + c\big) + \mathbb{E}_{x \sim \mu} \mathrm{KL}\left(P(\cdot \mid x) \,\|\, P'(\cdot \mid x)\right).$$

For lsc (hence computable) P this is $(K(P) + c)/n$. The expected excess 0–1 risk of $\arg\max_y \bar{q}_n(y \mid x)$ over the Bayes classifier is at most $\sqrt{2 \ln 2}$ times the square root of the right side.

(c) Frozen noise. *If F has the product law of a computable conditional probability P (one draw per input) and $y_t = F(x_t)$, then $\sum_{t\le n} \mathbb{E} d_t \le K(P) + c$, with d_t as in Theorem 4.3, for every input rule that sees only the past.*

Part (a) is online-to-batch conversion (Littlestone, 1989; Cesa-Bianchi et al., 2004): as x_{t+1} is independent of D_t, $\mathbb{E}R(Q_t)$ is the expected online loss at step $t+1$, Theorem 4.1(b) bounds the sum, and convexity moves the average inside. Averaging over a random order of a finite set instead gives Proposition 4.10; Haussler et al. (1994) use a permutation argument of this kind (not checked). In (b) the stamps make inputs distinct; (c) is Theorem 4.3 for a product law (proofs: Appendix C.2). The bounds are for the Cesàro average $\bar{Q}_n$, not for Q_n (Section 4.4).

Proposition 4.6 (random-function SFI cannot model fresh label noise)**.** [proved; refuted (the brief's fresh-noise claim, as stated); computed (foundations c5, r9(b))]

foundations Prop 6.8; F31 *Let $\mu(x_0) = \alpha > 0$, let $P(\cdot \mid x_0)$ give positive probability to two labels, and $p_{\max} := \max_y P(y \mid x_0) < 1$. Draw the pairs (x_t, y_t) i.i.d. from $\mu \times P$, with a fresh label at every draw, and predict with $\nu(\cdot \mid D, x)$ for a weak semimeasure ν, raw or normalised, with the repeated-input convention of Definition 2.1. This covers M_F, W, M_{prod} and M^w. (a) For every $n \ge 2$, $\mathbb{P}[D_n \text{ inconsistent}] > 0$, so the expected cumulative log loss is infinite, in both conventions. (b) $\mathbb{P}[D_n \text{ consistent}] \le (1-\alpha)^n + p_{\max}^{-1}(1 - \alpha + \alpha p_{\max})^n \to 0$. (c) The statement "for every input distribution and every computable conditional probability P, SFI's expected cumulative log loss exceeds that of P by at most $K(P) + O(1)$" is false for random-function SFI. For P with finite expected log loss it holds for all n exactly when $P(\cdot \mid x)$ is a point mass for every x with $\mu(x) > 0$; then it is Theorem 4.5(a).*

(Proof: Appendix C.2.) Random-function SFI models the label as a function of the input; a fresh-noise source is not one. With 10 equally likely inputs and fair-coin labels the data stay consistent with probability 0.95, 0.113, $5 \cdot 10^{-4}$ at $n = 2, 10, 20$ (foundations c5). This refutes the fresh-noise form of the supervised-learning hypothesis H7 for random-function SFI (Table 1). The finite-loss proviso in (c) is ours (for infinite loss the inequality is trivial). The repairs are stamps and frozen noise (Theorem 4.5(b),(c); example: Proposition 6.27) and mixing per query:

Proposition 4.7 (the per-query mixture)**.** [proved; computed (foundations c7, r7)]

foundations Prop 6.9 *Let $M_{\mathrm{pq}}(y_{1:n} \| x_{1:n}) = \sum_i \mathbf{m}(i) \prod_{t\le n} P_i(y_t \mid x_t)$ on all finite sequences of pairs (Definition 3.1). (a) M_{pq} is an lsc chronological semimeasure: $M_{\mathrm{pq}}(\epsilon\|\epsilon) \le 1$ and $\sum_y M_{\mathrm{pq}}(y_{1:n}y \| x_{1:n}x) \le M_{\mathrm{pq}}(y_{1:n}\|x_{1:n})$. It depends only on the multiset of pairs; it equals $M_{\mathrm{prod}}(D_n)$ on sequences with distinct inputs, is at most $M_{\mathrm{prod}}(D_n)$ on consistent sequences with repeats (often strictly), and is positive on inconsistent sequences. (b) On every sequence, consistent or not, $-\log M_{\mathrm{pq}}(y_{1:n}\|x_{1:n}) \le \sum_{t\le n} -\log P(y_t \mid x_t) + K(P)$ for every lsc conditional semimeasure P. (c) Let (x_t, y_t) be i.i.d. from $\mu \times P$, μ arbitrary and P any conditional probability, $q_t(\cdot \mid x) := M_{\mathrm{pq}}(y_{<t} \cdot \| x_{<t}x)/M_{\mathrm{pq}}(y_{<t}\|x_{<t})$ and $\bar{q}_n := \frac{1}{n}\sum_{t\le n} q_t$. For every lsc conditional semimeasure P', $\sum_{t\le n} \mathbb{E}\,\mathbb{E}_\mu \mathrm{KL}(P\|q_t) \le K(P') + n\,\mathbb{E}_\mu \mathrm{KL}(P\|P')$, so $\mathbb{E}\,\mathbb{E}_\mu \mathrm{KL}(P\|\bar{q}_n) \le K(P')/n + \mathbb{E}_\mu \mathrm{KL}(P\|P')$. (d) With the stamps of Theorem 4.5(b), $M_F(\tilde{D}_n) \ge 2^{-c} M_{\mathrm{pq}}(y_{1:n}\|x_{1:n})$.*

So the regret bound holds under fresh noise for M_{pq}, with no stamps (proofs: Appendix C.3). As secondary sources describe it, operator induction is M_{pq} (Remark 3.2); $\mathrm{SFI}^{\mathrm{hist}}$ dominates it (Proposition D.12); Proposition 8.4 is a time-bounded analogue.

4.4 Learning curves and in-distribution error

Proposition 4.8 (the expected risk of a Bayes predictor need not decrease)**.** [proved; computed (foundations c11(a), r12)] foundations Prop 6.10 *Let $X = \{a, b\}$, μ uniform, target $\varphi \equiv 0$, and a Bayes mixture of three total functions: φ with prior weight w, and $(1, 0)$ and $(0, 1)$ with weight $\frac{1-w}{2}$ each. The expected log risk r_t of the posterior predictive after t i.i.d. samples is $r_0 = -\log \frac{1+w}{2}$, $r_1 = \frac{1}{2} \log \frac{1+w}{2w}$ and $r_t = 2^{1-t} r_1$ for $t \geq 1$. So $r_1 > r_0$ whenever $(1 + w)^3 > 8w$, in particular for all $w \leq \frac{1}{5}$; for $w = \frac{1}{16}$, $r_0 = 0.912$ and $r_1 = 1.544$ bits. The sum $\sum_t r_t$ increases to $\log(1/w)$.*

The mixture is a random total function, so both conventions agree, but it is not M_F: "more data never hurts in expectation" fails for Bayes predictors with a fixed target; for M_F this is not established. 63 of 300 random finite instances had non-monotone curves (foundations r12); see also Remark C.2.

Remark 4.9 (last iterate and leave-one-out). [identity proved; computed (foundations c11(b)); bound open] foundations Rem 6.11, §8 item 7 In the setting of Theorem 4.5(a), exchangeability gives $\mathbb{E}R(Q_n) = \mathbb{E}\big[\frac{1}{n+1} \sum_{i \leq n+1} - \log Q^{(-i)}(\varphi(x_i) \mid x_i)\big]$, where $Q^{(-i)}$ is SFI's predictive given the data set of the samples $j \neq i$ (a held-out input that occurs among the others costs 0). Whether $\mathbb{E}R(Q_n) \leq (K(\varphi) + c)/(n + 1)$ is thus a leave-one-out question (cf. Conjecture 4.12); it is open. In 300 random finite instances the largest $(n + 1)r_n/K$ was 0.873 (foundations r12).

Errors. For $\nu \in \{W, M_F\}$, a target f, training inputs A with $D = f|_A$, and a test input x^*: $e(x^*) := 1 - \nu(f(x^*) \mid D, x^*)$ is the posterior mass wrong or undefined at x^* (0 if $x^* \in A$); $e^{\mathrm{wr}}(x^*) := \sum_{y \neq f(x^*)} \nu(D + (x^*, y))/\nu(D)$ is the mass defined and wrong; $e_{\mathrm{norm}}(x^*)$ is the error of the normalised predictor. Since the normaliser is at most $\nu(D)$, $e^{\mathrm{wr}} \leq e_{\mathrm{norm}} \leq e$: lower bounds on e^{wr} and upper bounds on e hold in both conventions.

Proposition 4.10 (random splits)**.** [proved; technique known; computed (context c3)] context Prop 2.9 *Let $\nu \in \{W, M_F\}$, B any finite input set (possibly adversarial), f any labelling of B. Train on the first $k - 1$ inputs of a uniformly random order of B and test on the k-th, with k uniform on $\{1, \ldots, |B|\}$ and independent. The expected test loss is $-\log(\nu(f|_B)/\nu(\emptyset))/|B| \leq L_\nu(f|_B)/|B| \leq (K_F(f|_B) + c)/|B|$, with $c = 0$ for W, and the expected test error is at most $(\ln 2) L_\nu(f|_B)/|B|$.*

Proposition 4.11 (leave-one-out)**.** [proved; computed (context c7)]
context Prop 2.10 *For $\nu \in \{W, M_F\}$, a labelling f of a finite B, and $e_{\mathrm{LOO}}(x) := 1 - \nu(f|_B)/\nu(f|_{B \setminus \{x\}})$: $\sum_{x \in B} e_{\mathrm{LOO}}(x) \leq (\nu(\emptyset) - \nu(f|_B))/\nu(f|_B) < 2^{L_\nu(f|_B)} \leq 2^{K_F(f|_B) + c}$.*

Proof ideas (Appendix C.4). Splits: in every order the step losses sum to the order-free total. Leave-one-out: the hypotheses correct on $B \setminus \{x\}$ and not at x form disjoint sets for different x.

Conjecture 4.12 (leave-one-out for SFI)**.** [conjecture]
context Conj 2.11; context §7 item 1 *For SFI, $\frac{1}{|B|} \sum_{x \in B} e_{\mathrm{LOO}}(x) = O(K_F(f|_B)/|B|)$, up to logarithmic factors.*

For general prior weights Proposition 4.11 is tight within a factor of about 2 (Remark C.2; heuristic in Appendix C.7).

Proposition 4.13 (separating properties are typical even in distribution)**.** [proved; computed (context c4)] context Prop 2.12 *Let $f \equiv 0$ on $\{0, 1\}^d$, $x_1, \ldots, x_n$ i.i.d. uniform training inputs and x^* an independent uniform test input, $d \geq \ell := \lceil \log n \rceil + 1$. Let G_u be the set of strings that start with the ℓ-bit prefix u of x^*. Then (1) G_u separates x^* from the training inputs with probability $(1 - 2^{-\ell})^n \geq \frac{1}{2}$; (2) $K(G_u) \leq \ell + 2 \log(\ell + 1) + c$; (3) for $\nu \in \{W, M_F\}$,*

$\mathbb{E}\, e^{\mathrm{wr}}(x^*) \geq \frac{1}{2} \cdot 2^{-\ell-2\log(\ell+1)-c} \geq 2^{-c}/(8n(\log n+3)^2)$, *in both conventions. For computable binary* f *the same argument gives* $\mathbb{E}\, e^{\mathrm{wr}}(x^*) \geq 2^{-K(f)-c}/(8n(\log n+3)^2)$.

Proof idea. On the event of (1), answering 1 on G_u and running a training-consistent program elsewhere costs $K(G_u)+c$ extra bits (a case of Proposition 7.1; Appendix C.5). So "no simple property separates test from train" typically holds only at scales of about $\log n$ bits. Reading this as "matching the Cesàro bound up to $\log^2 n$" is refuted (Remark C.3). Per step:

Corollary 4.14 (per-step in-distribution error)**.** [proved; computed (context c7)] context Cor 2.17 *Let* μ *be any distribution on* X*,* f *partial computable with* $\mu(\mathrm{dom}\, f)=1$*,* $x_1,\dots,x_n,x^*$ *i.i.d. from* μ*,* $\nu \in \{W, M_F\}$*, and* e *the semimeasure-convention error (so the bounds hold normalised too). (a) For every* $\varepsilon \in (0,1)$*,* $\mathbb{E}e(x^*) \leq \varepsilon + 2^{K(f)+c}e^{-\varepsilon n}$*; hence* $\mathbb{E}e(x^*) \leq ((K(f)+c)\ln 2 + \ln n + 1)/n$*. With probability at least* $1-\delta$ *over the training sample,* $\mathbb{E}_{x^*}[e(x^*) \mid D_n] \leq \varepsilon + 2^{K(f)+c}e^{-\varepsilon n}/\delta$*. The posterior-majority predictor has expected 0–1 risk at most* $2\mathbb{E}e(x^*)$*. (b)* $\mathbb{E}e(x^*) \leq \mathbb{E}\big[\frac{\nu(\emptyset)-\nu(f|_S)}{(n+1)\nu(f|_S)}\big] \leq 2^{K(f)+c}/(n+1)$*, where* S *is the set of the* $n+1$ *sampled values. For* $f \equiv 0$ *and uniform inputs on* $\{0,1\}^d$*,* $d \geq \lceil \log n \rceil + 1$*:* $2^{-c}/(8n(\log n+3)^2) \leq \mathbb{E}e(x_{n+1}) \leq 2^c/(n+1)$ *in both conventions, so the per-step error is* $\Theta(1/n)$ *up to the* $\log^2 n$ *factor.*

Proof idea (Appendix C.5). (a) A hypothesis with error above ε survives with probability at most $e^{-\varepsilon n}$; the normaliser is at least $2^{-K(f)-c}$. (b) By exchangeability the error is an average of leave-one-out errors. Part (a) is the Markov half of the slides' relevance-based bound (Vaintrob and Hänni, 2026b) for W and M_F (Remark C.4).

4.5 Comparison, and the sense of "ideal"

Proposition 4.15 (Occam bound in description-length form)**.** [proved (standard); references known, unverified] foundations §6.6; F32 *Let* x_t *be i.i.d. from* μ*,* φ *partial computable with* $\mu(\mathrm{dom}\, \varphi)=1$*,* $y_t = \varphi(x_t)$*. With probability at least* $1-\delta$*, every program* $p \in \mathrm{dom}\, V$ *consistent with* D_n *has* $\mu\{x : \varphi_{V(p)}(x) \neq \varphi(x)\} \leq (|p|\ln 2 + \ln(1/\delta))/n$*, where "*$\neq$*" includes "undefined". So the shortest consistent program, of length* $K_F(D_n) \leq K(\varphi)$*, has 0–1 risk at most* $(K(\varphi)\ln 2 + \ln(1/\delta))/n$.

So for deterministic targets SFI (Theorem 4.5(a)) and the Kolmogorov version learn at the same rate, though their code lengths can differ (Theorem 3.9). Appendix C.7 compares with Occam, PAC-Bayes, resolvability and expert-advice bounds (Blumer et al., 1987; McAllester, 1999; Barron and Cover, 1991; Vovk, 1990; Cesa-Bianchi and Lugosi, 2006).

Remark 4.16 (in what sense SFI is ideal supervised learning). [remark (each item cited)] foundations §6.7 Established: (1) SFI is the Bayes predictor for the prior "a random lower-semicomputable labeller", machine laws weighted by $\mathbf{m}(i) \geq 2^{-K(i)}$ (Theorem 2.9), and it is universal for that class pathwise (Theorem 4.1). (2) With i.i.d. inputs from any distribution and computable labels, the averaged predictor has expected log and 0–1 risk at most $(K(\varphi)+c)/n$; with frozen noise, or fresh noise on stamped inputs or for M_{pq}, the cumulative bound $K(P)+c$ holds (Theorem 4.5 and Proposition 4.7). This is the Occam rate with K for $\log|H|$, without a union bound or confidence parameter. (3) Inputs are never predicted, so the input distribution may be arbitrary at no cost (chosen inputs: Proposition C.1; positive examples and interventions: Remarks C.5 and C.6). Limits: *loss*: the guarantees are for log loss on labels, and 0–1 bounds follow (K/n without noise, $\sqrt{K/n}$ excess with noise); *averaged predictor*: Proposition 4.8 and Remark 4.9; *function semantics*: Proposition 4.6; *computability*: Proposition 2.24, and Section 8 for computable versions; *constants*: machine constants, and $K(\alpha,\beta)$ for encodings (Theorem 2.20);

distribution shift: Section 7; *comparator class*: simple history-aware predictors have loss 0 where SFI's loss is unbounded (Theorems 5.3, 5.4 and 6.3; the swap data of Example 2.22 do not show this), and $\mathrm{SFI}^{\mathrm{hist}}$ removes this limit (Corollary 5.18).

Remark 4.17 (prior work on the framing). [known; unverified] foundations §6.6 last item; history §4.6 item 2 A table-of-contents snippet shows a section "Supervised Learning from Examples (EX)" in Hutter (2007), and the abstracts of Hutter (2000) and of that chapter list supervised learning among the AIXI model's problem classes; Hutter (2005) is the book-length treatment. We have not read these sections; the universal mixture as ideal learner from examples is prior work in substance.

5 History-aware prediction

SFI sees the past only as a set: its hypotheses answer each query without seeing the other inputs, and its loss $L_{\mathrm{SFI}}(D_n)$ in the semimeasure convention ignores the order of the data (Proposition 2.16). Hänni's notes (Hänni, 2026b) give two examples where seeing the history helps without bound, and a dominance theorem ("Thm 1") for simple reveal orders; his request conjectures that SFI whose hypotheses see the list of previous inputs "just becomes as good as SI" (Hänni, 2026c). We prove the examples, decide Thm 1 in three readings (some cases stay open), and prove the conjecture in a precise form. Lower bounds on L_{SFI} are in the semimeasure convention unless stated; c_ω depends on the machines and on a Martin-Löf random ω. Proofs and some secondary statements are in Appendix D.

5.1 History-aware predictors

Definition 5.1 (History-aware predictors)**.** history §1.3, Lemma 1.0; foundations Prop 7.2 A *history-aware predictor* P gives sub-probabilities $P(y \mid h, x)$; its loss is $L_P := -\log P(y_{1:n} \| x_{1:n})$, $P(y_{1:n} \| x_{1:n}) := \prod_{i \le n} P(y_i \mid h_{i-1}, x_i)$. A *computable sampler* is a probabilistic machine that, given (h, x), reads random bits and halts with a label or diverges (*defective* if with positive probability); $P(y \mid h, x)$ is its probability of output y, lsc in the arguments. An *input sampler* (reveal law) Δ outputs an input given h. A *chronological semimeasure* is a map $\rho(y_{1:n} \| x_{1:n}) \ge 0$ with $\rho(\epsilon \| \epsilon) \le 1$ and $\sum_y \rho(y_{1:n} y \| x_{1:n} x) \le \rho(y_{1:n} \| x_{1:n})$ for all x; $L_\rho := -\log \rho(y_{1:n} \| x_{1:n})$.

These are Hutter's chronological semimeasures, with inputs as his actions and labels as his percepts (Hutter, 2000, 2005). Hänni's class C is the computable samplers; one that halts with probability 1 has computable probabilities ($1 - \sum_{y' \ne y} P(y' \mid h, x)$ is upper semicomputable).

Lemma 5.2 (SFI is a chronological semimeasure)**.** [proved] history Lemma 1.2; foundations Prop 7.2(b),(c) *$\rho_F(y_{1:n} \| x_{1:n}) := \mathbb{U}\{\omega : f_\omega(x_i) = y_i \ \forall i \le n\}$ is an lsc chronological semimeasure, equal to $M_F(D_n)$ on consistent sequences and 0 otherwise. Hence the loss L_H of $\mathrm{SFI}^{\mathrm{hist}}$ (Definition 5.16) satisfies $L_H \le L_{\mathrm{SFI}}(D_n) + c$ in every order. The lsc chronological semimeasures that depend only on D_n and vanish on inconsistent sequences are exactly the lsc weak function semimeasures.*

5.2 Hänni's counterexamples

Hänni's notes reveal "a deterministic sequence of increasingly long random strings" as $(x_1, 0), (x_2, x_1), \ldots$ and write: "I'm pretty sure Solomonoff induction gets unbounded loss on this sequence. I don't actually have a proof"; a "cleaner construction" gives "0 to everything except one string you've skipped among all strings of a certain length" (Hänni, 2026b). Both claims hold. (We take ϵ as the chain's first label.)

Theorem 5.3 (Hänni's chain)**.** [proved] history Thm 2.1 *Let $x_1, \dots, x_n$ ($n \geq 2$) be distinct nonempty strings, $D_n := \{(x_1, \epsilon)\} \cup \{(x_{i+1}, x_i) : i < n\}$, revealed as $(x_1, \epsilon), (x_2, x_1), \dots$. (a) $K(x_1, \dots, x_{n-1} \mid x_n) - c \leq L_{\mathrm{SFI}}(D_n) \leq K(x_1, \dots, x_{n-1}) + c$ (semimeasure convention). (b) If $1 \leq \ell_1 < \ell_2 < \cdots$ are computable, x_i is the i-th block, of length ℓ_i, of a Martin-Löf random ω, and $\bar{\ell}_k := \sum_{i \leq k} \ell_i$, then $\bar{\ell}_{n-1} - K(\ell_n) - c_\omega \leq L_{\mathrm{SFI}}(D_n) \leq \bar{\ell}_{n-1} + K(n) + c$ (semimeasure convention); for $\ell_i = i$, $L_{\mathrm{SFI}}(D_n) = n(n-1)/2 \pm (2 \log n + c_\omega)$. (c) The predictor "$\epsilon$ on the empty history; otherwise the unique input of the history that is not one of its labels" has loss 0; it is deterministic, set-based and of complexity $O(1)$.*

Proof idea. Iterating f_ω from x_n recovers $x_{n-1}, \dots, x_1$, so the chain events are disjoint (Lemma D.1); apply the conditional coding theorem.

For the second example fix $s_\ell \in \{0,1\}^\ell$ for each ℓ and let $f_s(x) := \mathbf{1}[x = s_{|x|}]$. Block ℓ reveals the other strings of length ℓ in lexicographic order, then s_ℓ; $D^{(\ell)}$ is the data after blocks $1, \dots, \ell$.

Theorem 5.4 (Hänni's skipped strings)**.** [proved]
history Thm 2.2 *(a) "Predict 1 at x iff every other string of length $|x|$ is already an input" has loss 0; it is deterministic, set-based and of complexity $O(1)$. (b) $K(s_1, \dots, s_\ell \mid \ell) - c \leq L_{\mathrm{SFI}}(D^{(\ell)}) \leq \ell(\ell+1)/2 + c$ (semimeasure convention). (c) If $s_1 s_2 \cdots$ are the consecutive blocks of a Martin-Löf random ω, then $L_{\mathrm{SFI}}(D^{(\ell)}) \geq \ell(\ell+1)/2 - K(\ell) - c_\omega$; for each ℓ, all but a 2^{-k} fraction of tuples $(s_1, \dots, s_\ell)$ give $L_{\mathrm{SFI}}(D^{(\ell)}) \geq \ell(\ell+1)/2 - k - c$ (semimeasure convention).*

The lower bound is the domain bound of Theorem 2.21(a). In the length-lexicographic order $\mathrm{SFI}^{\mathrm{hist}}$ pays as much as SFI (Corollary 5.25), in Hänni's order $O(1)$: the order carries the information.

Proposition 5.5 (Every order-free predictor pays)**.** [proved] history Prop 2.3 *Let ν be a weak function semimeasure (Definition 2.2), e.g. M^w, M_{prod} or W. (a) For every x, $\sum_z \nu(D(x,z)) \leq 1$ over all chains $D(x,z) := \{(x, z_1), (z_1, z_2), \dots, (z_k, \epsilon)\}$ ($k \geq 0$) of distinct nonempty strings $z_i \neq x$. (b) If ν is lsc, the lower bounds of Theorems 5.3 and 5.4 hold for $-\log \nu$ with $K(\nu)$ subtracted. (c) For every ν, with independent uniform blocks, $\mathbb{E}[-\log \nu(D_n)] \geq \bar{\ell}_{n-1}$ and $\mathbb{E}[-\log \nu(D^{(\ell)})] \geq \ell(\ell+1)/2$.*

So every predictor that sees the past as a set *and* is order-independent pays the cost (Hänni's predictors are not order-independent). The normalised $\bar{M}_F$ is not order-free, but its normaliser is at least $2^{-K(B)-c}$ times the probability that a total machine B extends the data (Lemma D.2). For the chain, the *broom* B_j reads blocks $v_1, \dots, v_j$ of lengths $\ell_1, \dots, \ell_j$, sets $f(v_1) := \epsilon$, $f(v_i) := v_{i-1}$, and sends every other input to v_j; it extends D_{j+1} with probability $2^{-\bar{\ell}_j}$ at a cost of $K(j) + c$ bits (as in Proposition 6.6).

Proposition 5.6 (Normalised SFI on both examples)**.** [proved; finite proxy computed]
history Prop 2.4, Lemma 2.4a; check c9 *(a) In the setting of Theorem 5.3(b), $-\log \bar{M}_F(y_t \mid D_{t-1}, x_t) \geq \ell_{t-1} - K(\ell_t) - K(t-2) - c_\omega$ for $t \geq 2$; for $\ell_i = i$, $\bar{L}_{\mathrm{SFI}} \geq n(n-1)/2 - n(4 \log n + c_\omega)$. With independent uniform blocks the expected normalised loss at step t is at least $\ell_{t-1} - K(t-2) - c$, so $\mathbb{E}\bar{L}_{\mathrm{SFI}} \geq n(n-1)/2 - n(2 \log n + c)$ for $\ell_i = i$. (b) In the setting of Theorem 5.4(c), the normalised loss at s_ℓ is at least $\ell - 2K(\ell) - c_\omega$, so $\bar{L}_{\mathrm{SFI}} \geq \ell(\ell+1)/2 - \ell(4 \log \ell + c_\omega)$ after block $\ell \geq 2$; with independent uniform s_j, at least $\ell - K(\ell) - c$ in expectation.*

The method needs more information per step than the $O(\log t)$ bits of the broom, so it gives nothing for Proposition 5.19 and Theorem 5.26.

5.3 Hänni's Theorem 1

Hänni's Thm 1: if f is drawn and the inputs are drawn from Δ, "each predictor P in C can only beat solomonoff function induction in expected log loss by at most the complexity C_Δ of sampling from the reveal sequence distribution Δ plus a constant C_P", in the case where "the reveal sequence distribution is simple to sample from conditional on any particular choice of function f"; a preceding remark says it fails if one puts "the randomization inside the reveal sequence distribution" of the chain (Hänni, 2026b).

Definition 5.7 (Three readings)**.** history §3 Let $y_i = f(x_i)$. *Reading A*: one reveal law $\Delta(x_i \mid h_{i-1})$ that does not see f (private randomness and memory allowed), f arbitrary. *Reading C*: for each f a reveal law $\Delta_f(x \mid h)$ that sees only the history, with $C_\Delta := K(\Delta_f)$ as a standalone sampler; A is the case $\Delta_f = \Delta$. *Reading B*: $x_i \sim \Delta(\cdot \mid f, h_{i-1})$, with C_Δ the complexity of Δ relative to an oracle for f.

Hänni's text fixes Reading C: his example of failure has $C_\Delta = O(1)$ in Reading B but is excluded in Reading C (Theorem 5.12), and his sketch simulates Δ in a function machine, which cannot see f. Let $I(f \to x_{1:n}) := \sum_{i \le n} I(f; x_i \mid h_{i-1})$ (*directed information*). A reveal law that does not see f may keep private memory; it enters only through its *reveal-sequence law* $\Delta(x_{1:n} \| y_{1:n-1})$, and $I(f \to x) = 0$ (Lemma D.3).

Proposition 5.8 (The proof sketch fails)**.** [computed; proved] history Prop 3.1; check c1 *Let $X = \{a, b\}$, $f(a) = f(b) = \beta$ for a uniform bit β, and let Δ show a, then b. The predictor with y_1 uniform and $y_2 = y_1$ has loss 1 bit; the simulator of the proof sketch, run with separate randomness for each input, has loss 2 bits.*

The sketch "internally simulates" Δ and P and then uses "jensen's inequality" (Hänni, 2026b). A simulator with private randomness per input loses the cross-input correlations that P uses, and the Jensen step averages over *simulated* histories, whose labels come from P, not f. (SFI itself pays at most $1 + c$ bits here.) One simulation shared by all inputs works, at the price of the reveal sequence's log loss:

Theorem 5.9 (Shared-randomness simulator)**.** [proved] history Thm 3.2 *For computable samplers Δ (inputs) and P (labels), possibly defective and with private memory, and every consistent sequence, $M_F(D_n) \ge 2^{-K(\Delta,P)-c} \Delta(x_{1:n} \| y_{1:n-1}) P(y_{1:n} \| x_{1:n})$; so $L_{\mathrm{SFI}}(D_n) \le L_P + L_\Delta + K(\Delta) + K(P) + c$ with $L_\Delta := -\log \Delta(x_{1:n} \| y_{1:n-1})$. For distinct inputs the product may be summed over all orders of D_n.*

Theorem 5.10 (Bayes form)**.** [proved] history Thm 3.5; checks c2, r1 *Let $f \sim \nu$, a computable random function (a machine law of a random total function, Lemma A.2), and let the inputs see f only through revealed labels: $x_i = \mathrm{sel}_i(\Gamma, h_{i-1})$, arbitrary maps, Γ independent of f. Then the Bayes predictor $\nu(y \mid D_h, x)$ (D_h the data of h; Definition 2.1) is the true conditional law, with loss $-\log \nu(D_n)$, and $L_{\mathrm{SFI}}(D_n) \le -\log \nu(D_n) + K(\nu) + c$ on every path. So $\mathbb{E} L_{\mathrm{SFI}}(D_n) \le \mathbb{E} L_P + K(\nu) + c$ for every predictor P, computable or not.*

Theorem 5.11 (Directed information)**.** [proved; identity computed] history Thm 3.6; checks c2, r1 *Let $f \sim \nu$, let the inputs have any joint law with f, $y_i = f(x_i)$, finite alphabets up to time n, and $L_{\mathbb{P}} := -\sum_{i \le n} \log \mathbb{P}(y_i \mid h_{i-1}, x_i)$. (a) For any law ν, $\mathbb{E}[-\log \nu(D_n)] - \mathbb{E} L_{\mathbb{P}} = I(f \to x_{1:n}) - \mathbb{E}\,\mathrm{KL}(\mathbb{P}(f \in \cdot \mid h_n) \| \nu(f \in \cdot \mid f \supseteq D_n)) \le I(f \to x_{1:n})$. (b) For computable ν and every P, $\mathbb{E} L_{\mathrm{SFI}}(D_n) \le \mathbb{E} L_P + I(f \to x_{1:n}) + K(\nu) + c$.*

Theorem 5.12 (Hänni's randomized chain; Reading B is false)**.** [proved; refutes Reading B] history Thm 3.7; check r1 *Let x_i be independent and uniform on $\{0,1\}^i$, $f(x_{i+1}) := x_i$ and $f := \epsilon$ elsewhere; reveal $x_1, x_2, \ldots$. Then f is a computable random function with $K(\nu) = O(1)$; an $O(1)$ program computes the reveal sequence from an oracle for f (x_i is the unique string of length $i \geq 2$ with $f(x_i) \neq \epsilon$, and $x_1 = f(x_2)$); the chain predictor has loss 0; $I(f \to x_{1:n}) = n(n+1)/2$; and $\mathbb{E}L_{\mathrm{SFI}}(D_n) \geq n(n-1)/2$, $\mathbb{E}\bar{L}_{\mathrm{SFI}} \geq n(n-1)/2 - n(2\log n + c)$.*

This is Hänni's own example, and his remark is correct. It refutes only Reading B: in Reading C, $K(\Delta_f) \geq n(n+1)/2 - k - c$ for all but a 2^{-k} fraction of chains. His question whether Thm 1 "can be improved to a version that talks about the algorithmic mutual information of Δ and f" thus has a negative answer in Reading B.

For arbitrary f assume (A1) Y is finite and (A2) a computable map $j \mapsto A_j$ to finite, listed subsets of X, of complexity $K(A)$, has all inputs up to step n in A_n. Let $C_{\mathrm{dir}}(\Delta, n) := \sup_\nu I_\nu(f \to x_{1:n})$ over laws ν on Y^{A_n} (0 if Δ does not see f).

Theorem 5.13 (Minimax theorem)**.** [(a) proved; (b) proved for reveal laws that do not see f, proof sketch for those that do; computed] history Thm 3.8; checks c3, c8, r2 *Fix n, assume (A1), (A2), and let the reveal law see the history (with private memory) and possibly $f|_{A_n}$. Assume that $P(y \mid h, x)$ and the reveal-sequence law have (a) rational values computable from their arguments, or (b) computable real values (this includes all computable samplers that halt with probability 1; Lemma D.3). Then for every $f : A_n \to Y$, and on average over any law of f independent of the reveal law's private randomness, $\mathbb{E}_{x|f} L_{\mathrm{SFI}}(D_n) \leq \mathbb{E}_{x|f} L_P + C_{\mathrm{dir}}(\Delta, n) + K(P) + K(\Delta) + K(n) + K(A) + c$.*

Proof idea. Sion's theorem (Sion, 1958) and Theorem 5.11(a) give a law on Y^{A_n} within $C_{\mathrm{dir}}+2$ bits of P for every f, found by a search computable from (P, Δ, n, A). For reveal laws that see f, case (b) needs C_{dir} to within 1, not written out.

Corollary 5.14 (Hänni's Theorem 1 in Reading C)**.** [proved under the stated hypotheses] history Cor 3.9 *(a) Under (A1), (A2), with P and every Δ_f of computable real values (e.g. samplers halting with probability 1, private memory allowed): $\mathbb{E}_{x|f} L_{\mathrm{SFI}}(D_n) \leq \mathbb{E}_{x|f} L_P + K(P) + K(\Delta_f) + K(n) + K(A) + c$ for every f; for random f, with $\mathbb{E}_f K(\Delta_f)$. (b) If $x_i = \delta_f(h_{i-1})$ with δ_f computable: for every f, sampler P (defective allowed) and n, $L_{\mathrm{SFI}}(D_n) \leq L_P + K(\delta_f) + K(P) + c$ on every path (Corollary D.4). (c) For all samplers P, Δ_f, with no finiteness assumption, $\mathbb{E}_{x|f} L_{\mathrm{SFI}} \leq \mathbb{E}_{x|f} L_P + H_f(x_{1:n}) + K(\Delta_f) + K(P) + c$, H_f the entropy of the reveal sequence given f.*

So Thm 1 is true in Hänni's reading, with an extra $K(n) + K(A)$ for random reveal laws; only the proof sketch fails. Open cases are listed in Remark D.6; no pathwise version with $K(\Delta)$ alone holds (Proposition D.7). The Shannon form of his mutual-information question holds: if the reveal law is indexed by a random Θ and does not see f, then $I(f \to x) \leq I(f; \Theta)$, with equality in the randomized chain (Proposition D.8). The algorithmic form is untested:

Conjecture 5.15 (Algorithmic form in Reading C)**.** [conjecture (untested)] history Conj 3.11; Q8 *With $I(a : b) := K(b) - K(b \mid a)$ and the hypotheses of Corollary 5.14(a): $\mathbb{E}_{x|f} L_{\mathrm{SFI}} \leq \mathbb{E}_{x|f} L_P + I(\Delta_f : f|_{A_n}) + K(P) + O(\log K(\Delta_f)) + K(n) + K(A) + c$.*

5.4 SFI with access to the previous inputs

Hänni's request: "if you further give the function oracle access to a list of previous x (but not previous y), then it just becomes as good as SI" (Hänni, 2026c).

Definition 5.16 ($\mathrm{SFI}^{\mathrm{hist}}$)**.**
history §4.1 Let X_H be the set of nonempty finite input lists. A *history-oracle function machine* is a function machine T on X_H (Definition 2.7), queried at time i at $x_{1:i}$; it never sees labels but can recompute its own. Its law $M_T^{\mathrm{hist}}(y_{1:n}\|x_{1:n}) := \mathbb{U}\{\omega : T^\omega(x_{1:i}) = y_i \ \forall i \le n\}$ ($n \ge 1$; 1 at the root) is an lsc chronological semimeasure (Lemma D.9). $\mathrm{SFI}^{\mathrm{hist}}$ is SFI over X_H: its law is $M_H := M^{\mathrm{hist}}_{T_U^H}$, T_U^H the universal function machine for X_H (Theorem 2.9), and its loss is $L_H := -\log M_H(y_{1:n}\|x_{1:n})$. We *define* the universal chronological semimeasure as M_H.

Theorem 5.17 (Representation)**.** [proved; construction computed] history Thm 4.2; checks c4, r5 *For every lsc chronological semimeasure ρ there is a history-oracle function machine T_ρ with $M^{\mathrm{hist}}_{T_\rho} = \rho$ at every non-root node and $K(T_\rho) \le K(\rho) + c$.*

The proof allocates dyadic intervals as Zvonkin and Levin (1970) do for sequences. For sequences, Wood et al. (2013) state that machine-based and semimeasure-based universal priors form the same class (abstract seen only); we have not checked whether Theorem 5.17 appears in this form in Hutter's work, and claim no novelty for it.

Corollary 5.18 (Universality of $\mathrm{SFI}^{\mathrm{hist}}$)**.** [proved; identification known (unverified)]
history Cor 4.3; foundations Prop 7.2(a),(d) *(a) $M_H \ge 2^{-K(\rho)-c}\rho$ for every lsc chronological semimeasure ρ; so M_H is a universal element of this class and agrees with every other one up to constant factors. (b) For every lsc history-aware predictor P (e.g. every computable sampler, defective or not) and* every *sequence, consistent or not: $L_H \le L_P + K(P) + c$. (c) $L_H \le L_{\mathrm{SFI}}(D_n) + c$ on every consistent sequence, in every order.*

So Hänni's conjecture holds in this sense. A trimming construction as in Theorem 2.12 gives the same law up to constants. Agreement with Hutter's universal chronological semimeasure (Hutter, 2000, 2005) is known, but not checked against the sources (he uses finite alphabets), and is not used; his "supervised learning from examples" is prior work in substance (Remark 4.17). Oracles that see the previous pairs, the time i, or the set of previous inputs define $\mathrm{SFI}^{\mathrm{pairs}}$, $\mathrm{SFI}^{\mathrm{count}}$ (SFI on time-stamped inputs) and $\mathrm{SFI}^{\mathrm{set}}$, with laws $M_{\mathrm{pairs}}, M_{\mathrm{count}}, M_{\mathrm{set}}$ and losses $L_{\mathrm{pairs}}, L_{\mathrm{count}}, L_{\mathrm{set}}$; on distinct inputs $\mathrm{SFI} \preceq \mathrm{SFI}^{\mathrm{count}} \preceq \mathrm{SFI}^{\mathrm{set}} \preceq \mathrm{SFI}^{\mathrm{hist}}$ up to constant factors, and M_{pairs} agrees with M_H up to constant factors (Proposition D.10).

Proposition 5.19 (The hierarchy is strict)**.** [proved; lemma computed] history Prop 4.5; checks c6, r4 *Semimeasure convention. (i) Every set S of $n \ge 2$ strings has an order such that, with labels $y_i = \mathbf{1}[i \text{ even}]$, $L_{\mathrm{SFI}} \ge n - \log(n+1) - c$ while $L_{\mathrm{count}} \le c$. (ii) On the chain of Theorem 5.3(b), $L_{\mathrm{count}} \ge \bar{\ell}_{n-1} - K(\ell_n) - c_\omega$ while $L_{\mathrm{set}} \le c$. (iii) Every such S has an order such that, with labels $y_1 = \epsilon$, $y_i = x_{i-1}$, $L_{\mathrm{set}} \ge \log n! - \log n - c$ while $L_H \le c$. Single infinite sequences with linearly growing gaps also exist (Proposition D.11).*

In (iii) a tape determines the order backwards from its last input. So the oracle must give the list, not the set (also Proposition 6.8). Operator induction, read as M_{pq}, ignores the history: $\mathrm{SFI}^{\mathrm{hist}}$ dominates it, and on distinct inputs it pays the lower bounds of Theorems 5.3 and 5.4 (Proposition D.12).

For deterministic programs on X_H, the program mixture W_{hist} pays at most $K^{\mathrm{tail}}(n) := -\log \sum_{N \ge n} 2^{-K(N)}$ extra bits against fair-coin labels, and $K^{\mathrm{tail}}(n) \le g(n) + K(g) + c$ for every computable nondecreasing unbounded g, while the single shortest program pays more than $\log n + \log\log n$ extra bits infinitely often (Proposition D.13): as for SFI (Theorem 3.9), the Kolmogorov and mixture versions differ unboundedly.

5.5 As good as Solomonoff induction?

Let M be the universal monotone semimeasure (Proposition 2.23), $\mathrm{code}(h_n) := \bar{x}_1\bar{y}_1 \cdots \bar{x}_n\bar{y}_n$ with $\bar{s} := 1^{|s|}0s$, and $L_M^{\mathrm{joint}} := -\log M(\mathrm{code}(h_n))$; SFI's loss is at most $L_M^{\mathrm{joint}} + c$ (Corollary D.5). Let L_M^x be the sum over i of $-\log$ of M's conditional probability of $\bar{x}_i$ given $\mathrm{code}(h_{i-1})$ (the loss on the inputs), and $L_M^y := L_M^{\mathrm{joint}} - L_M^x$, Solomonoff induction's unnormalised conditional label loss. Let M_H^{in} be the universal chronological semimeasure for inputs given previous labels (roles exchanged) and $L_{\mathrm{in}}^x := -\log M_H^{\mathrm{in}}(x_{1:n} \| y_{1:n-1})$.

Theorem 5.20 (Solomonoff induction on the joint sequence)**.** [proved] history Thm 4.6, Lemma B *On every sequence: (a) $L_H \le L_M^{\mathrm{joint}} + c$ and $L_{\mathrm{in}}^x \le L_M^{\mathrm{joint}} + c$; (b) $L_M^{\mathrm{joint}} \le L_H + L_{\mathrm{in}}^x + c$; (c) $L_H - L_M^y \le L_M^x + c$ and $L_M^y - L_H \le L_{\mathrm{in}}^x - L_M^x + c$; (d) if $x_i = \delta(h_{i-1})$ for a partial computable δ, then $L_M^{\mathrm{joint}} \le L_H + K(\delta) + c$. This holds for every prefix-free block code with nonempty codewords.*

Proposition 5.21 (Input-loss identity)**.** [proved; toy computed] history Prop 4.7; check s3 *For a partial computable δ and every sequence with $x_i = \delta(h_{i-1})$ for all i: $L_M^x - K(\delta) - c \le L_H - L_M^y \le L_M^x + c$.*

So for deterministic input policies, "$L_H \le L_M^y + O(1)$" holds iff M predicts the inputs with bounded loss. Lattimore et al. (2011) state that "the unnormalised version can fail to predict very simple recursive sub-patterns", such as each odd bit repeated at the next even position (abstract seen only). With $X = Y = \{0,1\}$ and the identity code, let (LHG-even) say that for some ω the loss of unnormalised M on positions $2, 4, \ldots$ of $\omega_1\omega_1\omega_2\omega_2\cdots$ is unbounded, and (LHG-odd) the same on positions $3, 5, \ldots$ of $0\omega_1\omega_1\omega_2\omega_2\cdots$.

Proposition 5.22 (The conditional label loss)**.** [conditional (on LHG transfer); LHG known, unverified beyond the abstract] history §4.4, Q3 *With $X = Y = \{0,1\}$ and the identity code: under (LHG-even), $L_M^y \le L_H + O(1)$ fails on some sequence (labels copy inputs); under (LHG-odd), $L_H \le L_M^y + O(1)$ fails on some sequence (inputs copy the previous label, Proposition 5.21). For the code $\bar{s} = 1^{|s|}0s$ the hypotheses must also transfer to framed blocks.*

This does not contradict Corollary 5.18: M's conditionals need not be lsc. For normalised M both directions are open.

Remark 5.23 (Verdict on Hänni's conjecture). [remark (cited items)] history §7 "H4" With the list of previous inputs SFI becomes the universal chronological semimeasure (Corollary 5.18); the set is not enough (Proposition 5.19(iii)). "As good as SI" holds against Solomonoff induction's joint loss (Theorem 5.20(a)); against its conditional label loss the difference is M's loss on the inputs (Proposition 5.21), and for unnormalised M it fails both ways, conditionally on Proposition 5.22; the normalised case is open.

5.6 What SFI can and cannot exploit

For D with distinct inputs let D_σ be D in order σ, x_σ its input list, and $K^{\mathrm{pred}}(x_{1:n}) := \min\{|p| : V(p, x_i) = \langle x_1, \ldots, x_{i-1}\rangle\ \forall i \le n\}$, the complexity of recovering each input's predecessors from the input.

Theorem 5.24 (Sandwich)**.** [proved]
history Thm 5.2 (with Prop 5.1) *For every data set D: $\max_\sigma L_H(D_\sigma) - c \le L_{\mathrm{SFI}}(D) \le \min_{\sigma'}[L_H(D_{\sigma'}) + K^{\mathrm{pred}}(x_{\sigma'})] + c$.*

Corollary 5.25 (Initial segments)**.** [proved] history Cor 5.3 *If $\mathrm{dom}\, D = \{e_1, \ldots, e_k\}$ for a computable injective sequence e, then $|L_{\mathrm{SFI}}(D) - L_H(D \text{ in } e\text{-order})| \le K(e) + c$, and the e-order is the worst for $\mathrm{SFI}^{\mathrm{hist}}$ up to $K(e) + c$.*

This covers the skipped strings and the full cyclic permutation ($K(e) \le K(d) + c$). The K^{pred} term can be needed beyond order information: identify $\{0,1\}^q$ with $\mathrm{GF}(2^q)$ (polynomial basis modulo the lexicographically least irreducible polynomial) and let $D_{abr} := \{(a, r), (b, r + ab)\}$, $a \neq b$.

Theorem 5.26 (A $\mathrm{GF}(2^q)$ example)**.** [proved; computed] history Thm 5.4, Lemma 5.5; checks c5, r3, s4 *(a) In both orders $L_H(D_{abr}) \le q + c$, with c independent of q. (b) $L_{\mathrm{SFI}}(D_{abr}) \ge K(a, b, r \mid q) - 3q/2 - 1 - c$; so for all but a 2^{1-k} fraction of triples, $L_{\mathrm{SFI}}(D_{abr}) \ge 3q/2 - k - c \ge \max_\sigma L_H(D_\sigma) + q/2 - k - c'$. (c) For uniform (a, b, r), $a \neq b$, $\mathbb{E}L_{\mathrm{SFI}}(D_{abr}) \ge 3q/2 - 2$ for every $q \ge 1$. These lower bounds are in the semimeasure convention; for $\bar{L}_{\mathrm{SFI}}$ they are open.*

$\mathrm{SFI}^{\mathrm{hist}}$ predicts $y_2 = y_1 + x_1 x_2$; SFI's hypotheses never see the first input. The key step is a character-sum bound (Lemma D.15). A direct computation (code in the accompanying repository) gives the optimum over *all* laws of random functions, $\min_w \mathbb{E}[-\log w(D_{abr})] = 2.708$ bits for $q = 2$ and 4.348 for $q = 3$, against q for $\mathrm{SFI}^{\mathrm{hist}}$.

Theorem 5.27 (Expected gap)**.** [proved]
history Thm 5.6 *Let $f \sim \nu$ computable and the inputs drawn by any kernel $\Delta(\cdot \mid f, h_{i-1})$, with finite alphabets up to time n. Then $-c \le \mathbb{E}L_{\mathrm{SFI}}(D_n) - \mathbb{E}L_H \le I(f \to x_{1:n}) + K(\nu) + c$, and the gap is at most $K(\nu) + c$ if the input choice does not see f.*

The upper bound is attained up to n in the randomized chain, but not in general: the GF triple revealed in random order has $I(f \to x) \approx 2q$ but a gap of at most $q + c$ (and fails Reading B: q bits for $\mathrm{SFI}^{\mathrm{hist}}$, $\mathbb{E}L_{\mathrm{SFI}} \ge 3q/2 - 2$).

On individual sequences, $L_{\mathrm{SFI}}(D) \le L_H(D_\sigma) + I(x_\sigma : D) + c$, with $I(a : b) := K(b) - K(b \mid a)$; the natural strengthening with $I(x_\sigma : y_\sigma)$ in place of $I(x_\sigma : D)$ is false, by a margin linear in n (Proposition D.16). Informally, shared randomness lets a label of SFI correlate with a *guess* about the other inputs, not with the inputs present (Remark D.17).

Open questions. (i) The open cases of Reading C (Remark D.6). (ii) Do (LHG-even), (LHG-odd) hold? How does L_H compare with normalised M's label loss? (iii) Normalised SFI on Propositions 5.19 and D.11 and Theorem 5.26. (iv) Is $L_M^{\text{joint}} = L_H + L_{\text{in}}^x \pm O(1)$ ("$\le$" is proved)? (v) The growth of $\max_G N(G)$ in Lemma D.15 (local search suggests about $2^{1.35q}$; an observation only). (vi) Conjecture 5.15. (vii) Which individual-sequence quantity measures $L_{\mathrm{SFI}}(D) - L_H(D_\sigma)$ tightly? (viii) The exact rate in Proposition D.13(d).

6 Worked examples

Hänni (2026c) proposed two examples in which SFI "does much worse than SI": a random cyclic permutation revealed backwards along its cycle, and a coin-flip example due to Sam Eisenstat, which Hänni reported with an argument from his e-mail to Eisenstat. We work both out and add examples in which SFI does well (his out-of-distribution examples are in Section 7). Losses are in the semimeasure convention, $L_{\mathrm{SFI}}(D_n) = -\log M_F(D_n)$ in every order (Proposition 2.16), unless stated otherwise; upper bounds also hold for the normalised predictor $\bar{M}_F$, lower bounds only where we say so (Remark 2.18). c is a machine constant, $j \ge 0$ a confidence parameter, and a finite function is identified with its graph. Proofs are in Appendix E.

6.1 Tools

Lower bounds come from the coding theorem given the domain, $L_{\mathrm{SFI}}(D) \geq K(D|_T \mid T) - c$ for $T \subseteq \operatorname{dom} D$ (Theorem 2.21(a)), and from a counting fact:

Lemma 6.1 (complexity is at least surprisal)**.** [proved]
examples Lemma 1.2 *Let Z take values in a countable set, with law P (computable or not), and let w be fixed. For every $j \geq 0$, $\mathbb{P}[K(Z \mid w) < -\log P(Z) - j] < 2^{-j}$, and $\mathbb{E}K(Z \mid w) \geq H(Z)$.*

Proof. The probability is $\sum_z P(z)$ over the z with $P(z) < 2^{-K(z|w)-j}$, hence less than $\sum_z 2^{-K(z|w)-j} \leq 2^{-j}$. The expectation bound is Gibbs' inequality for the sub-probability $2^{-K(\cdot|w)}$. □

SI. SI sees the history coded as $\mathrm{code}(h_t) = \bar{x}_1\bar{y}_1 \cdots \bar{x}_t\bar{y}_t$ with $\bar{s} := 1^{|s|}0s$ (Theorem 5.20). For $\rho \in \{M, M_{\mathrm{norm}}\}$, with $M_{\mathrm{norm}} \geq M$ the normalised Solomonoff measure, SI's label loss is

$$L^y_\rho(k) := \textstyle\sum_{t \leq k} -\log\big[\rho(\mathrm{code}(h_{t-1})\bar{x}_t\bar{y}_t)/\rho(\mathrm{code}(h_{t-1})\bar{x}_t)\big],$$

so $L^y_M = L^y_M$. If $\rho(\mathrm{code}(h)) \geq 2^{-k}\mathbb{P}(h)$ for the true law $\mathbb{P}$ of the pairs, L^y_ρ exceeds the Bayes-optimal history-aware label loss by at most k in expectation, and by at most $k + j$ at all times with probability at least $1 - 2^{-j}$, even if the inputs are incompressible (Lemma E.1).

6.2 The cyclic permutation

Definition 6.2 (the cyclic permutation)**.** examples §2.1; Hänni's request Fix $d \geq 1$, $N := 2^d$ and a cyclic permutation f of $\{0,1\}^d$ (a single N-cycle). Put $x_0 := 0^d$ and $x_k := f^{-1}(x_{k-1})$ for $1 \leq k \leq N$. The k-th revealed pair is (x_k, x_{k-1}), and $D_k := \{(x_i, x_{i-1}) : i \leq k\}$. We write $(a)_k := a(a-1)\cdots(a-k+1)$, $(a)_0 := 1$.

Hänni expected that this "will have very low loss under SI but high loss under SFI" (Hänni, 2026c). The map $f \mapsto (x_1, \ldots, x_{N-1})$ is a bijection onto the orderings of the nonzero strings (its inverse closes the cycle $0^d \mapsto x_{N-1} \mapsto \cdots \mapsto x_1 \mapsto 0^d$), so for uniform f the ordering is uniform. For $k < N$, D_k is a chain $x_k \to \cdots \to x_1 \to 0^d$ of distinct nonzero strings. The labels add nothing to the inputs ($y_k = x_{k-1}$): all of f is in the order.

Theorem 6.3 (the full cycle)**.** [proved; computed (examples e3, e6(d))]
examples Thm 2.1 *Semimeasure convention. (a) For every total $f : \{0,1\}^d \to \{0,1\}^d$, $|L_{\mathrm{SFI}}(f) - K(f \mid d)| \leq c$. (b) For every cyclic permutation f of $\{0,1\}^d$, $L_{\mathrm{SFI}}(f) \leq \log((N-1)!) + c$. (c) For uniform cyclic f, $\mathbb{P}[L_{\mathrm{SFI}}(f) < \log((N-1)!) - j - c] < 2^{-j}$, and $\mathbb{E}L_{\mathrm{SFI}}(f) \geq \log((N-1)!) - c$. (d) $\log((N-1)!) = (d - \log e)N - \frac{d}{2} + \frac{1}{2}\log(2\pi) + \epsilon_N$ with $\frac{1}{(12N+1)\ln 2} < \epsilon_N < \frac{1}{12N \ln 2}$. The same bounds hold for $L_{\mathrm{SFI}}(D_{N-1})$.*

So SFI pays $d - \log e \approx d - 1.443$ bits per pair over the cycle (Table 3). In (a) one function machine reads a single program for the whole table from the shared tape; in (b) one machine draws a uniform cyclic permutation of each length; (d) is Stirling's formula with Robbins' remainder bounds (Robbins, 1955).

Theorem 6.4 (k steps)**.** [proved; computed (examples e6(a),(b),(c))]
examples Thm 2.3 *Semimeasure convention; $1 \leq k \leq N-1$. (a) On every chain, $L_{\mathrm{SFI}}(D_k) \leq \log(N-1)_{k-1} + \min\{K(k), \log(N-k)\} + c$. (b) For uniform cyclic f, $\mathbb{P}[L_{\mathrm{SFI}}(D_k) < \log(N-1)_{k-1} - j - c] < 2^{-j}$. (c) For uniform cyclic f, with probability at least $1-2^{-j}$, $L_{\mathrm{SFI}}(D_k) \geq \log(N-1)_{k-1} - d - j - c$ for all $k < N$ at once.*

So $L_{\mathrm{SFI}}(D_k) = \log(N-1)_{k-1} \pm O(K(k))$ with high probability: SFI pays about $\log(N-t+1)$ bits for the label at step $t \geq 2$, and the newest input is free up to $K(k)$. The upper bound uses a *broom* B_k, a function machine that draws, for each length ℓ, a uniform injective sequence $(v_1, \ldots, v_{k-1})$ of nonzero ℓ-bit strings and maps $v_1 \mapsto 0^\ell$, $v_i \mapsto v_{i-1}$, and every other input to v_{k-1}; it extends D_k with probability $1/(N-1)_{k-1}$. It randomises the upper-bound program of Hänni's chain example (Theorem 5.3). The lower bounds rest on a depth count (Lemma E.3). The cost is intrinsic to order-free prediction:

Proposition 6.5 (order-free predictors)**.** [proved; sharper weak case open]
examples Prop 2.4 (general lemma: history Prop 2.3) *Let f be uniform cyclic and $k < N$. (a) For every law ν of a random partial function, computable or not (so every machine law, with shared or private randomness), $\mathbb{E}[-\log \nu(D_k)] \geq \log(N-1)_{k-1}$. (b) For every weak function semimeasure ν (every order-independent predictor, Proposition 2.3), $\mathbb{E}[-\log \nu(D_k)] \geq \log(N-1)_k - \log(N-1)$ and $\mathbb{E}[-\log \nu(f)] \geq \log((N-1)!)$.*

For $k \leq N/2$ the two bounds differ by at most one bit. So no random-function law, and for $k \leq N/2$ no order-independent predictor, does better than SFI by more than $O(K(k))$ bits in expectation.

Open question. Does the depth count, $\sum_{x,z} \nu(D(x,z)) \leq N-k$ in the notation of Lemma E.2, and hence (a), hold for every weak ν? A linear program confirms it for $N = 4$ only.

The normalised predictor is not order-free, and a set-based history-aware predictor has loss 0 here (Remark E.7), so information-theoretic arguments alone cannot bound it below. The structure of M_F can:

Proposition 6.6 (normalised SFI)**.** [proved; proxy computed (examples e6(e))]
examples Prop 2.9 *Let f be uniform cyclic and $\bar{\ell}_t := -\log \bar{M}_F(y_t \mid D_{t-1}, x_t)$. For $2 \leq t \leq N-1$, $\mathbb{E}\bar{\ell}_t \geq \log(N-t+1) - K(t-1) - c$, and $\bar{\ell}_t \geq \log(N-t+1) - K(t-1) - j - c$ with probability at least $1-2^{-j}$. Hence $\mathbb{E}\sum_{t \leq k} \bar{\ell}_t \geq (k-1)(d - 1 - \log k - 2\log\log(k+1) - c)$ for $k \leq N/2$, which is at least $(k-1)(\delta d - O(\log d))$ for $k \leq 2^{(1-\delta)d}$.*

Proof idea. $\bar{\ell}_t = L_{\mathrm{SFI}}(D_t) + \log \mathcal{Z}(D_{t-1}, x_t)$, and $\mathbb{E}L_{\mathrm{SFI}}(D_t) \geq \log(N-1)_{t-1}$ by Proposition 6.5(a). The broom B_{t-1} is total, consistent with the past, and sends x_t to the wrong label x_{t-2}, so it keeps the normaliser at least $2^{-K(t-1)-c}/(N-1)_{t-2}$ (Lemma D.2), as in Proposition 5.6. Over the whole cycle these bounds sum to at most about $2N$ bits (Section 6.3).

Proposition 6.7 ($\mathrm{SFI}^{\mathrm{hist}}$)**.** [proved, given Corollary 5.18] examples Prop 2.5 (= history Prop 4.8) *For every d, every cyclic f and every $k \leq N$, $L_H \leq c$ on $(x_1, x_0), \ldots, (x_k, x_{k-1})$, with c independent of d.*

Proof. "Answer $0^{|x|}$ first, then the previous input" is computable and has loss 0; apply Corollary 5.18. □

For Hänni's conjecture that seeing the previous inputs makes SFI "as good as SI" (Section 5.4), the hypotheses must see the *list* of previous inputs; their set is not enough:

Proposition 6.8 ($\mathrm{SFI}^{\mathrm{set}}$)**.** [proved]
examples Prop 2.11 *Let L_{set} be the loss of $\mathrm{SFI}^{\mathrm{set}}$, whose hypotheses see the set of previous inputs (Proposition D.10). For uniform cyclic f on the full cycle, $\mathbb{P}[L_{\mathrm{set}} < \log((N-1)!) - j - c] < 2^{-j}$, $\mathbb{E}L_{\mathrm{set}} \geq \log((N-1)!)$, and always $L_{\mathrm{set}} \leq \log((N-1)!) + c$.*

Proposition 6.9 (SI)**.** [proved; computed (examples e6(f), an illustration)]
examples Prop 2.10 *Let $\rho \in \{M, M_{\mathrm{norm}}\}$ and f uniform cyclic. There is c, independent of d, with $\mathbb{E}L^y_\rho(k) \le c$ for every $k \le N$ and $\mathbb{P}[\exists k \le N :\ L^y_\rho(k) > j + c] \le 2^{-j}$.*

SI pays about $K(d)$ bits on the header 1^d0 of the first input, which tells it d; then the labels are free. With raw d-bit blocks the bound becomes $K(d) + c$ (Proposition E.4), probably the right order (Conjecture E.5). So Hänni's description holds: SFI pays about $(d - \log e)2^d$ bits over the cycle (semimeasure convention) and at least $\delta d - O(\log d)$ bits per step for $t \le 2^{(1-\delta)d}$ (normalised), while SI and $\mathrm{SFI}^{\mathrm{hist}}$ pay $O(1)$, uniformly in d.

Remark E.7 locates the information and gives a separation in the other direction.

6.3 Eisenstat's example

Definition 6.10 (Eisenstat's example)**.** examples §3.1; Hänni's request Let $b_0, b_1, \dots$ be i.i.d. Bern($\frac{1}{3}$) (index i is a *head* if $b_i = 1$, else a *tail*) and $\zeta_1, \zeta_2, \dots$ i.i.d. Bern($\frac{2}{3}$), independent of b. At step t, reveal the smallest unrevealed head i as $(i, 1)$ if $\zeta_t = 1$, else the smallest unrevealed tail as $(i, 0)$. After n steps: N^1_n, N^0_n count the revealed heads and tails; τ_n (η_n) is one more than the last revealed tail (head), or 0; g_n is the smallest unrevealed index; and $UB_n := -\log P_{1/3}(b_{<\tau_n}) = \tau_n \log 3 - N^0_n$, with $P_{1/3}$ the Bern($\frac{1}{3}$) product measure.

Hänni's e-mail claims that SFI's "log loss ... is the loss given by our threshold function, up to $o(n)$ terms" (Hänni, 2026c). We take as threshold function the *threshold law* ν^{thr}_s: the product law with $\mathbb{P}[y = 1 \mid i] = \frac{1}{3}$ for $i < s$ and $= 1$ for $i \ge s$, a machine law with index computable from s (Proposition 2.10(a)).

Lemma 6.11 (shape of the revealed set)**.** [proved] examples Lemma 3.1 *The revealed tails (heads) are exactly the tails (heads) below τ_n (η_n). No unrevealed index lies below $\min(\tau_n, \eta_n)$; an unrevealed index below η_n is a tail, one below τ_n a head. If $\tau_n \le \eta_n$, then $g_n \ge \tau_n$ and $[\tau_n, g_n)$ consists of revealed heads.*

Proof. The process reveals the first unrevealed head (tail), so the revealed heads (tails) are the first N^1_n heads (N^0_n tails) of b. □

So the revealed set is about $[0, n/2)$ together with the heads in $[n/2, 2n)$.

Proposition 6.12 (upper bound)**.** [proved]
examples Prop 3.2 *For every b, every reveal sequence of this kind (each choice of head or tail may be arbitrary) and every n: $L_{\mathrm{SFI}}(D_n) \le UB_n + K(\tau_n) + c \le UB_n + \log(\tau_n + 1) + 2\log\log(\tau_n + 2) + c$, also for $\bar{L}_{\mathrm{SFI}}$.*

Indeed $M_F \ge 2^{-K(\tau_n)-c}\nu^{\mathrm{thr}}_{\tau_n}$ (Theorem 2.9); under $\nu^{\mathrm{thr}}_{\tau_n}$ the revealed indices at or above τ_n are heads (Lemma 6.11) and cost nothing, and those below τ_n cost at most UB_n; finally $K(i) \le \log(i + 1) + 2\log\log(i + 2) + c$.

Theorem 6.13 (pathwise sandwich)**.** [proved] examples Thm 3.3 *Semimeasure convention. For every b, every reveal sequence and every n, $L_{\mathrm{SFI}}(D_n) \ge K(b_{<g_n} \mid g_n) - c$. If b is Martin-Löf random for $P_{1/3}$, then at every n with $\tau_n \le \eta_n$,*

$$UB_n - K(g_n) - c_b \ \le\ L_{\mathrm{SFI}}(D_n) \ \le\ UB_n + K(\tau_n) + c.$$

In Eisenstat's process, almost surely $|L_{\mathrm{SFI}}(D_n) - UB_n| \le \log n + O(\log\log n) + c_b$ for all large n.

Proof idea. $[0, g_n)$ is revealed with its true labels; monotonicity and the coding theorem given the domain give the first bound, and the Levin–Schnorr theorem turns it into the surprisal $-\log P_{1/3}(b_{<g_n}) \geq UB_n$. Almost surely the heads are eventually ahead (Lemma E.8).

Proposition 6.14 (the threshold loss)**.** [proved; computed (examples e1, e2, e7(d))]
examples Prop 3.4 *Let* $a := \frac{3}{2}\log 3 - 1$ *and* $\sigma^2 := \frac{2}{9}a^2 + \frac{1}{4}(\log 3)^2 = 1.0496602$. *For every* n, $\mathbb{E}UB_n = \frac{H(1/3)}{2}n = 0.4591479\,n$ *and* $\mathrm{Var}(UB_n) = \sigma^2 n$ *exactly;* $UB_n/n \to \frac{H(1/3)}{2}$ *almost surely, and* $(UB_n - \frac{H(1/3)}{2}n)/\sqrt{n} \to \mathcal{N}(0, \sigma^2)$ *in law.*

Theorem 6.15 (expected loss)**.** [proved; computed (examples e2)]
examples Thm 3.6 *Semimeasure convention. For all* $n \geq 2$, $\frac{H(1/3)}{2}n - \log n - c \leq \mathbb{E}L_{\mathrm{SFI}}(D_n) \leq \frac{H(1/3)}{2}n + \log n + 2\log\log n + c$, *and the upper bound holds for* $\mathbb{E}\bar{L}_{\mathrm{SFI}}$. *The product law with* $\mathbb{P}[y = 1] = \frac{1}{3}$ *at every input pays* $(\log 3 - \frac{1}{3})n = 1.2516\,n$.

Proposition 6.16 (exact law; order-free predictors)**.** [proved; computed (examples e1, s2(d))] examples Prop 3.7 *Let* D *be a possible value of* D_n, $S := \operatorname{dom} D$, $s^+ := 1 + \max S$, N^1 *the number of* 1*-labels of* D, *and* $r := \frac{2}{3}$ *if the label at* $\max S$ *is* 1, $r := \frac{1}{3}$ *otherwise. (a)* $\mathbb{P}[D_n = D] = \binom{n}{N^1}(\frac{2}{9})^n r^{s^+ - n}$; *so the labelling given* $\operatorname{dom} D_n = S$ *depends on the label at* $\max S$, *not only on* N^1. *(b) Every weak function semimeasure* ν, *computable or not, has* $\mathbb{E}[-\log \nu(D_n)] \geq \frac{H(1/3)}{2}n - \log n - O(1)$.

So SFI is optimal among order-independent predictors up to $O(\log n)$ in expectation; as in the cyclic example, the gap to SI is a property of order-free prediction.

Remark 6.17 (Hänni's e-mail argument). [proved; normalised open] examples §3.6 The claim (quoted in Appendix E.3) is correct in the semimeasure convention, almost surely with $O(\log n)$ in place of $o(n)$ and the threshold at the tail frontier (Theorem 6.13). Its "easy direction" is Proposition 6.12. The "other direction" is correct in substance, with two repairs: the "pointing trick" is the coding theorem given the domain of the fully revealed segment $[0, g_n)$, which costs $K(g_n) = O(\log n)$ bits; and incompressibility must be measured by the empirical surprisal UB_n, not by $\tau_n H(1/3)$, which differs from it by $\Theta(\sqrt{n})$ (enough for $o(n)$, not for $O(\log n)$). For the normalised predictor the lower bound is open.

History-aware prediction and SI. Call a history *touching* if its revealed set is the initial segment $[0, \max(\tau, \eta))$. The Bayes-optimal history-aware predictor P^* is explicit (Lemma E.9): at a touching history it gives the input $\max(\tau, \eta) + k$ the label 1 with probability $2^k/(2^k + 1)$; at every other history it is certain and correct.

Proposition 6.18 (the burn-in)**.** [proved; computed (examples e1(e), e7(a),(b))]
examples Prop 3.9 *The number* N_{touch} *of touching steps in the infinite run has* $\mathbb{P}[N_{\mathrm{touch}} = k] = (\frac{2}{3})^{k-1}\frac{1}{3}$. *Each costs* P^* *an expected* $\vartheta = \sum_{k \geq 0} \frac{2}{9}[(\frac{2}{3})^k \log(1 + 2^{-k}) + (\frac{1}{3})^k \log(1 + 2^k)] = 0.7995167$ *bits, so* $\mathbb{E}L_{P^*}(n) \leq \mathbb{E}L_{P^*}(\infty) = 3\vartheta = 2.39855$ *bits.*

Theorem 6.19 ($\mathrm{SFI}^{\mathrm{hist}}$ and SI)**.** [proved] examples Thm 3.10 *(a) For every* b, *every coin sequence and every* n, $L_H(n) \leq L_{P^*}(n) + c$; *so* $\mathbb{E}L_H(n) \leq 3\vartheta + c$. *(b) Code indices as strings and labels as one-bit strings. For* $\rho \in \{M, M_{\mathrm{norm}}\}$, $\mathbb{E}L^y_\rho(n) \leq 3\vartheta + c$ *for all* n, *and* $\mathbb{P}[\exists n : L^y_\rho(n) > L_{P^*}(n) + j + c] \leq 2^{-j}$.

Proof. (a) Corollary 5.18 for the computable P^*. (b) The pair process has a computable law $\mathbb{P}$ whose label conditionals are those of P^*, so $M(\mathrm{code}(h)) \geq 2^{-c}\mathbb{P}(h)$ (Lemma D.14); apply Lemma E.1 and Proposition 6.18. □

A natural rule reads the labels off the history: *the new input is a tail if and only if it is the smallest unrevealed index.*

Corollary 6.20 (the rule)**.** [proved] examples Cor 3.11 *Almost surely there is a last touching step, and the rule is exact at every later step.*

Proposition 6.21 (errors of the rule)**.** [proved; computed (examples e2, e7)] examples Prop 3.12 *Over the infinite run the rule errs twice in expectation, and it makes no error with probability exactly* $\frac{1}{2}$*.*

In finite proxies (Table 4) order-free mixtures pay about 0.46 bits per step, history-aware and joint ones a bounded total.

Open question. Is the normalised loss of the same order as L_{SFI} over the whole cycle of Definition 6.2, and in Eisenstat's example? The method of Proposition 6.6 pays $O(\log t)$ bits per step, more than the information per step near the end of the cycle and in Eisenstat's example (0.46 bits), and set-based history-aware predictors have loss $O(1)$ in both. In finite mixtures with most prior on partial chains the full-cycle normalised loss is far below $\log((N-1)!)$ (19.2 against 40.25 bits at $d = 4$).

6.4 Positive examples

The bounds below hold in both conventions.

Proposition 6.22 (deterministic targets)**.** [proved] examples Prop 4.1 (cf. Corollary 4.4) *Let f be partial computable, $x_1, x_2, \cdots \in \operatorname{dom} f$ any inputs (adversarial, adaptive or non-computable; repeats allowed), $y_t = f(x_t)$. Then $L_{\mathrm{SFI}}(D_n) \le K(f) + c$, and at most $K(f) + c$ steps give $f(x_t)$ predictive probability at most $\frac{1}{2}$, raw or normalised.*

Proof. Theorem 4.1(b), Corollary 4.4, and $\mathcal{Z}(D, x) \le M_F(D)$. □

Example 6.23 (sparse parity)**.** [proved; computed (examples e4(a))]
examples Ex 4.2 For $S \subseteq \{0, \dots, d-1\}$ with $|S| = k$ and $\chi_S(x) := \bigoplus_{i \in S} x_i$ (bits beyond $|x|$ read as 0): $L_{\mathrm{SFI}} \le \log \binom{d}{k} + K(d) + K(k) + c$ on every input sequence. If S is uniform and the inputs form a fixed sequence whose labels determine S (e.g. containing the unit vectors), every learner, history-aware or not, has expected loss at least $\log \binom{d}{k}$.

Example 6.24 (thresholds)**.** [proved; computed (examples e4(b))] examples Ex 4.3 For $f_\theta(x) := \mathbf{1}[x \ge \theta]$ on $\mathbb{N}$, $L_{\mathrm{SFI}}(D) \le -\log \sum_{\theta' : f_{\theta'} \supseteq D} 2^{-K(\theta')} + c \le K(\theta) + c$. For θ uniform on $[0, 2^\ell)$, ℓ bisection queries cost every learner at least ℓ bits in expectation, and SFI at most $\ell + K(\ell) + c$.

Example 6.25 (sorting)**.** [proved] examples Ex 4.4 Sorting has complexity $O(1)$, so over any number of pairs (list, sorted list), in any order, SFI's total loss and its number of steps that give the sorted list probability at most $\frac{1}{2}$ are bounded by a constant.

Example 6.26 (integer polynomials)**.** [proved] examples Ex 4.5 For $g(x) = \sum_{i \le k} a_i x^i$ with all $a_i \in \{0, \dots, A-1\}$, $K(g) \le (k+1) \log A + K(k) + K(A) + c$; for uniform coefficients and $k+1$ distinct inputs every learner pays at least $(k+1) \log A$ in expectation. For arbitrary integer coefficients, $K(g) \le \sum_i K(a_i) + K(k) + c$.

So SFI is optimal in these examples up to $O(\log d)$, $O(\log \ell)$ and $O(\log k + \log A)$.

Proposition 6.27 (frozen label noise)**.** [proved; computed (examples e4(c))] examples Prop 4.6 *Let $g : X \to \{0,1\}$ be partial computable and $y_x := g(x) \oplus z_x$ with z_x i.i.d.* Bern(ε)*, drawn once per input. (a) For every data set D with $\operatorname{dom} D \subseteq \operatorname{dom} g$, $|\operatorname{dom} D| = n$, and e labels that disagree with g: $L_{\mathrm{SFI}}(D) \le \log \binom{n}{e} + \log(n+1) + K(g) + c$. (b) If the n distinct inputs are chosen without access to the noise at unrevealed inputs, $nH(\varepsilon) \le \mathbb{E}L_{\mathrm{SFI}}(D_n) \le nH(\varepsilon) + \log(n+1) + K(g) + c$; the lower bound holds for every learner.*

The machine behind (a) shares a flip rate across inputs, so it is a mixture of product laws (Appendix E.4). Fresh label noise is different: random-function SFI fails as soon as an input can repeat with another label (Remark E.10).

7 Out-of-distribution behaviour

Hänni's notes state that if "there is some simple property such that no input with that property showed up in training", SFI keeps "at least corresponding const probability of just outputting 0 on all inputs with that property", and ask whether "some log term" is lost (Hänni, 2026b). His draft adds a two-sided version, asks for a converse, and lists examples where SFI "plausibly" generalises well. This section makes these statements precise. His notes use plain complexity $C(\cdot)$; we use prefix complexity K throughout.

Conventions. ν ranges over $\{W, M_F\}$ (Definitions 2.15 and 3.1); for W the normalised conditional $\bar{\nu}$ is defined as $\bar{M}_F$ is for M_F. For a target f and inputs $x_1, x_2, \ldots$, $D_k := \{(x_j, f(x_j)) : j \le k\}$ and the *posterior error* $e_k := 1 - \nu(f(x_k) \mid D_{k-1}, x_k)$ is the posterior mass that is wrong *or undefined* at x_k. At a test point x^* with training labels $f|_A$: $e(x^*) := 1 - \nu(f(x^*) \mid f|_A, x^*)$; $e^{\mathrm{wr}}(x^*) := \sum_{y \neq f(x^*)} \nu(f|_A + (x^*, y))/\nu(f|_A)$ (defined and wrong); $e_{\mathrm{norm}}(x^*) := 1 - \bar{\nu}(f(x^*) \mid f|_A, x^*)$. Since $\sum_y \nu(f|_A + (x^*, y)) \le \nu(f|_A)$,

$$e^{\mathrm{wr}}(x^*) \;\le\; e_{\mathrm{norm}}(x^*) \;\le\; e(x^*), \tag{3}$$

so every lower bound below on e^{wr}, and every upper bound on e, holds in both conventions. A *property $Q \subseteq X$ separates x^** from A if $x^* \in Q$ and $Q \cap A = \emptyset$. For decidable Q and partial computable b, $K(Q, b)$ is the complexity of a pair $\langle e, e' \rangle$ with φ_e a total decider of Q and $\varphi_{e'}$ computing b on $Q \cap \operatorname{dom} b$; $K(Q, b) \le K(Q) + K(b) + c$.

7.1 Unseen simple properties

Proposition 7.1 (Hänni's proposition, with no log term)**.** [proved]
context Prop 2.1 *Let Q be decidable, b partial computable, $\operatorname{dom} D \cap Q = \emptyset$, $S \subseteq Q \cap \operatorname{dom} b$ finite and $E := b|_S$. Then (a) $\nu(D \cup E) \ge 2^{-K(Q,b)-c}\,\nu(D)$, and the hypotheses (programs for W, tapes for M_F) consistent with D that agree with b on all of $Q \cap \operatorname{dom} b$ have mass at least $2^{-K(Q,b)-c}\nu(D)$; (b) $K_F(D \cup E) \le K_F(D) + K(Q, b) + c$.*

Proof. For W: let V' read a V-program z for $\langle e, e' \rangle$, then a V-program p, and output an index of the function equal to $\varphi_{e'}$ on Q and to $\varphi_{V(p)}$ off Q; let $V(wu) = V'(u)$. The map $p \mapsto wzp$ is injective, preserves consistency with D (no input of D lies in Q), and its image agrees with b on $Q \cap \operatorname{dom} b$. Summing over consistent p with $|z| = K(Q, b)$ gives (a); a shortest p gives (b). For M_F: the function machine $T_{Q,b}$ that answers $b(x)$ on Q without reading the tape and runs T_U off Q has an index computable from $\langle e, e' \rangle$, and maps every tape with $f_\omega \supseteq D$ to a function extending $D \cup b|_{Q \cap \operatorname{dom} b}$; apply Theorem 2.9. □

So, whatever D is and however large, SFI behaves like b on all unseen Q-inputs with posterior probability at least $2^{-K(Q,b)-c}$; with $b \equiv 0$ this is Hänni's statement. For $|S| = 1$ and b wrong there, (a) bounds e^{wr}, so it holds in both conventions.

Remark 7.2 (where a log term would come from). [proved] context Rem 2.2 No log term arises, for W or for M_F: the modified program is a concatenation of self-delimiting codes, and the proof sums over all consistent programs. Plain complexity costs $K(Q) \le C(Q) + 2\log C(Q) + O(1)$. If Q is only semi-decidable the argument gives no bound at all, since the wrapper cannot decide Q; a time-truncated decidable subset of Q restores it there.

Hänni also worries about properties that are simple only relative to a good training program ("if this internal variable is set 0, then the if statement triggers"). Conditional complexity captures this.

Proposition 7.3 (properties simple given a training-consistent program)**.** [proved (corrected)] context Prop 2.3 *Let $p \in \operatorname{dom} V$ be consistent with D, and q, r indices with $\varphi_q(\langle p, \cdot\rangle)$ total; let $Q_p := \{x : \varphi_q(\langle p, x\rangle) = 1\}$ and $b_p(x) := \varphi_r(\langle p, x\rangle)$. If $Q_p \cap \operatorname{dom} D = \emptyset$ and $S \subseteq Q_p \cap \operatorname{dom} b_p$ is finite, then $\nu(D \cup b_p|_S) \ge 2^{-K(q,r|p)-c}\, 2^{-|p|}$; so the posterior probability of b_p on S is at least $2^{-K(q,r|p)-c}$ times $2^{-|p|}/\nu(D)$, the posterior weight of p.*

Proof idea. The program *wpz* reads p first, then a conditional program z for $\langle q, r\rangle$ given p; it outputs b_p on Q_p and p's values elsewhere. Reading z first would give the complexity of the pair instead. For M_F use $M_F \ge 2^{-c} W$ (Theorem 3.4).

Proposition 7.4 (dependence on the universal machine)**.** [proved; analogue known] context Prop 2.4 *Fix V, a decidable Q and a partial computable b. (a) Some universal prefix machine V' has $2^{-K(Q,b)-c} W' \le W \le 2W'$ and, for every D with $\operatorname{dom} D \cap Q = \emptyset$ and finite $S \subseteq Q \cap \operatorname{dom} b$, $W'(D) = W(D)$ and $W'(D \cup b|_S) \ge \frac{1}{2} W'(D)$. (b) Some universal prefix machine V'' yields an SFI M'_F with $2^{-K(Q,b)-c} M'_F \le M_F \le 2M'_F$ on nonempty data sets such that, for every nonempty D with $\operatorname{dom} D \cap Q = \emptyset$, $M'_F(D) = M_F(D)$ and the M'_F-posterior probability that $f_\omega = b$ on all of $Q \cap \operatorname{dom} b$ is at least $\frac{1}{2}$.*

Proof idea (Appendix F). V' runs V on programs $0u$ and the wrapper of Proposition 7.1 on programs $1u$; the halves agree off Q. So Hänni's remark that "there's a programming language in which [good behaviour] doesn't have a better const" is right. The machine is tailored to one property and behaviour, $K(Q, b) + O(1)$ bits from the reference machine. This is a supervised analogue of the dependence of AIXI on the reference machine (Leike and Hutter, 2015a).

7.2 The draft's two-sided result

For a property $P \subseteq X \times Y$ of input–output pairs ("y is a correct proof of x"), Hänni's draft states that "the probability of P-abiding generalization is at least $2^{-C(P)}$" and, if no training input has property Q, "at most $2^{-C(Q)}$ times 2 to the negative complexity of the smallest string not satisfying P on the given input times a const"; in sum, good OOD generalisation "has at least a const probability of happening but also at least some const probability of not happening" (Hänni, 2026b). The summary is correct:

Theorem 7.5 (two-sided bound on P-abiding generalisation)**.** [proved] context Thm 2.5 *Let Q be decidable, $\operatorname{dom} D \cap Q = \emptyset$, $S \subseteq Q$ finite and nonempty, and $K(P, Q)$ the complexity of a decider of Q paired with an enumerator of P (in (a)) or of its complement (in (b)). (a) If P is c.e. and every $x \in Q$ has some y with $(x, y) \in P$, then $\nu(\text{every } x \in S \text{ gets a } P\text{-abiding output} \mid D) \ge$*

$2^{-K(P,Q)-c}$. *(b) If the complement of P is c.e. and every $x \in Q$ has some y with $(x,y) \notin P$, then ν(some $x \in S$ gets a P-violating output $\mid D) \geq 2^{-K(P,Q)-c}$, so ν(all of S is P-abiding $\mid D) \leq 1 - 2^{-K(P,Q)-c}$. (c) For $x \in Q$ and $(x,z) \notin P$: $M_F(D + (x,z)) \geq 2^{-K(Q)-K(z|x)-c} M_F(D)$ and $W(D + (x,z)) \geq 2^{-K(Q)-K^{\mathcal{P}}(z|x)-c} W(D)$ (Remark B.1).*

Proof idea (Appendix F). (a), (b) apply Proposition 7.1 to the first abiding (violating) output found by enumeration; (c) runs a conditional program on half of the tape on Q and T_U on the other half elsewhere. The bounds hold for every amount of data. The draft's upper bound must be read as a lower bound on the probability of *violation*, as in (c). Literally, both bounds are false:

Proposition 7.6 (the draft's bounds as stated are false)**.** [refuted; proved]
context Prop 2.6 *Let $P = \{(x,0) : x \in X\}$. (a) If D labels inputs outside a decidable Q_d with 0 and $x^* \in Q_d$, then $\nu((x^*,0) \mid D) \geq 2^{-c}$, while the literal upper bound tends to 0 as $K(Q_d) \to \infty$. (b) If $x^* \in \{0,1\}^d$ has $K(x^* \mid d) \geq d$ and $D = \{(x,1) : x \in \{0,1\}^d \setminus \{x^*\}\}$, then $K(P) = O(1)$ but $\nu((x^*,0) \mid D) \leq 2^{-d+c}$, also normalised. With $Q = \{x^*\}$ Theorem 7.5(a) gives $2^{-d-2\log d-c}$, so its $K(Q)$ term is needed and nearly tight.*

7.3 OOD failure is a bounded resource

Theorem 7.7 (cumulative bound)**.** [proved] context Thm 2.7 *Let f be computable and $x_1, x_2, \ldots$ any distinct inputs (adversarial, adaptive or non-computable). For every n, with $c = 0$ for W,*

$$\sum_{k \leq n} -\log(1-e_k) \leq K(f)+c, \quad \sum_{k \leq n} e_k \leq (\ln 2)(K(f)+c), \quad \#\{k \leq n : e_k \geq \delta\} \leq \frac{(\ln 2)(K(f)+c)}{\delta}.$$

The same holds normalised, and with $K(f) + c$ replaced by $L_\nu(f|_B) \leq K_F(f|_B) + c$, $B = \{x_1, \ldots, x_n\}$.

Proof. By Proposition 2.16 the first sum is $-\log(\nu(D_n)/\nu(\emptyset)) \leq L_\nu(D_n) \leq K_F(f|_B) + c \leq K(f) + c$; use $-\ln(1-e) \geq e$, and count. Normalised errors are smaller by (3). □

This is the error form of the mistake bound (Corollary 4.4). Separating properties force errors, but only a bounded total:

Proposition 7.8 (novelty budget)**.** [proved] context Prop 2.8 *Let $Y = \{0,1\}$ and $\mathrm{sep}_k := \min\{K(Q) : Q$ decidable, $x_k \in Q$, $Q \cap \{x_1, \ldots, x_{k-1}\} = \emptyset\}$. Then $e_k \geq e_k^{\mathrm{wr}} \geq 2^{-\mathrm{sep}_k - c}$ for every k, and $\sum_k 2^{-\mathrm{sep}_k} \leq 1$.*

Proof. The first claim is Proposition 7.1 with $b \equiv 1 - f(x_k)$. A property separating x_k from its predecessors contains x_k, so it separates no other x_j; distinct k thus have distinct minimisers, with distinct shortest descriptions in $\mathrm{dom}\, V$; apply Kraft's inequality. □

Remark 7.9 (the novelty bound is lopsided). [refuted as stated; remark]
context Rem 2.8a Reading Theorem 7.7 and Proposition 7.8 as "the total OOD error lies between the novelty of the inputs and the complexity of f" is wrong: the novelty bound $2^{-c} \sum_k 2^{-\mathrm{sep}_k}$ never exceeds 2^{-c}. Separation forces constant error at single steps but at most $O(1)$ in total; each simple property surprises only once. The positive statement is relative (Theorem 4.1(a)): if a computable x-only predictor generalises correctly across all the shifts, SFI's total excess log loss over it is at most its complexity plus c. A human who generalises after seeing the data is history-aware, which needs Corollary 5.18; reading Proposition 7.8 as a formalisation of Hänni's "fake ood generalization" is an interpretation, not a theorem.

7.4 Converses

Proposition 7.10 (the naive converse fails for complex targets)**.** [proved]
context Prop 2.13 *Let $f : \{0,1\}^d \to \{0,1\}$ be uniformly random and $A = \{x_1, \dots, x_n\}$, x^* i.i.d. uniform, all independent. (a) Every learner has $\mathbb{E}[e_{\mathrm{norm}}(x^*) \mid x^* \notin A] \geq \frac{1}{2}$. (b) For every string z determined by f (e.g. a shortest program for f) and $s \geq 0$, the probability that some decidable Q with $K(Q \mid z) \leq s$ separates x^* from A is at most $2^{s+1}/(n+1)$. (c) (*A program fitted to the training data.*) Let $K(x^* \mid d) \geq d$, $A := \{0,1\}^d \setminus \{x^*\}$, $D := \{(x,1) : x \in A\}$, $f := D + (x^*, 0)$, and p_1 a shortest program for the constant 1 (consistent with D). Every decidable Q separating x^* from A has $K(Q \mid p_1) \geq d - c$, yet $e_{\mathrm{norm}}(x^*) \geq 1 - 2^{-d+c}$; here $K(f) \geq d - c$.*

So Hänni's converse ("works well if there isn't a simple distinguishing property given such a program") fails both for a program determined by the target (a, b) and for one fitted to the training data (c). The obstruction is $K(f)$. Allowing a factor depending on it, a converse holds once the property may depend on the labelled sample. For A finite, $x^* \notin A$, $B := A \cup \{x^*\}$ and $y \neq f(x^*)$, put $D^*_y := f|_A + (x^*, y)$.

Proposition 7.11 (sandwich for one test point)**.** [proved]
context Prop 2.14 *For every decidable Q separating x^* from A and partial computable b with $b(x^*) = y$, $2^{-K(Q,b)-c} \leq \nu(D^*_y)/\nu(f|_A) \leq 2^{L_\nu(f|_A) - K(D^*_y|B)+c} \leq 2^{K_F(f|_A) - K(D^*_y|B)+c'}$.*

Proof. Left: Proposition 7.1. Right: for B given as a list, ν restricted to labellings of B is a semimeasure (disjoint programs or tapes), lsc uniformly in B, so the conditional coding theorem gives $\nu(D^*_y) \leq 2^{-K(D^*_y|B)+c}$; and $\nu(f|_A) = 2^{-L_\nu(f|_A)} \geq 2^{-K_F(f|_A)-c}$. □

SFI errs at x^* only if a wrong completion is, *given all inputs*, not much more complex than the training labels are for a program that does not know the inputs. This gap is the one between x-only and history-aware prediction (Theorem 5.24).

Proposition 7.12 (converse given the labelled sample)**.** [proved; disjointness computed (context c7)]
context Prop 2.16 *Let f be a labelling of a finite B, $x^* \in B$, $A := B \setminus \{x^*\}$, and $\mathrm{sep}(x^*; A \mid f)$, $\mathrm{sep}(x^*; A)$ the least $K(Q \mid f)$, $K(Q)$ of a decidable Q separating x^* from A. Then*

$$e^{\mathrm{wr}}(x^*) \leq e_{\mathrm{norm}}(x^*) \leq 2^{\,L_\nu(f) - K(x^*|f)+c} \leq 2^{\,K_F(f) - K(x^*|f)+c'}, \qquad |K(x^* \mid f) - \mathrm{sep}(x^*; A \mid f)| \leq c,$$

and $e^{\mathrm{wr}}(x^*) \geq 2^{-\mathrm{sep}(x^*;A) - K(y) - c}$ for every $y \neq f(x^*)$. Also $e_{\mathrm{norm}}(x^*) \leq 2^{L_\nu(f) + K_F(f) - K(x^*|B)+c}$, and $e_{\mathrm{norm}}(x^*) \leq 2^{|p| - K(x^*|B,p)+c}$ for every $p \in \mathrm{dom}\, V$ that computes f on B.

Proof idea (Appendix F). The hypotheses that agree with f on $B \setminus \{x\}$ and are defined and wrong at x form disjoint sets for distinct x, with masses lsc uniformly in (f, x); the coding theorem bounds the mass for x^* by $2^{-K(x^*|f)+c}$, and the normaliser at x^* is at least $\nu(f)$.

So if f is simple and no property simple *given the labelled sample* separates the test point, the error there is small. The two sides use the separating complexity without and with the sample; the difference is the value of knowing the other inputs. The mass undefined at x^* is not lsc, so there is no pointwise bound on $e(x^*)$; Corollary 4.14 bounds it in distribution.

Example 7.13 (easy-to-hard generalisation; shortest proofs)**.** [proved] context Ex 2.15, 2.15a Let the inputs be sentences of a consistent theory $\mathcal{T}$ with decidable proof checking, labelled 1 if $\mathcal{T}$ proves them and 0 if it refutes them. Let every training sentence have a proof or refutation of length at most L, and the test sentence x^* none of length at most $L' := 2^{\lceil \log L \rceil} \leq 2L$. The property "no proof or refutation of length $\leq L'$" is decidable, separates x^* from the training set, and has

complexity at most $K(\mathcal{T}) + K(\lceil \log L \rceil) + c \le K(\mathcal{T}) + 2\log\log L + c$ for large L. With $b \equiv 0$ and $b \equiv 1$, Proposition 7.1 gives the wrong label probability at least $2^{-K(\mathcal{T})-2\log\log L-c}$, however many easy sentences were seen. Proof search, of length $K(\mathcal{T}) + c$, fits every correctly labelled training set, so the correct label has probability at least $2^{-K(\mathcal{T})-c}$. If instead each provable sentence is labelled by its shortest proof, Theorem 7.5(c) with the violating output ε ($K(\varepsilon \mid x^*) = O(1)$) gives probability at least $2^{-K(\mathcal{T})-2\log\log L-c}$ that x^* does not get its shortest proof.

Both ends are independent of the amount of data. The failure floor is not independent of the difficulty threshold: it decays, slowly, as L grows.

Remark 7.14 (the strongest true positive statements). [remark] context §2.5 (i) Over any shifts, SFI's total loss exceeds that of every computable x-only predictor P by at most $K(P) + c$, and its total error is at most $(\ln 2)(K(f) + c)$ (Theorems 4.1 and 7.7). (ii) A random train/test split of any input set B costs at most $L_\nu(f|_B)/|B|$ in expectation (Proposition 4.10). (iii) For simple f the error at a test point lies between $2^{-\operatorname{sep}(x^*;A)-O(1)}$ and $2^{L_\nu(f)-\operatorname{sep}(x^*;A|f)+O(1)}$; for random f no learner has a converse. So "no simple separating property implies small error" fails only for bounds independent of $K(f)$, or for properties simple unconditionally. (iv) In distribution the per-step error is at most $\min\{((K(f) + c)\ln 2 + \ln n + 1)/n,\ 2^{K(f)+c}/(n + 1)\}$, and for uniform inputs at least of order $2^{-K(f)}/(n \log^2 n)$ (Proposition 4.13 and Corollary 4.14).

Open question. Is there a positive statement for properties simple given a program fitted to the training data only (as in Proposition 7.10(c)), with a factor depending on $K(f)$?

7.5 Hänni's examples

Hänni's examples are sentences of a formal theory labelled by provability ("assuming the riemann hypothesis is decidable, it seems plausible that solomonoff function induction gives it the intended label"), sentences labelled by their shortest proofs, and "correctly counted the lengths of odd strings", tested on even lengths (Hänni, 2026b).

Proposition 7.15 (the intended answer). [proved] examples Prop 5.1 *If the labels follow a partial computable g and $x \in \operatorname{dom} g$ is new, SFI gives $g(x)$ predictive probability at least $2^{L_{\mathrm{SFI}}(D)-K(g)-c}$, in both conventions.*

Proof. $M_F(D + (x, g(x))) \ge 2^{-K(g)-c}$ (Theorem 4.1(b)); divide by $M_F(D)$; the normalised conditional is larger. □

This is near 1 only if $L_{\mathrm{SFI}}(D) \approx K(g)$, i.e. if g-like programs carry most of the posterior.

Example 7.16 (Hänni's three examples). [proved; computed (examples e5)] examples §5.1–5.3; context Ex 2.15, 2.15a Let $\mathcal{T}$ be a consistent theory with decidable proof checking, and $p_{\mathcal{T}}$ its proof search, of length $K(\mathcal{T}) + c$. (a) *Provability labels.* $p_{\mathcal{T}}$ fits every correctly labelled training set, so $L_{\mathrm{SFI}}(D) \le K(\mathcal{T}) + c$, Proposition 7.15 applies, and over any sequence of sentences the correct label gets probability at most $\frac{1}{2}$ at most $K(\mathcal{T}) + c$ times. A hard test sentence keeps a wrong-label probability of at least $2^{-K(\mathcal{T})-2\log\log L-c}$ however many easy sentences were seen (Example 7.13). This makes precise Hänni's weakly held hypothesis that for Solomonoff induction "ood generalization is a thing, but easy-to-hard generalization is not". Relative to proof search in the form "fixed searcher, then a description of $\mathcal{T}$" (Proposition 7.3), the floor is $2^{-K(L'|p_{\mathcal{T}})-c} \ge 2^{-c'}/(\log L(\log\log L)^2)$ times the posterior weight of $p_{\mathcal{T}}$. (b) *Shortest proofs as labels.* The same bounds hold, with the floor of Example 7.13. (c) *Lengths of odd strings.* The

correct answer has probability at least 2^{-c} on every even-length input and SFI makes at most c online errors; "$|x|$ even" and "output 0" cost $O(1)$ bits, so a wrong answer keeps probability at least $2^{-c'}$. Both constants depend on the machine. An exact computation over a small enumerated grammar of programs (Appendix F; code in the accompanying repository) gives the intended answer probability at least 0.979, 0.997 or 0.9997 on every even-length test string, depending on one prior weight, and about 0.11 when one extra program tilts the grammar.

Table 6 (Appendix F) summarises what is proved, what is only plausible, and natural strengthenings of Hänni's statements that are false.

Conjecture 7.17 (natural machines generalise as intended)**.** [conjecture] examples §5.1–5.3, §7 item 4 *For natural universal machines and natural training sets, the intended answers of Example 7.16 get posterior probability close to* 1*; for example, a decidable Riemann hypothesis gets its correct label with high probability.*

This is not a precise statement, and by Proposition 7.4 it fails for some universal machines; what is machine-invariant is the bounded number of errors (Theorem 7.7). We know no route to a version for a fixed reference machine.

8 Computable versions

SFI is not computable (Proposition 2.24). Hänni's notes sketch two computable relatives: a time-bounded Solomonoff predictor with constant regret against all predictors of a given running time, and induction over Boolean circuits, where "regret and complexity are polynomially related" (Hänni, 2026b). This section states what holds and shows that Occam (simplest-consistent) learning, unlike Bayes, fails exponentially out of distribution.

8.1 Time-bounded SFI

Definition 8.1 (time-bounded predictors; length-activated mixture)**.** context §3.1 Let Y be finite, $|Y| \geq 2$. A *probability predictor* is a RAM program A that on input x halts with rationals $(A(y \mid x))_{y \in Y}$, nonnegative and summing to 1, in time $T_A(x)$. For a nondecreasing $t : \mathbb{N} \to \mathbb{N}_{\geq 1}$, $\mathcal{C}_t := \{A : \exists c_A \, \forall x \; T_A(x) \leq c_A t(|x|)\}$. Fix RAM programs $A_1, A_2, \ldots$ with computable weights $w_i > 0$, $\sum_i w_i \leq 1$, activation lengths $n_1 < n_2 < \cdots$, and a nondecreasing unbounded clock factor $g \geq 1$, with t and $g \cdot t$ time-constructible. Let $q_i(\cdot \mid x) := A_i(\cdot \mid x)$ if $|x| \geq n_i$ and A_i halts on x within $g(|x|)t(|x|)$ steps with a valid vector, and the uniform vector u otherwise. The *length-activated mixture* is the mixture of product laws $M_{\mathrm{len}}(D) := \sum_i w_i \prod_{(x,y) \in D} q_i(y \mid x)$.

Theorem 8.2 (time-bounded SFI)**.** [(a) proved (corrected); (b) proof sketch; computed (context c5, c9)] context Thm 3.1 *(a) Let $A = A_i \in \mathcal{C}_t$ and $n_A := \max(n_i, \min\{n : g(n) \geq c_A\})$. On every sequence of* distinct *inputs, with any labels, the regret of M_{len} against A after k rounds is*

$$\sum_{j \leq k} \log \frac{A(y_j \mid x_j)}{M_{\mathrm{len}}(y_j \mid D_{j-1}, x_j)} \;\leq\; \log \tfrac{1}{w_i} + \#\{j \leq k : |x_j| < n_A\} \log |Y| \;\leq\; \log \tfrac{1}{w_i} + (2^{n_A} - 1) \log |Y|.$$

(b) With $n_i = 2^{2^i}$ and $g(n) = \lceil \log \log(n+4) \rceil$, round k costs at most $\log \log |x_k| + 1$ simulations of $g(|x_k|)t(|x_k|)$ steps plus as many arithmetic operations on stored weights; weights stored with relative precision k^{-2} change the cumulative loss by $O(1)$.

Proof idea (Appendix G). (a) The loss of M_{len} is at most $\log(1/w_i)$ plus that of $\prod q_i$; $q_i = A$ on inputs of length at least n_A, and on shorter ones $q_i \in \{A, u\}$ costs at most $\log|Y|$ more than A. (b) Only active machines change their weights, and rounding errors telescope. A direct computation on a toy instance (code in the accompanying repository) matches the brute-force mixture to $2.5 \cdot 10^{-16}$, with regret 132.4 bits against the proved 184. The constant in (a) is large, triple exponential in the index i (Remark G.1).

Remark 8.3 (distinct inputs are a real restriction). [refuted; computed (context c9)] context §3.1 One might hope that restricting (a) to distinct inputs loses nothing, since a repeated input costs nothing in the semimeasure convention. For noisy labels it does not: a repeated input with a new label has probability 0 under M_{len}, as under SFI (Proposition 4.6); and even if the product hypotheses are mixed round by round, length clocking leaves a comparator over budget on short inputs forever. On 4000 rounds that repeat 12 short inputs with labels drawn from a comparator in $\mathcal{C}_t$, a direct computation gives regret 469 bits, growing linearly.

Proposition 8.4 (round-activated mixture)**.** [proved; running time proof sketch; computed (context c9)] context Prop 3.2a *Let $q_i^{(k)} := A_i(\cdot \mid x_k)$ if $k \geq k_i$ and A_i halts on x_k within $g(k)t(|x_k|)$ steps with a valid vector, and u otherwise; let M_{rd} be the Bayes mixture, with weights w_i, of the per-round products $\prod_{j \leq k} q_i^{(j)}(y_j)$. For every $A = A_i \in \mathcal{C}_t$, every stream (repeated inputs and conflicting labels allowed) and every k, the regret of M_{rd} against A is at most $\log(1/w_i) + (J_A - 1)\log|Y|$, $J_A := \max(k_i, \min\{k : g(k) \geq c_A\})$. Round k costs $\#\{i : k_i \leq k\}$ simulations of $g(k)t(|x_k|)$ steps plus as many arithmetic operations.*

Proof. For $j \geq J_A$, machine i is active and $T_A(x_j) \leq c_A t(|x_j|) \leq g(j)t(|x_j|)$, so $q_i^{(j)} = A(\cdot \mid x_j)$; for $j < J_A$, $q_i^{(j)} \in \{A(\cdot \mid x_j), u\}$ and $-\log q_i^{(j)}(y_j) \leq \log|Y| - \log A(y_j \mid x_j)$. The mixture loses at most $\log(1/w_i)$ to $\prod_j q_i^{(j)}$. □

This is the time-bounded analogue of SFI on time-stamped inputs (Theorem 4.5(b)), and a mixture of the per-query kind (Remark 3.2). On the stream of Remark 8.3 its regret is 10.0 bits, against the bound 20.

Corollary 8.5 (all polynomial-time predictors)**.** [proved]
context Cor 3.2b *Let g be computable, nondecreasing and unbounded, and the clock $\tau_k(x) := (|x|+2)^{g(k)}$ time-constructible. With $\tau_k(x_k)$ in place of $g(k)t(|x_k|)$ in Proposition 8.4, the regret against every x-only predictor with $T_A(x) \leq c_A(|x|+2)^{e_A}$ is bounded by a constant depending on A, on every stream; round k costs $\#\{i : k_i \leq k\}$ simulations of $(|x_k|+2)^{g(k)}$ steps.*

Proof. $c_A(|x|+2)^{e_A} \leq (|x|+2)^{e_A + \log c_A}$, so A is within budget once $g(k) \geq e_A + \log c_A$; argue as for Proposition 8.4. □

This is Hänni's "Remark 4": constant regret against all polynomial-time predictors in time $|x|^{\omega(1)}$ per round.

Remark 8.6 (Hänni's "apples to oranges" caveat). [remark] context Rem 3.2 In Hänni's sequence theorem the learner reuses state across rounds while the comparator recomputes, which he calls comparing "apples to oranges". For x-only comparators, which see one input and have nothing to reuse, Theorem 8.2 and Proposition 8.4 compare like with like. History-aware comparators need a chronological learner: no x-only learner competes with them, since the chain and skipped-string predictors run in linear time while SFI's loss on those data is unbounded (Theorems 5.3 and 5.4).

Proposition 8.7 (time-bounded history-aware predictors)**.** [regret proved; running time proof sketch] context Prop 3.2c *Let $\mathcal{C}_t^{\mathrm{hist}}$ be the RAM programs A that keep a state, receive x_k, output $A(\cdot \mid h_{k-1}, x_k)$, then receive y_k, with round k taking at most $c_A t(\tilde{n}_k)$ steps, where $\tilde{n}_k$ is the length of the encoded h_{k-1} and x_k. Let the learner start machine i at round k_i and give it $2g(k)t(\tilde{n}_k)$ steps in round k, spent replaying the stored history and then on round k; it uses A_i's output if this finishes in time and u otherwise, and mixes the per-round products. Then the regret against every $A = A_i \in \mathcal{C}_t^{\mathrm{hist}}$ is at most $\log(1/w_i) + 2k_0 \log|Y|$ on every stream, $k_0 := \max(k_i, \min\{k : g(k) \geq c_A\})$; round k costs $O(\#\{i : k_i \leq k\}\, g(k)t(\tilde{n}_k))$ steps plus the replay memory.*

Proof idea (Appendix G). As $\tilde{n}_k$ grows, every past round of A fits into the current budget once $k \geq k_0$, so the replay catches up by round $2k_0 - 2$ and stays caught up. This is the function-setting form of Hänni's amortised warm-up; a budget depending only on $|x_k|$ could stall. The class contains the chain and skipped-string predictors; it is the time-bounded counterpart of $\mathrm{SFI}^{\mathrm{hist}}$ (Corollary 5.18).

Proposition 8.8 (samplers)**.** [proved (corrected)] context Prop 3.3 *Let $A(x; r)$ be a probabilistic RAM program that halts within $c_A t(|x|)$ steps on all coin sequences r, with output law $P_A(\cdot \mid x)$. For $|x| \geq n_i$ let $q_i^{\mathrm{samp}}(y \mid x)$ be the probability that $A_i(x; r)$ halts within $g(|x|)t(|x|)$ steps with output y, plus $1/|Y|$ times the probability that it does not; for $|x| < n_i$ let $q_i^{\mathrm{samp}} := u$. The mixture $\sum_i \frac{w_i}{2}(\prod_D q_i + \prod_D q_i^{\mathrm{samp}})$ satisfies Theorem 8.2(a) against every such sampler (regret against P_A) and every predictor in $\mathcal{C}_t$, with $\log(2/w_i)$ for $\log(1/w_i)$; Proposition 8.4 extends likewise. Computing q_i^{samp} exactly takes time $O(2^{g(|x|)t(|x|)} g(|x|)t(|x|))$.*

A sampler's program is not a predictor's program, so the mixture must contain both readings; the proof of Theorem 8.2(a) applies because $q_i^{\mathrm{samp}}(y \mid x) \geq P_A(y \mid x)/|Y|$ on short inputs.

Open question. Is there a learner with per-round time polynomial in $t(|x|)$ and constant regret against all time-$O(t)$ samplers? (Heuristic, not proved: Monte Carlo with s samples per round adds an expected excess of order $(|Y| - 1)/(2s \ln 2)$ bits per round, so constant regret would need $s_k \to \infty$ with $\sum_k 1/s_k < \infty$.) As secondary sources describe it, Impagliazzo and Levin (1990) tie efficient universal extrapolation to the non-existence of one-way functions. Speed priors (Schmidhuber, 2002; Filan et al., 2016) treat the sequence case.

8.2 Circuit SFI

Definition 8.9 (circuits and the circuit mixture)**.** context §3.2 Fix $d \geq 1$ and the basis of all 16 two-input Boolean gates. A circuit of size s is a list of s gates, gate j with a type and two inputs from $\{x_1, \ldots, x_d, 0, 1, \mathrm{gate}_1, \ldots, \mathrm{gate}_{j-1}\}$; gate s is the output. There are $N_s = \prod_{j=1}^{s} 16(d + 1 + j)^2$ of them. The prior $w(C) := 2^{-s}/N_s$ is the code length of "size in unary, then the gate list"; $\sum_C w(C) = 1$. Let $W_{\mathrm{circ}}(D) := \sum\{w(C) : C \text{ consistent with } D\}$, $\mathrm{size}(f)$ the least size of a circuit for f, and for $B \subseteq \{0,1\}^d$, $\mathrm{size}_B(f)$ the least size of one agreeing with f on B, and $L_B := -\log W_{\mathrm{circ}}(f|_B)$. The *exception gadget* appends $\gamma \leq d + 1$ gates computing $[x = a]$ and a new output gate, so that C' equals C off a and outputs a chosen bit at a; $C \mapsto C'$ is injective and $w(C')/w(C) \geq 2^{-\Gamma(s)}$, $\Gamma(s) := \gamma(5 + 2\log(d + 1 + s + \gamma))$.

For any ordering of B, let P_i be the posterior probability of the wrong label at step i (circuits are total, so both conventions agree) and $\mathcal{E} := \sum_i P_i$ the expected number of mistakes of posterior sampling (Gibbs mistakes).

Theorem 8.10 (circuit Bayes: mistakes and circuit size)**.** [proved; computed (context c1, c8)] context Thm 3.5 *For every $B \subseteq \{0,1\}^d$, every f and every ordering of B: (a) some circuit of size at most $\lceil L_B \rceil$ agrees with f on B, and $L_B \le \mathrm{size}_B(f) + \log N_{\mathrm{size}_B(f)} = O(\mathrm{size}_B(f)\log(\mathrm{size}_B(f)+d))$; (b) $\mathcal{E} \le (\ln 2) L_B$ and $L_B \le 2\mathcal{E}(\Gamma(\lceil L_B \rceil)+3)$; explicitly, for all $\mathcal{E} > 0$ and $d \ge 1$, $\mathrm{size}_B(f) \le \lceil L_B \rceil$ with $L_B \le 16\,\mathcal{E}\,(d+1)\log\big(2(\mathcal{E}+1)(d+2)\big)$.*

Proof idea (Appendix G). (a) Circuits larger than S have total weight 2^{-S}. (b) The first inequality is $P_i \le -\ln(1-P_i)$ with Proposition 2.16. For the second, the gadget maps the small wrong circuits at step i injectively into the right ones at weight cost $2^{-\Gamma}$, so $-\log(1-P_i) \le \Gamma + 2$; steps with $P_i \le \frac{1}{2}$ cost at most $2P_i$, and at most $2\mathcal{E}$ steps have $P_i > \frac{1}{2}$. A grid computation gives the largest ratio 10.85 in place of 16.

Remark 8.11 (relation to Hänni's sketch). [remark; computed (context c1)]
context Rem 3.6, 3.7 Hänni's direction, "expected regret m when learning the function f implies there is a circuit of size $O(m \log m \cdot \mathrm{inputdim})$", is (b) with (a), with $\log((\mathcal{E}+1)(d+2))$ for $\log m$; it holds for every single ordering, needs no Markov step (total log loss is order independent), and his flagged "$+O(\log m)$ term" is the $\log(d+s)$ in Γ. His step from "log loss $\le$ poly(m)" to "a circuit of size poly(m)" is a *circuit coding theorem*; (a) gives it for a size prior with exponential tail, while a polynomial tail gives only $\mathrm{size}(f) \le 2^{O(L)}$. The converse holds on every ordering: $\mathcal{E} \le (\ln 2) L \le (\ln 2)\,\mathrm{size}(f)(5 + 2\log(d+1+\mathrm{size}(f)))$. For formulas on $d = 3$ an exhaustive computation over all 256 functions and all orderings confirms inequalities of the same form.

Proposition 8.12 (Occam learning in distribution)**.** [(a) known; (b) proved] context Props 3.8, 3.9 *Order functions by shortest description length in a prefix-free language, then by a fixed tie rule; $\ell(f)$ is the shortest description length of f. The* Occam learner *predicts with the first function consistent with the data, so it uses only $H_f := \{h \text{ preceding or equal to } f\}$, $|H_f| < 2^{\ell(f)+1}$. For i.i.d. inputs from μ: (a) after $n \ge (\ln|H_f| + \ln(1/\delta))/\varepsilon$ samples its hypothesis has μ-error at most ε with probability at least $1-\delta$ (Blumer et al., 1987); (b) its expected number of mistakes is at most $6.7 + (\ln|H_f|)(1+\ln n_0) + \ln^2 n_0$, $n_0 := \lceil \ln|H_f|/p_{\min} \rceil$ with $p_{\min}$ the least positive point mass of μ; for uniform μ on $\{0,1\}^d$ this is $O(\ell(f)(d + \log \ell(f)) + d^2)$.*

(a) is a union bound over H_f; with $\varepsilon = \delta = 0.01$ it needs $70\ell(f) + 531$ samples, so Hänni's "$1000000 \cdot C(f)$ input-output pairs" suffice when $C(f)$ is description length. (b) is proved in Appendix G.

Remark 8.13 (the "physicist's proof" is a proof for weighted majority). [proved; known] context Rem 3.10 Hänni's "physicist's proof" that Occam makes $O(C(f))$ online mistakes on every ordering assumes that when the current guess is wrong, a constant fraction of the competing weight is wrong too. That is the guarantee of the *posterior-majority* (weighted-majority) predictor: when it errs, at least half of the posterior mass errs, so $W_{\mathrm{circ}}(D_i) \le W_{\mathrm{circ}}(D_{i-1})/2$, and it makes at most $L_B \le \ell(f)$ mistakes on every ordering (Littlestone, 1988; Littlestone and Warmuth, 1994). Under the circuit prior the maximum-a-posteriori *circuit* is a smallest consistent one, so the MAP-hypothesis learner is an Occam learner. For Occam the assumption can fail at every step:

8.3 Occam learning fails out of distribution

The construction below is from the Vaintrob–Hänni slides (Appendix K) (Vaintrob and Hänni, 2026b); we prove it for two languages and every tie rule. *Formulas* have leaves $x_j, \neg x_j, 0, 1$ and binary nodes AND, OR, XOR; size is the number of leaves; the prior is "size in unary, then uniform within the size". *Circuits* are as in Definition 8.9.

Theorem 8.14 (the conjunction ladder)**.** [proved; computed (context c6, c11); construction from the slides] context Thm 3.11 *Let $f = x_1 \wedge \cdots \wedge x_d$, and let Occam use any fixed tie rule among functions of equal least size, for formulas ($d \geq 2$) or circuits ($d \geq 3$). Present 1^d, then 0^d, then the inputs of Hamming weight $1, 2, \ldots, d-1$ level by level, inside a level an input on which the current hypothesis errs if there is one. Then Occam makes at least $2^d - 2$ mistakes (formulas) or $2^d - d - 2$ (circuits). Since $\ell(f) = \Theta(d \log d)$ in both languages, its worst-case number of mistakes is $2^{\Omega(\ell/\log \ell)}$; the trivial upper bound is $2^{\ell+1}$.*

Proof idea (Appendix G). At the start of stage k the data say $h(1^d) = 1$ and $h(x) = 0$ for $|x| < k$. A Möbius-inversion lemma (Lemma G.2) shows that the least-size consistent functions are then exactly the $\binom{d}{k}$ conjunctions of k variables (for circuits from $k = 2$). Each is wrong only at its own point 1_S, so every input of weight k is a mistake. The ordering is computable from d and the tie rule, so "structured orderings are safe" is false, as the slides note. A second adversary gives the circuit count without the lemma (Remark G.3). A direct computation ($d = 3, 4, 5$, seven tie rules) finds every input of every covered level to be a mistake.

Remark 8.15 (it is the selection rule, not the prior). [computed (context c6(e)); known; unverified (claimed in the slides)] context Rem 3.12 With the same formula prior the posterior-majority predictor makes at most $L(f) \leq \ell(f)$ mistakes on every ordering (Remark 8.13). An exact computation (code in the accompanying repository; ties count $\frac{1}{2}$) gives, for $d = 3$ and $d = 4$: Occam on the ladder 7 and 15 mistakes; posterior majority on the ladder 2.5 and 3.5, and on its worst ordering 5.5 and 8.5; Gibbs on its worst ordering 4.47 and 7.08; halving bound $L(f) = 10.83$ and 15.30. The slides tabulate four selection rules on one prior: deterministic argmax, $2^{\Theta(C/\log C)}$ worst-case mistakes; argmax with randomised tie-breaking, $\tilde{\Theta}(C^2)$; a uniformly random consistent hypothesis with a doubling budget, $O(C)$; weighted majority, at most C. They also state that a random ordering gives $O((C+d)d)$ expected mistakes in any language, and the descending-weight ordering exactly $d+1$. The first and last rows are proved here (Theorem 8.14 and Remark 8.13); the other claims are the slides' and we have not checked them.

Remark 8.16 (uniform tie-breaking on the ladder). [proved] context Rem 3.13 All formulas of one size have the same prior weight, so the limit $\lambda \to \infty$ of the power posterior $\propto w(\text{formula})^\lambda \mathbf{1}[\text{consistent}]$ (the direction of the companion paper on steeper priors, Hänni 2026a) is uniform over the least-size consistent formulas, and by symmetry uniform over the surviving k-conjunctions at stage k. On the ladder it makes at most $1 + \sum_{k=1}^{d-1} H_{\binom{d}{k}} \leq d + d^2 \ln 2$ Gibbs mistakes (H_m the harmonic number) and at most $\frac{3}{2}(d-1)+1$ majority mistakes. So Theorem 8.14 is a failure of deterministic tie-breaking; it does not show that steeper SFI fails on this ordering.

Open questions. A circuit coding theorem for size priors with polynomial tails: do large random circuits rarely compute functions without small circuits? Do few posterior-majority mistakes imply a small circuit (steps with P_i just below $\frac{1}{2}$ cost log loss but no majority mistakes)? Is there a natural language and target with $2^{\Omega(\ell)}$ worst-case Occam mistakes, and can a steeper posterior over a universal machine fail exponentially on some ordering? The machine-model details of the running times above (RAM versus multitape, the cost of the clocks) are not written out.

9 Neural-network Bayes and circuit SFI

Hänni describes Bayesian learning with neural networks (NN Bayes) as "kinda like a circuit version of solomonoff function induction" (Hänni, 2026c), referring to slides by Vaintrob and

Hänni (Vaintrob and Hänni, 2026b). This section states what is proved about this link, what the slides claim, and what is open. An *architecture* is a parameter space Θ with a prior ϖ and a map $\theta \mapsto f_\theta : \{0,1\}^d \to Y$.

Proposition 9.1 (NN Bayes is function-space Bayes)**.** [proved (trivial)] context Prop 4.1 *If the likelihood depends on θ only through the values of f_θ on the data (a 0–1 likelihood, a margin rule, any label-noise model), the network's posterior predictive equals the function-space Bayes predictive with prior $P_A(f) := \varpi\{\theta : f_\theta = f\}$.*

Proof. Group the parameter integral by the value of f_θ. □

So NN Bayes is function induction with the prior P_A, and the question is how P_A compares with 2^{-K}, with M_F, or with a circuit prior.

Proposition 9.2 (coding-theorem upper bound)**.** [proved (given the coding theorem)] context Prop 4.2 *If P_A, as a semimeasure on truth tables, is lower-semicomputed by a program a (encoding the architecture, the prior and d), then $P_A(f) \le 2^{-K(f|a)+c} \le 2^{-K(f)+K(a)+c'}$, with c independent of the architecture.*

This is the conditional coding theorem for the family indexed by a. P_A is lsc for finite parameter sets with rational probabilities; for Gaussian priors on real weights with sign outputs we only sketch it (up to a null boundary, $\{\theta : f_\theta = f\}$ is an open semialgebraic set, approximable from below by certified rational boxes). Hänni's version has an extra $\log\log(1/p)$ term (Hänni, 2026b), which is not needed. The bound is an exact, uncomputable form of the simplicity-bias bound of Dingle et al. (2018).

Proposition 9.3 (every prior nearly saturates the bound on its typical set)**.** [proved; known (standard)] context Prop 4.3 *For every semimeasure P on truth tables, every string a and $j \ge 0$: $P\{f : P(f) \le 2^{-K(f|a)-j}\} \le 2^{-j}$. Hence if P_A is lower-semicomputed by a and ϖ is a probability measure, then with probability at least $1-2^{-j}$ over $\theta \sim \varpi$, $K(f_\theta \mid a) - c \le -\log P_A(f_\theta) < K(f_\theta \mid a) + j$.*

Proof. The left side is at most $\sum_f 2^{-K(f|a)-j} \le 2^{-j}$ by Kraft's inequality; the lower bound is Proposition 9.2. □

This is the standard fact that $-\log P \approx K$ on P-most objects (Li and Vitányi, 2008) (cf. Lemma 6.1). It confirms Hänni's remark that the result "should just be true about any low complexity prior on functions — the NNs aren't doing anything here": P_A looks like 2^{-K} on its own typical set, which can omit most simple functions.

Proposition 9.4 (no general lower bound)**.** [proved] context Prop 4.4 *(a) Let F_d be the set of functions an architecture A_d can compute, with $|F_d| < 2^{2^d}$ and membership decidable given d (any finite parameter set; any real-weight threshold architecture, by Tarski's decision procedure). The first truth table $f_d \notin F_d$ has $K(f_d) \le K(d) + c$ and $P_{A_d}(f_d) = 0$. So no bound $P_A(f) \ge 2^{-aK(f)-b}$ with a, b independent of d holds for such a family. (b) The parameter–function map alone implies no simplicity bias: a lookup-table architecture with uniform parameters gives every f probability 2^{-2^d}.*

The slides make the point of (a) with functions that diagonalise against all small circuits. So the right comparison object for networks is *circuit* SFI, not program SFI. For the uniform prior over circuits of one size, one half of the comparison holds (Definition 8.9 for N_s and size):

Proposition 9.5 (uniform circuit priors contain the circuit simplicity prior)**.** [proved] context Prop 4.5 *Let $P_{\mathrm{unif},S}$ be uniform over circuits of size exactly S. For every f with $\mathrm{size}(f) < S$, $P_{\mathrm{unif},S}(f) \ge 1/(N_{\mathrm{size}(f)} \cdot 16(d+1+S)^2)$. Hence Bayes with this prior makes at most $(\ln 2)(\log N_{\mathrm{size}(f)} + 2\log(d+1+S) + 4) = O(\mathrm{size}(f)\log(\mathrm{size}(f)+d) + \log S)$ Gibbs mistakes on every ordering.*

Proof idea (Appendix H). Pad a minimal circuit with arbitrary dead gates and a final gate copying its output. This is one half of Hänni's "first main afaik-open problem" (Hänni, 2026b); the other half, that the uniform prior is not much *larger* than the simplicity prior on functions without small circuits, is open.

Remark 9.6 (what a bridge would give). [remark] context §4.3 If $P_A(f) \geq 2^{-a\ell(f)-b}$ for a description length ℓ, then on every ordering NN Bayes makes at most $(\ln 2)(a\ell(f)+b)$ Gibbs mistakes and at most $a\ell(f)+b$ posterior-majority mistakes (as in Theorem 7.7 and Remark 8.13); if also $P_A(f) \leq 2^{-a'\ell(f)+b'}$, the log loss on the full domain is at least $a'\ell(f)-b'$. Proposition 9.5 is the lower half for uniform circuit priors. The converse, "few mistakes implies a small circuit", which Hänni thought "true for NN bayes as well", is claimed by the slides in their setting and in distribution (Remark 9.10), and holds for circuit Bayes on every ordering (Theorem 8.10).

The Gaussian-process limits. Hänni suggests that in the infinite-width Gaussian limits (NNGP and NTK) the network is "doing sth more like a polyfit" than circuit SFI (Hänni, 2026b). On the cube $\{\pm 1\}^d$ with uniform measure, let $\chi_S(x) = \prod_{j\in S} x_j$ be the Walsh characters and, for a kernel $\mathcal{K}$, $(T_{\mathcal{K}}g)(x) := \mathbb{E}_{x'}\mathcal{K}(x,x')g(x')$.

Theorem 9.7 (Walsh diagonalisation; dimension bound)**.** [proved; (a) known, (b) standard; computed (context c2)] context Thm 4.6 *(a) If $\mathcal{K}(x,x') = \Psi(\langle x,x'\rangle/d)$, every χ_S is an eigenfunction of $T_{\mathcal{K}}$, with eigenvalue $\Lambda_{|S|}$ depending only on $|S|$. This covers the NNGP and NTK kernels of fully connected networks with i.i.d. Gaussian weights and biases, which are functions of $\langle x,x'\rangle$ on the cube. (b) If $\mathcal{V}$ is any space of functions on the cube of dimension at most n (for instance $\operatorname{span}\{\mathcal{K}(x_i,\cdot)\}_{i\leq n}$, which contains the GP posterior mean and the kernel ridge regression solution), then $\binom{d}{k}^{-1}\sum_{|S|=k}\min_{g\in\mathcal{V}}\|\chi_S-g\|^2 \geq 1-n/\binom{d}{k}$.*

Proof idea (Appendix H). (a) $\mathcal{K}(x,x')$ depends only on $x\odot x'$, so characters are eigenfunctions; permutation symmetry makes the eigenvalue depend on $|S|$ only. (b) The squared projections of all characters onto $\mathcal{V}$ sum to $\dim\mathcal{V}$. Part (a) is in Yang and Salman (2019) (as summarised in its abstract and in the slides) and is proved in the slides by the same argument; (b) is the standard dimension argument, cf. Kamath et al. (2020).

Corollary 9.8 (kernels versus SFI on sparse parities)**.** [proved; computed (context c2)] context Cor 4.7 *For $|S|=k$ and i.i.d. uniform inputs, SFI ($\nu\in\{W,M_F\}$) has cumulative posterior error at most $(\ln 2)(K(\chi_S)+c)$ on every sequence of inputs, with $K(\chi_S)\leq \log\binom{d}{k}+2\log(k+1)+2\log(d+1)+c$, and the squared error of its posterior-mean predictor averaged over the first n steps is at most $4(\ln 2)(K(\chi_S)+c)/n$. Every kernel method or GP posterior mean needs $n\geq(1-\varepsilon)\binom{d}{k}$ samples for average squared error at most ε over random k-sparse targets, while SFI needs $O(k\log d/\varepsilon)$ in this averaged sense.*

The separation is in squared error of the real-valued predictor; we do not claim a 0–1 version in general. For one kernel family a 0–1 failure is exact:

Remark 9.9 (bias-free two-layer ReLU kernels and odd parities). [proved; computed (context c2, c10, c12); known] context Rem 4.8(a)–(c) (a) For the bias-free two-layer ReLU NNGP, $\Psi_1(\rho) = (\sqrt{1-\rho^2}+\rho(\pi-\arccos\rho))/\pi$, and for the NTK, $\Psi_1(\rho)+\rho\Psi_0(\rho)$ with $\Psi_0(\rho)=(\pi-\arccos\rho)/\pi$, the function $\Psi(\rho)-a\rho$ is even ($a=\frac{1}{2}$, resp. $a=1$). Hence $\Lambda_{|S|}=0$ for every odd $|S|\geq 3$, $\mathcal{V}:=\operatorname{span}\{\mathcal{K}(x_i,\cdot)\}\perp\chi_S$, and $\|\chi_S-\hat{g}\|^2 = 1+\|\hat{g}\|^2\geq 1$ for every n: the kernel predictor never learns these parities. This is the cube version of the finding of Basri et al. (2019) that a shallow network without bias cannot learn odd frequencies (on the sphere, as stated in their abstract). Biases or a third layer make

the eigenvalues nonzero. (b) Let $z \sim \mathrm{GP}(0, \mathcal{K})$ for one of these kernels and $|S| \geq 3$ odd. Then $\mathbb{E}_x z(x)\chi_S(x)$ is Gaussian with variance $\Lambda_{|S|} = 0$, so it vanishes almost surely, while $\operatorname{sign} z = \chi_S$ would force it to equal $\mathbb{E}|z| > 0$. So the GP prior, and every posterior obtained by conditioning on an event of positive probability (sign or margin likelihoods), gives probability exactly 0 to the classifier χ_S. (c) For distinct training inputs $x_1, \ldots, x_n$ with labels $\chi_S(x_i)$ and the set U of the other $2^d - n$ points, every $g \in \mathcal{V}$ has $\sum_{x \in U} g(x)\chi_S(x) = -\sum_{i \leq n} g(x_i)\chi_S(x_i)$. For kernel ridge regression (the GP mean with noise variance $\beta > 0$) the training sum is $y^\top K(K + \beta I)^{-1} y \geq 0$, so the predictor is weakly anti-correlated with the target on the unseen points. A direct computation (Appendix H) gives kernel ridge regression a sign accuracy on unseen points of 0.05 to 0.07 at $n = 900$, $d = 10$, far below chance; this is computed, not proved.

Remark 9.10 (the Vaintrob–Hänni slides). [known; unverified (claimed in the slides); warm-up arithmetic re-checked (context c8)] context §4.5; slides 10–17, 29–37, App. A, F, J, K The slides warn that their appendices "have largely been generated by Claude" and have not been "meticulously reviewed", that the main results summarise "much more caveated and detailed versions", that only the strong-prior case is proved in the talk, and that Bayesian learning and SGD are "importantly different". They claim: (i) for programs with prior $\propto 3^{-|p|}$ and i.i.d. inputs, $O(C)$ samples (C the length of the shortest program for f) give average posterior probability above 99.9% on the correct answer, with probability $1 - \exp(-\Omega(C))$, and a "relevance-based" sample-complexity bound for any countable class; (ii) for an MLP of depth L and any width $N \geq C := C_L(f)$ (the least width and weight bound of a depth-L MLP implementing f), with bounded, Lipschitz, non-polynomial activation and Gaussian weights in the strong-prior limit, $\mathrm{poly}(d, C)$ samples suffice in the same sense, the polynomial hiding a factor exponential in depth; (iii) the orderings of functions by C_L, by a layer-weighted squared norm κ_N and by prior mass are "scrambled only polynomially", uniformly in width; (iv) an in-distribution mistake bound $\mathrm{poly}(d, C_L(f))$ in expectation, and "the circuit complexity C_L of the learned function is at most $\mathrm{poly}(d, \#\text{mistakes made while learning it})$"; (v) a mean-field analogue (Vaintrob and Hänni, 2026a), while the NNGP "does not have a good circuit prior"; (vi) out of distribution the guarantee "is simply false" (Theorem 8.14). We re-checked the proof and arithmetic of (i), the character argument of their Appendix J (Theorem 9.7(a)) and the ladder of Appendix K; the rest is unchecked. Corollary 4.14(a) is the Markov half of the relevance-based bound, applied to W and M_F. Claim (iii) concerns orderings, not two-sided bounds on prior mass; the slides list "a lower bound on the log prior of a function" as open.

Remark 9.11 (what can be said rigorously). [remark] context §4.6 Proved: NN Bayes is function-space Bayes; its prior is at most $2^{-K+O(K(a))}$, nearly equal on its typical set, with no general lower bound, so the comparison object is circuit SFI; uniform circuit priors contain the circuit simplicity prior; the Gaussian-process limits fail on sparse and odd parities where SFI succeeds; and for circuit Bayes, mistakes and circuit size are polynomially related on every ordering (Theorem 8.10), while Occam selection on the same prior fails exponentially out of distribution (Theorem 8.14). Claimed in the slides: the strong-prior theorems. Open: two-sided bounds on $P_A(f)$ at fixed σ, and whether NN Bayes generalises like SGD-trained networks at all, which Hänni doubts (Hänni, 2026b).

Remark 9.12 (simplicity-bias literature). [known (snippets; unverified)] context §4.7 Valle-Pérez et al. (2019) argue, following Dingle et al. (2018), that the parameter–function map of deep networks is exponentially biased towards simple functions. Hänni's objection (Hänni, 2026b) that the upper bound holds for every map while the needed lower bound fails is made rigorous by Propositions 9.3 and 9.4; the slides argue the same. Mingard et al. (2021) report that SGD samples functions with probabilities close to the Bayesian posterior, and Mingard et al. (2025) attribute much of generalisation to a prior bias towards simple functions; we take no position. Buzaglo et al. (2024)

show that random interpolators under a flat parameter prior generalise when a narrow teacher exists; the slides call this the closest prior work and note that its bound degrades with student width. Goldblum et al. (2024) relate no-free-lunch theorems to Kolmogorov complexity, and Grau-Moya et al. (2024) meta-train networks on data from universal machines to approximate Solomonoff induction. These descriptions follow the abstracts.

Open questions. The upper half of the uniform-circuit bridge (Proposition 9.5); a lower bound on the NN log prior at fixed σ; a check of the slides' main theorem and of "mistakes imply a small circuit"; the per-point test error of GP classification on odd parities; a proof that kernel-ridge sign accuracy on unseen points falls below $\frac{1}{2}$.

10 Steeper simplicity priors (summary of the companion paper)

Hänni's notes propose to penalise "the length of the program plus the negative log likelihood ... but with a higher coefficient on the length of the program", in function induction where "each input gets its own private randomness", so that the selected hypothesis is a simple model plus noise rather than a model plus memorised errors of the teacher (Hänni, 2026b; Hänni and Wyeth, undated). The companion paper (Hänni, 2026a) studies this proposal. It works with classes of hypotheses indexed by programs π of length $\ell(\pi)$, the objective $\lambda\ell(\pi) - \log P_\pi(D)$ with $\lambda \geq 1$, and the power posterior $\propto 2^{-\lambda\ell(\pi)} P_\pi(D)$. With λ proportional to the sample size n these are batch objectives at each n, not one Bayesian predictor. This section summarises the results; all of them are proved there.

The absorption objection. A standard point, which Hänni's note already makes for sequence prediction, is that under a steeper penalty a hypothesis that is itself a universal mixture absorbs everything. The companion paper's *absorption theorem* makes this exact: in any class that contains its own mixture U as a member of length ℓ_U (SI; SFI with shared randomness, $U = M_F$; the weak class, $U = M^w$; $\mathrm{SFI}^{\mathrm{hist}}$, $U = M_H$, which is universal by Corollary 5.18), every λ-MAP has length at most $(\lambda\ell_U + O(1))/(\lambda - 1)$ on every data set and fits almost as well as U, so a deterministic teacher's errors are reproduced. In the class of product hypotheses (private randomness per input, Hänni's x-only samplers) no member is universal: every fixed product hypothesis loses $2^d - 1$ bits on some parity on $\{0,1\}^d$ that has a program of length $d + 2\log d + O(1)$, and the universal conditional semimeasure never learns. So steeper SFI is defined over the product class, whose $\lambda \to 1$ limit is M_{prod}, not M_F. This restriction has a price. At $\lambda = 1$ it is the gap between M_{prod} and M_F, unbounded but at most $K(\mathrm{dom}\, D) + O(1)$ bits (Proposition 3.7 and Remark 3.8). At $\lambda > 1$, on labels that share a latent fair coin between pairs of inputs, the product class pays $\min(\lambda, 2)$ bits per coin against 1 bit for M_F. The protection ends where hypotheses see earlier *labelled* pairs: the in-context Bayes mixture is then one product hypothesis again. The paper presents the restriction to private randomness (per input; on data sets the same as per query, Proposition 4.7) as a modelling choice: the model is the part of the description reused across inputs.

The teacher's mistakes. Hänni's example has a 100-bit simple model and systematic errors on a 2^{-10} fraction of the inputs whose set takes 10,000 bits. The companion paper realises it rigorously with a random parity on $\{0,1\}^{100}$ and an error rule on the first 20 input bits (11,710 bits), quantifying over *all* product hypotheses by list decoding. For error rules that are incompressible

relative to the model, the *teacher's-mistakes theorem* gives a window of λ/n from $2^{-20} \approx 9.54 \cdot 10^{-7}$ up to $8.06 \cdot 10^{-3}$ (machine constant set to 0, an illustration) or $3.66 \cdot 10^{-3}$ (machine constant 30). Inside it every near-optimal hypothesis knows the model and does not reproduce the errors; at $\lambda/n = 10^{-4}$, with the machine constant set to 0, it gives the parity's label average probability at least 0.983, holds at most 37.7 of the 11,710 bits of the error rule, and gives the erroneous label average probability at most 0.199 on the error inputs. A few per cent below 2^{-20} and further down, near-optimal hypotheses reproduce the errors. The theorem uses a list-decodable model class, the product form of the error set, uniform inputs, an error density below 0.056, and a small randomness deficiency of the error rule. Plain Bayes (M_{prod}, M_F), SI, and the λ-MAP under every schedule with $\lambda/n \to 0$ learn the errors, so λ must grow linearly in n, and the learner is then deliberately not consistent for the teacher's label distribution. That a bounded λ overfits and consistency needs λ to grow is known for λ-weighted two-part MDL in noisy classification (Grünwald and Langford, 2007; Zhu and Srebro, 2025; Zhu et al., 2026), where the target is the data distribution rather than a norm that differs from it.

Limits. The trick sorts components by rate, not by validity. Density and description length do not suffice: with Hänni's density and an error set made of a cheap subcube of 128 points plus 896 random points (about 10,400 bits), no hypothesis that fits no better than the good model is near-optimal for $\lambda/n < 1.6 \cdot 10^{-5}$, about 31% of the window's logarithmic width (25% with machine constant 30). This refutes the extension of Hänni's example, which he called "somewhat cherry-picked", to every error rule with his density and description length; whether structured error rules leave a window is open. A cheap, frequent error rule defeats the trick for all input lengths above a threshold linear in the machine constant (rate blindness); λ cannot be chosen from imitation data, and the window depends on the input law. Out of distribution, λ proportional to n removes the constant lower bound of Proposition 7.1, which becomes $2^{-\lambda(K(Q,b)+O(1))}$, but only by sharpening the preference for the shortest hypotheses that fit the training data; no guarantee about unseen inputs follows.

The model–randomness split. Hänni also asked for "a preference for a simple sampler over a model where the sampled bits are filled in" and noted that Solomonoff induction has none, "both get similar posterior" (Hänni, 2026c). At $\lambda = 1$ this is confirmed with explicit factors for every computable sampler ν with a program of length k: in SFI the posterior odds of the deterministic fill-ins against the sampler lie in $[2^{-2\log(t_n+1)-O(1)}, 2^{k+\delta^*}]$, with t_n doubly logarithmic in the input length and δ^* a Ville deficiency with $\Pr[\delta^* \geq j] \leq 2^{-j}$, whatever the amount of sampled randomness; this covers non-dyadic samplers such as the Bern(1/3) threshold hypotheses of Eisenstat's example (Definition 6.10). SI behaves the same up to $2^{O(k+K(n))}$. For every fixed $\lambda > 1$ in every countable product class, the sampler wins: every λ-MAP has length at most $k + (k+\delta^*)/(\lambda-1)$, the fill-ins' posterior is at most $2^{\lambda k+\lambda\delta^*-(\lambda-1)(-\log\nu(D_n))}$, and with infinitely many distinct inputs the λ-MAPs are eventually equivalent to ν on the inputs seen (summable Hellinger distances, not equality). With shared randomness, or in SI, the universal mixture wins instead: no sampler with $(\lambda-1)k > \lambda\ell_U + O(1)$ is ever selected. In the language of algorithmic statistics, for a noisy random parity on the full truth table the set (shared-randomness) sophistication is $O(\log d)$, so the parity's parameter counts as noise, while the product sophistication is $d \pm O(\log d)$ for every flip rate below 1/4: the product structure function has the kink at the simple model that Hänni anticipated, where with shared randomness "1 bit pays for 1 bit".

11 Related work

Solomonoff's operator induction. Solomonoff (2008) describes three kinds of induction; the third concerns "an unordered set of ordered pairs", where one predicts the second element of a new pair from the first (we have seen only the abstract). Secondary sources describe its prediction as a mixture of computable conditional models weighted by $2^{-|O^j|}\prod_i O^j(a_i \mid q_i)$, with a convergence theorem bounding the total error by $\ln 2$ times the complexity of the source (Özkural, 2015, 2017). Read this way it is the per-query mixture M_{pq}, which equals M_{prod} on distinct inputs (Remark 3.2); x-only dominance on distinct inputs is then Solomonoff's, and M_{pq} is the natural object for fresh label noise (Proposition 4.7). SFI differs in sharing randomness across inputs (Proposition 3.7).

Hutter's chronological semimeasures. With inputs as actions and labels as percepts, $\mathrm{SFI}^{\mathrm{hist}}$ is a universal chronological semimeasure in the sense of AIXI (Hutter, 2000, 2005) (Corollary 5.18); that it agrees with Hutter's construction up to constant factors is expected but was not checked against his texts; Hutter lists supervised learning from examples among AIXI's problem classes (Hutter, 2007) (Remark 4.17). Lattimore et al. (2011) show, according to their abstract, that normalised Solomonoff induction learns computable sub-patterns of otherwise unstructured sequences while the unnormalised version can fail; this bears on SI applied to $x_1y_1x_2y_2\cdots$ (Proposition 5.22). Wood et al. (2013) state, in their abstract, that Solomonoff's and Levin's constructions define the same class of priors for sequences; Theorem 5.17 is the corresponding statement for chronological semimeasures. The observation that a history-aware universal mixture is the natural ideal learner from examples is therefore prior work in substance.

Universal priors for classification. Lattimore and Hutter (2013) use algorithmic information theory to argue against no-free-lunch pessimism and give a Solomonoff-inspired off-line classifier; whether their random-selection result coincides with Proposition 4.10 we have not checked. Poland and Hutter (2005) show that two-part MDL prediction can have loss bounds exponentially worse than Bayes mixtures; the slides compare this with the gap between Occam and weighted majority (Remark 8.15). Leike and Hutter (2015a) show that AIXI's optimality depends on the reference machine; Proposition 7.4 is a supervised analogue. Ryabko (2006) studies deterministic labels with conditionally independent objects.

Learning theory. The guarantees of Section 4 are of the Occam (Blumer et al., 1987), PAC-Bayes (McAllester, 1999), MDL (Barron and Cover, 1991; Grünwald, 2007), halving and weighted-majority (Littlestone, 1988; Littlestone and Warmuth, 1994), expert-advice (Vovk, 1990; Cesa-Bianchi and Lugosi, 2006) and online-to-batch (Littlestone, 1989; Cesa-Bianchi et al., 2004) kinds; Proposition 4.15 makes the comparison. Conformal prediction (Vovk et al., 2005) is an exchangeability-based supervised framework related to Proposition 4.10; Vovk's earlier universal forecasting algorithms (Vovk, 1992) we have not examined. Inductive inference of recursive functions (Gold, 1967; Barzdin and Freivalds, 1972; Case and Smith, 1983) is the deterministic, identification-in-the-limit ancestor, mostly for inputs in the order $0, 1, 2, \ldots$.

Computable versions and networks. Speed priors (Schmidhuber, 2002; Filan et al., 2016) are time-bounded universal priors for sequences; Section 8 gives function versions. The simplicity-bias literature on neural networks is discussed in Remark 9.12, and the steeper-prior material, with its relation to λ-weighted MDL, in Section 10.

Remark 11.1 (what exists and what appears new). [known (snippets; unverified)] context §6; history §4.6; foundations §6.6 At the level of a search that saw abstracts and secondary sources only, the following exist: the per-query product mixture for supervised pairs with its convergence theorem (operator induction, as secondary sources describe it); the universal chronological semimeasure; Solomonoff-style classifiers and discrete MDL for classification; Occam, PAC-Bayes and online-to-batch bounds, including the exchangeability argument of Proposition 4.10; the halving bound and its contrast with argmax selection on the same prior (the slides); the Walsh diagonalisation and dimension bound for kernels and the vanishing odd eigenvalues of bias-free ReLU kernels (Theorem 9.7 and Remark 9.9(a)); machine dependence of universal priors; and the conjunction ladder (the slides). We found no prior use of the name "Solomonoff function induction" and none of the following: the function-machine form of the universal chronological semimeasure, the separations between SFI and $\mathrm{SFI}^{\mathrm{hist}}$, and the comparisons with SI on the interleaved sequence (Section 5); the OOD results of Section 7; the length-activated, round-clocked and history-aware time-bounded mixtures (Section 8.1); the every-ordering form of circuit Bayes with explicit constants (Theorem 8.10; the statement itself is Hänni's claim); the uniform-circuit padding bound (Proposition 9.5); and the GP-classification and anti-correlation statements of Remark 9.9(b),(c). Each of these should be checked against Solomonoff (2008), Hutter (2005) and Lattimore and Hutter (2013) before any claim of novelty.

12 Discussion

SFI as ideal supervised learning. The positive results are what one asks of an ideal learner, and they need no assumption on how the inputs are chosen. SFI's cumulative log loss is within the complexity of any computable random labeller on every consistent data sequence (Theorem 4.1); when the labels come from a computable random function and the inputs do not depend on unrevealed labels, its predictions merge with the true ones (Theorem 4.3); and with i.i.d. inputs the averaged predictor has risk $O(K/n)$ (Theorem 4.5). These guarantees are relative: they compare SFI with computable labellers, in log loss, and they say nothing about which labeller the data come from. The out-of-distribution results have the same character (Remarks 7.9 and 7.14). On inputs that a simple property separates from the training data, SFI keeps a probability bounded below for every simple behaviour, whatever the amount of data and whatever the true labels; how large that probability is depends on the universal machine (Propositions 7.1 and 7.4). In exchange, for a computable target f the total posterior error over any sequence of distinct inputs is at most $(\ln 2)(K(f) + c)$ (Theorem 7.7). The limits listed in Remark 4.16—log loss, averaged predictors, function semantics, computability, machine constants, distribution shift and the comparator class—are each a precise statement, not a caveat.

What the hypotheses may see. The large gaps between SFI and Solomonoff induction in Sections 5 and 6 are properties of order-free prediction rather than of SFI. Every order-independent predictor pays them (Propositions 5.5, 6.5 and 6.16), and hypotheses that see the list of previous inputs remove them (Corollary 5.18, Proposition 6.7, and Theorem 6.19). Which of SFI and $\mathrm{SFI}^{\mathrm{hist}}$ is the right ideal therefore depends on whether the order of the data carries information about the labels. With i.i.d. inputs it does not, and SFI's guarantees apply as stated. When a teacher chooses the order, it can carry almost all of the information, as in the cyclic permutation, where the labels repeat the inputs. The set of previous inputs is not enough to recover this information (Propositions 5.19 and 6.8), and even with the list, "as good as Solomonoff induction"

holds against SI's joint loss, not automatically against its conditional label loss (Theorem 5.20 and Proposition 5.21).

Which version to call SFI. The versions of Section 3 agree up to $K(\operatorname{dom} D) + O(1)$ bits, and their gaps, although unbounded, are small on the known witnesses: about $\log d$ for the single shortest program on two points with d-bit inputs, and a padded tape-length cost for the program mixture (Corollary 3.10 and Proposition 3.6). For deterministic computable targets all of them learn at the same rate (Remark 3.13 and Proposition 4.15), so the choice matters mostly for random labels and for code lengths. There is no coding theorem for function programs (Theorem 3.9): the single shortest consistent program, which is what minimum description length with one hypothesis selects, and the Bayes mixture are different objects for functions, unlike for strings. We call M_F SFI because it is the object of Hänni (2026b), it has a posterior over functions, it dominates every machine law with constant $2^{-K(\nu)}$, it is invariant, and its semimeasure-convention loss does not depend on the order of the data (Remark 2.19). Where the randomness lives also matters for noise: a random function cannot produce fresh noise on a repeated input, while a per-query mixture can (Propositions 4.6 and 4.7). The companion paper shows that it matters for steeper priors as well (Section 10).

Computable versions. The same structure appears with bounded resources. Time-bounded mixtures keep constant regret against all predictors that see only the input and run within a time bound (Proposition 8.4 and Corollary 8.5). For circuits, Bayes mistakes and circuit size are polynomially related on every ordering (Theorem 8.10), while the simplest consistent hypothesis can fail exponentially out of distribution (Theorem 8.14): the selection rule matters more than the prior (Remark 8.15). For neural networks, the parameter–function map alone implies no simplicity bias (Proposition 9.4); any bridge to circuit SFI must come from the architecture and the prior, which is what the slides of Vaintrob and Hänni (2026b) claim for the strong-prior limit (Remark 9.10).

Relation to the literature. As secondary sources describe it, Solomonoff's operator induction is the per-query mixture M_{pq}, which equals the product mixture on distinct inputs; SFI differs from it by sharing randomness across inputs (Remark 3.2 and Proposition 3.7). The history-aware version is, in substance, Hutter's universal chronological semimeasure (we have not checked the identification against his texts), and Hutter lists supervised learning among AIXI's problem classes (Remark 4.17). The learning theorems are of the Occam, PAC-Bayes and online-to-batch kinds, with K in place of $\log |H|$ (Proposition 4.15). Section 11 gives details. Our literature search reached abstracts and secondary sources only, so we claim no novelty for anything listed there as existing (Remark 11.1).

Open problems. Collected from the sections.

- *Foundations.* Is every lower-semicomputable law of a random partial function dominated by M_F (Section 2.3)? Is there a two-sided characterisation of $-\log M_F(D)$ by standard complexities (Section 2.5)? What is the arithmetical level of SFI's conditionals (Section 2.5)? Is the normalised loss of M_F order-dependent, and which semimeasure-only lower bounds hold for it (Section 2.4)?
- *Versions.* Is $\log(M_F/W) = O(K(d, |D|))$, and what are the exact rates of the gaps of Table 2 (Section 3.4)? Does any computable family of data sets exhibit an unbounded gap

(Section 3.3)?

- *Learning.* Does the last iterate, not only the average, have expected risk $O(K/n)$ (Remark 4.9)? Is the leave-one-out error of SFI $O(K_F(f|_B)/|B|)$ up to logarithms (Conjecture 4.12)?
- *History.* Can the $K(n)+K(A)$ term of Corollary 5.14 be removed, and does the theorem extend to defective samplers and infinite alphabets (Remark D.6)? Can $K(\Delta_f)$ be replaced by an algorithmic mutual information (Conjecture 5.15)? How does $\mathrm{SFI}^{\mathrm{hist}}$ compare with normalised Solomonoff induction's label loss, and do the separations of Proposition 5.19 and Theorem 5.26 hold normalised (Section 5.6)?
- *Examples.* The normalised loss over the full cyclic permutation and in Eisenstat's example (Section 6.3); the depth count for weak semimeasures (Section 6.2); SI's label loss with raw undelimited blocks (Conjecture E.5).
- *Out of distribution.* A positive statement for properties that are simple given a program fitted to the training data (Section 7.4); a version of "natural machines generalise as intended" for a fixed reference machine (Conjecture 7.17).
- *Computable versions and networks.* Efficient constant regret against time-bounded samplers (Section 8.1); a circuit coding theorem for size priors with polynomial tails, and Occam failures beyond conjunctions (Section 8.3); the upper half of the uniform-circuit bridge, a lower bound on the network log prior, and a check of the slides' main theorem (Section 9).

The companion paper. Hänni (2026a) asks what a steeper simplicity prior does in function induction with private randomness per input. In a window of λ/n it keeps a learner from reproducing a teacher's incompressible systematic errors, and for fixed $\lambda > 1$ it prefers a simple sampler to a model with filled-in random bits; with shared randomness, or in sequence prediction, a universal hypothesis absorbs everything instead (Section 10). Whether structured error rules leave such a window is open there.

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Code and data

Every computation reported here was done by a Python script with a fixed seed or by a deterministic one. The scripts, their saved outputs, and the research notes with full proofs are in the folder

`solomonoff-function-induction/` of the repository `https://github.com/kaarelh/AI-works`: the scripts are in `research/tracks/*/checks/`, `research/tracks/*/referee_code/` and `code/`. Each script saves its output next to itself, and re-running the scripts reproduces the saved outputs, apart from timing lines and one time-limited optimisation run.

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A Function semimeasures and SFI: proofs

Notation is that of Section 2. Tapes are $\omega = \omega_1\omega_2\cdots$; a finite union of cylinders $[w]$ is identified with the corresponding finite union of half-open dyadic intervals of $[0,1)$, which changes nothing up to a $\mathbb{U}$-null set.

A.1 Function semimeasures

Proof of Proposition 2.3. (a) Along any order the conditionals telescope to $\nu(D)/\nu(\emptyset) = \nu(D)$; if an intermediate value vanishes, so does $\nu(D)$, by monotonicity. They sum to at most 1 by the weak inequality. (b) The empty product is 1. Revealing D first and x last gives $\nu_q(D + (x, y)) = \nu_q(D)\, q(y \mid D, x)$; summing over y gives the weak inequality, and dividing gives the formula for q. □

Proposition A.1 (Normalised predictors are random total functions)**.** [proved] context Prop 5.1(b) *If every conditional of an order-independent predictor q (Proposition 2.3) sums to* 1*, then ν_q is the containment functional of a unique law of a random total function $F \in Y^X$.*

Proof of Proposition A.1. Now $\sum_y \nu_q(D + (x, y)) = \nu_q(D)$ for all D and $x \notin \operatorname{dom} D$ (also where $\nu_q(D) = 0$, by monotonicity). By induction, for each finite $A \subseteq X$ the map $l \mapsto \nu_q(l)$ on labellings $l \in Y^A$ is a probability distribution, and these distributions are consistent under marginalisation. Y is countable, so Kolmogorov's extension theorem gives a law on Y^X; it is unique because its cylinder probabilities are determined. □

Proof of Proposition 2.5. Convergence. For each $R \subseteq T$, $\sum_{E:\operatorname{dom} E=R} \nu(D \cup E) \le \nu(D)$, by the second consequence of Definition 2.2 applied to the weak semimeasure $E \mapsto \nu(D \cup E)$ on data sets with domain disjoint from $\operatorname{dom} D$. So $\sum_{E:\operatorname{dom} E\subseteq T} \nu(D \cup E) \le 2^{|T|}\nu(D)$, and $\nabla_T \nu(D)$ converges absolutely.

Necessity. Let $\nu = \nu_P$ and $A_t := \{F(t) \neq \bot\}$ for $t \in T$. By inclusion–exclusion,

$$P\Big(F \supseteq D,\ \bigcap_{t\in T} A_t^c\Big) = \sum_{R\subseteq T} (-1)^{|R|} P\Big(F \supseteq D,\ \bigcap_{t\in R} A_t\Big),$$

and $P(F \supseteq D, \bigcap_{t\in R} A_t) = \sum_{E:\operatorname{dom} E=R} \nu(D \cup E)$, a countable disjoint union over the labellings of R. Collecting terms gives $\nabla_T \nu(D)$, which is therefore a probability.

Sufficiency. For finite $S \subseteq X$ and $v \in Y_\bot^S$ let v^{def} be the data set of the defined entries of v and $S_\bot(v)$ the set where $v = \bot$, and put $q_S(v) := \nabla_{S_\bot(v)} \nu(v^{\mathrm{def}}) \ge 0$. For $x \notin S$, split the sum defining $q_{S\cup\{x\}}(v, \bot)$ by whether $x \in \operatorname{dom} E$; absolute convergence allows the rearrangement, and

$$q_{S\cup\{x\}}(v, \bot) = q_S(v) - \textstyle\sum_{a\in Y} q_{S\cup\{x\}}(v, a).$$

So $\sum_{a\in Y_\bot} q_{S\cup\{x\}}(v, a) = q_S(v)$, and $q_\emptyset = \nu(\emptyset) = 1$. By induction each q_S is a probability distribution on the countable set $Y_\bot^S$, and the family is consistent under marginalisation. Kolmogorov's extension theorem gives a law P on $Y_\bot^X$ with these marginals, and $P(F \supseteq D) = q_{\operatorname{dom} D}(D) = \nabla_\emptyset \nu(D) = \nu(D)$. The cylinder probabilities of any such P are the Möbius sums, by necessity; so P is unique. □

A.2 Machine laws and filling-in processes

Proof of Proposition 2.8. Each coordinate of f_ω^T is determined by countably many cylinders, so $\omega \mapsto f_\omega^T$ is measurable into $Y_\bot^X$. The set $\{\omega : f_\omega^T(x) = y\} = \bigcup_{(x,w,y)\in T}[w]$ is effectively open, uniformly in (x, y) and an index of T; a finite intersection of effectively open sets is effectively open uniformly; and the measure of an effectively open set is lsc uniformly in its description. $\{f_\omega \supseteq \emptyset\}$ is the whole space.

Enumeration. Let (W_i) enumerate all c.e. sets of triples. Let T_i be obtained by enumerating W_i and accepting each triple unless it conflicts (same input, comparable strings, different labels) with a triple already accepted. Each T_i is c.e. and consistent, and $T_i = W_i$ whenever W_i is consistent. So (T_i) lists exactly the function machines. □

Proof of Theorem 2.9. T_U is c.e. If $(x, pw, y), (x, p'w', y') \in T_U$ with $pw, p'w'$ comparable, then p, p' are comparable elements of the prefix-free set $\operatorname{dom} V$, so $p = p'$, and $(x, w, y), (x, w', y')$ are triples of the consistent $T_{V(p)}$ with w, w' comparable; so $y = y'$. For $\omega = p\omega'$ with $p \in \operatorname{dom} V$, $f_\omega^{T_U} = f_{\omega'}^{T_{V(p)}}$, and ω' is uniform and independent of p. A tape with no prefix in $\operatorname{dom} V$ gives the empty function, which extends only $\emptyset$. Summing over p gives (1), and $\mathbf{m}(i) \geq 2^{-K(i)}$ gives the bound. □

Lemma A.2 (Laws of random total functions)**.** [proved] foundations Lemma 6.5a *For countable Y, every lsc law ν of a random total function ($\nu(\emptyset) = 1$, $\nu(D) = \sum_y \nu(D + (x, y))$) is computable, uniformly in an index. So the machine laws of random total functions are exactly the computable laws of random total functions.*

Proof of Lemma A.2. Induction on $|D|$. $\nu(\emptyset) = 1$ is computable. If $\nu(D)$ is computable, then $\nu(D + (x, y)) = \nu(D) - \sum_{y' \neq y} \nu(D + (x, y'))$ is a computable real minus an lsc (possibly infinite) sum, hence upper semicomputable; it is also lsc, hence computable. The induction is uniform. The converse is Proposition 2.10(c), last clause. □

Corollary A.3 (Universality among normalised order-independent predictors)**.** [proved] context Cor 5.2(c), extended by foundations Lemma 6.5a *For countable Y: if ν is lsc and weak, $\nu(\emptyset) = 1$, and all its conditionals sum to 1, then $M_F \geq 2^{-K(\nu)-O(1)}\nu$.*

Proof. By Proposition A.1, ν is the law of a random total function, hence computable (Lemma A.2), hence a machine law whose index is computable from that of ν (Proposition 2.10(c)); apply Theorem 2.9. □

So order-independence alone gives the weak semimeasures, where M^w is universal (Theorem 2.12); adding normalised predictions gives the random total functions, where M_F is universal.

Remark A.4 (Computable filling-in processes). [proved] foundations Rem 2.7 A *computable filling-in process* is a Markov chain $F_0 = \emptyset \subseteq F_1 \subseteq F_2 \subseteq \cdots$ of finite partial functions whose transition kernel $\Pr[F_{t+1} = g \mid F_t = f]$ is supported on finitely many extensions $g \supseteq f$, given as an explicit list with rational probabilities computable from (t, f). The law of its limit $F_\infty := \bigcup_t F_t$ is a machine law, with an index computable from the kernel. Conversely, every machine law is the law of the limit of the nested process $F_t :=$ "the part of f_ω certified by triples enumerated within t steps with $|w| \leq t$", which depends on $\omega_{1:t}$; this process is Markov in $\omega_{1:t}$, not in F_t. So machine laws are exactly the limits of computable nested processes driven by a hidden finite state.

Proof. The machine reads the tape bit by bit and samples $F_1, F_2, \ldots$ in turn: to sample from a finitely supported law with rational probabilities it reads bits until the dyadic interval named by the bits read so far lies inside one of the consecutive half-open intervals whose lengths are those probabilities, and takes that outcome. This fails to terminate only when the tape's real is an endpoint, a null set. On query x it runs the process until the first t with $x \in \operatorname{dom} F_t$ and outputs $F_t(x)$; it diverges if there is none. So T contains (x, w, y) when the process, run on the bits of w, produces $F_1, \ldots, F_t$ with t the first index such that $x \in \operatorname{dom} F_t$, and $F_t(x) = y$. The process reads its bits in order, so on comparable strings it produces the same initial segment of the chain; with $F_t \subseteq F_{t+1}$ this makes T consistent. Off the null set the sampled chain has the law of the process, and $f_\omega = F_\infty$. The converse is a restatement of Definition 2.7. □

A.3 Realising lower-semicomputable tree semimeasures

Lemma A.5 (Realisation)**.** [proved (binary case known: Zvonkin–Levin 1970, unverified)] foundations Lemma 3.1; check r2 *Let A be a countable alphabet with a fixed enumeration and ψ an lsc semimeasure on A^*: $\psi(\varepsilon) \le 1$ and $\psi(s) \ge \sum_{a\in A} \psi(sa)$. There are finite unions of cylinders $I_t(s)$, increasing in t and computable from (t, s) and an index of ψ, such that $I(s) := \bigcup_t I_t(s)$ satisfies $I(sa) \subseteq I(s)$, $I(sa) \cap I(sb) = \emptyset$ for $a \ne b$, and $\mathbb{U}(I(s)) = \psi(s)$ for every s; the cylinders enumerated for strings of one length are pairwise disjoint. Consequently $\omega \mapsto$ "the longest s with $\omega \in I(s)$" is a monotone machine whose output extends s with probability exactly $\psi(s)$, and the event "the output extends s" is certified by a finite prefix of ω.*

Proof. For binary sequences this is due to Zvonkin and Levin (1970) (not checked against the source); the proof below covers countable alphabets. Let ψ_t be computable dyadic lower approximations, nondecreasing in t, with $\psi_t \to \psi$ and $\psi_t(s) = 0$ unless $|s| \le t$ and s uses only the first t letters (round a rational approximation down to a multiple of 2^{-t}; this keeps monotonicity in t). On this finite tree define ψ'_t from the leaves up by $\psi'_t(s) := \max\{\psi_t(s),\ \sum_a \psi'_t(sa)\}$. Then:

- $\psi'_t \le \psi$, by induction from the leaves, since $\sum_a \psi'_t(sa) \le \sum_a \psi(sa) \le \psi(s)$;
- ψ'_t is nondecreasing in t (by induction from the leaves of the tree of stage t, using $\psi_{t+1} \ge \psi_t$), and $\psi'_t \ge \psi_t \to \psi$, so $\psi'_t \to \psi$;
- $\psi'_t(s) \ge \sum_a \psi'_t(sa)$, and $\psi'_t(\varepsilon) \le \psi(\varepsilon) \le 1$.

Build I_t from the root down. Let $I_t(\varepsilon) \supseteq I_{t-1}(\varepsilon)$ be a finite union of cylinders of measure $\psi'_t(\varepsilon)$. Given $I_t(s) \supseteq I_{t-1}(s)$ of measure $\psi'_t(s)$, the room $I_t(s) \setminus \bigcup_a I_{t-1}(sa)$ has measure $\psi'_t(s) - \sum_a \psi'_{t-1}(sa) \ge \sum_a (\psi'_t(sa) - \psi'_{t-1}(sa))$. All quantities are dyadic, so after refining to a common cylinder length we can add to each $I_{t-1}(sa)$ fresh cylinders inside the room, pairwise disjoint and of total measure $\psi'_t(sa) - \psi'_{t-1}(sa)$. The invariants are preserved, and $\mathbb{U}(I(s)) = \lim_t \psi'_t(s) = \psi(s)$. Siblings stay disjoint because the I_t increase in t. Each $I_t(s)$ is a finite union of cylinders, so $\omega \in I(s)$ is witnessed by a finite prefix. □

A.4 What machine laws include

Proof of Proposition 2.10. (a) Apply Lemma A.5 with alphabet Y, depth one and $\psi(\varepsilon) := 1$, $\psi(y) := P(y \mid x)$, uniformly in x. This gives c.e. unions of cylinders $G_{x,y}$, disjoint in y, with $\mathbb{U}(G_{x,y}) = P(y \mid x)$. The private stream of x is $\omega^{(x)}_i := \omega_{\langle j(x),i\rangle}$. Let T contain (x, w, y) when w contains the first L bits u of $\omega^{(x)}$, for some L, and $[u] \subseteq G_{x,y}$ is certified. If $(x, w, y), (x, w', y') \in T$ with $w \sqsubseteq w'$, the two stream prefixes u, u' are comparable, so $[u] \cap [u'] \ne \emptyset$ and $G_{x,y} \cap G_{x,y'} \ne \emptyset$; hence $y = y'$. For distinct inputs the streams use disjoint tape coordinates, so they are independent and uniform, and $\nu_T(D) = \prod_{(x,y)\in D} \mathbb{U}(G_{x,y}) = \nu_P(D)$. The index of T is computable from one of P. For the claim on samplers: the output probabilities of a probabilistic machine are lsc (enumerate halting computations); the construction just given is the converse.

(b) $T := \{(x, \varepsilon, \varphi(x)) : x \in \operatorname{dom}\varphi\}$.

(c) Enumerate X as $x_1, x_2, \ldots$ and $Y_\perp$ with $\perp$ last. Build nested partitions of $[0, 1)$: the interval $J(v)$ of $v \in Y^k_\perp$ is split, from left to right, into the intervals $J(va)$ of lengths $q_{k+1}(va)$, $a \in Y$ in order, and then $J(v\perp)$. Each left endpoint is $\mathrm{left}(v) + \sum_{a'<a} q_{k+1}(va')$, and $J(v\perp)$ starts at $\mathrm{left}(v) + q_k(v) - q_{k+1}(v\perp)$; all are computable reals. Let T contain (x_k, w, y) when the dyadic interval of w is certified, by rational approximations of the endpoints, to lie in the interior of some $J(v)$ with $v \in Y^k_\perp$ and $v_k = y \ne \perp$. Intervals of one level are disjoint and the levels are nested, so T is consistent. Off the null set of endpoints a tape lies in the interior of exactly one $J(v)$ at each

level, so $\mathbb{U}\{f_\omega(x_i) = v_i,\ i \le k\} = \mathbb{U}(J(v)) = q_k(v)$ for every v, and the law of f_ω is P. If Y is finite, each $q_S(v) = \nabla_{S_\perp(v)}\nu_P(v^{\mathrm{def}})$ is a finite alternating sum of computable reals. For a random total function $q_S(v) = \nu_P(v)$ if v has no $\perp$, and 0 otherwise.

(d) Realise w by Lemma A.5 (depth one, alphabet $\mathbb{N}$). This enumerates pairwise disjoint cylinders $[u]$, each labelled by some i, those labelled i having total measure w_i; disjointness makes the set of strings u prefix-free. Let $T := \{(x, uw', y) : [u] \text{ enumerated with label } i,\ (x, w', y) \in T_i\}$. As in Theorem 2.9, T is consistent, the tape after u is uniform and independent of u, and $\nu_T(D) = \sum_i w_i\nu_{T_i}(D)$ for $D \neq \emptyset$; tapes outside all enumerated cylinders give the empty function. □

A.5 Weak universality and its gap

Proof of Theorem 2.12. Let (ϕ_i) enumerate all lsc maps from data sets to $[0, 1]$, with computable approximations $\phi_{i,t}$ nondecreasing in t. At stage t let U_t be the finite set of data sets whose inputs and labels are among the first t inputs and labels. Define $\phi'_{i,t}$ on U_t (and 0 outside) by decreasing $|\operatorname{dom} D|$:

$$\phi'_{i,t}(D) := \max\Big\{\phi_{i,t}(D),\ \max_{x\le t,\ x\notin\operatorname{dom} D}\ \textstyle\sum_{y\le t}\phi'_{i,t}(D + (x, y))\Big\}.$$

If $\phi'_{i,t}(\emptyset) > 1$, freeze: $\phi'_{i,s} := \phi'_{i,t-1}$ for all $s \ge t$. Then $\phi'_{i,t}$ is nondecreasing in t (by induction from the largest domains) and satisfies the weak inequality everywhere: inside U_t by construction; outside U_t both sides vanish, since a data set outside U_t has no extension inside it; a frozen $\phi'_{i,t-1}$ is weak for the same reason. Let $\nu_i := \lim_t \phi'_{i,t}$. For $x \notin \operatorname{dom} D$, monotone convergence gives $\sum_y \nu_i(D + (x, y)) = \lim_t \sum_y \phi'_{i,t}(D + (x, y)) \le \lim_t \phi'_{i,t}(D) = \nu_i(D)$, and $\nu_i(\emptyset) \le 1$; so ν_i is an lsc weak semimeasure. If ϕ_i is already weak, induction from the largest domains gives $\phi'_{i,t} \le \phi_i$; so there is no freezing, and $\phi_{i,t} \le \phi'_{i,t} \le \phi_i$ gives $\nu_i = \phi_i$. Put $M^w := \sum_i \mathbf{m}(i)\nu_i$. The weak inequalities are linear and $\sum_i \mathbf{m}(i) \le 1$, so M^w is an lsc weak semimeasure, and it dominates each ν_i with factor $\mathbf{m}(i) \ge 2^{-K(i)}$. □

Proof of Theorem 2.13. (i) M_F is an lsc weak semimeasure (Proposition 2.8); apply Theorem 2.12.

(ii) Fix $S = \{s_1 < \cdots < s_k\}$ and put $\psi_S(y_1 \cdots y_j) := \nu(\{(s_1, y_1), \ldots, (s_j, y_j)\})$ for $j \le k$ and $\psi_S := 0$ on longer strings. The weak inequality at s_{j+1} is exactly the semimeasure inequality for ψ_S, and ψ_S is lsc uniformly in (ν, S). Let $I(\cdot)$ realise ψ_S (Lemma A.5), and let $T_{\nu,S}$ contain (s_j, w, y) when $[w]$ lies in an enumerated cylinder of some $I(u)$ with $|u| = j$ and $u_j = y$. The sets $I(u)$ with $|u| = j$ are pairwise disjoint, so $T_{\nu,S}$ is consistent; and a tape lies in $I(u^{(j)})$ for each $j \le k$ exactly when it lies in $I(u^{(k)})$ and $u^{(j)}$ is the length-j prefix of $u^{(k)}$. So for $\operatorname{dom} D = S$, $\nu_{T_{\nu,S}}(D) = \psi_S(y_{s_1} \cdots y_{s_k}) = \nu(D)$. (For proper subsets of S only "$\le$" holds; check c6 confirms both facts on examples.) The index of $T_{\nu,S}$ is computable from (ν, S), so Theorem 2.9 gives $M_F(D) \ge 2^{-K(\nu,S)-O(1)}\nu(D)$, and $K(\nu, S) \le K(\nu) + K(S) + O(1)$. Take $\nu = M^w$, of complexity $O(1)$, for the second claim.

(iii) Use $\nu_\triangle$ of Example 2.6 on the blocks $B_j := \{3j, 3j + 1, 3j + 2\}$, $j < k$ (inputs as natural numbers). Put $\nu_\triangle^{\otimes k}(D) := \prod_{j<k}\nu_\triangle(D|_{B_j})$ if $\operatorname{dom} D \subseteq \bigcup_{j<k} B_j$, and 0 otherwise. A one-point inequality at $x \in B_j$ involves only the j-th factor, and at x outside the blocks both sides vanish; so $\nu_\triangle^{\otimes k}$ is weak (c1 checks $k = 2$ exhaustively, 1458 inequalities; r1 also $k = 3$). $\tilde\nu := \sum_k \mathbf{m}(k)\nu_\triangle^{\otimes k}$ is an lsc weak semimeasure, so $M^w \ge 2^{-O(1)}\tilde\nu$. For $\vec\imath \in \{1, 2, 3\}^k$ let $E_{\vec\imath}$ be the union over j of the copy of E_{i_j} in block j. These 3^k data sets are pairwise inconsistent, each has $|\operatorname{dom} E_{\vec\imath}| = 2k$ and $\nu_\triangle^{\otimes k}(E_{\vec\imath}) = 2^{-k}$. M_F is the containment functional of a random partial function, so $\sum_{\vec\imath} M_F(E_{\vec\imath}) \le 1$, and some $D_k := E_{\vec\imath}$ has $M_F(D_k) \le 3^{-k}$. But $M^w(D_k) \ge 2^{-O(1)}\mathbf{m}(k)2^{-k} \ge 2^{-k-K(k)-O(1)}$.

(iv) Each $-\log M^w(E_{\vec{\imath}}) \le k + K(k) + O(1)$ as in (iii). By concavity of log,

$$3^{-k}\textstyle\sum_{\vec{\imath}} -\log M_F(E_{\vec{\imath}}) \;\ge\; -\log\left(3^{-k}\sum_{\vec{\imath}} M_F(E_{\vec{\imath}})\right) \;\ge\; k\log 3.$$

Subtract.

(v) The argument of (iii) shows $\min_{\vec{\imath}} P(F \supseteq E_{\vec{\imath}}) \le 3^{-k}$ for every law P of a random partial function, so the ratio to $\nu_\triangle^{\otimes k}(E_{\vec{\imath}}) = 2^{-k}$ is at most $(2/3)^k$. Let P_1 be the uniform law on the three total functions of Example 2.6(b); it satisfies $P_1 \ge \frac{2}{3}\nu_\triangle$ on all ten data sets of the support of $\nu_\triangle$ (checked by hand and by c9(b)). Its k-fold product over independent blocks has $P_1^{\otimes k}(F \supseteq D) = \prod_j P_1(F|_{B_j} \supseteq D|_{B_j}) \ge (2/3)^k \nu_\triangle^{\otimes k}(D)$, with equality on the $E_{\vec{\imath}}$ (c9(b) for $k \le 4$; r1). □

On the family of (iii), $\operatorname{dom} E_{\vec{\imath}}$ is determined by $(\vec{\imath}, k)$, so $K(\operatorname{dom} D_k) \le k\log 3 + K(k) + O(1)$. Larger label sets might give gaps closer to $K(\operatorname{dom} D)$; we have not worked this out.

A.6 The open question on random-function laws

Reformulation (proof sketch; foundations Problem 3.6). Suppose an lsc ν_P has approximations $\nu_{P_t} \uparrow \nu_P$, where each P_t is a law of a random partial function on a finite domain with finitely many labels and rational probabilities, computable from t, and consecutive laws admit a coupling with $F_t \subseteq F_{t+1}$. The couplings form a polytope given by rational linear constraints, so a rational coupling can be found by search, and Remark A.4 then realises ν_P exactly as a machine law. By Strassen's theorem for finite partially ordered sets (Strassen, 1965) (quoted, not checked against the source), such a coupling exists if and only if $P_t(U) \le P_{t+1}(U)$ for every up-set U of the extension order. The containment events $\{F \supseteq D\}$ are only the principal up-sets, so $\nu_{P_t} \le \nu_{P_{t+1}}$ does not give this. The question becomes whether every lsc ν_P admits, at a constant-factor loss, approximations that increase on all up-sets.

Two steps are not written out: the Strassen step is quoted, and the reduction from general lower approximations of ν_P to finite rational laws P_t is not carried out. An adversary argument in the spirit of Example 2.6, forcing repeated couplings on fresh blocks, was tried and settled neither direction.

A.7 The normalised convention

Proposition A.6 (Two-point criterion)**.** [proved; computed] foundations Prop 5.4 (= its referee's Prop B); checks c10(a), r13 *Let P be the law of a random partial function on inputs a, b, with $A := \{F(a) \ne \bot\}$, $B := \{F(b) \ne \bot\}$ and $P(A), P(B) > 0$. Consider*

$$(\star) \qquad P(B \mid F(a) = \alpha) = P(B \mid A) \quad \textit{and} \quad P(A \mid F(b) = \beta) = P(A \mid B)$$

for all α, β of positive probability. (i) If $(\star)$ holds, the normalised probability of $\{(a,\alpha),(b,\beta)\}$ is the same in both reveal orders, for every (α,β). (ii) Conversely, if every two-point probability $P(F(a) = \alpha, F(b) = \beta)$, $\alpha, \beta \in Y$, is positive (as for M_F) and the two orders agree for every (α,β), then $(\star)$ holds. (iii) Without positivity (ii) can fail.

Proof. Put $p(\alpha,\beta) := P(F(a) = \alpha, F(b) = \beta)$, $u(\alpha) := P(B \mid F(a) = \alpha)/P(B)$ and $v(\beta) := P(A \mid F(b) = \beta)/P(A)$. Revealing a first gives the normalised product

$$\frac{P(F(a) = \alpha)}{P(A)} \cdot \frac{p(\alpha,\beta)}{P(F(a) = \alpha,\, B)} = \frac{p(\alpha,\beta)}{P(A)P(B)\,u(\alpha)},$$

and revealing b first gives $p(\alpha,\beta)/(P(A)P(B)\,v(\beta))$. When $p(\alpha,\beta)>0$ the two agree exactly when $u(\alpha)=v(\beta)$; when $p(\alpha,\beta)=0$ both are 0. (i) $(\star)$ gives $u\equiv P(A\cap B)/(P(A)P(B))\equiv v$. (ii) Agreement on all pairs forces $u(\alpha)=v(\beta)$ for all α,β, so $u\equiv v\equiv\gamma$ for a constant γ. Averaging u over α with weights $P(F(a)=\alpha)/P(A)$ gives $\gamma=P(A\cap B)/(P(A)P(B))$, which is $(\star)$. (iii) The law with $(F(a),F(b))$ equal to $(0,0)$ with probability $\frac{1}{4}$, $(1,1)$, $(1,\perp)$, $(\perp,1)$ with probability $\frac{1}{8}$ each, and $(\perp,\perp)$ with probability $\frac{3}{8}$ has $P(B\mid F(a)=0)=1$, $P(B\mid F(a)=1)=\frac{1}{2}$, $P(B\mid A)=\frac{3}{4}$; it puts positive mass only on the diagonal defined pairs and is symmetric in a,b, so its normalised probabilities are order-independent (c10(a4)). □

The law of Proposition 2.17(b) violates $(\star)$: $P(B\mid F(a)=0)=\frac{1}{2}\neq\frac{2}{3}=P(B\mid A)$. Total laws always satisfy it. Check c10 found agreement between $(\star)$ and order independence on 400 of 400 random laws with positive two-point cells (r13: 300 of 300).

Proposition A.7 (What transfers to the normalised convention)**.** [proved; (b) uses the relativised coding theorem (known); computed]

foundations Prop 5.5; checks c10(b,c) *Let $Z_t:=\mathcal{Z}(D_{t-1},x_t)/M_F(D_{t-1})$, the posterior probability that $f_\omega(x_t)$ is defined ($Z_t:=1$ at a repeated input). (a) $\bar{L}_{\mathrm{SFI}}=L_{\mathrm{SFI}}(D_n)-\sum_t\log(1/Z_t)$. As $Z_1\geq 2^{-c}$, single points transfer: $-\log\bar{M}_F(y\mid\emptyset,x)=K(y\mid x)\pm O(1)$. (b) $\bar{L}_{\mathrm{SFI}}\geq K^{\emptyset'}(y_{1:n}\mid x_{1:n})-O(1)$ for all inputs and labels, $K^{\emptyset'}$ relative to the halting problem. (c) For any joint law of inputs and labels (inputs may depend on past labels), $\mathbb{E}\bar{L}_{\mathrm{SFI}}\geq\sum_t\mathbb{E}H(y_t\mid h_{t-1},x_t)$, the loss of the chronological Bayes predictor.*

Proof of Proposition A.7. (a) Each normalised factor is the raw factor divided by Z_t. At the first step $Z_1=\mathbb{U}\{f_\omega(x_1)\neq\perp\}\geq 2^{-c}$, because a constant function is a machine law of complexity $O(1)$; combine with Theorem 2.21(c).

(b) Each normalised conditional is a ratio of two positive lsc reals, hence computable relative to $\emptyset'$, uniformly. For a fixed input list, $y_{1:n}\mapsto\prod_t\bar{M}_F(y_t\mid D_{t-1},x_t)$ is a probability distribution on Y^n: sum out y_n, then y_{n-1}, and so on; each factor sums to 1, and a repeated input contributes a point mass on its stored label (c10(b) checks this exactly on 16,200 cases, repeated inputs included). Apply the conditional coding theorem relative to $\emptyset'$.

(c) The normalised predictor is a history-aware predictor issuing probability vectors. At each step, conditionally on (h_{t-1},x_t), Gibbs' inequality gives $\mathbb{E}[-\log\bar{M}_F(y_t\mid D_{t-1},x_t)\mid h_{t-1},x_t]\geq H(y_t\mid h_{t-1},x_t)$; take expectations and sum. □

A.8 Invariance, coding bounds, sequence induction, computability

Proof of Theorem 2.20. (a) Each of M_F,M'_F is a machine law in the other's enumeration, so each dominates the other up to a constant factor by Theorem 2.9; the same argument applies to any two function machines whose laws dominate all machine laws up to constants.

(b) α^{-1} and β^{-1} are partial computable (by search). The set of triples $(\alpha(x),w,\beta(y))$ with $(x,w,y)\in T_U$ is a function machine over X',Y' (consistent because α is injective), whose random function is $\beta\circ f_\omega\circ\alpha^{-1}$, undefined off the range of α. Its law ν' satisfies $\nu'(D^{\alpha\beta})=\mathbb{U}\{\beta(f_\omega(x))=\beta(y)\ \forall(x,y)\in D\}=M_F(D)$, since β is injective. Its index is computable from indices of α,β, so $M'_F(D^{\alpha\beta})\geq 2^{-K(\alpha,\beta)-O(1)}M_F(D)$. Conversely, the machine with random function $\beta^{-1}\circ f'_\omega\circ\alpha$ on X has law ν with $\nu(D)=M'_F(D^{\alpha\beta})$, so $M_F(D)\geq 2^{-K(\alpha,\beta)-O(1)}M'_F(D^{\alpha\beta})$.

(c) For W: a program for φ becomes a program for $\beta\circ\varphi\circ\alpha^{-1}$ by a prefix of length $K(\alpha,\beta)+O(1)$, and conversely. For M_{prod}: transform each conditional semimeasure in the same way. For M^w: given a weak ν on $X\times Y$, put $\nu'(D'):=\nu(D)$ if $D'=D^{\alpha\beta}$ and $\nu'(D'):=0$ otherwise; ν' is lsc (search for preimages). If $D'=D^{\alpha\beta}$ and $x'\notin\operatorname{dom}D'$, the extensions of D' at x' of the form

$E^{\alpha\beta}$ have $x' = \alpha(x)$ and labels $\beta(y)$, so the one-point inequality transfers; if D' is not an image, neither is any extension of it. The converse transformation is symmetric. □

Proof of Theorem 2.21. (a) $P(E \mid S) := M_F(E)$ for $\operatorname{dom} E = S$ is lsc uniformly in (E, S) and sums to at most 1 over E (second consequence of Definition 2.2). The conditional coding theorem gives $K(E \mid S) \leq -\log M_F(E) + O(1)$. Take $E = D|_T$, $S = T$, and use $M_F(D) \leq M_F(D|_T)$.

(b) First term: the shortest consistent program p gives the machine law $\mathbf{1}[\varphi_{V(p)} \supseteq D]$, whose complexity is at most $|p| + O(1)$, so $M_F(D) \geq 2^{-K_F(D)-O(1)}$ (equivalently, $M_F \geq 2^{-O(1)}W$ by Theorem 3.4). Second term: $\mathbf{m}(y \mid x) := \sum_{q:V(q,x)=y} 2^{-|q|}$, with $V(\cdot, x)$ the conditional universal prefix machine, is an lsc conditional semimeasure with $\mathbf{m}(y \mid x) \geq 2^{-K(y|x)}$; its product law is a machine law of complexity $O(1)$ (Proposition 2.10(a)). Last inequality: $K_F(D) \leq K(D) + O(1)$ (a program computes the lookup table of D from a shortest description of D) and $K(D) \leq K(\operatorname{dom} D) + K(D \mid \operatorname{dom} D) + O(1)$.

(c) Combine (a) with $T = \{x\}$ and (b). □

Proof of Proposition 2.23. ($\geq$) Let T_1 contain (i, w, b) when the universal monotone machine, run on w, has printed more than i bits and its bit $i + 1$ is b. Outputs on comparable strings are comparable, so T_1 is a function machine, and $\nu_{T_1}(D_x) = M(x)$; apply Theorem 2.9. ($\leq$) The monotone machine that prints $f_\omega(0), f_\omega(1), \ldots$ in order, as long as each value is certified along ω and lies in $\{0, 1\}$, prints a string extending x exactly when $f_\omega \supseteq D_x$. So $x \mapsto M_F(D_x)$ is the output semimeasure of a monotone machine, and M dominates it (Zvonkin and Levin, 1970). □

Proof of Proposition 2.24. (a) Proposition 2.8 and the remark after Definition 2.15.

(c) Take $X = \mathbb{N}$ and suppose the raw conditional is computable. Define f by recursion: with $D_t := \{(s, f(s)) : s < t\}$, let $f(t)$ be the first $y \in \{0, 1\}$ for which approximation certifies $M_F(y \mid D_t, t) < 0.6$. The two conditionals sum to at most 1, so one of them is at most $\frac{1}{2}$, and the search halts. f is total computable, and $M_F(D_n) = \prod_{t<n} M_F(f(t) \mid D_t, t) < 0.6^n$, which contradicts $M_F(D_n) \geq 2^{-K(f)-O(1)}$ for large n. For the normalised conditional the same construction gives normalised loss at least $n \log(1/0.6)$, while it is at most $L_{\mathrm{SFI}}(D_n) \leq K(f)+O(1)$ (Proposition 2.17(a)).

(b) If M_F were computable, the raw conditional would be a ratio of positive computable reals, hence computable.

(d) Lsc reals are computable relative to $\emptyset'$; the conditionals are ratios of positive lsc reals ($M_F(D)$ and, for the normalised one, $\mathcal{Z}(D, x)$). □

B Kolmogorov and semimeasure versions: proofs

Notation is that of Section 3. $j(x)$ is the length-lexicographic index of $x \in \{0, 1\}^*$, so $j(x) \leq 2^{d+1} - 2$ for $|x| \leq d$.

B.1 The variants and the chain

Remark B.1 (Why W's programs delimit themselves before reading the input). [proved] foundations remark after Def 4.1; context §1.1 With this choice W is the containment functional of the random partial function "draw p with probability $2^{-|p|}$, output $\varphi_{V(p)}$": a mixture of fixed functions with input-independent weights. Summing $2^{-|p|}$ over the programs of a conditional prefix machine $V(p, x)$ is a different object: for each x the halting programs form a prefix-free set, but their union over x need not, so the weights define no prior over functions. Reading a conditional program

lazily from one shared tape does give a random function, the function machine $x \mapsto V(\omega, x)$; it is a machine law of complexity $O(1)$, not a mixture of fixed functions, and on single points it gives $2^{-K(y|x)\pm O(1)}$ (Theorem 2.21(c)). The two notions of conditional complexity are close: with $K^{\mathcal{P}}(z \mid x) := \min\{|h| : h \in \operatorname{dom} V,\ \varphi_{V(h)}(x) = z\}$,

$$K(z \mid x) - c \ \le\ K^{\mathcal{P}}(z \mid x) \ \le\ K(z \mid x) + K(K(z \mid x)) + c,$$

since every $h \in \operatorname{dom} V$ is a conditional program, and a conditional program prefixed with a self-delimiting code of its length delimits itself without reading x.

Proof of Theorem 3.4. (1) The shortest consistent program is one term of W. (2) $P_e(y \mid x) := \mathbf{1}[\varphi_e(x) = y]$ is a conditional semimeasure, lsc uniformly in e; let g be computable with $P_{g(e)} = P_e$. Then $i \mapsto \sum_{e:g(e)=i} \mathbf{m}(e)$ is an lsc discrete semimeasure, so $\mathbf{m}(i) \ge 2^{-O(1)} \sum_{e:g(e)=i} \mathbf{m}(e)$, and $M_{\mathrm{prod}}(D) \ge 2^{-O(1)} \sum_e \mathbf{m}(e)\mathbf{1}[\varphi_e \supseteq D] = 2^{-O(1)} W(D)$. (3) M_{prod} is an lsc mixture of product laws, a machine law (Proposition 2.10(a,d)); apply Theorem 2.9. (4) Theorem 2.13(i). (5) $K_F(D) \le K(D) + O(1)$ (lookup table), $K(D) \le K(\operatorname{dom} D) + K(D \mid \operatorname{dom} D) + O(1)$, and $K(D \mid \operatorname{dom} D) \le -\log M^w(D) + O(1)$ by the conditional coding theorem applied to $E \mapsto M^w(E)$ on data sets with domain $\operatorname{dom} D$, whose total is at most 1 (Definition 2.2). □

B.2 Programs that cannot separate inputs

Proof of Proposition 3.5. $\Omega_V^{\ge \ell} \to 0$ because $\sum_{p \in \operatorname{dom} V} 2^{-|p|} \le 1$.

(a) Let E_ℓ be the set of indices $V(p)$ with $|p| < \ell$; it has fewer than 2^ℓ elements. Colour $x \in \{0,1\}^d$ by $(\chi_e(x))_{e \in E_\ell}$, where $\chi_e(x) = \varphi_e(x)$ if this is 0 or 1 and $\chi_e(x) = *$ otherwise. There are at most 3^{2^ℓ} colours, so if $2^d > 3^{2^\ell}$ two distinct strings x_1, x_2 share a colour. No $e \in E_\ell$ has $\varphi_e(x_1) = 0$ and $\varphi_e(x_2) = 1$, so $W(\{(x_1, 0), (x_2, 1)\}) \le \Omega_V^{\ge \ell}$. Choose ℓ with $\Omega_V^{\ge \ell} \le \varepsilon$, then d with $2^d > 3^{2^\ell}$. For the lower bounds, the uniform product law $P(0 \mid x) = P(1 \mid x) = \frac{1}{2}$ has complexity $O(1)$ and gives every two-point binary data set probability $\frac{1}{4}$; so $M_{\mathrm{prod}}(D) \ge 2^{-O(1)}/4$, and $M_F \ge 2^{-O(1)} M_{\mathrm{prod}}$ by Theorem 3.4.

(b) Fewer than 2^{d-1} programs have length at most $d-2$, and each gives at most one value at 0^d. So at least 2^{d-1} strings $y \in \{0,1\}^d$ receive W-mass only from programs of length at least $d-1$. Each program contributes to at most one y, so these $W(\{(0^d, y)\})$ sum to at most $\Omega_V^{\ge d-1}$, and one of them is at most $\Omega_V^{\ge d-1}/2^{d-1}$. For the lower bound, $P(y \mid x) := 2^{-|x|}\mathbf{1}[|y| = |x|]$ is a computable conditional probability of complexity $O(1)$, so $M_{\mathrm{prod}}(\{(0^d, y)\}) \ge 2^{-O(1)} 2^{-d}$. □

Remark B.2 (How fast the gap of W grows). [proved; rate open]

foundations Rem 4.3a Let $\ell_d := \lceil \log(d/\log 3) \rceil - 1$, the largest ℓ with $3^{2^\ell} < 2^d$. By Proposition 3.5(a), some two-point binary D on d-bit inputs has $-\log W(D) \ge \log(1/\Omega_V^{\ge \ell_d})$. By Proposition 3.6(b), every such D has $-\log W(D) \le 2 + K^{\mathrm{pad}}(2^{d+1}) + O(1)$. Both bounds are unbounded and both are of padded type, since $\log(1/\Omega_V^{\ge \ell}) \le K^{\mathrm{pad}}(\ell) + O(1)$. By Proposition 3.6(c) the gap is at most $2\log(h(d)+2) + O_h(1)$ for every computable, nondecreasing, unbounded h. Its exact rate is open.

Proof of the claims in Remark B.2. ℓ_d is the largest ℓ with $2^\ell \log 3 < d$, i.e. with $3^{2^\ell} < 2^d$ (c8(a)). $\log(1/\Omega_V^{\ge \ell}) \le K^{\mathrm{pad}}(\ell) + O(1)$: choose $L \ge \ell$ with $K(L) = K^{\mathrm{pad}}(\ell)$. The prefix machine that reads a V-program for L, then L more bits, and halts, is simulated by V with a fixed prefix r. The resulting elements of $\operatorname{dom} V$ have length at least $L \ge \ell$ and total weight $2^L \cdot 2^{-|r|-K(L)-L}$, so $\Omega_V^{\ge \ell} \ge 2^{-K^{\mathrm{pad}}(\ell)-|r|}$. *Rate*: apply Proposition 3.6(c) with $g(N) := h(\max(0, \lfloor \log N \rfloor - 1))$, which is computable, nondecreasing and unbounded; then $K^{\mathrm{pad}}(2^{d+1}) \le 2\log(h(d)+2) + O_h(1)$. □

B.3 The tape-length bound and the hashing bound

Proof of Proposition 3.6. (a) *Monotonicity and limit.* A prefix of $\omega_{1:L}$ is a prefix of $\omega_{1:L'}$ for $L' \geq L$. The event $f_\omega \supseteq D$ holds exactly when each of the finitely many pairs of D has a certifying triple along ω, and then all of them lie within the longest one; so $\{f_\omega \supseteq D\} = \bigcup_L \{\text{certified within } L \text{ bits}\}$, an increasing union.

The bound. Fix $L' \geq L$ with $K(L') = K^{\mathrm{pad}}(L)$. For $w \in \{0,1\}^{L'}$ let $\varphi_{T,w}(x) := y$ if some $(x, w', y) \in T$ has w' a prefix of w, and undefined otherwise. This is well defined because T is consistent and any two prefixes of w are comparable, and it is partial computable (enumerate T). Let V' be the prefix machine that reads a V-program for an index i of T, then a V-program for L', then L' further bits w, and outputs an index of $\varphi_{T_i,w}$; its domain is prefix-free. By universality $V(rs) = V'(s)$ for a fixed string r. With shortest programs $q_i, q_{L'}$, the strings $r\, q_i\, q_{L'}\, w$, $w \in \{0,1\}^{L'}$, are distinct elements of $\operatorname{dom} V$, so

$$W(D) \;\geq \sum_{w \in \{0,1\}^{L'}} 2^{-|r|-K(T)-K(L')-L'}\, \mathbf{1}[\varphi_{T,w} \supseteq D] \;=\; 2^{-|r|-K(T)-K^{\mathrm{pad}}(L)}\, \nu^{(L')}(D),$$

and $\nu^{(L')}(D) \geq \nu^{(L)}(D)$. The consequence for $L_{1/2}$ is immediate, and $K(T_U) = O(1)$.

(b) Apply (a) to the machine $F(x) := \omega_{j(x)+1}$, of complexity $O(1)$, whose law gives every binary-labelled D probability $2^{-|D|}$. For $|x| \leq d$ its certificates lie within the first $2^{d+1} - 1$ tape bits, and $K^{\mathrm{pad}}(2^{d+1} - 1) \leq K^{\mathrm{pad}}(2^{d+1})$ since K^{pad} is nondecreasing. For the second inequality put $s := \lceil \log(d+1) \rceil$ and $L'' := 2^{2^s} \geq 2^{d+1}$; then $K^{\mathrm{pad}}(2^{d+1}) \leq K(L'') \leq K(s) + O(1)$.

(c) K^{pad} is nondecreasing because the minimum runs over a shrinking set, and unbounded because only finitely many L have $K(L) \leq c$ for each c. $K^{\mathrm{pad}}(2^{2^s}) \leq K(2^{2^s}) \leq K(s) + O(1)$. For g: let $N^* := \min\{N' : g(N') \geq g(N) + 1\}$. It exists because g is unbounded, it exceeds N because g is nondecreasing, and it is computable from $g(N) + 1$. So $K^{\mathrm{pad}}(N) \leq K(N^*) \leq K(g(N) + 1) + c_g \leq 2\log(g(N) + 2) + c'_g$, using $K(m) \leq 2\log(m+1) + O(1)$. □

Remark B.3 (The hashing bound, superseded). [proved; computed] foundations Rem 4.4a; checks c2, r10 For every D with labels in $\{0,1\}$, $|D| \leq k$ and $\operatorname{dom} D \subseteq \{0,1\}^{\leq d}$: $W(D) \geq 2^{-|D|-K(d,k)-c}$. Proposition 3.6(b) is stronger, since $K(d,k) \geq K(d) - O(1) \geq K^{\mathrm{pad}}(2^{d+1}) - O(1)$, and it does not depend on $|D|$. The natural guess of "a log factor via k-wise independent hashing" describes the single shortest program, not W: for K_F a logarithmic cost is real (Theorem 3.9), and Proposition 3.11 bounds it by a covering argument rather than by hashing.

Proof. We may assume $k < 2^{d+1}$, since $|\{0,1\}^{\leq d}| < 2^{d+1}$ (replacing k by $\min(k, 2^{d+1} - 1)$ changes $K(d,k)$ by $O(1)$). Put $q := d + 1$ and fix, computably from q, an irreducible polynomial of degree q over GF(2). Map $x \in \{0,1\}^{\leq d}$ injectively to the element z_x of $\mathrm{GF}(2^q)$ whose coefficient vector is the binary string $1x$. For a seed $s \in \mathrm{GF}(2^q)^k$ let $h_s(z)$ be the lowest bit of $\sum_{i<k} s_i z^i$. For distinct $z_1, \ldots, z_k$ the Vandermonde map $s \mapsto (\sum_i s_i z_l^i)_{l \leq k}$ is a bijection of $\mathrm{GF}(2^q)^k$, so for uniform s the values at any $j \leq k$ distinct points are uniform and independent, and so are their lowest bits (c2 checks this exhaustively for $(q,k) \in \{(3,2),(3,3),(4,2),(4,3)\}$). Hence $\Pr_s[h_s(z_x) = y\ \forall (x,y) \in D] = 2^{-|D|}$. Let V' read a V-program for (d,k), then kq further bits s, and output an index of $x \mapsto h_s(z_x)$ on $\{0,1\}^{\leq d}$; its domain is prefix-free. With $V(rs') = V'(s')$, summing over the 2^{kq} seeds gives $W(D) \geq 2^{-|r|-K(d,k)} \Pr_s[\cdots] = 2^{-|r|-K(d,k)-|D|}$. □

B.4 Shared randomness

Proof of Proposition 3.7. ν_{len} is the law of the function machine $\{(x,w,b) : |w| = |x|+1,\ w_{|x|+1} = b\}$, of complexity $O(1)$. Suppose $M_{\mathrm{prod}} \geq \gamma\, \nu_{\mathrm{len}}$ for some $\gamma \in (0,1]$. Put $\varepsilon := \gamma/8$, write

$M_{\mathrm{prod}} = \sum_i w_i \nu_{P_i}$ with $w_i = \mathbf{m}(i)$, and choose J with $\sum_{i>J} w_i \le \varepsilon$. For $n \ge 1$ let $a_n := 0^n$, $b_n := 1^n$, and $s_i(n,y) := P_i(y \mid a_n) P_i(y \mid b_n)$ for $y \in \{0,1\}$.

Compactness. Choose an infinite set of lengths along which $P_i(y \mid a_n) \to \alpha_i(y)$ and $P_i(y \mid b_n) \to \beta_i(y)$ for all $i \le J$ and $y \in \{0,1\}$. Let $S_i := \alpha_i(0)\beta_i(0) + \alpha_i(1)\beta_i(1) \le \max_y \alpha_i(y) \cdot (\beta_i(0) + \beta_i(1)) \le 1$. If $S_i = 1$, then $\max_y \alpha_i(y) = 1$, so $\alpha_i = \delta_{y_i}$ for some y_i, and then $\beta_i(y_i) = 1$, so $\beta_i = \delta_{y_i}$. Let $I_1 := \{i \le J : S_i = 1\}$, $I_0 := \{i \le J : S_i < 1\}$ and $\varrho := \max_{i \in I_0}(1 + S_i)/2 < 1$.

Subsequence. Choose lengths $n_1 < n_2 < \cdots$ from that set such that, for every j, $s_i(n_j, 0) + s_i(n_j, 1) \le \varrho$ for $i \in I_0$, and $s_i(n_j, 1 - y_i) \le \varepsilon' 2^{-j}$ for $i \in I_1$, where $\varepsilon' := \varepsilon/2$.

Data. For $b \in \{0,1\}^k$ let $D_b := \{(a_{n_j}, b_j), (b_{n_j}, b_j) : j \le k\}$. Then $\nu_{\mathrm{len}}(D_b) = 2^{-k}$. All D_b have the same domain, so $\sum_b \nu_{P_i}(D_b) = \prod_j \left(s_i(n_j,0) + s_i(n_j,1)\right) \le 1$ for every i.

Count. Let B_k be $\{0,1\}^k$ without the two constant vectors. For $i \in I_1$ and $b \in B_k$ the set $T_b := \{j : b_j \ne y_i\}$ is nonempty, so $\prod_j s_i(n_j, b_j) \le \prod_{j \in T_b} \varepsilon' 2^{-j}$, and $b \mapsto T_b$ is injective; hence $\sum_{b \in B_k} \prod_j s_i(n_j, b_j) \le \prod_j (1 + \varepsilon' 2^{-j}) - 1 \le e^{\varepsilon'} - 1 \le 2\varepsilon'$. Therefore

$$\textstyle\sum_{b \in B_k} M_{\mathrm{prod}}(D_b) \;\le\; \sum_{i>J} w_i + \sum_{i \in I_0} w_i \varrho^k + \sum_{i \in I_1} w_i \, 2\varepsilon' \;\le\; \varepsilon + \varrho^k + \varepsilon \;\le\; 3\varepsilon$$

once $\varrho^k \le \varepsilon$. But $\sum_{b \in B_k} \gamma \, \nu_{\mathrm{len}}(D_b) = \gamma(1 - 2^{1-k}) \ge \gamma/2 = 4\varepsilon$ for $k \ge 2$, a contradiction. ($M_F \ge 2^{-O(1)} \nu_{\mathrm{len}}$ by Theorem 2.9, so M_F / M_{prod} is unbounded.) □

For Remark 3.8: the machine above certifies the labels of all inputs of length at most d within $\omega_{1:d+1}$, so $\nu_{\mathrm{len}}^{(d+1)} = \nu_{\mathrm{len}}$ on such data, and Proposition 3.6(a) with Theorem 3.4 gives $M_{\mathrm{prod}}(D) \ge 2^{-O(1)} W(D) \ge 2^{-K^{\mathrm{pad}}(d+1) - O(1)} \nu_{\mathrm{len}}(D)$.

B.5 No coding theorem; the covering bound

Proof of Theorem 3.9. (i) Let $g := \ell_d = \lceil \log(d / \log 3) \rceil - 1$, so that $3^{2^g} < 2^d$ (c8(a) checks the arithmetic for all $d \le 2 \cdot 10^5$ and along powers of 2 up to 2^{200}). Fewer than 2^g programs have length less than g; let E be the set of their indices $V(p)$. Colour $x \in \{0,1\}^d$ by $(\chi_e(x))_{e \in E} \in \{0,1,*\}^E$ as in Proposition 3.5. There are at most $3^{|E|} < 2^d$ colours, so two distinct x_1, x_2 share a colour, and no $e \in E$ has $\varphi_e(x_1) = 0$, $\varphi_e(x_2) = 1$. So $K_F(D) \ge g \ge \log d - \log\log 3 - 1 > \log d - 2$. (ii) Proposition 3.6(b) with $|D| = 2$. (iii) The uniform product law, as in Proposition 3.5. For the last claim: by (i), $\log d \le K_F(D) + 2$, so $\lceil \log(d+1) \rceil \le K_F(D) + 4$; and $K(m) \le \log m + 2 \log\log m + O(1)$ for $m \ge 2$. □

In terms of sets of programs: with $S_D := \{e : \varphi_e \supseteq D\}$, $W(D) = \mathbf{m}(S_D)$ and $K_F(D) = \min_{e \in S_D} K(e)$. Levin's bound on the lowest complexity of the elements of a set relates exactly these two quantities to the information the set carries about the halting problem (Levin, 2016; Epstein and Levin, 2011) [known; unverified]; we have not checked its statement, nor whether it applies to infinite c.e. sets such as S_D, and nothing here depends on it. For binary sequences, monotone complexity and $-\log M$ also differ unboundedly (Gács, 1983; Day, 2011) (statements not checked); we make no comparison of rates.

Proof of Corollary 3.10. Lower bound: Theorem 3.9(i),(iii). Upper bound: $L_{\mathrm{SFI}}(D) \ge 0$, and Proposition 3.11 with $k = 2$ gives $K_F(D) \le 2 + \log(2(d+2)) + K(\lceil \log(d+2) \rceil, 2) + O(1)$. □

Proof of Proposition 3.11. Put $s := \lceil \log(d+2) \rceil$ and $d' := 2^s - 2 \ge d$; then $d' + 2 < 2(d+2)$ (c8(d)). There are at most $\binom{2^{d'+1}-1}{k} 2^k \le 2^{k(d'+2)}$ binary-labelled data sets of size k on $\{0,1\}^{\le d'}$. Put $N_{\mathrm{cov}} := \lfloor 2^k k (d'+2) \ln 2 \rfloor + 1$. N_{cov} independent uniform truth tables on $\{0,1\}^{\le d'}$ all miss

a fixed such D with probability $(1-2^{-k})^{N_{\mathrm{cov}}} < e^{-N_{\mathrm{cov}}2^{-k}} \le 2^{-k(d'+2)}$. By the union bound the expected number of uncovered data sets is below 1, so some family of N_{cov} truth tables covers all of them. The lexicographically first such family is computable from (d',k), hence from (s,k), by exhaustive search. A program consists of a V-program for (s,k) followed by a $\lceil \log N_{\mathrm{cov}} \rceil$-bit index (its length is computable from (s,k), so the program is self-delimiting); it outputs an index of the chosen truth table, undefined off $\{0,1\}^{\le d'}$. Its length is at most $K(s,k) + \log N_{\mathrm{cov}} + 1 + O(1)$, and $\log N_{\mathrm{cov}} \le k + \log(k(d'+2)) \le k + \log(k(d+2)) + 1$. For the consequence, $L_{\mathrm{SFI}}(D) \ge K(D \mid \operatorname{dom} D) - O(1) = k - \delta_D - O(1)$ by Theorem 2.21(a). □

Check c8(c) computes, for $d \le 3$ and $k \le 3$, the exact expected number of uncovered data sets (< 1 in all nine cases) and finds that random families of size N_{cov} cover everything in at least two thirds of the trials in every case. Explicit two-point coverings also exist: the $\lfloor N/2 \rfloor$-subsets of $\{0,\dots,N-1\}$ that contain 0, read as N functions on $\binom{N-1}{\lfloor N/2 \rfloor - 1}$ inputs, show all four label patterns on every pair of inputs (c8(b), $N \le 14$). This is the construction of Kleitman and Spencer (1973), reported to be optimal [known; unverified]; we did not check optimality against the source, and the checks confirm it only for $N \le 6$ (c8(b)) and $N \le 8$ (r11). Covering all inputs of length at most d this way takes at most $d+6$ functions for $d \le 40$ (c8(b)). Nothing above depends on optimality.

B.6 Structured outputs and history-aware SFI

Proposition B.4 (Single queries and structured outputs)**.** [proved] context Prop 5.4(3),(4) *Let $M(\cdot \mid x)$ be the universal monotone semimeasure given x, over the alphabet $\{0,1,\$\}$: Solomonoff induction predicting an output y symbol by symbol after x. Then $2^{-K(y|x)} \le 2^c M(y \mid x)$; but $M(0^d \mid \varepsilon) \ge 2^{-c}$, while $2^{-K(0^d)} = 2^{-K(d)\pm O(1)}$. With an end marker, $M(y\$ \mid x) = 2^{-K(y|x)\pm O(1)}$.*

Since $L_{\mathrm{SFI}}(\{(x,y)\}) = K(y \mid x) \pm O(1)$ (Theorem 2.21(c)), single-query SFI treats an output as a complete object: it commits to where the output ends and pays for its length, while sequence induction on an undelimited output does not. With delimiters the two agree up to constants; across many examples they differ in how information is shared (Section 5).

Proof of Proposition B.4. A conditional prefix program for y given x is a monotone program, given x, that prints y; so $M(y \mid x) \ge 2^{-c} \sum_{q:V(q,x)=y} 2^{-|q|} \ge 2^{-K(y|x)-c}$. A constant program prints 0^∞, so $M(0^d \mid \varepsilon) \ge 2^{-c}$, while $K(0^d) = K(d) \pm O(1)$. For the end marker, "$\ge$" is the first claim applied to $y\$$, with $K(y\$ \mid x) = K(y \mid x) \pm O(1)$. For "$\le$": the strings $y\$$, $y \in \{0,1\}^*$, are pairwise incomparable, so their cylinders are disjoint and $\sum_y M(y\$ \mid x) \le 1$; $y \mapsto M(y\$ \mid x)$ is lsc uniformly in x, and the conditional coding theorem gives $M(y\$ \mid x) \le 2^{-K(y|x)+c}$. □

Remark B.5 (The same gap for history-aware SFI). [proved]
history Prop 4.10(d) For $\mathrm{SFI}^{\mathrm{hist}}$ the two versions also differ unboundedly: on fair-coin labels W_{hist} pays at most $K^{\mathrm{tail}}(n) + c \le g(n) + K(g) + c$ extra bits for every computable, nondecreasing, unbounded g, while the single shortest program pays more than $\log n + \log\log n$ extra bits infinitely often along every input sequence (Proposition D.13).

C Proofs and remarks for the learning theorems

Notation is that of Section 4; KL is in bits. For a probability p and a sub-probability q on Y: $\mathrm{KL}(p\|q) \ge -\log \sum_y q(y) \ge 0$ (Jensen); and adding to q one atom that carries its defect $1 - \sum_y q(y)$

leaves $\mathrm{KL}(p\|q)$ unchanged and does not decrease $\|p-q\|_1$, so Pinsker's inequality gives

$$\|p-q\|_1^2 \le 2\ln 2 \cdot \mathrm{KL}(p\,\|\,q). \tag{4}$$

In proofs about $\nu \in \{W, M_F\}$ we treat each hypothesis as a deterministic partial function: for W a program p with weight $2^{-|p|}$ and function $\varphi_{V(p)}$; for M_F a tape ω with measure $\mathbb{U}$ and function f_ω. Then $\nu(D)$ is the mass of the hypotheses that extend D.

C.1 Merging

Proof of Theorem 4.3. Posterior given the past. Fix t and a history h of length $t-1$ with positive probability (histories take countably many values). The event $\{h_{t-1} = h\}$ is $\{F \supseteq D_h\}$ intersected with an event that depends only on $(\xi_s)_{s<t}$: that the rule, fed the labels of h, produces the inputs of h. As (ξ_s) is independent of F, given $h_{t-1} = h$ and $(\xi_s)_{s\le t}$ the law of F is $\nu(\cdot \mid F \supseteq D_h)$. Since $x_t = s_t(h, \xi_t)$, $\mathbb{P}[y_t = y \mid h_{t-1}, (\xi_s)_{s\le t}] = \nu(y \mid D_{t-1}, x_t)$.

One step. Let $p := \nu(\cdot \mid D_{t-1}, x_t)$, a probability since ν is the law of a random total function, $q := M_F(\cdot \mid D_{t-1}, x_t)$, and $Z_t := \log(p(y_t)/q(y_t))$. At a repeated input both are point masses at the stored label and $Z_t = d_t = 0$. Otherwise the conditional expectation of Z_t given the past is $d_t \ge 0$.

Integrability. With $\log u \le (u-1)\log e$, the conditional expectation of Z_t^- is

$$\textstyle\sum_y p(y) \max\left(0, \log \frac{q(y)}{p(y)}\right) \le \log e \sum_y (q(y) - p(y))^+ \le \log e.$$

On every path the conditionals telescope, since $\nu(\emptyset) = M_F(\emptyset) = 1$, and Theorem 4.1 gives

$$\textstyle\sum_{t\le n} Z_t = \log\left(\nu(D_n)/M_F(D_n)\right) \le K(\nu).$$

Hence $\sum_t Z_t^+ \le K(\nu) + \sum_t Z_t^-$, each $\mathbb{E}Z_t^+$ is finite, and linearity with the tower property gives $\sum_{t\le n} \mathbb{E}d_t = \mathbb{E}\log(\nu(D_n)/M_F(D_n)) \le K(\nu)$.

Consequences. $\sum_t d_t$ has finite expectation, so it is finite almost surely and $d_t \to 0$ almost surely. Summing (4) and taking expectations gives the ℓ_1 bound. □

C.2 Online-to-batch and noise

Proof of Theorem 4.5. (a) Labels are defined almost surely since $\mu(\operatorname{dom}\varphi) = 1$. As $x_{t+1} \sim \mu$ is independent of D_t, $\mathbb{E}R(Q_t) = \mathbb{E}[-\log M_F(y_{t+1} \mid D_t, x_{t+1})]$ (a repeated input contributes 0). Summing over $t < n$ gives $\mathbb{E}L_{\mathrm{SFI}}(D_n) \le K(\varphi) + c$ (Theorem 4.1(b)). By convexity of $-\log$, $R(\bar{Q}_n) \le \frac{1}{n}\sum_{t<n} R(Q_t)$. If $\bar{Q}_n(\varphi(x) \mid x) > \frac{1}{2}$, then $\varphi(x)$ is the unique maximiser; elsewhere the loss is at least one bit, so Markov's inequality bounds that set's μ-mass by $R(\bar{Q}_n)$.

(b) Stamped inputs are distinct, so $M_F(\tilde{D}_n) = \prod_{t\le n} q_t(y_t \mid x_t)$. Given $\tilde{D}_{t-1}$ and x_t, $y_t \sim P(\cdot \mid x_t)$; so $Z_t := \log(P(y_t \mid x_t)/q_t(y_t \mid x_t))$ has conditional expectation $\mathrm{KL}(P(\cdot \mid x_t)\|q_t(\cdot \mid x_t))$, and, $x_t \sim \mu$ being independent of the past, $\mathbb{E}Z_t = \mathbb{E}\,\mathbb{E}_\mu \mathrm{KL}(P\|q_t)$. The stamp-ignoring product law $\tilde{P}'(y \mid \langle x, t\rangle) := P'(y \mid x)$ is a machine law of complexity at most $K(P')+c$ (Proposition 2.10(a)), so on every path $\sum_{t\le n} Z_t \le K(P') + c + \sum_{t\le n} W_t$ with $W_t := \log(P(y_t \mid x_t)/P'(y_t \mid x_t))$. *Integrability* (not written out in the notes for (b)). If $\mathbb{E}_\mu \mathrm{KL}(P\|P') = \infty$ there is nothing to prove. Otherwise, as for Z_t^- above, $\mathbb{E}Z_t^- \le \log e$ and $\mathbb{E}W_t^- \le \log e$; with $\mathbb{E}W_t = \mathbb{E}_\mu \mathrm{KL}(P\|P') < \infty$ this gives $\mathbb{E}W_t^+ < \infty$ and then $\mathbb{E}Z_t^+ < \infty$. Linearity gives $\sum_{t\le n} \mathbb{E}\,\mathbb{E}_\mu \mathrm{KL}(P\|q_t) \le K(P') + c + n\,\mathbb{E}_\mu \mathrm{KL}(P\|P')$, and convexity of KL in its second argument (also for sub-probabilities) gives the bound for $\bar{q}_n$. An lsc conditional probability is computable, since $P(y \mid x) = 1 - \sum_{y'\ne y} P(y' \mid x)$ is also upper

semicomputable (as in Lemma A.2); take $P' = P$. For 0–1 loss, let y^* be a Bayes label and $\hat{y}$ the maximiser of $\bar{q}_n(\cdot \mid x)$. As $\bar{q}_n(y^* \mid x) \le \bar{q}_n(\hat{y} \mid x)$, $P(y^* \mid x) - P(\hat{y} \mid x) \le [P(y^* \mid x) - \bar{q}_n(y^* \mid x)] + [\bar{q}_n(\hat{y} \mid x) - P(\hat{y} \mid x)] \le \|P(\cdot \mid x) - \bar{q}_n(\cdot \mid x)\|_1$. Apply (4) and Jensen's inequality for the square root.

(c) The product law of a computable conditional probability is a machine law of complexity at most $K(P) + c$ and the law of a random total function; apply Theorem 4.3. □

Proof of Proposition 4.6. (a) Two draws of x_0 with different labels have positive probability, and the second gets probability 0, because the stored label gets probability 1 in both conventions. (b) The labels at x_0 agree with probability $\sum_{j\ge 0} \binom{n}{j} \alpha^j (1-\alpha)^{n-j} \sum_y P(y \mid x_0)^j$. For $j \ge 1$, $\sum_y P(y \mid x_0)^j \le p_{\max}^{j-1}$, so the sum is at most $(1-\alpha)^n + p_{\max}^{-1}(1 - \alpha + \alpha p_{\max})^n$. (c) If some x with $\mu(x) > 0$ has two labels of positive probability, (a) gives infinite expected loss for $n \ge 2$, while P's is finite. Otherwise the label $\varphi(x)$ with $P(\varphi(x) \mid x) > \frac{1}{2}$ can be found by search, so φ is partial computable with $K(\varphi) \le K(P) + c$ and $\mu(\operatorname{dom} \varphi) = 1$, and Theorem 4.5(a) applies. □

C.3 The per-query mixture

Proof of Proposition 4.7. (a) Each term $\prod_t P_i(y_t \mid x_t)$ satisfies the two inequalities because $\sum_y P_i(y \mid x) \le 1$; mixing with $\sum_i \mathbf{m}(i) \le 1$ preserves them; lower semicomputability is clear. The product is symmetric in the pairs, and on distinct inputs it is the definition of $M_{\mathrm{prod}}(D_n)$. On consistent sequences the factors at repeats are at most 1, which gives $M_{\mathrm{pq}} \le M_{\mathrm{prod}}(D_n)$, strictly when some term that is positive on D_n has a factor below 1 at a repeat. A positive computable conditional probability (for $Y = \{0,1\}^*$, for instance $P(y \mid x) = 2^{-2|y|-1}$) is a term with positive weight, so every sequence gets positive mass. (b) $\mathbf{m}(i) \ge 2^{-K(i)}$ for an index i of P. (c) The product of the predictions telescopes to $M_{\mathrm{pq}}(y_{1:n} \| x_{1:n}) / M_{\mathrm{pq}}(\epsilon \| \epsilon) \ge M_{\mathrm{pq}}(y_{1:n} \| x_{1:n})$. Given the past, $y_t \sim P(\cdot \mid x_t)$ and $x_t \sim \mu$ is independent of it, so the expected log ratio $\log(P(y_t \mid x_t)/q_t(y_t \mid x_t))$ is $\mathbb{E}_\mu \mathrm{KL}(P \| q_t) \ge 0$. Summing and applying (b) to P' bounds $\sum_t \mathbb{E}\, \mathbb{E}_\mu \mathrm{KL}(P \| q_t)$ by $K(P') + \sum_t \mathbb{E} \log \frac{P(y_t|x_t)}{P'(y_t|x_t)} = K(P') + n\, \mathbb{E}_\mu \mathrm{KL}(P \| P')$, with integrability as in Theorem 4.5(b). Convexity of KL gives the averaged bound. (d) Stamping makes all inputs distinct, so on every sequence the set mixture of the stamp-ignoring laws $\tilde{P}_i(y \mid \langle x, t \rangle) := P_i(y \mid x)$ equals $\sum_i \mathbf{m}(i) \prod_t P_i(y_t \mid x_t) = M_{\mathrm{pq}}(y_{1:n} \| x_{1:n})$. The $\tilde{P}_i$ have indices computable from i, so this mixture is a machine law of complexity $O(1)$ (Proposition 2.10(a),(d)); apply Theorem 2.9. □

C.4 Learning curves, splits and leave-one-out

Proof of Proposition 4.8. Before data, the predictive probability of 0 at either input is $w + \frac{1-w}{2} = \frac{1+w}{2}$. After $(a, 0)$ the consistent hypotheses are φ and $(0, 1)$, and the predictive probability of 0 at b is $\frac{w}{w+(1-w)/2} = \frac{2w}{1+w}$; symmetrically after $(b, 0)$. After $t \ge 1$ samples both inputs have been seen with probability $1 - 2^{1-t}$ (risk 0) and one with probability 2^{1-t} (risk $\frac{1}{2} \log \frac{1+w}{2w}$, as the unseen input has probability $\frac{1}{2}$). $r_1 > r_0$ is equivalent to $(1+w)^3 > 8w$; the left side minus the right is decreasing on $[0, \frac{1}{5}]$ and equals 0.128 at $\frac{1}{5}$. Finally $\sum_t r_t = r_0 + 2r_1 = \log \frac{1}{w}$. □

Identity of Remark 4.9. With x_{n+1} an independent sample, $\mathbb{E}R(Q_n) = \mathbb{E}[-\log Q_n(\varphi(x_{n+1}) \mid x_{n+1})]$. SFI sees its data only as a set and $(x_1, \ldots, x_{n+1})$ is exchangeable, so the same expectation results with any sample i held out; average over i.

Proof of Proposition 4.10. In every order the step losses sum to $-\log(\nu(f|_B)/\nu(\emptyset)) \le L_\nu(f|_B)$ (Proposition 2.16, $\nu(\emptyset) \le 1$). The split point is uniform and independent of the order; divide by

$|B|$. A shortest program consistent with $f|_B$ is a term of $W(f|_B)$, so $L_W \le K_F$, and $L_{M_F} \le L_W + c$ (Theorem 3.4). For the error use $e \le -\ln(1-e)$. □

Proof of Proposition 4.11. $e_{\mathrm{LOO}}(x)\,\nu(f|_{B\setminus\{x\}}) = \nu(f|_{B\setminus\{x\}}) - \nu(f|_B)$ is the mass of hypotheses correct on $B \setminus \{x\}$ and wrong or undefined at x. A hypothesis in the sets of two points $x \ne x'$ would be correct and not correct at x', so the sets are disjoint, and none contains a hypothesis correct on all of B. So the masses sum to at most $\nu(\emptyset) - \nu(f|_B)$. Divide by $\nu(f|_{B\setminus\{x\}}) \ge \nu(f|_B)$; the last inequality is as in Proposition 4.10. □

C.5 Typical separating properties and per-step error

Separation step (the special case of Proposition 7.1 needed here, proved so as not to rely on a later section). Let G be decidable, b total computable, $\operatorname{dom} D \cap G = \emptyset$ and $x^* \in G$. Then for $\nu \in \{W, M_F\}$, with $K(G)$ the least complexity of an index of a total decider of G,

$$\nu(D + (x^*, b(x^*))) \;\ge\; 2^{-K(G)-K(b)-c}\,\nu(D). \tag{5}$$

For M_F: $\{(x, \epsilon, b(x)) : x \in G\} \cup \{(x, w, y) \in T_U : x \notin G\}$ is c.e. and consistent, hence a function machine with an index computable from those of the decider and of b; every tape whose f_ω extends D makes it extend $D + (x^*, b(x^*))$. Apply Theorem 2.9. For W: let V' read a V-program z for the two indices, then a V-program p, and output an index of "$b(x)$ if $x \in G$, else $\varphi_{V(p)}(x)$". Then $V(wu) = V'(u)$ for a fixed w; the map $p \mapsto wzp$ is injective and sends programs consistent with D to programs consistent with $D + (x^*, b(x^*))$.

Proof of Proposition 4.13. (1) The prefix u depends only on x^*, which is independent of the training inputs, and $x^* \in G_u$. Each training input lies in G_u with probability $2^{-\ell} \le 1/(2n)$, independently, so the probability is $(1-2^{-\ell})^n \ge 1 - n2^{-\ell} \ge \frac{1}{2}$. (2) A decider needs u: a self-delimiting code of ℓ, then ℓ bits. (3) On the event of (1), (5) with $G = G_u$ and $b \equiv 1$ gives $e^{\mathrm{wr}}(x^*) \ge 2^{-K(G_u)-c}$. As $\ell \le \log n + 2$, $2^{-\ell} \ge 1/(4n)$ and $(\ell+1)^2 \le (\log n + 3)^2$. For computable binary f take $b = 1 - f$, with $K(b) \le K(f) + c$. □

Proof of Corollary 4.14. (a) Let $\varepsilon_h := \mu\{x : h(x) \ne f(x) \text{ or undefined}\}$. For a fresh x^*, $e(x^*)$ is the posterior mass of hypotheses not correct at x^* (zero at a training input, where all survivors are correct). By Fubini,

$$\mathbb{E}[e(x^*) \mid D_n] = \frac{\int_{h \supseteq D_n} \varepsilon_h \, d\nu(h)}{\nu(D_n)} \le \varepsilon + \frac{\nu\{h \supseteq D_n : \varepsilon_h > \varepsilon\}}{\nu(D_n)}.$$

The denominator is at least $2^{-K(f)-c}$ on every sample, as a shortest program for f is consistent with D_n. A fixed hypothesis survives the n independent inputs with probability $(1-\varepsilon_h)^n$, so $\mathbb{E}\,\nu\{h \supseteq D_n : \varepsilon_h > \varepsilon\} \le e^{-\varepsilon n}\nu(\emptyset) \le e^{-\varepsilon n}$. With $\varepsilon := ((K(f)+c)\ln 2 + \ln n)/n$ the second term is $1/n$ (for $\varepsilon \ge 1$ the bound is trivial). Markov's inequality on the surviving bad mass gives the high-probability form. The posterior-majority predictor errs at x only if $e(x) \ge \frac{1}{2}$. (b) Condition on the multiset of the $n+1$ sampled values; by exchangeability x^* is a uniformly random one of the $n+1$ positions. If its value occurs more than once it is a training input and $e(x^*) = 0$. Otherwise the training set is $S \setminus \{x^*\}$ and $e(x^*) = e_{\mathrm{LOO}}(x^*; S)$, since SFI sees its data as a set. So the conditional expectation is at most $\frac{1}{n+1}\sum_{x \in S} e_{\mathrm{LOO}}(x; S)$; apply Proposition 4.11. For $f \equiv 0$ the lower bound is Proposition 4.13, as $e \ge e^{\mathrm{wr}}$, and the upper bound is (b) with $K(f) = O(1)$. □

C.6 Occam bound and chosen inputs

Proof of Proposition 4.15. Put $\varepsilon_p := (|p| \ln 2 + \ln(1/\delta))/n$. A fixed p whose disagreement with φ exceeds ε_p is consistent with D_n with probability at most $(1-\varepsilon_p)^n \le e^{-n\varepsilon_p} = 2^{-|p|}\delta$. Sum over $p \in \operatorname{dom} V$ and use $\sum_p 2^{-|p|} \le 1$. A shortest program for φ is consistent, so $K_F(D_n) \le K(\varphi)$. □

The following proposition is stated here rather than in Section 4, for the page cap.

Proposition C.1 (chosen inputs)**.** [proved] context Prop 5.3 *(1) For every weak semimeasure ν and every query set A, chosen adaptively or not, the total semimeasure-convention log loss on the labels of A is $-\log(\nu(f|_A)/\nu(\emptyset))$ in every order: for log loss, choosing the order is worthless. (2) Small total loss does not give efficient identification. For $f_a(x) := \mathbf{1}[x = a]$ on $\{0,1\}^d$, SFI's total loss on any query sequence, even all of $\{0,1\}^d$, is at most $K(f_a) + c \le d + 2\log d + c$, yet every query strategy, adaptive or randomised, needs at least $(2^d - 1)/2$ queries in expectation over a uniform a to identify a.*

Proof. (1) is Proposition 2.16. (2) f_a is computable from a and $K(a) \le d + 2\log d + c$; apply Theorem 4.1(b). Fix the strategy's coins. Until a is queried every answer is 0, so the queries are those made on $f \equiv 0$. Identifying a requires querying a or all other $2^d - 1$ points, so at least $\min(\mathrm{pos}(a), 2^d - 1)$ queries, where $\mathrm{pos}(a)$ is the time of the first query of a on $f \equiv 0$. These times are distinct, so their mean over uniform a is at least $2^{-d}\big(\sum_{k=1}^{2^d-1} k + (2^d - 1)\big) \ge \frac{2^d-1}{2}$. Average over the coins. □

So for active learning the log-loss guarantee must be paired with another objective, such as expected information gain (Orseau et al., 2013) (content not checked).

C.7 Further remarks

Heuristic for Conjecture 4.12 (not proved). The natural one-exception hypotheses "f except at x" have weight about $2^{-K(f)-K(x|f)}$, and $\sum_x 2^{-K(x|f)} \le 1$. A counterexample needs many more than $K(f)$ points x whose one-point flips f^x have $K(f^x) \lesssim K(f)$; natural attempts (halting-set indicators, pseudo-random seeds, thresholds) give only $O(1)$ or $O(K(f))$ such points.

Remark C.2 (learning curves can increase). [computed (context c3 Part 2)] context Rem 2.11 Take a Bayes mixture of the truth f (weight w_f) and $|B|$ hypotheses each wrong at exactly one point of B (weight w each). Sampling without replacement, the loss at step k is $\log \frac{w_f + w(|B|-k+1)}{w_f + w(|B|-k)}$ in every order, and it increases with k. With $w = w_f = 1/(|B|+1)$ and $|B| = 16$, the last step (leave-one-out) costs 1 bit, while the random-split average is 0.255 bits. Each leave-one-out error is $\frac{1}{2}$, within a factor of about 2 of Proposition 4.11. By Kraft's inequality such weights $w = w_f = 2^{-K}$ need $|B| \le 2^K - 1$.

Remark C.3 (a refuted comparison). [refuted] context Prop 2.12, X13 An earlier version of the context notes read Proposition 4.13 as "matching the $O(1/n)$ Cesàro upper bound up to $\log^2 n$". The comparison is invalid: it sets a per-step lower bound against an average over steps, which says nothing about the per-step rate. Indeed $\sum_k \mathbb{E}e_k \le (\ln 2)(K(f) + c)$, so $\liminf_k k\,\mathbb{E}e_k = 0$. The valid per-step comparison is Corollary 4.14.

Remark C.4 (relations among the in-distribution bounds). [remark] context Rem 2.18 (i) For computable f and distinct inputs, $\sum_{k \le n} e_k \le (\ln 2)(K(f) + c)$ on every sequence (Theorem 4.1(b) with $e \le -\ln(1-e)$; this is Theorem 7.7). Corollary 4.14(a) gives a per-step rate in distribution at the price of an additive $\ln n$; Theorem 4.5(a) is the Cesàro statement

for the log loss of the averaged predictor. (ii) Corollary 4.14(a) is the Markov half of the relevance-based bound in the Vaintrob–Hänni slides (Vaintrob and Hänni, 2026b, Appendix A), specialised to W and M_F. The union-bound half and the slides' geometric length prior are not needed: long programs enter only multiplied by $e^{-\varepsilon n}$. (iii) Leave-one-out is different. For a fixed finite B and a random held-out point, the training set eliminates every hypothesis with two or more errors on B; what remains is the mass of one-exception hypotheses (Conjecture 4.12). If Conjecture 4.12 holds, the proof of Corollary 4.14(b) gives $\mathbb{E}e(x^*) = O(K(f)/(n+1))$ up to logarithmic factors, without the additive $\ln n$.

Comparison with other bounds (foundations §6.6; the cited works were not checked against their sources). *Occam* (Blumer et al., 1987): Proposition 4.15 is a description-length form of the Occam bound. *PAC-Bayes* (McAllester, 1999): risk bounds with a $\sqrt{\mathrm{KL}(Q\|\mathrm{prior})/n}$ term, or a KL/n term in the realisable case; with prior 2^{-K} and a posterior concentrated on the target the term is about $K(\varphi)\ln 2/n$, the same rate. The difference is in kind: PAC-Bayes bounds hold with high probability for a randomised classifier, Theorem 4.5 in expectation for the averaged Bayes predictive. *MDL and the index of resolvability* (Barron and Cover, 1991): Theorem 4.5(b) and Proposition 4.7(c) are Cesàro-average resolvability bounds $\min_{P'}[K(P')/n + \mathrm{KL}(P\|P')]$ in the conditional setting, for every input distribution. *Expert advice* (Vovk, 1990; Cesa-Bianchi and Lugosi, 2006): under log loss a Bayes mixture has regret $\log(1/w_i)$ against expert i on every sequence; Theorem 4.1 is this fact with machine laws as experts, each predicting with its own posterior given past labels, so shared randomness is allowed.

Remark C.5 (positive-example induction). [proof sketch; known (snippet)] context Rem 5.5 Hänni lists "positive example induction", learning a set or a distribution from samples, among the ideal inductions he wants to define (Hänni, 2026b). In SFI terms positive labels $(x, 1)$ alone are useless, because the constant-1 program explains them. The right formalisation is a sampling likelihood: the samples are the labels of the index inputs $i \mapsto x_i$, under index-ignoring hypotheses with private randomness per sample. The mixture is then $\sum_P w_P \prod_i P(x_i)$ over computable distributions P on X, the private-randomness analogue of the universal semimeasure on the sample sequence. It appears to be the object of the ideal learner from positive evidence of Chater and Vitányi (2007); their exact model was not checked.

Remark C.6 (interventions). [remark]
context Rem 5.6 If the inputs are interventions set by the experimenter independently of the system's randomness, x-only dominance (Theorem 4.1(a)) applies to the interventional law $P(y \mid \mathrm{do}(x))$. With observational inputs SFI targets $P(y \mid x)$ like any conditional learner. Nothing specific to SFI is proved here.

Computations. All scripts are deterministic or seeded and store their output next to themselves; the numbers quoted in Section 4 and here are taken from these outputs. Foundations: c4 (dominance, order independence and the worst adaptive rule on finite mixtures over four inputs), r6 (a rule that sees F), c5 (seed 20261011; fresh noise), c7 and r7 (the per-query mixture on a class of 133 laws), r9 and r12 (exact finite analogues of Theorem 4.5(a) and of the last iterate), c11 (seed 20261012; Proposition 4.8, the identity of Remark 4.9 to $1.6\cdot10^{-15}$, and the integrability inequality, largest value $0.51 < \log e$). Context: c3 (seed 2; random splits and Remark C.2), c4 (seed 3; separating prefixes at $d = 48$, $n \le 10^4$, matching the exact probabilities), c7 (seed 7; Proposition 4.11 on all 31 subsets of a five-point domain and Corollary 4.14 exactly for $n \le 6$).

D Proofs for history-aware prediction

Notation is that of Section 5. We use: (F1) Theorem 2.9 for any countable decidable input set, $M_F(D) \geq 2^{-K(T)}\nu_T(D)$ for $D \neq \emptyset$, and $M_F(D) \geq 2^{-K(g)-c}$ if a partial computable g extends D (Proposition 2.10(b)); (F2) the conditional coding theorem of Section 2.1: if $\nu(z \mid w)$ is lsc uniformly in w and $\sum_z \nu(z \mid w) \leq 1$, then $K(z \mid w) \leq -\log \nu(z \mid w) + K(\nu) + c$; (F3) Levin–Schnorr: ω is Martin-Löf random iff $K(\omega_{1:\ell}) \geq \ell - c_\omega$ for all ℓ (Li and Vitányi, 2008); (F4) $K(a,b) \leq K(b) + K(a \mid b) + c$ and $K(s) \leq |s| + K(|s|) + c$ (Li and Vitányi, 2008); (F5) $M \geq 2^{-K(\nu)-c}\nu$ for every lsc semimeasure ν on $\{0,1\}^*$ (Zvonkin and Levin, 1970; Li and Vitányi, 2008); (F6) Sion's minimax theorem (Sion, 1958), for compact convex sets in $\mathbb{R}^m$ and a continuous payoff convex in one variable and affine in the other.

D.1 The counterexamples

Proof of Lemma 5.2. The events for different next labels are disjoint subsets of the event for $y_{1:n}$, and at a repeated input only the stored label survives; lower semicomputability is as for M_F. The second claim follows from Corollary 5.18(a) with $K(\rho_F) = O(1)$. For the third, at a new input the chronological inequality is the weak inequality, and at a repeated input only one term is nonzero. □

Lemma D.1 (Chain semimeasure)**.** [proved]
history Lemma 2.0 *For a string x and a sequence $z = (z_1, \dots, z_k)$ of nonempty strings ($k \geq 0$) let $\nu(z \mid x) := \mathbb{U}\{\omega : f_\omega(x) = z_1,\ f_\omega(z_1) = z_2, \dots, f_\omega(z_k) = \epsilon\}$. Then $\sum_z \nu(z \mid x) \leq 1$, and ν is lsc uniformly in x.*

Proof. Fix ω, x, and $z \neq z'$ that both satisfy the condition; put $z_0 = z'_0 := x$. If they first differ at $j \geq 1$, then $z_j = f_\omega(z_{j-1}) = z'_j$. If z is a proper prefix of z', of length k, then $\epsilon = f_\omega(z_k) = z'_{k+1} \neq \epsilon$. Both are contradictions, so the events are disjoint; they are effectively open, uniformly. □

Proof of Theorem 5.3. (a) Every ω with $f_\omega \supseteq D_n$ satisfies the condition of Lemma D.1 with $x = x_n$, $z = (x_{n-1}, \dots, x_1)$, so $M_F(D_n) \leq 2^{-K(z|x_n)+c}$ by (F2). The program with $x_1, \dots, x_{n-1}$ built in that outputs ϵ at x_1, x_{i-1} at x_i ($2 \leq i \leq n-1$) and x_{n-1} elsewhere extends D_n; apply (F1). (b) $x_{1:n}$ determines $\omega_{1:\bar\ell_n}$, so $K(x_{1:n}) \geq \bar\ell_n - c_\omega$ by (F3); by (F4), $K(x_{1:n}) \leq \ell_n + K(\ell_n) + K(x_{1:n-1} \mid x_n) + c$. For the upper bound, $x_{1:n-1}$ is determined by n and $\omega_{1:\bar\ell_{n-1}}$, and $\bar\ell_{n-1}$ is computable from n. For $\ell_i = i$ use $K(n) \leq 2\log n + c$. (c) At step $i \geq 2$ the history's inputs are $x_1, \dots, x_{i-1}$ and its labels $\epsilon, x_1, \dots, x_{i-2}$. □

Proof of Theorem 5.4. (a) In block ℓ the strings other than s_ℓ come first, so s_ℓ is unseen when they are queried, and all its length-ℓ siblings are seen when it is. (b) $\operatorname{dom} D^{(\ell)}$ and ℓ are computable from each other, and given the domain the labelling determines $s_{1:\ell}$; apply Theorem 2.21(a). A function machine that reads, for each length j, its own block of j tape bits as a guess s'_j and answers 1 at x iff $x = s'_{|x|}$ extends $D^{(\ell)}$ with probability $2^{-\ell(\ell+1)/2}$; apply (F1). (c) $K(s_{1:\ell} \mid \ell) \geq K(s_{1:\ell}) - K(\ell) - c \geq \ell(\ell+1)/2 - K(\ell) - c_\omega$ by (F3); and fewer than 2^{m-k} tuples have $K(\cdot \mid \ell) < m - k$, $m = \ell(\ell+1)/2$. □

Proof of Proposition 5.5. (a) For distinct nonempty $z_1, \dots, z_j \neq x$ let

$$O_j(z_{1:j}) := \{(x, z_1), (z_1, z_2), \dots, (z_{j-1}, z_j)\}, \qquad O_0 := \emptyset.$$

As z_j (or x, if $j = 0$) is not in $\operatorname{dom} O_j$, the weak inequality gives $\sum_y \nu(O_j \cup \{(z_j, y)\}) \le \nu(O_j)$; the term $y = \epsilon$ is $\nu(D(x, z_{1:j}))$, each nonempty $y \notin \{x, z_1, \dots, z_j\}$ gives $\nu(O_{j+1}(z_{1:j}, y))$, and the other terms are dropped. Induction on J gives $\sum_{|z|\le J} \nu(D(x, z)) \le \nu(\emptyset) \le 1$. (b) $D_n = D(x_n, (x_{n-1}, \dots, x_1))$, and $z \mapsto \nu(D(x, z))$ is a semimeasure by (a), lsc uniformly in x; apply (F2) and continue as for Theorem 5.3. For the skipped strings, the labellings of a finite set have total ν-mass at most 1, so $-\log \nu(D) \ge K(D \mid \operatorname{dom} D) - K(\nu) - c$. (c) Gibbs' inequality given x_n against the sub-probability $z \mapsto \nu(D(x_n, z))$; given x_n, $(x_{n-1}, \dots, x_1)$ is uniform on $\bar\ell_{n-1}$ bits. For the skipped strings, against the labellings of the domain. □

Lemma D.2 (Normaliser lower bound)**.** [proved] history Lemma 2.4a *For every D, $x \notin \operatorname{dom} D$ and function machine B: $\mathcal{Z}(D, x) \ge 2^{-K(B)-c}\, \mathbb{U}\{\omega : f^B_\omega \supseteq D,\ f^B_\omega(x) \ne \bot\}$, which is $2^{-K(B)-c}\nu_B(D)$ if f^B_ω is always total.*

Proof. Apply (F1) to each $D + (x, y)$ and sum over y; the events $\{f^B_\omega \supseteq D,\ f^B_\omega(x) = y\}$ are disjoint. □

Proof of Proposition 5.6. Let $\beta_t := -\log \bar M_F(y_t \mid D_{t-1}, x_t) = -\log M_F(D_t) + \log \mathcal{Z}(D_{t-1}, x_t) \ge 0$. (a) The broom B_j of Section 5.2 is well defined (the v_i have distinct lengths), total, and $K(B_j) \le K(j) + c$. On $\{v_i = x_i\ \forall i \le j\}$, of probability $2^{-\bar\ell_j}$, it extends D_{j+1}: $f(x_{j+1}) = v_j = x_j$ since $\ell_{j+1} \notin \{\ell_1, \dots, \ell_j\}$. At step $t \ge 2$, Lemma D.2 with B_{t-2} gives $\mathcal{Z}(D_{t-1}, x_t) \ge 2^{-K(t-2)-c-\bar\ell_{t-2}}$, so $\beta_t \ge -\log M_F(D_t) - \bar\ell_{t-2} - K(t-2) - c$. Insert Theorem 5.3(b) at horizon t, or, for uniform blocks, Proposition 5.5(c) in expectation. Sum the positive parts; for $\ell_i = i$ use $K(t), K(t-2) \le 2\log n + c$ for $t \le n$. (b) T_ℓ reads blocks g_j of length j for $j < \ell$ and outputs 1 at x iff $|x| < \ell$ and $x = g_{|x|}$, and 0 elsewhere; it is total and $K(T_\ell) \le K(\ell) + c$. Before the query s_ℓ the data are $D' := D^{(\ell-1)} \cup \{(x, 0) : |x| = \ell,\ x \ne s_\ell\}$, which T_ℓ extends iff $g_j = s_j$ for all $j < \ell$, with probability $2^{-\ell(\ell-1)/2}$. So $\mathcal{Z}(D', s_\ell) \ge 2^{-K(\ell)-c-\ell(\ell-1)/2}$; insert Theorem 5.4(c), or Proposition 5.5(c) in expectation, and use $2K(j) \le 4\log \ell + c$ for $j \le \ell$. □

D.2 Hänni's Theorem 1

Lemma D.3 (Private memory)**.** [proved] history Lemma 3.0 *Let a reveal mechanism that does not see f consist of a random element Γ (private memory and coins) independent of f and maps $x_i = \mathrm{sel}_i(\Gamma, h_{i-1})$, with reveal-sequence law $\Delta(x_{1:n} \| y_{1:n-1}) := \mathbb{P}[\mathrm{sel}_i(\Gamma, h_{i-1}) = x_i\ \forall i \le n]$. (a) Given h_{i-1}, f and x_i are conditionally independent; so $I(f \to x_{1:n}) = 0$ for finite-valued inputs and labels. (b) Given f, the input sequence has probability $\Delta(x_{1:n} \| f(x_1), \dots, f(x_{n-1}))$, and the effective kernel $\Delta(x_{1:i} \| y_{1:i-1}) / \Delta(x_{1:i-1} \| y_{1:i-2})$ depends neither on f nor on its law. (c) If the mechanism is a computable sampler with private memory that halts with probability 1 at each step whatever labels it is fed, and its inputs up to time n lie in a finite listed set A_n, then $\Delta(x_{1:n} \| y_{1:n-1})$ is a computable real, uniformly in its arguments.*

Proof. The joint law of (f, Γ, h_n) factorises (on countable spaces; in general via densities with respect to $\mathrm{law}(f) \otimes \mathrm{law}(\Gamma)$) as $[\mathrm{law}(f) \prod_i \mathbf{1}[y_i = f(x_i)]] \cdot [\mathrm{law}(\Gamma) \prod_i \mathbf{1}[x_i = \mathrm{sel}_i(\Gamma, h_{i-1})]]$. (a) Given (h_{i-1}, x_i) the law of (f, Γ) is proportional to a function of f times a function of Γ, and the f-factor does not involve x_i. (b) Sum the Γ-factor over Γ. (c) $\Delta(x\|y)$ is lsc (enumerate halting computations fed the labels y). Feed the sampler the labels y_j while its inputs agree with $x_{1:j}$, and a fixed label otherwise; its input sequences in A_n^n then have total probability 1, and $x_{1:n}$ has probability $\Delta(x\|y)$. So $\Delta(x\|y) = 1 - \sum_{x' \ne x} \mathbb{P}[x']$, a finite sum of lsc terms subtracted from 1. □

Proof of Proposition 5.8. P pays 1 bit for y_1 and none for y_2. The simulator at a outputs P's uniform y_1; at b it simulates a then b on fresh bits and outputs $y_2' = y_1'$, a uniform bit independent of the run at a; so (β, β) gets probability $\frac{1}{4}$. The mixture of the two constant functions gives it $\frac{1}{2}$. □

Proof of Theorem 5.9. The function machine T uses ω for one run of the alternating process: $x_1' \sim \Delta(\cdot \mid \emptyset)$, $y_1' \sim P(\cdot \mid \emptyset, x_1')$, $x_2' \sim \Delta(\cdot \mid h_1')$, …, each call reading fresh consecutive bits (continuing from the stored state if there is private memory). On query x it runs until the first j with $x_j' = x$ and outputs y_j', diverging if there is none. Its answer depends on a finite prefix of ω, so T is a function machine, with $K(T) \le K(\Delta, P) + c$. If the trajectory begins with $x_1 y_1 \cdots x_n y_n$, each $x \in \operatorname{dom} D_n$ is first visited with its label in D_n (the sequence is consistent), so $f_\omega^T \supseteq D_n$. The tapes producing this beginning are the disjoint union of the cylinders $[w_1 v_1 \cdots w_n v_n]$, w_i ranging over the bits read by halting runs of Δ on h_{i-1} with output x_i and v_i over those of P on (h_{i-1}, x_i) with output y_i; each set is prefix-free, so the union has measure $\prod_i \Delta(x_i \mid h_{i-1}) P(y_i \mid h_{i-1}, x_i)$. For distinct inputs, different orders give different trajectory prefixes of equal length, hence disjoint events. Apply (F1). □

Corollary D.4 (Deterministic reveal policies)**.** [proved]
history Cor 3.3 *If $x_i = \delta(h_{i-1})$ for a computable δ, then for every f, every sampler P and every n, $L_{\mathrm{SFI}}(D_n) \le L_P + K(\delta) + K(P) + c$ on every path.*

Corollary D.5 (Against Solomonoff induction on the joint sequence)**.** [proved] history Cor 3.4 *With M and* code *as in Section 5.5: $L_{\mathrm{SFI}}(D_n) \le -\log M(\mathrm{code}(h_n)) + c$ on every consistent sequence; for distinct inputs, $L_{\mathrm{SFI}}(D_n) \le -\log \sum_\sigma M(\mathrm{code}(D_n \text{ in order } \sigma)) + c$.*

Proof of Corollaries D.4 and D.5. For Corollary D.4, take the input sampler δ in Theorem 5.9; $L_\Delta = 0$ on the realised path. For Corollary D.5, run the universal monotone machine on ω, parse its output (uniquely) into blocks $\bar{x}_1' \bar{y}_1' \bar{x}_2' \cdots$, and answer x with the label after the first input block equal to $\bar{x}$. If the output extends $\mathrm{code}(h_n)$ then $f_\omega \supseteq D_n$, and that event has probability $M(\mathrm{code}(h_n))$. For distinct inputs the codes of different orders have equal length and differ, so the events are disjoint. □

Proof of Theorem 5.10. The joint law of (f, Γ, h_n) is $\nu(df)\mathbb{P}(d\Gamma) \prod_i \mathbf{1}[x_i = \mathrm{sel}_i(\Gamma, h_{i-1})] \mathbf{1}[f \supseteq D_n]$, a function of f times a function of Γ for fixed h_{i-1}. So given h_{i-1} and x_i the law of f is $\nu(\cdot \mid f \supseteq D_{i-1})$, whose prediction for y_i is the Bayes predictor; the product telescopes to $\nu(D_n)$, and (F1) gives $M_F \ge 2^{-K(\nu)}\nu$. Gibbs' inequality, valid for sub-probabilities, gives $\mathbb{E}L_P \ge \mathbb{E}[-\log \nu(D_n)]$ step by step. □

Proof of Theorem 5.11. (a) Let Q_f be a regular conditional law of $x_{1:n}$ given f. Then $\mathbb{P}(h_n \mid f) = Q_f(x_{1:n}) \mathbf{1}[f \supseteq D_n]$ and $\mathbb{P}(h_n) = \nu(D_n) \bar{Q}_D(x_{1:n})$ with $\bar{Q}_D := \mathbb{E}_{f' \sim \nu}[Q_{f'} \mid f' \supseteq D_n]$; the posterior of f has density $Q_f / \bar{Q}_D$ with respect to $\nu(\cdot \mid f \supseteq D_n)$. Hence $I(f; h_n) = \mathbb{E}\,\mathrm{KL}(\mathbb{P}(f \in \cdot \mid h_n) \| \nu(\cdot \mid f \supseteq D_n)) + \mathbb{E}[-\log \nu(D_n)]$. By the chain rule $I(f; h_n) = \sum_i [I(f; x_i \mid h_{i-1}) + I(f; y_i \mid h_{i-1}, x_i)] = I(f \to x) + \mathbb{E}L_{\mathbb{P}}$, because y_i is a function of (f, x_i). All terms are finite ($I(f; h_n) \le H(h_n)$). (b) (F1), then (a), then $\mathbb{E}L_P \ge \mathbb{E}L_{\mathbb{P}}$ by Gibbs' inequality. □

Proof of Theorem 5.12. The machine compares a query of length i with tape block i and returns block $i-1$ if they agree and $i \ge 2$, and ϵ otherwise. Given h_{i-1}, x_i is uniform on $\{0,1\}^i$; given f it is determined; so $I(f; x_i \mid h_{i-1}) = i$. The losses are Proposition 5.5(c) and Proposition 5.6(a). □

Proof of Theorem 5.13. Write $Y^A := Y^{A_n}$, $\mathcal{U}$ for the uniform law on it, and $\mathcal{G} := \{\frac{1}{2}G_0 + \frac{1}{2}\mathcal{U}\}$ over laws G_0 on Y^A; $\mathcal{G}$ is compact and convex, and $G(D) \geq \frac{1}{2}|Y|^{-n}$ when $|D| \leq n$. Let $\mathcal{F}_{\text{fin}} := \{f : \mathbb{E}_{x|f} L_P < \infty\}$ (other f are trivial), $\Phi_f(G) := \mathbb{E}_{x|f}[-\log G(D_n)] - \mathbb{E}_{x|f} L_P$, and $\Phi(G, \nu) := \sum_f \nu(f)\Phi_f(G)$ for laws ν on $\mathcal{F}_{\text{fin}}$. *Step 1.* Φ is continuous, convex in G, affine in ν; by (F6), $\min_G \max_\nu \Phi = \max_\nu \min_G \Phi$. For $G = \frac{1}{2}\nu + \frac{1}{2}\mathcal{U}$, $-\log G(D) \leq 1 - \log \nu(D)$, so Theorem 5.11(a) and Gibbs' inequality give $\Phi(G, \nu) \leq 1 + I_\nu(f \to x)$. Hence $v^* := \min_G \max_{f \in \mathcal{F}_{\text{fin}}} \Phi_f(G) \leq 1 + C_{\text{dir}}$. *Step 2.* For $G, G' \in \mathcal{G}$ and $|D| \leq n$, $|\log G(D) - \log G'(D)| \leq |G(D) - G'(D)|/(\ln 2 \min(G(D), G'(D))) \leq |Y|^n \|G_0 - G_0'\|_1 / \ln 2$. So a rational G_0' within ℓ^1-distance $|Y|^{-n} \ln 2$ of a minimiser has $\max_{f \in \mathcal{F}_{\text{fin}}} \Phi_f(G') \leq v^* + 1 \leq 2 + C_{\text{dir}}$. *Step 3, case (a).* Path probabilities given f are computable rationals, $\mathcal{F}_{\text{fin}}$ is decidable ($f \notin \mathcal{F}_{\text{fin}}$ iff some path of positive probability has $P(y_i \mid h, x) = 0$), and each Φ_f is computable on rational points. With the Lipschitz constant of Step 2, evaluating $\max_f \Phi_f$ to within $\frac{1}{4}$ on a fine rational grid yields a grid point G' with $\max_f \Phi_f(G') \leq v^* + 1$. *Step 3, case (b).* Let γ be rational with $C_{\text{dir}} \leq \gamma \leq C_{\text{dir}} + 1$ ($\gamma = 0$ if Δ does not see f). Enumerate rational G_0' and stop at the first G' certified to satisfy ($\star$): $\mathbb{E}_{x|f}[-\log G'(D_n)] < \mathbb{E}_{x|f} L_P + 3 + \gamma$ for every $f \in Y^A$. The left side is a computable real. The right side is a finite sum of products of a computable nonnegative real $\mathbb{P}[\text{path} \mid f]$ and an lsc value $L_P(\text{path}) \in [0, \infty]$, hence lsc in $[0, \infty]$ (with $0 \cdot \infty = 0$). A true strict inequality "computable $<$ lsc" is certified in finite time, also when the right side is $+\infty$; dovetail over candidates. By Step 2 some rational G' satisfies ($\star$), so the search halts. If Δ sees f, γ must be computed: C_{dir} is the maximum over a simplex of finitely many terms $u \log(u/v)$, with u, v polynomial in ν and in computable values of Δ, and should be computable to within 1 via a modulus of continuity of entropy on a finite alphabet; this is not written out, so this case is a proof sketch. *Step 4.* The procedure is uniform in (P, Δ, n, A), so G' has rational probabilities computable from these, and rejection sampling (undefined outside A_n) makes it a function machine with $K(G') \leq K(P) + K(\Delta) + K(n) + K(A) + c$. (F1) gives $\mathbb{E}_{x|f} L_{\text{SFI}} \leq \mathbb{E}_{x|f}[-\log G'(D_n)] + K(G') + c \leq \mathbb{E}_{x|f} L_P + 4 + C_{\text{dir}} + K(G') + c$; average over f for the last claim. An almost surely halting reveal sampler with private memory has $K(\Delta)$ at most its complexity plus $K(A) + c$ (Lemma D.3(c) uses the list A_n). □

Proof of Corollary 5.14. (a) Theorem 5.13(b) with $\Delta := \Delta_f$, which does not see f; then average. (b) Corollary D.4. (c) Take expectations in Theorem 5.9; given f the law of $x_{1:n}$ is $\Delta_f(x\|y)$, and $\mathbb{E}_{x|f}[-\log \Delta_f(x\|y)] = H_f(x_{1:n})$. □

Remark D.6 (What is open in Reading C). [open] history remarks after Cor 3.9; Q1, Q7; checks r6, r7, c3, r2 (i) Whether $K(n) + K(A)$ can be dropped: the minimax law depends on the horizon, and in one small instance a direct computation certifies that one law for horizons 1–4 needs at least 0.026 bits (each single horizon: at most 0.005); no growth was found up to horizon 6, weak evidence that $K(n)$ is an artefact. (ii) Defective samplers with $K(P)$ alone ($\mathbb{E}L_P$ is then only upper semicomputable, so ($\star$) cannot be certified). (iii) Infinite alphabets at a fixed horizon. (iv) Reveal laws that see f, in case (b). The computed minimax value without the uniform component is at most $1.8 \cdot 10^{-15}$ bits in all tested instances where Δ does not see f.

Proposition D.7 (No pathwise version of Reading A)**.** [proved]
history Prop 3.12 *No bound $L_{\text{SFI}}(D_n) \leq L_P + K(P) + K(\Delta) + c$ holds on every path for every fixed f and every reveal law Δ that does not see f.*

Proof. Let ω be Martin-Löf random with blocks b_i of length i, $f(b_{i+1}) := b_i$ and $f := \epsilon$ elsewhere; let Δ draw independent uniform strings of lengths $1, 2, \ldots$, and P be the chain predictor. The path $x_i = b_i$ has probability $2^{-n(n+1)/2} > 0$; on it $L_P = 0$ and $L_{\text{SFI}}(D_n) \geq n(n-1)/2 - 2\log n - c_\omega$ (Theorem 5.3(b)). □

Proposition D.8 (Shannon form of the mutual-information question)**.** [proved; computed] history Prop 3.10; check c7 *Let (f,Θ) be jointly random and, given (f,Θ), let $x_i \sim \Delta_{\Theta,\Gamma}(\cdot \mid h_{i-1})$, a kernel that does not see f, with private memory Γ independent of f given Θ; finite alphabets up to time n. (a) If $I(f;\Theta)<\infty$, then $I(f \to x_{1:n}) = I(f;\Theta) - I(f;\Theta \mid h_n) - \sum_i I(y_i;\Theta \mid h_{i-1},x_i) \le I(f;\Theta)$. (b) For $f \sim \nu$ computable, $\mathbb{E}L_{\mathrm{SFI}}(D_n) \le \mathbb{E}L_P + I(f;\Theta) + K(\nu) + c$ for every P. (c) In Reading C, $\Theta := \Delta_f$ restricted to n steps is a function of f, $I(f;\Theta) = H(\Delta_f)$, and in Theorem 5.12 $I(f \to x) = H(\Delta_f) = n(n+1)/2$.*

Proof. (a) Given $\Theta = k$, the law of (f,Γ,h_{i-1}) is $p(df \mid k)p(d\Gamma \mid k)\prod_{j<i}\Delta_{k,\Gamma}(x_j \mid h_{j-1})\mathbf{1}[f(x_j)=y_j]$, so $f \perp \Gamma \mid (\Theta,h_{i-1})$, and the law of x_i given (f,Θ,h_{i-1}) does not involve f: $I(f;x_i \mid \Theta,h_{i-1}) = 0$. The chain rule gives $I(f;\Theta,h_n) = I(f;\Theta) + \sum_i H(y_i \mid \Theta,h_{i-1},x_i)$, and also $I(f;\Theta,h_n) = I(f;h_n) + I(f;\Theta \mid h_n)$ with $I(f;h_n) = I(f \to x) + \sum_i H(y_i \mid h_{i-1},x_i)$ (proof of Theorem 5.11). Equate, using $H(y_i \mid h_{i-1},x_i) - H(y_i \mid \Theta,h_{i-1},x_i) = I(y_i;\Theta \mid h_{i-1},x_i)$. (b) Theorem 5.11(b) and (a). (c) $I(f;\Theta) = H(\Theta)$ when Θ is a function of f; in the randomized chain, Δ_f restricted to n steps determines the chain, uniform on $n(n+1)/2$ bits. □

D.3 $\mathrm{SFI}^{\mathrm{hist}}$

Lemma D.9. [proved] history Lemma 4.1 *M_T^{hist} is an lsc chronological semimeasure, uniformly in an index of T.*

Proof. The events $\{T^\omega(x_{1:i}) = y_i \ \forall i \le n,\ T^\omega(x_{1:n}x) = y\}$ for different y are disjoint subsets of the event for $(y_{1:n},x_{1:n})$ and are effectively open, uniformly (Proposition 2.8); the label at time i depends only on $x_{1:i}$. □

Proof of Theorem 5.17. Nodes are $z = (x_{1:k},y_{1:k})$ with parent $p(z)$; the *family* $F(z)$ of a non-root node consists of the nodes with the same parent and the same last input. Take dyadic, finitely supported $\rho_t \uparrow \rho$, computable and nondecreasing in t, and a computable enumeration $z_1,z_2,\dots$ of the non-root nodes with every node after its parent. Each node carries a finite union $S(z)$ of dyadic intervals (cylinders), initially empty; $S(\text{root}) := [0,1)$. At stage t process $z_1,\dots,z_t$ in order: with $\mathrm{free}(z) := S(p(z)) \setminus \bigcup_{z' \in F(z)} S(z')$ (current sets, z included), if $a := \min(\rho_t(z) - \mathbb{U}(S(z)), \mathbb{U}(\mathrm{free}(z))) > 0$, add to $S(z)$ a dyadic part of $\mathrm{free}(z)$ of measure a. By construction: (I1) $S(z) \subseteq S(p(z))$; (I2) family members are disjoint; (I3) $\mathbb{U}(S_t(z)) \le \rho_t(z)$ after stage t; (I4) $S_t(z)$ grows with t. *Limit.* We show $\mathbb{U}(S(z)) = \rho(z)$, $S(z) := \bigcup_t S_t(z)$, by induction along the enumeration. With $p := p(z)$, the hypothesis (or $\mathbb{U}(S(\text{root})) = 1$) and the chronological inequality give $\mathbb{U}(S(p)) \ge \sum_{z' \in F(z)} \rho(z')$. When z is processed at stage t, $S(p) = S_t(p)$ (p is processed earlier), and each sibling has measure at most $\rho(z')$ by (I3). So afterwards $\mathbb{U}(S_t(z)) \ge \min(\rho_t(z), \mathbb{U}(S_t(p)) - \sum_{z' \in F(z)\setminus\{z\}} \rho(z'))$; only "$\ge$" holds, since a later sibling may still grow. Letting $t \to \infty$ gives $\liminf_t \mathbb{U}(S_t(z)) \ge \rho(z)$, and (I3) gives equality. *Machine.* $T_\rho^\omega(x_{1:k})$ recomputes $y_i := T_\rho^\omega(x_{1:i})$ for $i<k$, then searches for t,y with $\omega \in S_t((x_{1:k},y_{1:k-1}y))$ and outputs the first y found; by (I2) at most one y is ever found, and membership depends on a finite prefix of ω. By induction with (I1), (I2), $\{\omega : T_\rho^\omega(x_{1:i}) = y_i \ \forall i \le n\} = S((x_{1:n},y_{1:n}))$, of measure $\rho(y_{1:n}\|x_{1:n})$. The construction is uniform in an index of (ρ_t). □

Proof of Corollary 5.18. (a) Theorem 5.17 and (F1) over X_H at non-root nodes; at the root $M_H = 1 \ge \rho(\epsilon\|\epsilon)$; two universal elements dominate each other. (b) $P(y\|x)$ is an lsc chronological semimeasure of complexity at most $K(P)+c$. (c) Lemma 5.2. □

Proposition D.10 (Other oracles)**.** [proved]
history Prop 4.4 *Let the inputs be distinct. (a) M_{pairs} and M_H agree up to constant factors. (b)* $\mathrm{SFI}^{\mathrm{count}}$

is SFI on stamped inputs (i, x_i) (as in Theorem 4.5(b)), and $L_{\mathrm{count}} \le -\log M(\bar{y}_1 \cdots \bar{y}_n) + c$. (c) SFI $\preceq \mathrm{SFI}^{\mathrm{count}} \preceq \mathrm{SFI}^{\mathrm{set}} \preceq \mathrm{SFI}^{\mathrm{hist}}$, where $\nu \preceq \nu'$ means $\nu \le 2^c \nu'$.

Proof. (a) A pairs-machine T gives the history-oracle machine

$$T'^{\omega}(x_{1:i}) := T^{\omega}(x_1, T'^{\omega}(x_1), \ldots, x_{i-1}, T'^{\omega}(x_{<i}), x_i);$$

on the event that T' outputs $y_{1:n}$ the recursion reproduces the true history, and conversely by induction on i. (b) The hypothesis that outputs the i-th $\bar{y}$-block of the universal monotone machine's output. (c) Each oracle is computable from the next (the count is the size of the set plus one). □

Proof of Proposition 5.19. (i) $A := \{x_i : i \text{ even}\}$ and D are computable from each other given S, so $L_{\mathrm{SFI}} \ge K(A \mid S) - c$ (Theorem 2.21(a)); choose A of size $\lfloor n/2 \rfloor$ with $K(A \mid S) \ge \log \binom{n}{\lfloor n/2 \rfloor} - 1 \ge n - \log(n+1) - 1$. The count hypothesis $\mathbf{1}[i \text{ even}]$ gives $L_{\mathrm{count}} \le c$. (ii) With U the universal count-machine, $\nu(z \mid x, n) := \mathbb{U}\{\omega : U^{\omega}(n, x) = z_1,\, U^{\omega}(n-1, z_1) = z_2, \ldots, U^{\omega}(1, z_{n-1}) = \epsilon\}$ sums to at most 1 over z (iteration determines z), so $L_{\mathrm{count}} \ge K(x_{1:n-1} \mid x_n, n) - c$; n is computable from $|x_n|$, and the proof of Theorem 5.3(b) finishes. "Output the longest element of the set" gives $L_{\mathrm{set}} \le c$. (iii) For a tape ω of the universal set-machine U, an order is consistent iff $U^{\omega}(\emptyset, x_1) = \epsilon$ and $U^{\omega}(\{x_{<i}\}, x_i) = x_{i-1}$. Then x_n determines $x_{n-1} = U^{\omega}(S \setminus \{x_n\}, x_n)$, and so on backwards, so at most n orders are consistent with ω and $\sum_{\sigma} M_{\mathrm{set}}(\sigma) \le n$. (F2) for this semimeasure divided by n, lsc uniformly in S, gives $L_{\mathrm{set}} \ge K(\sigma \mid S) - \log n - c$; choose $K(\sigma \mid S) \ge \log n! - 1$. "$y_1 = \epsilon$, $y_i = x_{i-1}$" gives $L_H \le c$. □

Proposition D.11 (Single infinite sequences)**.** [proved] history Prop 4.5′ *Let ω be Martin-Löf random and u_j, v_j the strings with indices $2j-1, 2j$ in length-lexicographic order; block j reveals u_j, v_j in this order if $\omega_j = 0$, reversed otherwise. (i) With labels $\mathbf{1}[i \text{ even}]$, $L_{\mathrm{SFI}}(D_{2k}) \ge k - K(k) - c_\omega$ for all k, while $L_{\mathrm{count}} \le c$. (ii) The chain of Theorem 5.3(b) is a single sequence for Proposition 5.19(ii). (iii) With labels $y_1 = \epsilon$, $y_i = x_{i-1}$, $L_{\mathrm{set}} \ge k - K(k) - c_\omega$ at time $2k$, while $L_H \le c$. So each gap grows linearly along one sequence (semimeasure convention).*

Proof. (i) $\mathrm{dom}\, D_{2k}$ is computable from k, and the labelling determines $\omega_{1:k}$; use Theorem 2.21(a), (F4) and (F3). (iii) By the backward reconstruction, a tape fixes the order given its last input (u_k or v_k), so at most two of the 2^k block orders σ_b fit one tape; apply (F2) to $b \mapsto M_{\mathrm{set}}(\sigma_b)/2$ and take $b = \omega_{1:k}$. □

Proposition D.12 (Operator induction)**.** [proved, given the unverified description] history Prop 4.9 *Read as M_{pq} (Remark 3.2), operator induction (Solomonoff, 2008) is an lsc chronological semimeasure that ignores the history, so $L_H \le -\log M_{\mathrm{pq}} + c$ on every sequence. On distinct inputs $M_{\mathrm{pq}} = M_{\mathrm{prod}}(D_n)$, a weak semimeasure, so it pays the lower bounds of Theorems 5.3 and 5.4 (semimeasure convention).*

Proof. Proposition 4.7(a), Corollary 5.18(a), and $M_{\mathrm{prod}} \le 2^c M_F$ (Theorem 3.4); M_{prod} is weak, so Proposition 5.5(b) applies. □

Proposition D.13 (Deterministic programs for $\mathrm{SFI}^{\mathrm{hist}}$)**.** [proved; toy computed] history Prop 4.10; checks s2, c10 *Let $W_{\mathrm{hist}}(y\|x) := \sum_{p \in \mathrm{dom}\, V} 2^{-|p|} \mathbf{1}[\varphi_{V(p)}(x_{1:i}) = y_i \ \forall i \le n]$ and $K_{\mathrm{hist}}(y\|x)$ the least such $|p|$, with (φ_e) the partial computable functions on X_H; let $K^{\mathrm{tail}}(n) := -\log \sum_{N \ge n} 2^{-K(N)}$ and $\{0,1\} \subseteq Y$. For every input list and $y \in \{0,1\}^n$, $W_{\mathrm{hist}}(y\|x) \ge 2^{-n - K^{\mathrm{tail}}(n) - c}$, where (a) $K^{\mathrm{tail}}(n) \le K^{\mathrm{pad}}(n) \le K(n)$, (b) $K^{\mathrm{tail}}(n) \le g(n) + K(g) + c$ for every computable nondecreasing unbounded*

g, and (c) $K^{\mathrm{tail}}(n) \to \infty$*. (d) But* $n - c \le \max_y K_{\mathrm{hist}}(y\|x_{1:n}) \le n + K(n) + c$*, and along every input sequence* $\max_y K_{\mathrm{hist}}(y\|x_{1:n}) - n > \log n + \log\log n$ *for infinitely many* n*.*

Proof. W_{hist} is an lsc chronological semimeasure (programs consistent with different next labels differ). Let $\varphi_{N,w}(x_{1:i}) := w_i$ for $i \le N$, $w \in \{0,1\}^N$. A prefix machine reads a V-program for N and then N bits w and outputs an index of $\varphi_{N,w}$; by universality, with a fixed string r and q_N a shortest program for N, the strings rq_Nw are distinct elements of $\operatorname{dom} V$ ($q_N, q_{N'}$ are incomparable for $N \ne N'$), of length $|r| + K(N) + N$. For $N \ge n$, $\varphi_{N,w}$ is consistent iff $w_{1:n} = y$, for 2^{N-n} strings w, so $W_{\mathrm{hist}}(y\|x) \ge \sum_{N\ge n} 2^{N-n-|r|-K(N)-N} = 2^{-n-|r|-K^{\mathrm{tail}}(n)}$. (a) One term of the sum. (b) $q(N) := 2^{-g(N)} - 2^{-g(N+1)}$ is a computable discrete measure with $\sum_{N\ge n} q(N) = 2^{-g(n)}$, and (F2) gives $2^{-K(N)} \ge 2^{-K(g)-c} q(N)$. (c) The tails of $\sum_N 2^{-K(N)} \le 1$ tend to 0. (d) The upper bound is the program rq_ny; the lower bound holds since a consistent program and $x_{1:n}$ determine y. Fix the input sequence, let $e(n) := \max_y K_{\mathrm{hist}}(y\|x_{1:n}) - n$, $e^*(n) := \max_{m\le n} e(m)$, and let $C(L)$ and k_L count the $p \in \operatorname{dom} V$ with $|p| \le L$ and $|p| = L$. Distinct labellings need distinct programs, so $C(n + e^*(n)) \ge 2^n$. By summation by parts and Kraft, $\sum_{L\le m} C(L)2^{-L-1} \le \sum_L k_L 2^{-L} \le 1$; as $n + e^*(n)$ is strictly increasing, $\sum_n 2^{-e^*(n)} \le 2$. If $e(n) \le \log n + \log\log n$ for all $n \ge n_0$, then $\sum_n 2^{-e^*(n)} \ge 2^{-O(1)} \sum_n 1/(n \log n) = \infty$. □

D.4 Solomonoff induction on the joint sequence

Block sequences are $b = (x_1, y_1, x_2, \dots)$, possibly ending after an input; b^- is b without its last block.

Lemma D.14 (Block codes)**.** [proved] history Lemma B; check r4 *Let* ν *be an lsc semimeasure on block sequences* ($\nu(\emptyset) \le 1$, $\sum_\beta \nu(b\beta) \le \nu(b)$)*, and* $s \mapsto \bar{s}$ *a prefix-free code with nonempty codewords. Let* $B(z)$ *be the set of* b *with* $\mathrm{code}(b) \sqsupseteq z$ *and* ($b = \emptyset$ *or* $\mathrm{code}(b^-) \not\sqsupseteq z$)*. Then* $\nu_{\mathrm{bin}}(z) := \sum_{b\in B(z)} \nu(b)$ *is an lsc semimeasure on* $\{0,1\}^*$ *with* $\nu_{\mathrm{bin}}(\mathrm{code}(b)) = \nu(b)$*, so* $M(\mathrm{code}(b)) \ge 2^{-K(\nu)-c}\nu(b)$*.*

Proof. $B(\mathrm{code}(b)) = \{b\}$: codes are prefix-free and nonempty, so $\mathrm{code}(b') \sqsupseteq \mathrm{code}(b)$ forces b' to extend b blockwise, and if $b' \ne b$ then $\mathrm{code}(b'^-) \sqsupseteq \mathrm{code}(b)$. $B(\epsilon) = \{\emptyset\}$. Let $b \in B(za)$. Either $\mathrm{code}(b^-) \not\sqsupseteq z$, and $b \in B(z)$ with $\mathrm{code}(b) \ne z$; or $\mathrm{code}(b^-) \sqsupseteq z$, and since $\mathrm{code}(b^-)$ and za are prefixes of $\mathrm{code}(b)$ with $\mathrm{code}(b^-) \not\sqsupseteq za$, $\mathrm{code}(b^-) = z$, so b extends by one block the element b^- of $B(z)$ with code z. Conversely each $b \in B(z)$ with $\mathrm{code}(b) \ne z$ lies in exactly one $B(za)$. Hence $\nu_{\mathrm{bin}}(z0) + \nu_{\mathrm{bin}}(z1) \le \sum_{b\in B(z),\mathrm{code}(b)\ne z} \nu(b) + \sum_{b'\in B(z),\mathrm{code}(b')=z} \sum_\beta \nu(b'\beta) \le \nu_{\mathrm{bin}}(z)$. Apply (F5). □

Proof of Theorem 5.20. (a) $M(\mathrm{code}(h_n))$ is an lsc chronological semimeasure: with $z := \mathrm{code}(h_n)$, $\sum_y M(z\bar{x}\bar{y}) \le M(z\bar{x}) \le M(z)$, as $\{\bar{y}\}$ is prefix-free. Apply Corollary 5.18(a). Likewise $M(\mathrm{code}(h_{n-1})\bar{x}_n) \ge M(\mathrm{code}(h_n))$ is one for inputs given labels. (b) $\nu(h_n) := M_H^{\mathrm{in}}(x_{1:n}\|y_{1:n-1})M_H(y_{1:n}\|x_{1:n})$ and $\nu(h_{n-1}x_n) := M_H^{\mathrm{in}}(x_{1:n}\|y_{1:n-1})M_H(y_{1:n-1}\|x_{1:n-1})$ define an lsc semimeasure on block sequences (sum over y_n with M_H's chronological inequality, over x_n with M_H^{in}'s); apply Lemma D.14. (c) Combine with $L_M^{\mathrm{joint}} = L_M^x + L_M^y$. (d) Proposition 5.21. □

Proof of Proposition 5.21. $\nu(h_n) := \mathbf{1}[x_i = \delta(h_{i-1})\ \forall i \le n]M_H(y_{1:n}\|x_{1:n})$ and $\nu(h_{n-1}x_n) := \mathbf{1}[x_i = \delta(h_{i-1})\ \forall i \le n]M_H(y_{1:n-1}\|x_{1:n-1})$ define an lsc semimeasure on block sequences with $K(\nu) \le K(\delta) + c$: the indicator is c.e., summing over the next input leaves only $\delta(h_{n-1})$, and summing over the next label uses M_H's chronological inequality. Lemma D.14 gives $L_M^{\mathrm{joint}} \le L_H + K(\delta) + c$; with $L_M^{\mathrm{joint}} = L_M^x + L_M^y$ this is the left inequality. The right one is Theorem 5.20(c). □

On δ-consistent sequences also $L^x_{\text{in}} \le K(\delta)+c$, so $L_H + L^x_{\text{in}} - K(\delta) - c \le L_M^{\text{joint}} \le L_H + L^x_{\text{in}} + c$.

Proof of Proposition 5.22. (Even.) $x_i := \omega_i$, $y_i := x_i$: $L_H \le c$ by the copy predictor (Corollary 5.18(b)), while L^y_M is M's loss on the copied bits. (Odd.) $\delta(\emptyset) := 0$, $\delta(h) :=$ the last label of h, $y_i := \omega_i$: the joint sequence is $0\omega_1\omega_1\omega_2\omega_2\cdots$, the inputs $x_2, x_3, \ldots$ are the copied bits, so L^x_M is unbounded, and Proposition 5.21 with $K(\delta) = O(1)$ gives $L_H - L^y_M \to \infty$. Non-convergence of the copied bits' conditional probabilities to 1 already makes the cumulative loss unbounded. □

D.5 What SFI can and cannot exploit

Proof of Theorem 5.24 and Corollary 5.25. The left inequality is Lemma 5.2. For the right one fix σ' and p attaining $K^{\text{pred}}(x_{\sigma'})$. The function machine T, which does not depend on p, on query x enumerates the prefix-free set $\operatorname{dom} V(\cdot, x)$ and takes the first element q found that is a prefix of the tape; checking a candidate reads the tape only up to the first mismatch, which comes within $|q|$ bits, so T reads exactly q. It computes $V(q,x) = \langle x'_1, \ldots, x'_k\rangle$ and runs T^H_U on the rest of the tape at $(x'_1, \ldots, x'_k, x)$. On tapes $p\omega'$, every $x \in \operatorname{dom} D$ finds p (as $p \in \operatorname{dom} V(\cdot, x)$, a prefix-free set), so all queries share ω', and $\nu_T(D) \ge 2^{-|p|} M_H(y_{\sigma'} \| x_{\sigma'})$; apply (F1). For Corollary 5.25, $K^{\text{pred}}(e\text{-order}) \le K(e) + c$: given x, enumerate e until x appears. □

Lemma D.15 (Character-sum bound)**.** [proved; computed] history Lemma 5.5; checks c5, s4 *Let $F = \mathrm{GF}(2^q)$. Every partial $G : F \to F$ has $N(G) := \#\{(a,b) : a \ne b,\ G(a) + G(b) = ab\} < 2^q + 2^{3q/2}$ (pairs where G is defined).*

Proof. Extending G to a total function and including $a = b$ does not decrease the count. With $\chi_u(z) := (-1)^{\mathrm{Tr}(uz)}$, $\sum_u \chi_u(z) = 2^q \mathbf{1}[z = 0]$, so in characteristic 2

$$N \le 2^{-q} \textstyle\sum_u \sum_{a,b} \chi_u(G(a))\chi_u(G(b))\chi_u(ab).$$

The term $u = 0$ is 2^{2q}. For $u \ne 0$, $(\mathbf{C}_u)_{ab} := \chi_u(ab)$ has $\mathbf{C}_u\mathbf{C}_u^{\mathsf{T}} = 2^q I$, so $|v^{\mathsf{T}}\mathbf{C}_u v| \le 2^{q/2}|v|^2 = 2^{3q/2}$ for $v_a := \chi_u(G(a))$. Hence $N \le 2^{-q}(2^{2q} + (2^q - 1)2^{3q/2}) < 2^q + 2^{3q/2}$. □

Proof of Theorem 5.26. (a) P predicts y_1 uniformly on $\{0,1\}^{|x_1|}$ and then $y_2 = y_1 + x_1x_2$; in the order (b,a) it predicts $(r + ab) + ba = r$. Apply Corollary 5.18(b). (b) For fixed ω, $f_\omega \supseteq D_{abr}$ iff $r = f_\omega(a)$ and $f_\omega(a) + f_\omega(b) = ab$, so $\sum_{a\ne b,r} M_F(D_{abr}) = \mathbb{E}_\omega N(f_\omega|_F) < 2^{3q/2+1}$ by Lemma D.15 (both sides count ordered pairs). So $M_F(D_{abr})2^{-3q/2-1}$ is a semimeasure on triples, lsc uniformly in q; apply (F2). Fewer than 2^{3q-k} of the at least 2^{3q-1} triples have $K(a,b,r \mid q) < 3q - k$. (c) Jensen: $\mathbb{E}L_{\text{SFI}} \ge -\log \mathbb{E}M_F(D_{abr}) \ge \log(2^{2q}(2^q - 1)) - 3q/2 - 1 \ge 3q/2 - 2$. □

The optima 2.708 and 4.348 bits mix the maximal sets of triples one function extends (4 at $q = 2$, 42 at $q = 3$); $\mathbb{E}L_{\text{SFI}}$ is at least the optimum, as SFI restricted to the field is a random partial function. Exhaustive search gives $\max_G N(G) = 8$ ($q = 2$) and 22 ($q = 3$); for $q = 4$ it lies in $[56, 80]$.

Proof of Theorem 5.27. The lower bound is Lemma 5.2, pathwise; the upper bound is Theorem 5.11(b) with P the predictor of M_H (sub-probability conditionals), and Lemma D.3(a). In the GF process, uniform labels give $\mathbb{E}L_{\text{SFI}} \le 2q + c$ and $\mathbb{E}L_H$ is at least the Bayes loss q; x_1 has entropy q and about 1 bit given f, and x_2 is uniform on $2^q - 1$ points given h_1 but determined by f, so $I(f \to x) \approx 2q$. □

Proposition D.16 (An individual-sequence information bound)**.** [(a) proved; (b) refuted; computed] history Prop 5.7; check s1 *With $I(a:b) := K(b) - K(b \mid a)$: (a) $L_{\mathrm{SFI}}(D) \le L_H(D_\sigma) + I(x_\sigma : D) + c$ for every D and order σ. (b) The natural strengthening with $I(x_\sigma : y_\sigma)$ in place of $I(x_\sigma : D)$ is false, even up to $O(\log n)$: on the sequences of Proposition 5.19(i), $L_{\mathrm{SFI}}(D) - L_H(D_\sigma) - I(x_\sigma : y_\sigma) \ge n - \log(n+1) - K(n) - c$ (semimeasure convention).*

Proof. (a) The lookup table gives $L_{\mathrm{SFI}}(D) \le K(D) + c$ (F1). For fixed $x_{1:n}$, $y_{1:n} \mapsto M_H(y_{1:n} \| x_{1:n})$ is a semimeasure on Y^n, lsc uniformly in $x_{1:n}$, so (F2) gives $K(D \mid x_\sigma) \le L_H(D_\sigma) + c$; and $K(D) = I(x_\sigma : D) + K(D \mid x_\sigma)$. (b) $L_{\mathrm{SFI}}(D) \ge n - \log(n+1) - 1 - c$ (proof of Proposition 5.19(i)); $L_H(D_\sigma) \le c$, as $\mathbf{1}[i \text{ even}]$ knows i; and $I(x_\sigma : y_\sigma) \le K(y_\sigma) + c \le K(n) + c$, as $y_\sigma = 0101\cdots$. With A uniform, the events $\{f_\omega \supseteq D_A\}$ are disjoint, so Gibbs' inequality gives $\mathbb{E}_A L_{\mathrm{SFI}}(D_A) \ge \log \binom{n}{\lfloor n/2 \rfloor} \ge n - \log(n+1)$ exactly. □

Remark D.17 (What SFI can and cannot exploit). [remark (informal part not a theorem)] history §5.1 remark, §5.2 "Gap bounds", §5.5; check r3 *Informally* (not a theorem): SFI's hypotheses answer each query without seeing the other inputs, coupled only through the shared tape, so a label can correlate with a *guess* about the other inputs (as the brooms and the optimal law for Theorem 5.26 do) but not depend on which inputs are present. *Proved*: $\mathrm{gap}(\sigma) := L_{\mathrm{SFI}}(D) - L_H(D_\sigma)$ is at least $\max_{\sigma'} L_H(D_{\sigma'}) - L_H(D_\sigma) - c$ and at most $\min\{K^{\mathrm{pred}}(x_\sigma), L^x_{\mathrm{in}}(x_\sigma \| y_\sigma), I(x_\sigma : D)\} + c$, with $K^{\mathrm{pred}}(x_\sigma) \le K(x_{1:n-1}) + c$ (Theorems 5.20 and 5.24, Corollary D.5, and Proposition D.16); in expectation it is at most $I(f \to x) + K(\nu) + c$ (Theorem 5.27). The bound $I(x_\sigma : D)$ is weak: for constant labels it is about $K(\mathrm{dom}\, D)$, while $L_{\mathrm{SFI}} = O(1)$.

E Proofs for the worked examples

Notation is that of Section 6. The coding theorem is used in the form of Section 2.1: if $q(a \mid w)$ is lsc uniformly in (a, w) and $\sum_a q(a \mid w) \le 1$, then $K(a \mid w) \le -\log q(a \mid w) + K(q) + c$. "Gibbs" is $\mathbb{E}[-\log q(Z)] \ge H(Z)$ for every q with $\sum_z q(z) \le 1$.

E.1 SI's label loss

Lemma E.1 (SI's label loss)**.** [proved] examples Lemma 1.3 *Let $\mathbb{P}$ be a probability law on infinite sequences of pairs in countable sets, coded by a prefix-free code with nonempty codewords, and ρ a semimeasure on $\{0,1\}^*$ with $\rho(\mathrm{code}(h_t)) \ge 2^{-k}\mathbb{P}(h_t)$ for all t and histories. Let $L^y_{\mathbb{P}}(n) := \sum_{t \le n} -\log \mathbb{P}(y_t \mid h_{t-1}, x_t)$. Then (i) $\mathbb{E}L^y_\rho(n) \le \mathbb{E}L^y_{\mathbb{P}}(n) + k$ for every n, and (ii) $\mathbb{P}[\exists n : \ L^y_\rho(n) > L^y_{\mathbb{P}}(n) + k + j] \le 2^{-j}$.*

Proof. Put $a_t := \rho(\mathrm{code}(h_{t-1})\bar{x}_t)/\rho(\mathrm{code}(h_{t-1}))$. Telescoping, $\rho(\mathrm{code}(h_n)) = \rho(\epsilon)\prod_{t \le n} a_t \cdot 2^{-L^y_\rho(n)}$ with $\rho(\epsilon) \le 1$, and $-\log \mathbb{P}(h_n) = L^y_{\mathbb{P}}(n) + \sum_t -\log \mathbb{P}(x_t \mid h_{t-1})$. Hence, $\mathbb{P}$-almost surely,

$$L^y_\rho(n) \le -\log \rho(\mathrm{code}(h_n)) + \textstyle\sum_t \log a_t \le k + L^y_{\mathbb{P}}(n) + \log \Lambda_n, \qquad \Lambda_n := \prod_{t \le n} a_t / \mathbb{P}(x_t \mid h_{t-1}).$$

Given h_{t-1}, $\mathbb{E}[a_t / \mathbb{P}(x_t \mid h_{t-1})] \le \sum_x \rho(\mathrm{code}(h_{t-1})\bar{x})/\rho(\mathrm{code}(h_{t-1})) \le 1$, since $\sum_{w \in A} \rho(sw) \le \rho(s)$ for prefix-free A. So Λ is a nonnegative supermartingale with $\Lambda_0 = 1$. Jensen gives (i), and Ville's inequality $\mathbb{P}[\sup_n \Lambda_n \ge 2^j] \le 2^{-j}$ (Ville, 1939) gives (ii). □

The normalisation, $M_{\mathrm{norm}}(\epsilon) = 1$ and $M_{\mathrm{norm}}(zb) = M_{\mathrm{norm}}(z) M(zb)/(M(z0) + M(z1))$, is a probability measure with $M_{\mathrm{norm}} \ge M$, since

$$M_{\mathrm{norm}}(z) = \tfrac{M(z)}{M(\epsilon)} \textstyle\prod_{i < |z|} \frac{M(z_{1:i})}{M(z_{1:i}0) + M(z_{1:i}1)}$$

and each factor is at least 1. For computable $\mathbb{P}$, $M(\mathrm{code}(h)) \geq 2^{-K(\mathbb{P})-c}\mathbb{P}(h)$ (Lemma D.14), so the lemma applies to M and M_{norm} with $k = K(\mathbb{P}) + c$.

E.2 The cyclic permutation

For a string x and strings $z = (z_1, \ldots, z_i)$ put

$$D(x, z) := \{(x, z_1), (z_1, z_2), \ldots, (z_{i-1}, z_i), (z_i, 0^{|x|})\}, \qquad D(x, ()) := \{(x, 0^{|x|})\},$$

so that $D_k = D(x_k, (x_{k-1}, \ldots, x_1))$.

Lemma E.2 (chain semimeasure)**.** [proved] examples Lemma 2.2 (history Lemma 2.0 with terminal label $0^{|x|}$) *For every law ν of a random partial function F and every x, $\sum_z \nu(D(x, z)) \leq 1$, over the z for which $D(x, z)$ is a data set. Hence $K(z \mid x) \leq L_{\mathrm{SFI}}(D(x, z)) + c$.*

Proof. If $F \supseteq D(x, z)$, then z is the orbit of x up to its first visit to $0^{|x|}$; so for each F at most one z qualifies. $M_F(D(x, z))$ is lsc; apply the coding theorem. □

Lemma E.3 (depth count)**.** [proved; computed (examples e6(b))]
examples Lemma 2.2a *For a partial function F and $1 \leq k \leq N - 1$, let $A_k(F)$ be the set of nonzero $x \in \{0, 1\}^d$ such that $F(x), \ldots, F^{k-1}(x)$ are nonzero d-bit strings and $F^k(x) = 0^d$. Then $|A_k(F)| \leq N - k$, with equality for the broom B_k.*

Proof. If $x \in A_k(F)$, the strings $F(x), \ldots, F^{k-1}(x)$ first reach 0^d at times $k-1, \ldots, 1$, so they are distinct, nonzero and not in $A_k(F)$: $|A_k(F)| \leq (N - 1) - (k - 1)$. For the broom, every nonzero input off the spine is in A_k. □

No violation occurs over all 625 partial functions for $d = 2$, all 8^8 total functions for $d = 3$, and 10^5 random partial functions for $d = 3, 4$.

Proof of Theorem 6.3. (a) The lower bound is Theorem 2.21(a) with $T = \{0, 1\}^d$, computable from d. For the upper bound, the function machine T, on input x, runs $V(\cdot, \{0, 1\}^{|x|})$ on program bits read from the start of the tape and, if V halts with a table E of $\{0, 1\}^{|x|}$, outputs $E(x)$. For fixed x the halting prefixes are prefix-free, so T is consistent; all inputs of length d read the same program, so $\nu_T(f) \geq 2^{-K(f|d)}$; apply Theorem 2.9. (The same argument gives $L_{\mathrm{SFI}}(D) \leq K(D \mid \mathrm{dom}\, D) + K(e) + c$ whenever a partial computable e maps each $x \in \mathrm{dom}\, D$ to $\mathrm{dom}\, D$.) (b) T_{cyc} reads, for each length ℓ, its own sub-stream $(\omega_{\langle \ell, i \rangle})_i$ of the tape, draws a uniform ordering of the nonzero ℓ-bit strings by rejection sampling, closes it into a cyclic permutation $\hat{f}_\ell$ (the bijection after Definition 6.2), and outputs $\hat{f}_{|x|}(x)$. It halts with probability 1, each value is fixed by finitely many tape bits, $K(\nu_{T_{\mathrm{cyc}}}) = O(1)$, and $\nu_{T_{\mathrm{cyc}}}(f) = 1/(N-1)!$; apply Theorem 2.9. (c) By (a), $L_{\mathrm{SFI}}(f) \geq K(f \mid d) - c$; apply Lemma 6.1 with $Z = f$, $w = d$. (d) Robbins' bounds $\ln m! = m \ln m - m + \frac{1}{2}\ln(2\pi m) + r_m$, $\frac{1}{12m+1} < r_m < \frac{1}{12m}$ (Robbins, 1955), for $(N-1)! = N!/N$ give $\epsilon_N = r_N / \ln 2$; a 120-digit computation confirms the window for $d \leq 24$ and $d = 32, 40, 64$. D_{N-1} has the computable domain $\{0, 1\}^d \setminus \{0^d\}$ and determines f, so the bounds transfer. □

Proof of Theorem 6.4. (a) If $k - 1 > 2^\ell - 1$, B_k answers 0^ℓ on length ℓ; for $k = 1$ it maps everything to 0^ℓ. Each value is fixed by finitely many tape bits almost surely, B_k is total almost surely, and its index is computable from k. On the event $(v_1, \ldots, v_{k-1}) = (x_1, \ldots, x_{k-1})$, of probability $1/(N-1)_{k-1}$, it maps $x_1 \mapsto 0^d$, $x_i \mapsto x_{i-1}$, and x_k, which is off the spine, to $v_{k-1} = x_{k-1}$. Theorem 2.9 gives the term $K(k)$. A uniform cyclic permutation contains the k arcs $x_k \to \cdots \to$

$x_1 \to 0^d$, which do not close the cycle, with probability $(N-k-1)!/(N-1)! = 1/(N-1)_k$ (contract the path), so T_{cyc} gives $\log(N-1)_{k-1} + \log(N-k) + c$. (b) Let F have law ν. Each $x \in A_k(F)$ has exactly one z of length $k-1$ with $F \supseteq D(x,z)$, its orbit, and other nonzero x have none. So, summing over nonzero d-bit x and $|z| = k-1$,

$$\textstyle\sum_{x,z} \nu(D(x,z)) = \mathbb{E}|A_k(F)| \le N-k. \tag{6}$$

Thus $M_F(D(x,z))/(N-k)$ is a semimeasure in (x,z), lsc uniformly in (d,k), and $K(x,z \mid d,k) \le L_{\mathrm{SFI}}(D(x,z)) + \log(N-k) + c$. As $(x_k,\dots,x_1)$ is uniform over $(N-1)_k$ values, Lemma 6.1 with $w=(d,k)$ gives (b). (c) Use $j+d$ in (b) and a union bound over the $N-1 < 2^d$ values of k. □

Proof of Proposition 6.5. (a) Gibbs with $q := \nu(D(\cdot,\cdot))/(N-k)$ and (6). (b) First, for weak ν and nonzero x, $\sum_z \nu(D(x,z)) \le 1$ over sequences z of distinct nonzero strings other than x. Put $O_i(z) := \{(x,z_1),\dots,(z_{i-1},z_i)\}$, $O_0 := \emptyset$, whose last point z_i (x if $i=0$) is not in its domain. The weak inequality there, keeping the term $y = 0^d$, which is $\nu(D(x,z))$, and the terms with nonzero $y \notin \{x,z_1,\dots,z_i\}$, which are $\nu(O_{i+1}(z,y))$, gives by induction $\sum_{i\le J}\sum_{|z|=i}\nu(D(x,z)) + \sum_{|z|=J+1}\nu(O_{J+1}(z)) \le \nu(\emptyset) \le 1$ for every J (as for Proposition 5.5). Given x_k, $(x_{k-1},\dots,x_1)$ is uniform over $(N-2)_{k-1}$ values; Gibbs gives $\log(N-2)_{k-1} = \log(N-1)_k - \log(N-1)$. For the full cycle, $\sum_g \nu(g) \le 1$ over total g on $\{0,1\}^d$ (Definition 2.2); Gibbs. □

Proof of Proposition 6.6. $\bar\ell_t = -\log M_F(D_t) + \log \mathcal{Z}(D_{t-1}, x_t)$. (i) M_F is the law of a random partial function (Proposition 2.8), so $\mathbb{E}[-\log M_F(D_t)] \ge \log(N-1)_{t-1}$ by Proposition 6.5(a). (ii) B_{t-1} is total almost surely and $x_t \notin \operatorname{dom} D_{t-1}$, so Lemma D.2 and the proof of Theorem 6.4(a) give $\mathcal{Z}(D_{t-1},x_t) \ge 2^{-K(t-1)-c}\nu_{B_{t-1}}(D_{t-1}) \ge 2^{-K(t-1)-c}/(N-1)_{t-2}$ on every path. (iii) $\log(N-1)_{t-1} - \log(N-1)_{t-2} = \log(N-t+1)$. The high-probability form uses Theorem 6.4(b) for (i). For the sum use $\bar\ell_t \ge 0$, $K(t-1) \le \log t + 2\log\log(t+1) + c$ and $N-t+1 \ge N/2$. *Why the method stops.* Over the cycle these bounds sum to at most $\sum_{t<N}(d - K(t))^+ \le 2N$ bits, since fewer than 2^{d-i+1} integers have $K(t) \le d-i$. In the finite mixtures of Section 6.3 (cyclic law, brooms and *partial* chains, undefined off a random spine) the per-step inequality holds with margin at least 2.77 bits; the full-cycle normalised loss is 8748.7 against $\log((N-1)!) = 8759.0$ bits at $d=10$. □

Proof of Proposition 6.8. The labels are determined by the order $\sigma = (x_1,\dots,x_{N-1})$. Fix a tape ω of the universal set-machine, with answer $U^\omega(A,x)$ at x given the set A. If σ is consistent with ω, then $x_{N-1} = U^\omega(\{0,1\}^d \setminus \{0^d\}, 0^d)$, $x_{N-2} = U^\omega(\{0,1\}^d \setminus \{0^d, x_{N-1}\}, x_{N-1})$, and so on. So $\sum_\sigma M_{\mathrm{set}}(y(\sigma)\|(\sigma,0^d)) \le 1$, lsc uniformly in d. Gibbs gives the expectation bound; the coding theorem and Lemma 6.1 give the probability bound. The upper bound is $L_{\mathrm{set}} \le L_{\mathrm{SFI}} + c$ on distinct inputs (Proposition D.10) and Theorem 6.3(b). □

Proof of Proposition 6.9. Let $\mathbb{P}_d$ be the law of the pairs for bit length d (x_t uniform over unused nonzero strings, $x_N = 0^d$, $y_t = x_{t-1}$, continued after step N by a fixed computable rule). It is computable uniformly in d, so by Lemma D.14 applied to $\sum_d \mathbf{m}(d)\mathbb{P}_d$, $\rho(\mathrm{code}(h)) \ge M(\mathrm{code}(h)) \ge 2^{-c}\mathbf{m}(d)\mathbb{P}_d(h)$. Telescoping as in Lemma E.1, with label conditionals 1, $L^y_\rho(k) \le c - \log \mathbf{m}(d) + \log\rho(\epsilon) + \log\Lambda_k$. At the first step $\mathbb{E}\Lambda_1 = \sum_x \rho(1^d0x)/\rho(\epsilon) \le \rho(1^d0)/\rho(\epsilon)$, so $V_k := \Lambda_k\rho(\epsilon)/\rho(1^d0)$ is a nonnegative supermartingale with $V_0 := 1$, and $L^y_\rho(k) \le c + K(d) + \log\rho(1^d0) + \log V_k$. Finally $\log\rho(1^d0) \le -K(d) + c$: for M by the coding theorem, as $\sum_d M(1^d0) \le 1$ and $d \mapsto M(1^d0)$ is lsc; for M_{norm} because $M_{\mathrm{norm}}(1^d0) \le M_{\mathrm{norm}}(0 \mid 1^d) \le M(1^d0)/M(1^{d+1})$ and $M(1^{d+1}) \ge 2^{-c}$. Jensen and Ville finish. □

d	$\log((N-1)!)$	per pair	$d - \log e$	SFI proxy	history proxy	SI proxy
4	40.250	2.516	2.557	42.748	1.585	1.544
8	1675.996	6.547	6.557	1678.576	1.585	1.583
12	43238.047	10.556	10.557	43240.632	1.585	1.585

[computed] examples §2.6; checks e3, e6

Table 3: The cyclic permutation: constants and full-cycle losses (bits) of exact finite mixtures. SFI proxy: weight $\frac{1}{6}$ each on the uniform cyclic-permutation, permutation and function laws and the maps $x \mapsto 0^d$, $x \mapsto x$, $x \mapsto x+1 \bmod N$. History proxy: $\frac{1}{3}$ each on "copy the previous input", uniform labels, uniform unused labels. SI proxy: label loss of a mixture of the true process, true inputs with uniform labels, and uniform blocks.

Proposition E.4 (SI on raw blocks)**.** [proved] examples Prop 2.6 *Code the history as $x_1y_1x_2y_2\cdots$ in raw d-bit blocks without delimiters. For $\rho \in \{M, M_{\mathrm{norm}}\}$ and uniform cyclic f: $\mathbb{P}[\exists k \le N : L^y_\rho(k) > K(d) + j + c] \le 2^{-j}$, and $\mathbb{E}L^y_\rho(k) \le K(d) + c$.*

Proof. For fixed d the raw blocks are a prefix-free code and Lemma D.14 applies uniformly in d, so $M \ge 2^{-K(d)-c}\mathbb{P}_d$ on coded histories; the label loss of $\mathbb{P}_d$ is 0. Apply Lemma E.1. □

An exact Bayes mixture of the true processes for $d' \le 10$ and a uniform-bits model has full-cycle label loss 0.13–0.26 bits with the prefix code and 1.3–7.8 bits with raw blocks ($d = 2, \ldots, 6$).

Conjecture E.5 (raw blocks)**.** [conjecture] examples Conj 2.6a *For raw blocks, SI's expected label loss is of order $K(d)$ for typical d.*

Heuristic, not proved. SI does not know where the first block ends. The label 0^d is predicted well by models with block length d, and by models with block length $d' > d$ only at a cost $d' - d$. So the first label costs about $-\log(\mathbf{m}(d)/\sum_{d' \ge d}\mathbf{m}(d')) = K(d) - K^{\mathrm{tail}}(d) \pm c$ bits, and $K^{\mathrm{tail}}(d) \le K^{\mathrm{pad}}(d) + c = O(\log\log d)$.

Remark E.6 (a prior over d). [proved] examples Rem 2.6b If d is drawn from a computable prior w, then with raw blocks $\mathbb{E}L^y_M \le H(w) + K(w) + c$.

Proof. $M \ge 2^{-K(w)-c}w(d)\mathbb{P}^{\mathrm{bin}}_d$, with $\mathbb{P}^{\mathrm{bin}}_d$ the law of the raw coding. Telescoping gives $L^y_M(k) \le K(w) + c - \log w(d) + \log M(\epsilon) + \log \Lambda_k$, and $\mathbb{E}[\Lambda_1 \mid d] \le \sum_{|x|=d} M(x)/M(\epsilon) \le 1$, since strings of one length are prefix-free. (Raw blocks of different lengths are not, so Lemma E.1 does not apply to the mixture directly.) □

Remark E.7 (where the information is). [proved; the other-direction part conditional] examples Rem 2.7, 2.8, 2.12 A history-aware predictor that sees the previous *pairs* as a set predicts perfectly: the most recent input is the unique input of D_{t-1} that is not a label. So the separation is between scoring a data set as a whole and predicting with the past in view, not between sets and sequences; the previous *inputs* as a set are not enough (Proposition 6.8). In the other direction, take $X = Y = \{0, 1\}$ and $y_i = x_i$: $L_{\mathrm{SFI}} \le c$ on every input sequence, while, if unnormalised M has unbounded loss on the copied bits of some $\omega_1\omega_1\omega_2\omega_2\cdots$ (a form of the failure reported by Lattimore et al. (2011)), SI's unnormalised label loss with the identity code is unbounded on the inputs $\omega_1\omega_2\cdots$ (Proposition 5.22).

n	runs	SFIproxy$/n$	UB_n/n	iid$/n$	sd$(UB_n)/\sqrt{n}$	SFIproxy$-UB_n$	L_{P^*}	SIproxy	touching
100	400	0.5785	0.4498	1.2532	0.989	12.9	2.18	3.32	2.83
1000	200	0.4840	0.4630	1.2502	1.113	21.0	2.30	3.43	2.88
10000	40	0.4627	0.4597	1.2510	1.113	29.4	2.17	3.29	2.85
limit			0.4591	1.2516	1.0245		2.3986		3

[computed] examples §3.8; check e2, seed 20261011

Table 4: Eisenstat's example: means over seeded runs, in bits. SFIproxy: exact order-free Bayes mixture of an i.i.d. law and two families of threshold laws, with Laplace biases; iid: the product law with $\mathbb{P}[y=1]=\frac{1}{3}$; SIproxy: label loss of an exact mixture of the true process, the true inputs with Laplace labels, and a null model. SFIproxy$-UB_n$ grows like $2.5\log n$ (cf. Proposition 6.12). Last row: the limits of Propositions 6.14 and 6.18.

E.3 Eisenstat's example

Hänni's e-mail argument, as quoted in his request (Hänni, 2026c), reads:

> "Claim. log loss in your solomonoff function induction coinflip question is the loss given by our threshold function, up to o(n) terms
>
> Pf sketch.
>
> easy direction: We know log loss of solomonoff function induction L_S is at most the log loss of an appropriate threshold function L_T (up to a log n term for specifying the threshold), because this guy lives inside the solomonoff distribution.
>
> other direction: Say for a contradiction that L_S < 0.999 * L_T. Then SFI log loss on the initial mixed coinflip segment L'_S would be even lower than that. But then you can specify that coinflip sequence with only L'_S bits (by the standard pointing trick for going from a solomonoff measure lower bound to a complexity upper bound). This contradicts a random 1/3 coinflip sequence being with very high probability incompressible below h(1/3) per bit.
>
> So L_S approx L_T."

In Remark 6.17, $UB_n - \tau_n H(\frac{1}{3}) = \frac{2}{3}(G_{N_n^0} - \frac{1}{2}N_n^0)$, with G as in the proof of Proposition 6.14.

Lemma E.8 (the heads frontier is rarely behind)**.** [proved; computed (examples e7(a))] examples Lemma 3.5 $\mathbb{P}[\eta_n < \tau_n] \le 20e^{-n/18}$. *The exact values are* $6.48\cdot10^{-2}$, $1.55\cdot10^{-2}$, $1.11\cdot10^{-3}$, $7.31\cdot10^{-6}$, $4.27\cdot10^{-10}$ *at* $n = 10, 20, 40, 80, 160$.

Proof. If $\eta_n < \tau_n$, all revealed indices lie below τ_n, so $\tau_n \ge n$ and $b_{<\tau_n}$ has exactly N_n^0 tails (Lemma 6.11). So $N_n^0 \ge n/2$, or some $k \ge n$ has fewer than $k/2$ tails in $b_{<k}$. By Hoeffding's inequality (Hoeffding, 1963) these have probabilities at most $e^{-n/18}$ and $e^{-k/18}$; the sum is at most $e^{-n/18}(1 + 1/(1-e^{-1/18})) \le 19.6\, e^{-n/18}$. The exact values come from a dynamic program over the frontier process of Proposition 6.18. □

Proof of Theorem 6.13. $[0, g_n) \subseteq \operatorname{dom} D_n$ with labels $b_{<g_n}$, and $[0, g_n)$ and g_n are computable from each other; monotonicity and Theorem 2.21(a) give $L_{\mathrm{SFI}}(D_n) \ge K(b_{<g_n} \mid g_n) - c$. For b Martin-Löf random for the computable $P_{1/3}$, the Levin–Schnorr theorem (see Li and Vitányi, 2008) gives $K(b_{<g}) \ge -\log P_{1/3}(b_{<g}) - c_b$, so $K(b_{<g_n} \mid g_n) \ge -\log P_{1/3}(b_{<g_n}) - K(g_n) - c_b$; if $\tau_n \le \eta_n$, then $g_n \ge \tau_n$ and $-\log P_{1/3}(b_{<g_n}) \ge UB_n$. The upper bound is Proposition 6.12. By Lemma E.8 and Borel–Cantelli the heads are eventually ahead; by the strong law $\eta_n/n \to 2$ and $\tau_n/n \to \frac{1}{2}$, so $g_n \le \max(\tau_n, \eta_n) \le 3n$ for large n. Martin-Löf random b have probability 1. □

Proof of Proposition 6.14. Let G_i be the number of heads before the i-th tail. Then $\tau_n = N_n^0 + G_{N_n^0}$ and, using $H(\frac{1}{3}) = \log 3 - \frac{2}{3}$,

$$UB_n - \tfrac{H(1/3)}{2} n = a(N_n^0 - \tfrac{n}{3}) + \log 3\,(G_{N_n^0} - \tfrac{1}{2} N_n^0),$$

where $G_i = \sum_{l \le i} \gamma_l$ with γ_l i.i.d. geometric ($\mathbb{P}[\gamma = m] = \frac{2}{3}(\frac{1}{3})^m$, mean $\frac{1}{2}$, variance $\frac{3}{4}$), independent of $N_n^0 \sim \mathrm{Bin}(n, \frac{1}{3})$. The second term has conditional mean 0 given N_n^0, so the terms are uncorrelated, with variances $a^2 \frac{2n}{9}$ and $(\log 3)^2 \frac{3}{4} \frac{n}{3}$. The strong law follows from $N_n^0/n \to \frac{1}{3}$, $G_i/i \to \frac{1}{2}$. For the limit law, let $\psi(s) := \mathbb{E} e^{is(\gamma - 1/2)} = 1 - \frac{3}{8} s^2 + o(s^2)$; by independence the characteristic function at θ is $\mathbb{E}[e^{i\theta a (N_n^0 - n/3)/\sqrt{n}} \psi(\theta \log 3/\sqrt{n})^{N_n^0}]$, the second factor tends to $e^{-\theta^2 (\log 3)^2/8}$ in probability with modulus at most 1, and bounded convergence with the de Moivre–Laplace and Lévy continuity theorems gives $\mathcal{N}(0, \sigma^2)$. The exact laws give $\mathbb{E} UB_n/n = 0.459147917027$ and $\mathrm{Var}(UB_n)/n = 1.049660151594$ for every $n \le 200$ tested. □

Proof of Theorem 6.15. Upper. Take expectations in Proposition 6.12; $t \mapsto \log(t+1) + 2 \log\log(t+2)$ is concave and $\mathbb{E}\tau_n = \frac{n}{3} + \frac{n}{6} = \frac{n}{2}$. *Lower.* By Theorem 6.13 and Kraft, conditionally on each value of g_n, $\mathbb{E} L_{\mathrm{SFI}}(D_n) \ge H(b_{<g_n} \mid g_n) - c = H(b_{<g_n}) - H(g_n) - c$. Since $\mathbb{P}[b_{<g_n} = s] \le P_{1/3}(s)$, $H(b_{<g_n}) \ge \mathbb{E}[-\log P_{1/3}(b_{<g_n})] \ge \mathbb{E}[UB_n; \tau_n \le \eta_n]$, and $\mathbb{E}[UB_n; \eta_n < \tau_n] \le \log 3\,(\mathbb{E}\tau_n^2)^{1/2} \mathbb{P}[\eta_n < \tau_n]^{1/2} = O(n) e^{-n/36}$. Next, g_n is at most the first tail at or after τ_n; $\{\tau_n = k\}$ depends only on the coins and $b_{<k}$, so $\mathbb{E} g_n \le \frac{n}{2} + \frac{1}{2}$, and a variable on $\mathbb{N}$ with mean m has entropy at most $\log(m+1) + \log e$. The i.i.d. law pays $\mathbb{E}[N_n^1 \log 3 + N_n^0 \log \frac{3}{2}] = (\log 3 - \frac{1}{3}) n$. □

Proof of Proposition 6.16. (a) $D_n = D$ exactly when the coins contain N^1 ones (probability $\binom{n}{N^1} (\frac{2}{3})^{N^1} (\frac{1}{3})^{n - N^1}$) and the first N^1 heads and $n - N^1$ tails of b are the inputs of D labelled 1 and 0. By Lemma 6.11 this fixes b on $[0, s^+)$: each of the $s^+ - n$ unrevealed indices below s^+ is a tail if the top label is 1, a head otherwise; probability $(\frac{1}{3})^{N^1} (\frac{2}{3})^{n - N^1} r^{s^+ - n}$. Given $\mathrm{dom}\, D_n = S$, with $g := \min(\mathbb{N} \setminus S)$, the labelling has probability $\binom{n}{N^1} r^{s^+ - n} / Z(S)$, where $Z(S) = ((\frac{2}{3})^{s^+ - n} + (\frac{1}{3})^{s^+ - n}) \binom{n+g}{g}$ if $g < s^+$ and $Z(S) = \binom{2n}{n}$ if $S = [0, n)$. Indeed, a possible labelling with top label 1 is fixed by its tail frontier $\tau' \in [0, g]$ (arbitrary labels on $[0, \tau' - 1)$, 0 at $\tau' - 1$, 1 on $S \setminus [0, \tau')$); summing $\binom{n}{N^1}$ gives $\sum_i \binom{\tau' - 1}{i} \binom{n}{\tau' - i} = \binom{n + \tau' - 1}{\tau'}$, and $\sum_{\tau' \le g} \binom{n - 1 + \tau'}{\tau'} = \binom{n+g}{g}$; top label 0 is symmetric, and $\sum_m \binom{n}{m}^2 = \binom{2n}{n}$. On $S = \{0, 1, 3\}$ the labellings $(0, 1, 1)$ and $(1, 1, 0)$ both have $N^1 = 2$ but probabilities $\frac{1}{5}$ and $\frac{1}{10}$. (b) Weak semimeasures are monotone and $\sum_l \nu(l) \le 1$ over the labellings of $[0, g)$, so Gibbs given g_n yields $\mathbb{E}[-\log \nu(D_n)] \ge H(b_{<g_n} \mid g_n)$; continue as for Theorem 6.15. □

Lemma E.9 (the Bayes predictor)**.** [proved; computed (examples e1(c))] examples Lemma 3.8 *Let h be a history with frontiers τ, η, $F := \max(\tau, \eta)$ and smallest unrevealed index g, and x the next input. The conditional probability of label 1 at x is 0 if $x < \eta$ and 1 if $x < \tau$ (for $x < F$ exactly one holds); for $x \ge F$ it is 1 if $g < F$ and $\tau \le \eta$, 0 if $g < F$ and $\eta < \tau$, and $2^k/(2^k + 1)$, $k := x - F$, if $g = F$ (touching).*

Proof. The first case is Lemma 6.11. Let $x \ge F$. If $\tau \le \eta$ and $g < F$, the first unrevealed index in $[\tau, \eta)$ is the next tail, so a tail reveal would give an input below η; the case $\eta < \tau$ is symmetric. If h is touching, it depends only on $\zeta_{<t}$ and $b_{<F}$, so a head at $F + k$ has probability $\frac{2}{3}(\frac{2}{3})^k \frac{1}{3}$ and a tail there $\frac{1}{3}(\frac{1}{3})^k \frac{2}{3}$; the ratio is 2^k. □

A brute-force comparison over all 14178 reachable pairs (history, input) for $n \le 5$ finds no difference.

Proof of Proposition 6.18. After each step the process is touching, or in a state (A,u) (heads ahead, $u \geq 1$ unrevealed tails in $[\tau,\eta)$), or (B,u) (heads behind, $u \geq 1$ unrevealed heads in $[\eta,\tau)$); indices $\geq F$ are fresh. From touching, a head reveal skipping k tails (probability $\frac{2}{9}(\frac{2}{3})^k$) leads to (A,k), a tail reveal skipping k heads (probability $\frac{2}{9}(\frac{1}{3})^k$) to (B,k); $k=0$ means touching. From (A,u): $u-1$ with probability $\frac{1}{3}$, $u+m$ with probability $\frac{2}{9}(\frac{2}{3})^m$. From (B,u): $u-1$ with probability $\frac{2}{3}$, $u+m$ with probability $\frac{2}{9}(\frac{1}{3})^m$. The A-walk moves down by exactly one and has drift $+1$; its probability β of ever going one level down satisfies $\beta = \frac{1}{3} + \frac{2}{3}\sum_m \frac{1}{3}(\frac{2}{3})^m \beta^{m+1}$, i.e. $2\beta^2 - 3\beta + 1 = 0$, and $\beta = 1$ would contradict its convergence to $+\infty$; so $\beta = \frac{1}{2}$, and (A,k) returns with probability 2^{-k}. The B-walk has drift $-\frac{1}{2}$ and returns almost surely. The return probability from touching is $\frac{2}{9} + \frac{2}{9} + \sum_{k\geq 1}\frac{2}{9}(\frac{1}{3})^k + \sum_{k\geq 1}\frac{2}{9}(\frac{1}{3})^k = \frac{2}{3}$, and after a touching step the process starts afresh, so N_{touch} is geometric. By Lemma E.9, a touching step costs P^* an expected ϑ and other steps cost 0, so $\mathbb{E}L_{P^*}(n) = \vartheta \mathbb{E}N_{\text{touch}}(n)$; exact dynamic programming gives $\mathbb{E}N_{\text{touch}}(n) = 2.611, 2.907, 2.993, 3.000$ at $n = 10, 20, 40, 80$. □

Proof of Corollary 6.20. In (A,u) the unrevealed indices in $[\tau,\eta)$ are tails, so g is the next tail to be revealed and a head lands at an index $\geq \eta > g$. Almost surely $N_{\text{touch}} < \infty$; after the last touching step the process stays in A-states, since B-states return to touching almost surely and A-states are left only through touching. □

Proof of Proposition 6.21. At a touching step ($g = F$, input $F + k$) the rule errs on a head with $k = 0$ (probability $\frac{2}{9}$) and on a tail with $k \geq 1$ (probability $\frac{1}{9}$). It never errs in A-states and always in B-states, where g is the next head. A B-excursion from (B,k) moves down by exactly one with i.i.d. increments of mean $-\frac{1}{2}$; optional stopping for the martingale $u_t + t/2$ shows that its length S is integrable, and Wald's identity gives $\mathbb{E}S = 2k$. So a touching step is followed by $\sum_{k\geq 1}\frac{2}{9}(\frac{1}{3})^k 2k = \frac{1}{3}$ B-steps on average, and a cycle (a touching step and its excursion) has $\frac{2}{9} + \frac{1}{9} + \frac{1}{3} = \frac{2}{3}$ expected errors. Cycles are i.i.d., N_{touch} is a stopping time for them, and Wald's identity gives $3 \cdot \frac{2}{3} = 2$. For the second claim, the probability p of no error from a touching history satisfies $p = \frac{2}{9}p + \sum_{k\geq 1}\frac{2}{9}(\frac{2}{3})^k(2^{-k}p + 1 - 2^{-k}) = \frac{1}{3}p + \frac{1}{3}$ (an error-free step is a tail at $k = 0$ or a head at $k \geq 1$), so $p = \frac{1}{2}$. Exact dynamic programming gives 1.575, 1.891, 1.992, 2.000 expected errors in the first 10, 20, 40, 80 steps; in 1000 simulated runs of length 1000 the rule erred 1.98 times on average and never in 48.8% of runs. □

E.4 Positive examples

Proof of Proposition 6.27. (a) T_g reads a uniform real ε' bit by bit from a shared part of the tape and, for each input x, a private uniform real u_x, and outputs $g(x) \oplus \mathbf{1}[u_x < \varepsilon']$ once the comparison is decided (after finitely many bits, with probability 1). Given ε' the flips are independent, so $\nu_{T_g}(D) = \int_0^1 \varepsilon'^e (1-\varepsilon')^{n-e} d\varepsilon' = 1/((n+1)\binom{n}{e})$, with $K(\nu_{T_g}) \leq K(g) + c$. So T_g is a mixture of product laws, not a product law. Among product laws, "g flipped independently with probability e/n" has complexity $K(g) + O(\log n)$ and loss $nH(e/n) \leq \log\binom{n}{e} + \log(n+1)$, which gives (a) up to $O(\log n)$. (b) Each new input's flip is Bern(ε) and independent of the past, so $e \sim \text{Bin}(n,\varepsilon)$; use $\log\binom{n}{e} \leq nH(e/n)$ and concavity. Given the past and a new input, the label has conditional entropy $H(\varepsilon)$, so Gibbs at each step bounds every predictor below by $H(\varepsilon)$ per step. □

Finite versions. For sparse parity ($d = 16$, $k = 3$, prior $2^{-|S|-1}/\binom{d}{|S|}$ on $|S| \leq 4$) the loss saturates at $\log\binom{16}{3} + 4 = 13.13$ bits after 16 to 24 random examples, in any order. For frozen noise (parities of at most 3 of 12 bits, $\varepsilon = 0.1$, Laplace flip rate) the regret against the noise-aware predictor is 14.9–17.3 bits for $n = 100$–3200, close to $-\log w(S^*) + \frac{1}{2}\log n$.

Remark E.10 (fresh noise). [proved (citing)]
examples Rem 4.7 Let the pairs be i.i.d. from $\mu \times P(y \mid x)$ with fresh noise at each draw. If inputs never repeat, fresh and frozen noise coincide and Theorem 4.1(a) bounds the regret against P by $K(P) + c$ on every path. If some input has positive probability and $P(\cdot \mid x)$ is not a point mass there, $\mathbb{E}L_{\mathrm{SFI}} = \infty$ in both conventions (Proposition 4.6); time stamps or the per-query mixture repair this (Theorem 4.5(b), Proposition 4.7).

E.5 Computations

The numbers in Section 6 come from seeded or deterministic Python scripts in the accompanying repository, whose saved outputs re-runs reproduce: `e1_eisenstat_exact` (exhaustive enumeration in exact rationals, $n \le 5$, for Proposition 6.16 and Lemma E.9), `e2_eisenstat_sim` (Table 4, the rule's errors), `e3_cyclic` (Table 3), `e4_positive` (Section 6.4), `e5_dsl_odd_lengths` (Section 7), `e6_cyclic_sharp` (broom and chain laws against closed forms, 1092 exact comparisons; depth count; Robbins' window; normalised and SI mixtures) and `e7_eisenstat_dp` (exact dynamic program over the frontier process, cross-checked by exact rationals, enumeration and simulation). Separately written implementations agree on the broom laws, the depth count, Robbins' window, the SI codings and the dynamic program. The mixtures are finite, not SFI or SI; statements about K, M, M_F and M_H are proved by hand.

F Proofs for Section 7

Notation and conventions are those of Section 7; c, c' are machine constants. We use the conditional coding theorem as stated in Section 2: a discrete semimeasure $\mu(\cdot \mid z)$ that is lsc uniformly in z satisfies $\mu(a \mid z) \le 2^{-K(a|z)+c_\mu}$.

Proof of Proposition 7.4. (a) Let $V'(0u) := V(u)$, and let $V'(1u)$ be an index of the function that is b on Q and $\varphi_{V(u)}$ off Q (defined when $u \in \operatorname{dom} V$). $\operatorname{dom} V'$ is prefix-free, and V' is universal because it simulates V with one extra bit. If $\operatorname{dom} D \cap Q = \emptyset$, the programs $0u$ and $1u$ agree on $\operatorname{dom} D$, so each half of $\operatorname{dom} V'$ contributes $W(D)/2$: $W'(D) = W(D)$. Every program $1u$ outputs b on S, so $W'(D \cup b|_S) \ge W(D)/2 = W'(D)/2$. For all D: $W(D) \le 2W'(D)$ by the programs $0u$, and $W(D) \ge 2^{-K(Q,b)-c}W'(D)$ because V simulates $1u$ by the wrapper wzu of Proposition 7.1 and $0u$ by u.

(b) For a function machine T_i let $T_i^{Q,b} := \{(x, \varepsilon, b(x)) : x \in Q \cap \operatorname{dom} b\} \cup \{(x, w, y) \in T_i : x \notin Q\}$: it answers b on Q without reading the tape and runs T_i elsewhere. It is c.e. because Q is decidable, and consistent because on Q all triples carry the label $b(x)$ and off Q they are those of T_i. Its index is computable from i and $\langle e, e' \rangle$. Let $V''(0u) := V(u)$ and let $V''(1u)$ be an index of $T_{V(u)}^{Q,b}$, and build M'_F from V'' as in Theorem 2.9. For $D \ne \emptyset$,

$$M'_F(D) = \tfrac{1}{2} \sum_{u \in \operatorname{dom} V} 2^{-|u|} \nu_{T_{V(u)}}(D) + \tfrac{1}{2} \sum_{u \in \operatorname{dom} V} 2^{-|u|} \nu_{T_{V(u)}^{Q,b}}(D).$$

If $\operatorname{dom} D \cap Q = \emptyset$, the functions $f_\omega^{T_i^{Q,b}}$ and $f_\omega^{T_i}$ agree on $\operatorname{dom} D$ for every ω, so both halves equal $\frac{1}{2}M_F(D)$ and $M'_F(D) = M_F(D)$. Every tape in the second half gives a function equal to b on $Q \cap \operatorname{dom} b$, which gives the posterior bound $\frac{1}{2}$. The first half gives $M'_F \ge \frac{1}{2}M_F$ on nonempty sets, so M'_F dominates every machine law. Conversely M'_F is a machine law with an index computable from $\langle e, e' \rangle$, so $M_F \ge 2^{-K(Q,b)-c}M'_F$. The decomposition holds only for nonempty data sets; for $D = \emptyset$ the same computation gives $M'_F(b|_S) \ge \frac{1}{2}\Omega_V$ with $\Omega_V := \sum_{p \in \operatorname{dom} V} 2^{-|p|}$. □

Proof of Theorem 7.5. (a) Let $b_P(x)$ be the first y found in a fixed enumeration of P with first coordinate x. It is partial computable, total on Q, P-abiding, and $K(Q, b_P) \le K(P, Q) + c$. Apply Proposition 7.1 with S. (b) The same with $b_{\neg P}$ from an enumeration of the complement. For the consequence: the events "every $x \in S$ is P-abiding" and "some $x \in S$ is P-violating" concern disjoint sets of labellings of S; for W they are disjoint sets of programs and for M_F disjoint sets of tapes, so their conditional probabilities sum to at most 1. (c) For W: let $h \in \operatorname{dom} V$ be shortest with $\varphi_{V(h)}(x) = z$, so $|h| = K^{\mathcal{P}}(z \mid x)$; apply Proposition 7.1 with $b := \varphi_{V(h)}$ and $K(Q, b) \le K(Q) + |h| + c$. For M_F: let $T(x', \omega)$ split ω into its even and odd bits; if $x' \in Q$, run the conditional universal prefix machine on input x', reading program bits lazily from the odd bits; otherwise run T_U on the even bits. T is a function machine. The two streams are independent and uniform and no input of D lies in Q, so $\nu_T(D + (x, z)) = M_F(D) \cdot \sum\{2^{-|u|} : u \text{ outputs } z \text{ on } x\} \ge 2^{-K(z|x)} M_F(D)$. T has an index computable from a decider of Q, so universality costs $K(Q) + c$. □

Proof of Proposition 7.6. (a) The constant-0 program p_0 is consistent with $D + (x^*, 0)$, so $\nu((x^*, 0) \mid D) \ge 2^{-|p_0|-c}/\nu(D) \ge 2^{-|p_0|-c}$, a fixed positive constant. The violating output 1 has $K(1) = O(1)$, so the literal bound $c\, 2^{-K(Q_d)} 2^{-K(z)}$ tends to 0. (b) For $x \in \{0,1\}^d$ let $\mu_d(x) := \nu(\{(x', 1) : x' \ne x\} \cup \{(x, 0)\})$. These are labellings of $\{0,1\}^d$, so the events are disjoint and $\sum_x \mu_d(x) \le 1$; μ_d is lsc uniformly in d. The conditional coding theorem gives $\mu_d(x^*) \le 2^{-K(x^*|d)+c} \le 2^{-d+c}$. The constant-1 program is consistent with D and with $D + (x^*, 1)$, so $\nu(D) \ge 2^{-c_1}$ and $\sum_y \nu(D + (x^*, y)) \ge 2^{-c_1}$. Dividing $\mu_d(x^*) = \nu(D + (x^*, 0))$ by either gives the bound in both conventions. Finally $K(\{x^*\}) \le d + 2 \log d + c$. □

Proof of Proposition 7.10. (a) Given $(A, f|_A, x^*)$ with $x^* \notin A$, the label $f(x^*)$ is a fair coin independent of everything the learner sees. A normalised prediction (q_0, q_1) has expected error $1 - \frac{q_0+q_1}{2} = \frac{1}{2}$. (b) Condition on f, hence on z; then A and x^* are still i.i.d. uniform. Fewer than 2^{s+1} properties have $K(Q \mid z) \le s$. For one of them, with density δ in $\{0,1\}^d$, $\mathbb{P}[x^* \in Q,\ A \cap Q = \emptyset] = \delta(1-\delta)^n \le \frac{1}{n+1}$. Take a union bound. (c) p_1 is a fixed string, so $K(Q \mid p_1) \ge K(Q) - c$. A separating Q meets $\{0,1\}^d$ exactly in $\{x^*\}$, so x^* is computable from d and a decider of Q: $d \le K(x^* \mid d) \le K(Q) + c$. The error bound is Proposition 7.6(b), which holds normalised. Finally x^* is the unique 0 of f among strings of length d, so $d \le K(x^* \mid d) \le K(f) + c$. □

Proof of Proposition 7.12. For $x \in B$ let $\mu_f(x) := \sum_{y \ne f(x)} \nu(f|_{B \setminus \{x\}} + (x, y))$: the mass of hypotheses (programs or tapes) that agree with f on $B \setminus \{x\}$ and give a defined wrong label at x. A hypothesis counted for x and for $x' \ne x$ would be wrong at x (first set) and right at x (second set); so the sets are disjoint and $\sum_{x \in B} \mu_f(x) \le \nu(\emptyset) \le 1$. (A direct computation on a finite mixture with deterministic, partial, product and shared-randomness hypotheses confirms the disjointness on all subsets of a five-point domain; code in the accompanying repository.) $\mu_f(x)$ is lsc uniformly in the finite object f and x (enumerate programs, or tape cylinders), so $\mu_f(x^*) \le 2^{-K(x^*|f)+c}$. The normaliser at x^* satisfies $\sum_y \nu(f|_A + (x^*, y)) \ge \nu(f) = 2^{-L_\nu(f)}$, and $L_\nu(f) \le K_F(f) + c$. Hence

$$e_{\mathrm{norm}}(x^*) = \frac{\mu_f(x^*)}{\sum_y \nu(f|_A + (x^*, y))} \le 2^{L_\nu(f) - K(x^*|f) + c}.$$

Separating properties. $Q = \{x^*\}$ is decidable from x^*, so $\mathrm{sep}(x^*; A \mid f) \le K(x^* \mid f) + c$. Conversely, given f (hence the list B) and a decider of a separating Q, x^* is the unique element of $B \cap Q$, so $K(x^* \mid f) \le \mathrm{sep}(x^*; A \mid f) + c$. *Lower bound.* Proposition 7.1 with $b \equiv y$ and $K(Q, b) \le K(Q) + K(y) + c$. *Conditioning on the input set.* $K(x^* \mid B) \le K(x^* \mid f) + K(f \mid B) + c$, and $K(f \mid B) \le K_F(f) + c$ (run a shortest consistent program on the list B). *Conditioning on*

the input set and a program. From $\langle B, p\rangle$ one computes f, so $K(x^* \mid B, p) \le K(x^* \mid f) + c$; and $L_\nu(f) \le K_F(f) + c \le |p| + c$. As for f, $K(x^* \mid B, p)$ equals up to c the least $K(Q \mid B, p)$ of a decidable separating Q. This form is never sharper than the main bound by more than a constant. □

The grammar computation of Example 7.16(c). The script `e5_dsl_odd_lengths.py` (deterministic; code in the accompanying repository) computes an exact Bayes mixture over an enumerated grammar. A plain program is an arithmetic expression of depth at most 2 over the leaves $|x|$, $\#_1(x)$, $\#_0(x)$, 0, 1, 2 with $+, -, \times$; an `if` program is "if C then E else E'" with one of 80 simple conditions C (parity of a leaf, comparisons and equalities of leaves, first and last bit). Programs with the same values on all strings of length 1 to 8 are merged into 1345 classes. The prior gives weight p_{plain} to plain programs and $1 - p_{\text{plain}}$ to `if` programs, with fixed grammar weights inside each. Training: all 170 strings of length $1, 3, 5, 7$, labelled by their length; test: all 340 strings of length $2, 4, 6, 8$. Only one plain class ("$|x|$") fits the training data. Table 5 gives the posterior predictive. The "tilted" rows add one program "if $|x|$ even then 0 else $|x|$" with prior mass equal to (or ten times) that of all plain programs.

prior	P(correct) at length 2 / 4 / 6 / 8				min over test strings
$p_{\text{plain}} = 0.5$	0.9824	0.9835	0.9834	0.9834	0.9790
$p_{\text{plain}} = 0.9$	0.9975	0.9977	0.9977	0.9977	0.9971
$p_{\text{plain}} = 0.99$	0.99977	0.99978	0.99978	0.99978	0.99972
tilted $\times 1$	0.557	0.557	0.557	0.557	0.555
tilted $\times 10$	0.114	0.114	0.114	0.114	0.113

Table 5: Posterior predictive of the intended answer $|x|$ on even-length strings, for the grammar of Example 7.16(c).

Online, with $p_{\text{plain}} = 0.5$: after all strings of length 2 are revealed with their labels, the probability of a wrong answer at lengths 4 to 8 drops to about $2.7 \cdot 10^{-3}$; after length 4, to about $1.5 \cdot 10^{-8}$; after length 6, to 0 within the class. A single revealed string ("00", label 2) leaves about $6.7 \cdot 10^{-3}$, because programs such as "if $|x|$ even then 2 else $|x|$" survive it. A finite grammar is not a universal machine, so this only illustrates Proposition 7.4, which is the proof that the answer depends on the machine.

Summary of Hänni's examples. Table 6 collects what Example 7.16 proves.

G Proofs for Section 8

Proof of Theorem 8.2(a). M_{len} is a weak semimeasure with $M_{\text{len}}(\emptyset) = \sum_i w_i \le 1$ (each term is a product of probability vectors). On distinct inputs Proposition 2.16 gives cumulative loss $-\log(M_{\text{len}}(D_k)/M_{\text{len}}(\emptyset)) \le \log(1/w_i) + \sum_{j \le k} -\log q_i(y_j \mid x_j)$. If $|x_j| \ge n_A$, machine i is active and $T_A(x_j) \le c_A t(|x_j|) \le g(|x_j|)t(|x_j|)$, so $q_i(\cdot \mid x_j) = A(\cdot \mid x_j)$. If $|x_j| < n_A$, either $q_i(\cdot \mid x_j) = A(\cdot \mid x_j)$ (the machine is active and happens to finish in time) or $q_i(\cdot \mid x_j) = u$; in both cases $-\log q_i(y_j \mid x_j) \le \log|Y| - \log A(y_j \mid x_j)$, because $-\log A \ge 0$. Subtract A's loss. Distinctness bounds the number of inputs shorter than n_A by $2^{n_A} - 1$. □

Proof sketch of Theorem 8.2(b). Store $v_i := w_i |Y|^k \prod_{j \le k} q_i(y_j \mid x_j)$ for the machines with $n_i \le \max_{j \le k} |x_j|$; every other machine has $v_i = w_i$. Then $M_{\text{len}}(D_k) = |Y|^{-k} \sum_i v_i$. A machine

example	proved	plausible only	false strengthening
provability labels	correct label $\geq 2^{L_{\mathrm{SFI}}(D)-K(\mathcal{T})-c}$; $\leq K(\mathcal{T})+c$ online errors; wrong label $\geq 2^{-K(\mathcal{T})-2\log\log L-c}$, and $\geq 2^{-K(L'|p_{\mathcal{T}})-c}$ times the weight of proof search	a hard sentence (e.g. the Riemann hypothesis) labelled correctly with probability near 1	at fixed difficulty L of the training sentences, the wrong-label probability tends to 0 as more are added
shortest proofs	the same form	a hard sentence gets its shortest proof w.h.p.	the same, for shortest proofs
odd lengths	correct $\geq 2^{-c}$, wrong $\geq 2^{-c'}$, $\leq c$ online errors	correct w.h.p. on natural machines (at least 0.979 on the grammar)	correct w.h.p. on every universal machine (Prop. 7.4; 0.11 tilted)

Table 6: Hänni's OOD examples (Ex. 7.16). He wrote "plausibly" and did not claim the last column. The floor $2^{-K(\mathcal{T})-2\log\log L-c}$ is tiny for a natural theory; the relative bound is the one that can bite.

inactive at x_k has $q_i = u$, and $|Y|u(y) = 1$, so its v_i does not change in round k; hence

$$M_{\mathrm{len}}(y \mid D_{k-1}, x_k) = \frac{\sum_{i \text{ active at } x_k} v_i q_i(y \mid x_k) + (V - \sum_{i \text{ active at } x_k} v_i)/|Y|}{V}, \qquad V := \sum_i v_i,$$

and an update touches only the $\#\{i : n_i \leq |x_k|\} \leq \log\log|x_k| + 1$ active weights, with V maintained by adding their changes to its start value $\sum_i w_i$. *Precision.* Let the learner store approximations $\hat{v}_i$ and use $\hat{V}_{k-1}$, computed exactly from them, with each active $\hat{v}_i$ replaced in round k by $\hat{v}_i|Y|q_i(y_k \mid x_k)$ up to relative error ε_k. Its prediction $\hat{p}_k$ is a probability vector, and $\hat{V}_k \leq (1+\varepsilon_k)|Y|\hat{p}_k(y_k)\hat{V}_{k-1}$, so the cumulative loss is at most $k\log|Y| + \log\hat{V}_0 - \log\hat{V}_k + \sum_{j\leq k}\log(1+\varepsilon_j)$. Since $\hat{V}_0 \leq 1$ and $\hat{V}_k \geq \hat{v}_i \geq w_i|Y|^k\prod_{j\leq k} q_i(y_j \mid x_j)(1-\varepsilon_j)$, the regret bound of (a) holds up to $\sum_j \log\frac{1+\varepsilon_j}{1-\varepsilon_j} = O(1)$ for $\varepsilon_j = j^{-2}$. The machine model (RAM versus multitape) and the cost of running the clocks are not written out; this part is a proof sketch.

Remark G.1 (size of the constant). [proved] context Rem 3.1a With $w_i = 2^{-i}$ and $n_i = 2^{2^i}$, the constant $\log(1/w_i) + (2^{n_A}-1)\log|Y|$ of Theorem 8.2(a) is of order $2^{2^{2^i}}$, triple exponential in the index i, and i can be exponential in the description length of A. Weights $w_A = 2^{-\ell(A)}$ from a prefix-free code make $\log(1/w_A)$ the description length, but the activation term remains. Hänni's sequence predictor starts machine i at the round ℓ_1 with $\log\log\ell_1 \geq i$ (Hänni, 2026b), so his regret, $\max(\ell_1, \ell_2)$ bits, is doubly exponential in i. Length activation adds one exponential, because up to 2^{n_i} distinct inputs are shorter than n_i; it buys a per-round cost that depends only on the current input, as an x-only learner's should. The round-activated Proposition 8.4 returns to the doubly exponential form, in i and in c_A. A hybrid that activates machine i at round k_i but clocks by input length has, on distinct inputs, regret at most $\log(1/w_i) + (k_i - 1)\log|Y| + \#\{j : |x_j| < n'_A\}\log|Y|$ with $n'_A := \min\{n : g(n) \geq c_A\}$, by the proof of (a); it removes the extra exponential in i but not in c_A, and its per-round cost grows slowly with the round count.

Proof of Proposition 8.7 (regret). $\tilde{n}_k$ and t are nondecreasing, so for $k \geq k_0$ and $j \leq k$ one round of A at time j costs at most $c_A t(\tilde{n}_j) \leq g(k)t(\tilde{n}_k)$. Hence in every learner round $k \geq k_0$ the budget $2g(k)t(\tilde{n}_k)$ completes at least two further rounds of A (counting a partly done one), unless the replay reaches the present. If r_k is the number of rounds of A completed after learner round k,

then $r_k \geq \min(k, r_{k-1}+2)$ for $k \geq k_0$, so $r_k \geq \min(k, 2(k-k_0+1)) = k$ for $k \geq 2k_0-2$. A caught-up replay stays caught up, since one round costs at most the budget. So $q_i^{(k)} = A(\cdot \mid h_{k-1}, x_k)$ for $k \geq 2k_0-2$; for the earlier rounds $q_i^{(k)} \in \{A(\cdot \mid h_{k-1}, x_k), u\}$ and $-\log q_i^{(k)}(y_k) \leq \log|Y| - \log A(y_k \mid h_{k-1}, x_k)$. Sum as in Proposition 8.4. If the budget depended only on $|x_k|$, a stream of short inputs after a long one would leave a budget too small to replay even that one round, and the catch-up could stall forever. □

Proof of Proposition 8.12(b). For step $n+1$ with $n \geq 1$ and every ε, $\mathbb{P}[\text{mistake}] = \mathbb{E}[\mathrm{err}_\mu(h_n)] \leq \varepsilon + |H_f|(1-\varepsilon)^n$. With $\varepsilon_n := \ln(|H_f|(n+1)^2)/n$ this is at most $\varepsilon_n + (n+1)^{-2}$. Summing over steps $1, \ldots, n_0$ (step 1 costs at most 1), with $\ln(n+1) \leq \ln n + 1/n$ and $\sum_{n<n_0} \frac{\ln n}{n} \leq \frac{1}{2}\ln^2 n_0 + \frac{1}{e}$, gives at most $5.7 + (\ln|H_f|)(1+\ln n_0) + \ln^2 n_0$. After step n_0, each mistake is made by the current hypothesis, which lies in H_f, is consistent with the first n_0 samples, is wrong at a point of positive mass (so has error at least $p_{\min}$), and is eliminated by the mistake. The expected number of such hypotheses is at most $|H_f|(1-p_{\min})^{n_0} \leq 1$. □

Proof of Theorem 8.10. (a) $\sum_{s>S} 2^{-s} = 2^{-S}$. If no circuit of size at most S agrees with f on B, then $W_{\mathrm{circ}}(f|_B) < 2^{-S}$, strictly because at every size some constant circuit disagrees with f on $B \neq \emptyset$; so $S < L_B$. Take $S = \lceil L_B \rceil$. Also $W_{\mathrm{circ}}(f|_B) \geq w(C_{\min}) = 2^{-\mathrm{size}_B(f)}/N_{\mathrm{size}_B(f)}$, and $\log N_s \leq s(4 + 2\log(d+1+s))$.

(b) First inequality: $P_i \leq -\ln(1-P_i)$, and $\sum_i -\log(1-P_i) = L_B$ by Proposition 2.16 with $W_{\mathrm{circ}}(\emptyset) = 1$. Second: fix i and $S := \lceil L_B \rceil$; let $\mathcal{R}$ and $\mathcal{W}$ be the circuits consistent with D_{i-1} that give x_i the right and the wrong label. The exception gadget at x_i maps $\mathcal{W}_{\leq S}$ (size at most S) injectively into $\mathcal{R}$, since earlier inputs differ from x_i, and Γ is increasing; so $w(\mathcal{W}_{\leq S}) \leq 2^{\Gamma(S)} w(\mathcal{R})$. Also $w(\mathcal{W}_{>S}) \leq 2^{-S} \leq 2^{-L_B} = W_{\mathrm{circ}}(f|_B) \leq w(\mathcal{R})$, because $\mathcal{R}$ contains every circuit that agrees with f on B. So $P_i/(1-P_i) \leq 2^{\Gamma(S)} + 1$ and $-\log(1-P_i) \leq \Gamma(S) + 2$. In $L_B = \sum_i -\log(1-P_i)$, terms with $P_i \leq \frac{1}{2}$ contribute at most $2P_i$ (the chord from $(0,0)$ to $(\frac{1}{2}, 1)$), and at most $2\mathcal{E}$ indices have $P_i > \frac{1}{2}$. So $L_B \leq 2\mathcal{E}(\Gamma(S)+3)$.

Explicit form. $\gamma \leq d+1$ and $d+1+S+\gamma \leq 2d+4+L_B$, so $L_B \leq R(L_B)$ with $R(L) := 2\mathcal{E}[(d+1)(5+2\log(2d+4+L))+3]$. R is concave and increasing with $R(0) > 0$, so $\{L \geq 0 : L \leq R(L)\}$ is an interval $[0, L^*]$, and it suffices to show $K > R(K)$ for $K := 16a\Lambda$, where $a := \mathcal{E}(d+1)$, $\Lambda := X+Y$, $X := \log(2(d+2)) \geq \log 6$, $Y := \log(\mathcal{E}+1)$. Since $6\mathcal{E} \leq 3a$, $R(K) \leq 13a + 4a\log(2d+4+K)$, and $2d+4+16a\Lambda \leq 2(d+2)(1+8\mathcal{E}\Lambda) \leq 2(d+2)(1+8\mathcal{E})(1+X)(1+Y)$. So it suffices that

$$\big[12X - 4\log(1+X) - 13\big] + \big[16Y - 4\log(1+8\mathcal{E}) - 4\log(1+Y)\big] > 0.$$

The first bracket is increasing in X and equals 10.65 at $X = \log 6$. For the second, $\log(1+Y) \leq Y + 0.087$ for all $Y \geq 0$, and $\log(1+Y) \leq Y$ for $Y \geq 1$. If $\mathcal{E} \leq 1$, then $(1+8\mathcal{E})/(1+\mathcal{E}) \leq \frac{9}{2}$ and the bracket is at least $12Y - 4\log 4.5 - 4(Y+0.087) \geq -9.03$. If $\mathcal{E} \geq 1$, then $(1+8\mathcal{E})/(1+\mathcal{E}) < 8$ and the bracket exceeds $12Y - 12 - 4Y \geq -4$. The sum is positive. A bisection on a grid ($\mathcal{E} \in [10^{-3}, 10^5]$, $d \in [1, 10^4]$; code in the accompanying repository) gives the largest ratio $L^*/(\mathcal{E}(d+1)\log(2(\mathcal{E}+1)(d+2))) = 10.85$, at $\mathcal{E} \approx 0.63$, $d = 1$. □

Lemma G.2 (Möbius lemma)**.** [proved; computed (context c6(a), $d \leq 4$)]
context Lemma 3.11a *Let $1 \leq k \leq d$ and $h : \{0,1\}^d \to \{0,1\}$ with $h(1^d) = 1$ and $h(x) = 0$ whenever $|x| < k$. Then (i) h depends on at least k variables; (ii) if it depends on exactly k variables, then $h = \mathrm{AND}_S := \bigwedge_{j \in S} x_j$ for some $|S| = k$.*

Proof. Write $h = \bigoplus_S c_S \prod_{j \in S} x_j$ over GF(2). Evaluating at the indicator 1_T of T gives $h(1_T) = \bigoplus_{S \subseteq T} c_S$, so $c_S = \bigoplus_{T \subseteq S} h(1_T)$. If $|S| < k$, every $T \subseteq S$ has $|T| < k$, so $c_S = 0$. Since $h \neq 0$,

some $c_S = 1$ with $|S| \geq k$; by uniqueness of this representation, h depends on every variable of such a monomial, which gives (i). For (ii), let h depend only on the variables in R, $|R| = k$. For $a \in \{0,1\}^R$ with $a \neq 1_R$, the input equal to a on R and 0 elsewhere has weight less than k, so $h = 0$ there; and $h(1_R, \cdot) = h(1^d) = 1$. So $h = \mathrm{AND}_R$. □

Proof of Theorem 8.14. Stage k ($1 \leq k \leq d-1$) begins when the data are $h(1^d) = 1$ and $h(x) = 0$ for $|x| < k$, that is, after 1^d, 0^d and all inputs of weight below k have been shown with the labels of f.

Formulas. A formula with s leaves depends on at most s variables. By Lemma G.2(i) every function consistent at the start of stage k has formula size at least k; one of size exactly k depends on exactly k variables and is a k-conjunction by Lemma G.2(ii). Every AND_S with $|S| = k$ is consistent and has a formula of size k. So the consistent functions of least size are exactly the $\binom{d}{k}$ k-conjunctions, and Occam's hypothesis is one of them, since description length increases with size.

Circuits, $k \geq 2$. In a circuit with s gates, let $t \leq s$ be the number of gates from which the output can be reached. Each of the $t-1$ reachable non-output gates occupies at least one of the $2t$ input slots of reachable gates, so at most $t+1$ slots hold variables, and the function depends on at most $s+1$ variables. By Lemma G.2(i) a consistent circuit has $s \geq k-1$. If $s = k-1$, then $t = s$ and exactly k slots hold variables, which must be k distinct ones; so the function depends on exactly k variables and is a k-conjunction. AND_S has a circuit of $k-1$ gates. So again the least-size consistent functions are the k-conjunctions. (For $k = 1$ this fails: the one-gate circuit $x_1 \wedge x_2$ is consistent at the start of stage 1.)

Every input of weight k is a mistake. For $|S| = |S'| = k$, $\mathrm{AND}_{S'}(1_S) = 1$ iff $S' \subseteq S$ iff $S' = S$. If Occam's hypothesis in stage k is AND_S, then 1_S is unrevealed (else AND_S would be inconsistent), and the ordering shows it: Occam predicts 1, the label is $f(1_S) = 0$, and exactly AND_S among the k-conjunctions is eliminated. So all $\binom{d}{k}$ inputs of weight k are mistakes, after which stage $k+1$ begins.

Count and description length. Formulas: $\sum_{k=1}^{d-1} \binom{d}{k} = 2^d - 2$; circuits: $\sum_{k=2}^{d-1} \binom{d}{k} = 2^d - d - 2$. At $k = d$ the argument shows that f has least size d (formulas) or $d-1$ (circuits), so $\ell(f) = d + \log(\mathrm{Cat}(d-1)\, 3^{d-1}(2d+2)^d)$ for formulas, with Cat the Catalan numbers counting tree shapes, and $\ell(f) = (d-1) + \log N_{d-1}$ for circuits. Both are $\Theta(d \log d)$: $\log((2d+2)^d)$ dominates $\mathrm{Cat}(d-1) \leq 4^d$, and $\log N_{d-1} = \sum_{j<d}(4 + 2\log(d+1+j))$. So $d = \Theta(\ell / \log \ell)$ and the number of mistakes is at least $2^{d-1} = 2^{\Omega(\ell/\log \ell)}$ for $d \geq 4$. The trivial upper bound is $\min(2^d, |H_f| - 1) \leq 2^{\ell+1}$. □

Remark G.3 (a second adversary). [proved; computed (context c11)] context Rem 3.11b For circuits the count $2^d - d - 2$ also follows without the stage lemma, for every tie rule. Reveal 1^d; then, while Occam's hypothesis h has $\mathrm{size}(h) < d-1$, let V_h be the set of variables h depends on and reveal 1_{V_h}. Each step is a mistake: by the slot count, $|V_h| \leq \mathrm{size}(h) + 1 < d$; h depends only on V_h, so $h(1_{V_h}) = h(1^d) = 1$, while $f(1_{V_h}) = 0$; and the point is new, or h would be inconsistent. The phase reveals a new point at each step, so it ends, and then $\mathrm{size}(h) \geq d-1$. Fix $2 \leq k \leq d-1$ and $|T| = k$, so $\mathrm{size}(\mathrm{AND}_T) = k-1$. While AND_T is consistent, Occam's hypothesis precedes it, so $\mathrm{size}(h) \leq k-1 < d-1$: the phase continues and reveals points of weight $|V_h| \leq k$. When the phase ends AND_T has been killed by a revealed point x with $\mathrm{AND}_T(x) = 1 \neq f(x)$, revealed while AND_T was consistent; so the ones of x contain T, $x \neq 1^d$ and $|x| \leq k$, hence $x = 1_T$, revealed as a mistake. So every point of weight $2, \ldots, d-1$ is a mistake. A direct computation under seven tie rules ($d = 3, 4, 5$) confirms this. (For formulas the slot count gives $|V_h| \leq \mathrm{size}(h)$, and every point of weight $1, \ldots, d-1$ is a mistake.)

A direct computation (code in the accompanying repository) checks Lemma G.2 for all functions with $d \le 4$, the least-size statement for formulas and circuits with $d = 3, 4, 5$, and simulates Occam on the ladder: $7, 15, 31$ mistakes (formulas) and $4, 11, 26$ (circuits) for $d = 3, 4, 5$, with ties broken by truth-table index. Under seven tie rules every input of every level the theorem covers is a mistake. The ratio of log(mistakes) to $\ell/\log \ell$ stays between 0.64 and 0.82 (formulas) and between 0.35 and 0.59 (circuits) for $d = 3, \dots, 40$.

H Proofs for Section 9

Proof of Proposition 9.5. Let C be a circuit for f of size $s = \mathrm{size}(f) < S$. The circuits of size S whose first s gates equal C, whose gates $s+1, \dots, S-1$ are arbitrary, and whose last gate is $\mathrm{AND}(\mathrm{gate}_s, \mathrm{gate}_s)$ all compute f. Their fraction among all circuits of size S is $\prod_{j=s+1}^{S-1} 16(d+1+j)^2 / \prod_{j=1}^{S} 16(d+1+j)^2 = 1/(N_s \cdot 16(d+1+S)^2)$. For the mistakes: a law of a random total function has order-independent cumulative log loss $-\log P_{\mathrm{unif},S}(f)$ on every ordering (Proposition 2.16), and the expected number of Gibbs mistakes is at most $\ln 2$ times the log loss, as in Theorem 8.10(b). □

Proof of Theorem 9.7. (a) Write $\mathcal{K}(x, x') = h(x \odot x')$ with $h(z) := \Psi(\sum_j z_j/d)$ and $\odot$ the coordinatewise product, using $\langle x, x' \rangle = \sum_j x_j x'_j$. For fixed x, $x' = x \odot z$ with z uniform, and $\chi_S(x \odot z) = \chi_S(x)\chi_S(z)$, so $(T_{\mathcal{K}}\chi_S)(x) = \mathbb{E}_z[h(z)\chi_S(z)]\,\chi_S(x)$. The eigenvalue $\Lambda_S := \mathbb{E}_z[h(z)\chi_S(z)]$ depends only on $|S|$, because h is invariant under permutations of the coordinates. For the NNGP and NTK recursions with i.i.d. Gaussian weights and biases, every layer's kernel depends on the inputs only through $\langle x, x \rangle$, $\langle x', x' \rangle$ and $\langle x, x' \rangle$, and the first two equal d on the cube. (b) The characters form an orthonormal basis of L^2 of the uniform measure. For each S, $\min_{g \in \mathcal{V}} \|\chi_S - g\|^2 = 1 - \|P_{\mathcal{V}}\chi_S\|^2$ with $P_{\mathcal{V}}$ the orthogonal projection, and $\sum_{\text{all } T} \|P_{\mathcal{V}}\chi_T\|^2 = \operatorname{tr} P_{\mathcal{V}} = \dim \mathcal{V} \le n$. So $\sum_{|S|=k} \|P_{\mathcal{V}}\chi_S\|^2 \le n$. □

Proof of Corollary 9.8. χ_S is total and computable, so Theorem 7.7 gives $\sum_t e_t \le (\ln 2)(K(\chi_S) + c)$ on every sequence of distinct inputs; repeated inputs have $e_t = 0$, so the bound holds for i.i.d. inputs too (Proposition 2.16). An enumerative code of S given d and k gives the bound on $K(\chi_S)$. Let $y^* = \chi_S(x) \in \{\pm 1\}$ and let $w \le e$ be the posterior mass of the wrong label at x; the posterior mean is $\hat{g} = y^*(1 - e - w)$, so $|y^* - \hat{g}| = e + w \le 2e$ and $(y^* - \hat{g})^2 \le 4e^2 \le 4e$. For i.i.d. inputs the expected test error after $t-1$ samples equals the expected online error at step t, which gives the averaged bound. The kernel side is Theorem 9.7(b) with $\mathcal{V} = \mathrm{span}\{\mathcal{K}(x_i, \cdot)\}$. Since $\log \binom{d}{k} \le k \log d$, the SFI side needs $n = O(k \log d/\varepsilon)$, while $\binom{d}{k} \ge (d/k)^k$. □

A direct computation ($d = 10$, 20 targets per cell, ridge 10^{-3}; code in the accompanying repository) finds, for two- and three-layer ReLU NNGP, NTK and RBF kernels, squared errors above $1 - n/\binom{d}{k}$ in every cell; for $k = 5$ they stay above 0.68 for all $n \le 640$. A Bayes mixture over parities with prior $2^{-\ell(S)}$ has error 0.000 from $n = 20$ ($k = 5$) and $n = 40$ ($k = 3$), and its cumulative Gibbs mistakes (mean 3.7 to 5.5, maximum 6.04) stay below $(\ln 2)\ell(S)$ (6.70 to 9.81).

Proof of Remark 9.9. (a) With $\arccos(-\rho) = \pi - \arccos\rho$ one gets $\Psi_1(\rho) - \Psi_1(-\rho) = \rho$ and $\rho\Psi_0(\rho) - (-\rho)\Psi_0(-\rho) = \rho$. So $\Psi_1(\rho) - \frac{1}{2}\rho$ and $\Psi_1(\rho) + \rho\Psi_0(\rho) - \rho$ are even. The function $z \mapsto \Psi(\rho(z)) - a\rho(z)$, $\rho(z) = \sum_j z_j/d$, is then invariant under $z \mapsto -z$, so its Walsh coefficients at odd $|S|$ vanish, while $a\rho(z)$ has Walsh coefficients only at $|S| = 1$. Hence $\Lambda_S = 0$ for odd $|S| \ge 3$. Then $\langle \mathcal{K}(x_i, \cdot), \chi_S \rangle = (T_{\mathcal{K}}\chi_S)(x_i) = 0$, so $\mathcal{V} \perp \chi_S$ and $\|\chi_S - \hat{g}\|^2 = 1 + \|\hat{g}\|^2$. (b) $\langle z, \chi_S \rangle := \mathbb{E}_x z(x)\chi_S(x)$ is a linear functional of a centred Gaussian process, hence centred Gaussian, with

variance $\mathbb{E}_{x,x'}\mathcal{K}(x,x')\chi_S(x)\chi_S(x') = \langle \chi_S, T_{\mathcal{K}}\chi_S\rangle = \Lambda_S = 0$. So $\langle z, \chi_S\rangle = 0$ almost surely. If $\operatorname{sign} z(x) = \chi_S(x)$ for all x, then $z(x)\chi_S(x) = |z(x)|$ and $\langle z, \chi_S\rangle = \mathbb{E}|z| > 0$. So this event has prior probability 0, and conditioning on an event of positive probability keeps it at 0. (c) From (a), $\mathcal{K}(x,x') = \sum_T \Lambda_T \chi_T(x)\chi_T(x')$, so $\sum_x \mathcal{K}(x_i,x)\chi_S(x) = 2^d\Lambda_S\chi_S(x_i) = 0$, and $\sum_x g(x)\chi_S(x) = 0$ for every $g \in \mathcal{V}$; split the sum into training and unseen points. For kernel ridge regression $g = \sum_i \alpha_i \mathcal{K}(x_i,\cdot)$ with $\alpha = (K+\beta I)^{-1}y$ and K the training Gram matrix, so $\sum_i g(x_i)y_i = y^\top K(K+\beta I)^{-1}y \geq 0$: K and $(K+\beta I)^{-1}$ are commuting positive semidefinite matrices. If some $g \in \mathcal{V}$ interpolates the training labels, its unseen sum is exactly $-n$. □

The computations behind Remark 9.9 ($d = 8$ and $d = 10$; code in the accompanying repository): the eigenvalues at odd $|S| \geq 3$ are below $3 \cdot 10^{-18}$ in absolute value for both bias-free two-layer kernels and nonzero with a bias or a third layer; the variance of $\langle z, \chi_S\rangle$ is at most $4.4 \cdot 10^{-18}$ for $|S| \in \{3,5\}$ against $1.5 \cdot 10^{-4}$ and $1.4 \cdot 10^{-3}$ at $|S| = 4$; the identity of (c) holds to relative error $6.2 \cdot 10^{-12}$, the training sum is at least 89.7 in every run, and the mean unseen product falls from -0.10 ($n = 100$) to between -0.81 and -0.92 ($n = 900$). The sign accuracy of kernel ridge regression on unseen points ($d = 10$, ten targets per cell, ridge 10^{-3} times the mean diagonal) is 0.46–0.51, 0.37–0.41, 0.18–0.20 and 0.05–0.07 for $n = 100, 300, 640, 900$, while controls with a nonzero eigenvalue ($|S| = 4$, or a three-layer NNGP with $|S| = 3$) reach 0.99–1.00 at $n = 640$.

I How the claims were checked

This appendix is an internal record of how the results were established. Its vocabulary ("track", "referee", the hypotheses H1–H10) refers to the research process, whose files are in `solomonoff-function-induction/research/` of the accompanying repository.

I.1 Process

Roles. Every role below was played by a separate instance of Claude (Anthropic); no human checked the mathematics. Hänni's request (Hänni, 2026c), his notes (Hänni, 2026b) and the slides (Vaintrob and Hänni, 2026b) were the inputs. A preliminary analysis turned them into ten working hypotheses, H1–H10 (Table 1), to be proved, corrected or refuted, never assumed.

1. *Five research tracks*, each one instance: *foundations* (H1, H2, H7, invariance, the later request on versions; Sections 2 to 4), *history* (H3, H4; Section 5), *examples* (H5, H6, positive examples, Hänni's out-of-distribution examples; Sections 6 and 7.5), *context* (H8, H10, literature, extras; Sections 4.4, 7 to 9 and 11) and *steeper* (H9; the companion paper, summarised in Section 10). Each wrote notes with a status on every claim (proved with a full proof, computed with a seeded script and saved output, known with a reference, conjecture, proof sketch, refuted with its counterexample) and its own check scripts.

2. *An adversarial referee* per track, a separate instance that had not written the notes. Each checked every proof marked proved line by line, re-ran the track's scripts on scratch copies, wrote independent code, searched the literature, and reported fatal, major and minor issues. The examples and steeper tracks had two referee runs; the foundations and context referees each built on an earlier pass whose scripts they re-ran.

3. *A revision* by each track, answering every referee item, re-running all scripts and comparing outputs byte for byte, and ending with a verification log and a cross-track consistency section.

These final notes (`notes-final.md`) are the only sources of claims for the paper; they were not refereed a second time.

4. *A planning editor* wrote the notation, the outline and a ledger of every result with its status and source, and resolved 25 cross-track conflicts (Appendix I.3). *Section writers*, one per group of sections, wrote the sections from the final notes, checking proofs as they adapted them. *A framing editor* wrote the introduction, the discussion and this appendix, and ran the full build.

No referee found a fatal issue in any track. The major issues they found (4 in foundations, 6 in history, 2 and 3 in the two examples runs, 6 in context, 3 and 4 in the two steeper runs) concerned what the notes claimed or missed, not false proofs; all were accepted.

Scripts. Table 7 lists the check scripts behind the computed statements. They check finite combinatorial, analytic or information-theoretic steps and finite analogues (mixtures over finite classes); statements about K, M, M_F, M_H and other uncomputable objects are proved by hand. Environment: Python 3.11.15, numpy 2.4.6, scipy 1.17.1, mpmath 1.3.0.

track	own scripts (seeds)	referee scripts	re-run in the revision
foundations	`c1`–`c11`; 20261011 (c5, c6), 20261012 (c8–c11), others deterministic	`r1`–`r10` (first pass), `r11`–`r13` (second pass, seed 20261012)	all 20 older outputs and the 5 new ones byte-identical
history	`c1`–`c10`; seeds 0, 20261011, 7, 11, 5, 0, 20261012, 20261013, 0, 20261014	`r1`–`r7` (earlier run), `s1`–`s4`	byte-identical, except `s4` (600-second MILP limit: same incumbent, dual bound 184 vs 186)
examples	`e1`–`e7`; 20261011 (e1–e4), e5 deterministic, 20261012 (e6, e7); modules in `code/`	`r1`–`r3` (run 1), `s1`–`s3` (run 2)	byte-identical except runtime lines; referee outputs compared, not re-run
context	`c1`–`c12`; seeds 0, 1, 2, 3, 4, none, 7, none, 9, 10, 11, 12	`r1`–`r7` (first pass), `ref_*` (second pass)	c2–c12 byte-identical with `python3 -I`; c1 changed only in output labels
steeper	`c1`–`c7` (deterministic or seeds 2–7); T5's `c1`–`c3` re-run in `t5_rerun/`	`r1`–`r6` (run 1), `q1`–`q4` (run 2)	c1–c6 byte-identical; c7 run in place

Table 7: Check scripts, in `research/tracks/<track>/checks/` and `referee_code/`. Seeds are listed in script order.

I.2 Corrections

Table 8 lists every correction recorded in the tracks' final notes (responses to the referees, claims tables, verification logs) and every correction made while writing this paper. "F", "Hi", "Ex", "Co", "St" are the five tracks; `notes.md` is a track's first version; Mk and mk are a referee's major and minor items. Corrections to Hänni's own claims are summarised in Section 1 and Table 1; the steeper track's corrections are recorded in the companion paper.

Table 8: Corrections.

earlier statement (source)	correction	here
Foundations		
Problem 4.6(a) open: are K_F and W within a constant? (F `notes.md`)	Settled: no coding theorem for function programs, rate $\log d \pm O(\log\log d)$ (referee's Prop A; M1)	Thm. 3.9 and Cor. 3.10
Hashing bound as the main bound for W (F `notes.md` Prop 4.4)	Superseded by the tape-length bound; no log factor (referee M2)	Prop. 3.6 and Rem. B.3
Operator induction $= M_{\mathrm{prod}}$ (F36, Co X53)	Per secondary sources it is the per-query mixture M_{pq}, equal to M_{prod} on distinct inputs (M3; C1)	Rem. 3.2
H7 refuted under fresh noise, stated generally (F Prop 6.8)	Refuted only for random-function SFI; holds for M_{pq} without stamps (M3)	Props. 4.6 and 4.7
"$K(\operatorname{dom} D)$ is necessary" for weak comparators	It cannot be $O(1)$; about 0.37 of it is needed on some data (M4(a))	Thm. 4.1(c)
"and no better" (answer to H1(a))	"not up to a constant" (M4(b))	Rem. 2.14
Normalisation "lowers" the loss and "makes it order-dependent" (said of SFI)	Does not increase it; can make it order-dependent for random partial functions; open for M_F (M4(c))	Prop. 2.17
Hashing verdict in the summary; stated reason for the gap of Hänni's §2.2	Removed; the reason applies to K_F, not W (M4(d),(e))	Rem. 3.14
Lsc laws of random total functions computable for finite Y	For every countable Y (m1; C3)	Lem. A.2
Swap data as evidence about history-aware prediction (F §6.7)	Withdrawn; tightness added (m2; C21)	Ex. 2.22
$(2/3)^k$ constant shown for small k	Exact for every k, by an explicit law (m3)	Thm. 2.13(v)
Merging proof without integrability step	Added, $\mathbb{E}[Z_t^-] \le \log e$ (m4)	App. C.1
History		
Theorem 1 "refuted" (Hi `notes.md`, oracle reading)	Reading C, the one Hänni's text supports, is true under stated hypotheses; only Reading B is false (M1; C22)	Cor. 5.14 and Thm. 5.12
Minimax theorem quoted without hypotheses	(A1), (A2), $K(A)$ stated; private memory; computable real values (M2)	Thm. 5.13 and Lem. D.3
Q2: $L_{\mathrm{SFI}} \le L_H(D_\sigma)+I(x_\sigma : y_\sigma)+O(\log)$	Refuted, gap $\ge n - \log(n+1) - K(n) - c$; a weaker bound with $I(x_\sigma : D)$ proved (M3)	Prop. D.16
$\mathrm{SFI}^{\mathrm{hist}}$ versus SI's conditional label loss	Input-loss identity; both directions fail for unnormalised M, conditional on LHG (M4)	Props. 5.21 and 5.22
Deterministic programs for $\mathrm{SFI}^{\mathrm{hist}}$ pay $\approx K(n)$	The mixture pays $\le K^{\mathrm{tail}}(n) + c$; the single shortest program more than $\log n + \log\log n$ infinitely often (M5; C6)	Prop. D.13
Novelty of the universal chronological semimeasure	Withdrawn: Hutter's work is prior work in substance; LHG venue corrected to LNAI 6925 (M6)	Rem. 4.17 and § 11
$K(n)$ in the hierarchy separation (ii)	Superfluous, dropped (m18)	Prop. 5.19
Examples		
k-step upper bound (Ex `notes.md` Thm 2.3)	About d bits too high; sharp form $\log(N-1)_{k-1} \pm O(K(k))$ via a broom (M1, run 2)	Thm. 6.4
Verdicts on H5, H6 without a convention	Semimeasure convention named; normalised H5 bound for $t \le 2^{(1-\delta)d}$ added (M2)	Prop. 6.6
SI pays $\approx K(d)$ on the cyclic permutation	Artefact of raw blocks; $O(1)$ with the prefix code (M3)	Props. 6.9 and E.4
Labels given the domain depend "only through the number of heads"	Also through the label at $\max \operatorname{dom} D$ (m2)	Prop. 6.16

earlier statement (source)	correction	here
Remark on the Stirling error	False; removed; Robbins' window checked at 120 digits (m3)	Thm. 6.3(d)
Other items (Ex `notes.md`)	$K(d)$ dropped from the full-cycle bound (m1); prior over d re-proved (m4); order-free bound for weak semimeasures weakened to what is proved (m5); polynomial bound sharpened (m8)	Thm. 6.3, Rem. E.6, Prop. 6.5, and Ex. 6.26
Script slips (Ex log)	e6 remark "55 bits" $\to$ 52.8; e6(e) broom weights misdescribed; e7 truncated mass at $n = 1500$ (checkpoints stop at 1000); an underflow; Lemma constant $19 \to 20$	Lem. E.8
Sign of the H6 fluctuation term (second referee)	The referee's sign is wrong; no statement uses it (C10)	—
Context		
Super-polynomial Occam failure out of distribution "open" (X33)	Proved by the conjunction ladder, every tie rule (M1)	Thm. 8.14 and Rem. G.3
Posterior majority called "MAP"	Renamed; the MAP-hypothesis learner is an Occam learner (M1; C18)	Rem. 8.13
Order independence characterises random-function predictors; M_F universal among them (X48)	Order-independent predictors are the weak semimeasures; M^w is universal; M_F only among normalised ones (M2)	Prop. 2.3 and Cor. A.3
"No bound of the form 'no simple separating property $\Rightarrow$ small error' holds" (X15)	Overbroad: a converse given the labelled sample holds (M3)	Prop. 7.12
"Restricting to distinct inputs loses nothing" (X22)	False: linear regret on repeated noisy inputs (M4)	Rem. 8.3
Slides treated as unavailable; "apparently new" list	Slides used with slide numbers (M5); prior sources added and the list corrected (M6)	Rems. 9.10 and 11.1
Time-bounded regret "$= \log \lvert Y \rvert$" on short inputs	An inequality, in two cases (m1)	Thm. 8.2
Mixture over predictors covers samplers	Both readings needed (m3; X26)	Prop. 8.8
"Total OOD error between novelty and complexity" (X8)	Lopsided: the novelty bound sums to at most 2^{-c} (m4)	Rem. 7.9
Per-step bound "matches the Cesàro bound up to $\log^2 n$" (X13)	Invalid comparison (m5; C19)	Rem. C.3 and Cor. 4.14
Relative OOD bound with the complexity of a pair (X2)	Conditional complexity $K(q, r \mid p)$ (m7; C15)	Prop. 7.3
Semi-decidable properties "cost a log term"	They lose the bound (m6)	Rem. 7.2
Kernel separation constant $4e$; 0–1 reading	$4e^2$; squared error only (m10)	Cor. 9.8
Circuit constant $52\,\mathcal{E} d \log(\mathcal{E} d + 2)$ (first pass)	Fails for $\mathcal{E} \ll 1/d$; replaced by $16\,\mathcal{E}(d+1)\log(2(\mathcal{E}+1)(d+2))$ (c8)	Thm. 8.10
Machine dependence proved for W only	M_F version added (examples referee m7)	Prop. 7.4
Heavy tail of a prefix prior blocks the relevance argument (draft)	Wrong: the tail enters multiplied by $e^{-\varepsilon n}$	Cor. 4.14
Quotation attributed to Yang–Salman; "Thm 19" of Kamath et al.	Withdrawn; content cited in paraphrase (M6, m14)	Thm. 9.7
Made while writing this paper		

earlier statement (source)	correction	here
Co §1.1: M_{prod} is operator induction; Co §5.1: 2/3 by a linear program; Co Cor 5.2(c) for finite Y; Co §1.3: referee's Prop A "not yet in foundations"	Stale: per-query reading; explicit law (F Ex 2.4); countable Y (F Lemma 6.5a); Prop A is F Thm 4.7 (C1–C4)	Rem. 3.2, Ex. 2.6, Cor. A.3, and Thm. 3.9
F §9, Co, St cite history `notes.md` numbering	Final numbering used (claims 4.9, 4.10 are Props 4.9, 4.10; C5)	—
Shared-coin law $F(x) = \omega_{\lvert x\rvert}$ with certificates in $\omega_{1:n+1}$ (F Prop 4.5, Rem 4.5a; NOTATION)	Consistent only for tapes indexed from ω_0; written $F(x) = \omega_{\lvert x\rvert+1}$ with tapes $\omega_1\omega_2\cdots$; no bound changes	Prop. 3.7
F Prop 4.9 writes the deficiency as d	Renamed δ_D (d is the input length)	Prop. 3.11
F Rem 4.4a: hashing proof correct but not reproduced	Proof adapted from F `notes.md`; claim and status from the final notes and checks c2, r10	Rem. B.3
F log: c8(c) covers "10 of 15 to 15 of 15 trials"	The output also has 30/40 and 34/40; the minimum fraction is 2/3	App. B.5
F Lemma 3.1 omits disjointness of cylinders at one depth	Stated explicitly (used for prefix-freeness)	Lem. A.5
F Prop 6.8: the fresh-noise bound holds "exactly when" labels are deterministic	Fails if P has infinite expected log loss (countable Y); proviso "finite expected log loss" added	Prop. 4.6(c)
F Thm 6.7(b): integrability not written out for stamped noise	Written out; needs finite expected KL of the comparator	App. C.2
F Rem 6.11: largest $(n+1)r_n/K$ is 0.87	Saved output 0.8731; stated as 0.873	Rem. 4.9
Co Rem 2.11: leave-one-out bound "tight up to constants"	Toy mixture checked by hand: tight within a factor 2	Prop. 4.11 and Rem. C.2
Co Prop 2.12 uses a later OOD result	Special case proved directly (no forward dependency)	App. C.5
Co line 431: $\mathrm{SFI}^{\mathrm{hist}}$ "equals Hutter's ξ"	Hedged: known but not checked against Hutter's texts (C13)	Cor. 5.18 and § 11
Hi Prop 3.10(c): $I(f;\Theta) = n(n+1)/2$	Holds for Θ restricted to the first n steps; the unrestricted law has infinite entropy	Prop. D.8
Hi Thm 5.6 tightness on the GF process: exact formula for $I(f \to x)$	The marker value can coincide with r or $r+ab$; exact only up to $O(q2^{-q})$; stated as "about $2q$"	Thm. 5.27
Ledger: $\mathbb{E}L_{\mathrm{SFI}} \geq n(n-1)/2 - c$ for the randomized chain	Hi Prop 2.3(c) gives $n(n-1)/2$ exactly	Thm. 5.12
Ex Conj 2.6a heuristic: first-label cost $\approx K(n) - \min_{L\geq n} K(L)$	$K(d) - K^{\mathrm{tail}}(d) \pm c$, with $K^{\mathrm{tail}}(d) \leq K^{\mathrm{pad}}(d) = O(\log\log d)$; conclusion unchanged	Conj. E.5
Ex Thm 2.1(b) relies on Sattolo's algorithm (snippet only)	Replaced by a self-contained construction	App. E.2
Co Ex 2.15: Hänni wrote "easy-to-hard generalization is not a thing"	His words: "ood generalization is a thing, but easy-to-hard generalization is not"	Ex. 7.16
Co Prop 4.2 marked proved for Gaussian priors	Lower semicomputability only sketched there; said so	Prop. 9.2
Ex §5.1 relative bound $2^{-K(L'\mid p_T)-c}$	Needs p_T in the form "fixed searcher, then a description of $\mathcal{T}$"; condition added	Ex. 7.16
Ex e5 summary "0.98–0.9998"	Mixed per-length and per-string values; per-string minima 0.979, 0.997, 0.9997	Ex. 7.16
Co Rem 3.12: the slides' table of selection rules	Only the argmax and weighted-majority rows are proved; the rest is the slides'	Rem. 8.15

earlier statement (source)	correction	here
Learning section: the abstracts of Hutter's 2000 report and 2005 book list supervised learning	Confirmed only for the 2000 report and the 2007 chapter; citation adjusted	Rem. 4.17
Related work: $\mathrm{SFI}^{\mathrm{hist}}$ "is" the universal chronological semimeasure of AIXI	Hedged as unchecked against Hutter's texts	§ 11

I.3 Cross-track conflicts

The planning stage found no result of one track contradicting a result of another. It found 25 conflicts of other kinds (CLAIMS.md §2), resolved as follows. C1: operator induction read per query everywhere. C2: the foundations referee's Prop A, A2, A3, R1 are F Thm 4.7, Cor 4.8, Prop 4.9, Prop 4.4. C3: countable Y (F Lemma 6.5a). C4: 2/3 for $\nu_{\triangle}$ by an explicit law. C5: final history numbering. C6: deterministic programs for $\mathrm{SFI}^{\mathrm{hist}}$ (Hi Prop 4.10). C7: the updated list of semimeasure-only lower bounds (Remark 2.18). C8: SI on the cyclic permutation with the prefix code (Proposition 6.9). C9: one code $\bar{s} = 1^{|s|}0s$ for SI on interleaved sequences. C10: the sign in H6. C11: the hashing citation in the steeper notes is F Rem 4.4a. C12: the cost of private randomness at $\lambda = 1$ sharpened by F Rem 4.5a. C13: one definition of the universal chronological semimeasure, M_H. C14–C15: the final forms of the OOD propositions. C16: one home for each of ten duplicated results. C17: W_{det} and W never identified. C18–C19: renamed predictor; invalid comparison withdrawn. C20: separate symbols for the time-bounded and per-query mixtures. C21: the swap data are not a history separation. C22: Theorem 1 in Reading C. C23: the steeper notes' two stream lemmas. C24: about thirty notation clashes, resolved in NOTATION.md. C25: a promised Wyeth–Hutter reference not cited (not relevant).

I.4 What is not fully established

Conjectures. Conjectures 4.12, 5.15, 7.17 and E.5; the last is not a precise statement and fails for some universal machines.

Proof sketches and conditional results. Theorem 5.13(b) for reveal laws that see f; the running times in Theorem 8.2(b), Propositions 8.4 and 8.7; the reformulation of Section 2.3 (Strassen's theorem quoted, not proved); lower semicomputability of the network prior for Gaussian weights (Proposition 9.2); positive-example induction (Remark C.5). Proposition 5.22 and one part of Remark E.7 are conditional on a transfer of the Lattimore–Hutter–Gavane phenomenon, which was seen only in an abstract.

Open problems. Collected in Section 12.

Results checked by their authors only. The final notes were not refereed again. Results added in revision follow a referee's own argument (for example Theorems 3.9 and 6.4 and Propositions 3.6, 5.21, 6.6 and 6.9) or were checked only by the track that wrote them: in history, Propositions 5.6 and D.11 and Lemmas D.2 and D.3, Proposition D.13(d), the private-memory form of Theorem 5.13, and the rewritten proofs of Theorems 5.17 and 5.24; in context, Propositions 7.12, 8.4 and 8.7 and Corollaries 4.14 and 8.5, the circuit case of Theorem 8.14, Remarks 8.16 and G.3 and Corollary A.3, Remark 9.9(b),(c), Proposition 7.4(b), Proposition 7.10(c), the constant of

Theorem 8.10 and the precision argument of Theorem 8.2(b). The section writers re-checked the proofs they adapted.

The slides. Only the warm-up arithmetic, the character argument of their Appendix J (Theorem 9.7(a)) and the ladder of their Appendix K (Theorem 8.14) were checked; their main theorem, equivalence theorem, mistake bounds and mean-field claim were not.

I.5 Literature

No cited source could be opened: arxiv.org, lesswrong.com and publisher sites did not resolve from the research environment, and a web-search tool returned excerpts only. Bibliographic data were checked against these excerpts; each entry in `bib/*.bib` says what was confirmed. Content was seen at the level of abstracts or secondary descriptions for every work cited for content, in particular: operator induction (Solomonoff, 2008), through Özkural (2015, 2017); Hutter's chronological semimeasures and his section on supervised learning from examples (Hutter, 2000, 2005, 2007); Lattimore et al. (2011); Wood et al. (2013); Lattimore and Hutter (2013); Poland and Hutter (2005); Leike and Hutter (2015a,b); Ryabko (2006); the simplicity-bias and kernel literature of Section 9; Impagliazzo and Levin (1990); and the classical results used as tools (coding theorem, Levin–Schnorr, Zvonkin–Levin, Sion, Kleitman–Spencer, Robbins, Strassen), which are standard and cited without checked section numbers. Fields that could not be confirmed were omitted: the volume and pages of Day (2011), the pages of Vovk (1990) and Basri et al. (2019), the series volumes of Orseau et al. (2013), Schmidhuber (2002) and Wood et al. (2013). Works whose attribution could not be confirmed are not cited (the vanishing odd eigenvalues attributed to Bietti and Mairal; the network–Gaussian-process correspondence papers of Lee et al. and Jacot et al. of 2018). The LessWrong thread (Hänni and Wyeth, undated) is cited from the text Hänni pasted. Vaintrob and Hänni (2026a) was seen as a title and abstract only.

I.6 Reproduction

From the folder `solomonoff-function-induction/research/tracks/` of the repository:

```
for f in */checks/*.py */referee_code/*.py; do
  d=$(dirname "$f"); b=$(basename "$f" .py)
  if grep -q "open(.*['\"]w['\"]" "$f"; then
    (cd "$d" && python3 -I "$b.py")              # writes its own output
  else
    (cd "$d" && python3 -I "$b.py" > "$b.out")   # prints its output
  fi
done
git status --short     # changed outputs show up as modifications
```

The scripts that print are the history scripts `c1`–`c6` and its referee code, and the steeper scripts. Expected differences are runtime lines (e.g. examples `e2`) and the time-limited history script `s4`. The longest runs take a few minutes (history `r3` about 265 seconds, `s4` 600 seconds, context `c7` about 140 seconds). The examples scripts import `code/sfi_examples.py` and `code/sfi_dsl.py` by relative path. The paper is built by `./build.sh` in `paper/` (bibliography merge, then `pdflatex`, `bibtex`, `pdflatex` twice).

I.7 Unresolved points

- *Length.* The main text (pages 3–52 in the current build) runs well beyond its target of about 38 pages, and the appendices (pages 57–103, including this one) beyond their target of about 35: the writers kept every ledger result with its statement and proof idea, moving secondary items to the appendices (33 items in all). Further cuts would drop proofs or results.
- *Conversion to arXiv form.* The status tags, source notes, this appendix, process vocabulary in a few sentences of the body (for example references to check scripts by name, "Hänni's request") and the hedges written as "unverified" are for the conversion pass, which must also decide how to cite the request (Hänni, 2026c) once Hänni is the sole author.
- *Status vocabulary.* Some items carry statuses outside the fixed set, as the ledger gives them: "remark", "remark (cited items)", "conditional (on LHG transfer)", "proved, given ... ".
- *Open questions as paragraphs.* Their labels sit on paragraphs, so a cross-reference prints the number of the enclosing subsection.
- *Notation.* NOTATION.md still writes the shared-coin law as $F(x) = \omega_{|x|}$; this paper uses $F(x) = \omega_{|x|+1}$ (Table 8). The companion paper should use the same indexing.
- *Shared bibliography.* The two papers' `bib/core.bib` files are meant to be identical; this paper's copy now names Hänni as author of both companion papers, and the companion's copy needs the same change.